

# Application of variational algorithms in quantum computation

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I, Hongxiang Chen, confirm that the work presented in this thesis is my own. Where information has been derived from other sources, I confirm that this has been indicated in the work.

# Abstract

Variational quantum algorithms are a class of techniques that are particularly suitable for near-term applications and borrow many ideas from classical variational methods in quantum chemistry and machine learning. We developed variational quantum algorithms to improve error-correcting codes as well as modelling quantum mechanical systems, hence looking at both the long-term and short-term perspectives in quantum computing. For error correction, we designed an algorithm that generates logical gates when encoding and decoding circuits are given as input. We demonstrated its capability in numerical simulations and generated non-trivial logical gates for small-size error-correcting codes. For modelling quantum mechanical systems, we considered an approach based on the Green function, which describes the response of the system to external perturbations. We adapted traditional formulae for the Green function so that they can be computed on near-term quantum hardware, and we conducted both numerical and experimental studies on IBM prototype quantum processors for proof-of-principle demonstrations. Our results highlighted the limitations of the approach, and we suggested potential alternatives.

# Impact Statement

Variational quantum algorithms are a type of quantum computing algorithms that have gained significant attention due to their potential for application in the near-term quantum hardware. These algorithms allow for experimental demonstrations of quantum algorithms and have seen iterative improvements over the years.

By developing an algorithm to generate logical gates for error-correcting codes, we have provided a tool that can help researchers design smaller codes, which can then be combined through code concatenations to create large-scale error-corrected quantum systems.

We also developed variational algorithms to facilitate studying quantum mechanical systems through Green function based methods. Green function based methods are commonly used in studying systems with strong electron correlation effects in material science and quantum chemistry. We provided improvements in two aspects: first, we addressed the computational bottleneck in related research based on classical computers, which is the computation of the Green function itself. Second, through our experimental demonstrations, we elucidated the path in which our algorithms can be executed on near-term quantum systems once they are available.

From a broader perspective, our methods, if further improved, have the potential to facilitate the development of quantum algorithms in general, and enable the discovery of new materials or more effective drugs in particular.

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*Should auld acquaintance be forgot,  
And never brought to mind?  
Should auld acquaintance be forgot,  
And auld lang syne?*

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# Reading notes

The thesis was written over an extended period, during which the writing style evolved. As a result, sometimes the information may be too dense, while other sections can be read quickly. To help, we often provide an overview at the beginning to guide the reader.

Throughout the thesis, the collective pronoun “we” is used, as is common in any collaborative scientific research. When emphasising the individual opinion or contribution of the author, we use the phrase “the author”.

The thesis includes many concepts from condensed matter physics and quantum chemistry. It is hard to remain self-contained in these fields. Thus, we attempt to be self-sufficient for reproducing the experiments described. Some sections may seem overly detailed or might be better explained by experts in those specific fields. These sections should be viewed as personal notes for the author, and original references are always provided. In other cases where the discussion is too brief, it is believed that such knowledge are tangential for the idea described in the thesis, and are thus better understood using the references provided.

The chapters in this thesis are largely independent, with three of them each covering a topic from their respective research paper. As a result, they can be read individually with minimal need for prior knowledge from the other chapters.

## Notations

We follow the conventions commonly used in the quantum computation community, and readers are referred to the popular reference book Nielsen and Chuang [1] for details. A glossary is attached at the end of the thesis to provide a quick reference for symbols, abbreviations, and certain key concepts.

Still, we explain a few symbols and definitions here that may be better detailed in the beginning, or may be confusing at times. These include the

symbol  $\sigma$ , the Pauli matrices and Pauli operators, and certain quantum gates.

$\sigma$  is a widely used notation in this thesis that can have different meanings in different contexts. To respect other literatures, we use the same symbol and hope the context will make the meaning clear.

For Pauli matrices, in quantum computation, we often write them as,

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

However, in physics, it is more popular to use  $\sigma_1 = X$ ,  $\sigma_2 = Y$ , and  $\sigma_3 = Z$ . This is useful to express the various rules relating products of Pauli matrices.

Unfortunately,  $\sigma$  is often used in condensed matter for spin orientations. Thus, we are careful in  $\sum_{\sigma}$  to be clear if  $\sigma \in \{X, Y, Z\}$  or  $\sigma \in \{\uparrow, \downarrow\}$ . Further,  $\sigma$  is sometimes used for a density matrix if the symbol  $\rho$  has been used. However, the context will ensure no confusion arises.

Sometimes the  $2 \times 2$  identity matrix is considered a special Pauli matrix, denoted by  $I$ . Adding  $I$  into the family of Pauli matrices is helpful in the multi-qubit case, where tensor products of Pauli matrices, sometimes called *Pauli operators*, are expressed as,

$$P = \sigma_1 \otimes \sigma_2 \otimes \cdots \otimes \sigma_n, \text{ where } \sigma_i \in \{X, Y, Z, I\}.$$

Notice the further abuse of the symbol  $\sigma_i$ , which means the  $i$ -th constituent in the tensor product above.

The wide usage of multi-qubit Pauli operator creates many other notations for them. For example, in computer code for quantum computation, we sometimes use the *Pauli string*: we denote the Pauli operator  $P$  by its components, using a string of characters from “XYZI”, each representing the corresponding one-qubit Pauli matrices forming the tensor product. As an example, the three-qubit Pauli operator  $P = X \otimes Y \otimes Z$  is denoted by the Pauli string “XYZ”.

In quantum error correction, another notation is popular. Here, the  $n$ -qubit Pauli operator is denoted by its non-identity matrix components, and each such component contains a subscript indicating the qubit on which this component acts. For example, when  $n = 3$ , under this notation, the Pauli string “IIX” is denoted by  $X_3$ , and “XXI” by  $X_1 X_2$ .

**Quantum gates.** We follow the textbook Nielsen and Chuang [1] for notations on quantum gates and circuits diagrams. Notably, the explicit matrices for most quantum gates are available in [1, Nomenclature and notation]. These include the Hadamard gate, the CNOT gate (the controlled-NOT gate), the CZ gate (the controlled-Z gate), and the Toffoli gate. Unfortunately, the “Nomenclature and notation” has not defined the rotation gates ( $R_x$ ,  $R_y$ , and  $R_z$ ). They are defined here in terms of Pauli matrices by  $R_\sigma(\theta) = e^{-i\sigma\theta/2}$ , where  $\sigma = X/Y/Z$ .

## Chapter 1

# Introduction and Overview

Quantum computation, first suggested by Richard Feynman in 1982 [2], has garnered significant attention and interest for its potential to solve real-world problems much faster than classical computers. For example, quantum algorithms can provide exponential speed-ups over their classical counterparts in areas such as factoring numbers [3, 4], simulating quantum systems [5], and solving linear systems of equations [6]. Considerable advancements have also been made in the manufacturing of quantum computers, resulting in widespread availability of online access to these machines through various cloud service providers [7–9]. At the same time, various groups argued the supremacy of quantum computers over classical supercomputers by demonstrating that quantum computers currently available have already surpassed the computation power of the latter [10, 11]. While these demonstrations are not conclusive [12, 13], they have nonetheless showcased the unique capabilities of quantum computers and stimulated the interest in developing quantum algorithms for quantum devices available today or in the near term.

A key limitation of these near-term devices, is that they will be noisy and of only intermediate scale with only a few hundred qubits at best. Hence, these devices are commonly referred to as Noisy Intermediate-Scale Quantum (NISQ) devices [14]. The inherent noise and limited scale pose significant challenges for the applications of quantum error correction (QEC) technology, which is a crucial requirement for implementing those quantum algorithms with exponential speed-ups. The lack of realistic and scalable QEC for the NISQ devices therefore prevents the application of such quantum algorithms to real-world problems.

We expect the era of NISQ devices to last for years or even decades. With the stimulated interest in quantum computation and the limitations pre-

sented by NISQ devices, researchers have been pushed to propose a variety of methods to make use of NISQ devices. Among many proposals, variational quantum algorithm, or variational algorithm for short, emerges as a prominent approach. Variational algorithms integrate classical optimisation methods with quantum computers to find approximate solutions to complex problems. In a variational algorithm, the quantum circuits are parameterised and adjusted to minimise a problem-specific loss function, which is calculated using a quantum computer. Typical examples of variational algorithms include the Variational Quantum Eigensolver (VQE), and the Quantum Approximate Optimisation Algorithm (QAOA) [15–17]. These algorithms showed promising results in many proof-of-principle demonstrations with an emphasis on addressing classical computation bottlenecks in quantum chemistry and optimisation [15, 16].

Variational algorithms offer significant promises due to the following factors. First, variational algorithms involve the optimisation of parameterised circuits which in many cases help keep the quantum circuit depth shallow, making them more suitable for the noisy hardware. This has been suggested theoretically [18] and confirmed in early experimental demonstrations [19]. A more popular approach employed by later experiments involves using an adaptive ansatz for the parameterised circuit to find a low-depth circuit solution for the problem at hand [20].

Another case of adaptations to near-term devices by variational algorithms is the creation of error mitigation techniques [21, 22]. Error mitigation technique improves the performance of quantum algorithms despite the high level of noise on currently available noisy hardware. The progress in error mitigation techniques has enabled demonstrations of quantum algorithms on problems of increasingly larger size and scale [22–24].

Furthermore, for specific quantum chemistry applications, techniques from traditional variational methods in quantum chemistry have been adapted naturally to variational algorithms, leading to a variety of interesting explorations. For instance, the coupled-cluster theory has been adapted to produce a series of quantum circuit ansatzes aiming at studying the energy spectrum of quantum mechanical systems on NISQ devices [25].

Lastly, there is a healthy synergy between the techniques developed by the machine learning community and those developed for variational algorithms. For one, variational algorithms often make use of established optimisation techniques [26, 27]. For another, gradient-based optimisation al-

gorithms common in machine learning were adapted for quantum machine learning and spurred the investigations into challenges in optimising the parameterised ansatz [28, 29].

The aim of this thesis is therefore to develop variational algorithms for quantum computation. Contributing to the ongoing research on variational algorithms, our focus is specifically directed towards exploring these two dimensions:

**Research Question 1.** *Development of variational algorithms for improving quantum error correction and for studying quantum mechanical systems.*

**Research Question 2.** *Proof-of-principle demonstration of variational algorithms we developed on quantum computers to assess their practicality.*

## Contribution and synopsis of the thesis

In Chapter 2, we explain the details of the variational quantum eigensolver algorithm, which serves as the fundamental template for constructing other variational algorithms.

Then, in Chapter 3 we explore a variational algorithm which generates logical gates for quantum error-correcting codes. This algorithm demonstrates the basic techniques that will be frequently encountered later in the thesis. It is different from other applications of variational algorithms in quantum error correction as it focuses on the problem of generating logical gates given an error-correcting code. We demonstrate the viability of this algorithm in Chapter 3 with numerical simulations resulting in promising results for small-sized quantum error-correcting codes.

In the following chapters, we turn to variational algorithms for the investigation of quantum mechanical systems. We focus on approaches based on the Green function to analyse strongly correlated electron systems, since its computation is a bottleneck for these approaches. Variational algorithms can potentially ease this bottleneck.

In Chapter 4, we show how to adapt a few traditional formulae for the Green function into variational algorithms suitable for near-term devices. We also demonstrate the viability of a correction-vector-based formula through numerical simulations for studying the excitation dynamics of  $H_2$  and  $LiH$ .

Then, we move to proof-of-principle experiments on quantum computers. In Chapter 5, we evaluate the capability of current quantum hardware in studying dynamics using dynamical mean-field theory (DMFT), which is a

quantum embedding theory based on the Green function. We focus the discussion on a toy model for Mott transition and present an experimental study of this model at a critical point on quantum computers available from IBM. We discuss the severe limitations presented by the hardware and the solutions we attempted to overcome these limitations.

The thesis will conclude in Chapter 6 by reviewing briefly the latest experimental progress in variational algorithms and suggesting research directions for enhancing these techniques. We start with a discussion of problems faced by variational algorithms, which is followed by a review of the progress and experimental demonstrations in tackling these problems. Finally, we touch on the so-called era of early fault-tolerant quantum computation and the role of variational algorithms at that stage.

## Statement of authorship

This thesis is based on the following research articles published from 2019 to 2022:

1. I. Rungger, N. Fitzpatrick, H. Chen, C. H. Alderete, H. Apel, A. Cowtan, A. Patterson, D. Munoz Ramo, Y. Zhu, N. H. Nguyen, E. Grant, S. Chretien, L. Wossnig, N. M. Linke, and R. Duncan. Dynamical mean field theory algorithm and experiment on quantum computers, 2019, arXiv:1910.04735. (Chapter 5).
2. Hongxiang Chen, Max Nusspickel, Jules Tilly, and George H. Booth. Variational quantum eigensolver for dynamic correlation functions. *Phys. Rev. A*, 104:032405, Sep 2021. URL <https://link.aps.org/doi/10.1103/PhysRevA.104.032405>. (Chapter 4).
3. Hongxiang Chen, Michael Vasmer, Nikolas P. Breuckmann, and Edward Grant. Automated discovery of logical gates for quantum error correction (with supplementary (153 pages)). *Quantum Information and Computation*, 22(11&12):947–964, aug 2022. URL <https://doi.org/10.26421/qic22.11-12-3>. (Chapter 3).

I have additionally contributed as a co-author to the following articles that are not included in this thesis:

1. Jules Tilly, Hongxiang Chen, Shuxiang Cao, Dario Picozzi, Kanav Setia, Ying Li, Edward Grant, Leonard Wossnig, Ivan Rungger, George H.

- Booth, and Jonathan Tennyson. The variational quantum eigensolver: A review of methods and best practices. *Physics Reports*, 986:1–128, 2022. ISSN 0370-1573. URL <https://www.sciencedirect.com/science/article/pii/S0370157322003118>. The Variational Quantum Eigensolver: a review of methods and best practices
2. Michael Broughton, Guillaume Verdon, Trevor McCourt, Antonio J. Martinez, Jae Hyeon Yoo, Sergei V. Isakov, Philip Massey, Ramin Halavati, Murphy Yuezhen Niu, Alexander Zlokapa, Evan Peters, Owen Lockwood, Andrea Skolik, Sofiene Jerbi, Vedran Dunjko, Martin Leib, Michael Streif, David Von Dollen, Hongxiang Chen, Shuxiang Cao, Roeland Wiersema, Hsin-Yuan Huang, Jarrod R. McClean, Ryan Babbush, Sergio Boixo, Dave Bacon, Alan K. Ho, Hartmut Neven, and Masoud Mohseni. Tensorflow quantum: A software framework for quantum machine learning, 2021, 2003.02989. URL <https://arxiv.org/abs/2003.02989>
  3. Andrew Patterson, Hongxiang Chen, Leonard Wossnig, Simone Severini, Dan Browne, and Ivan Rungger. Quantum state discrimination using noisy quantum neural networks. *Phys. Rev. Res.*, 3:013063, Jan 2021. URL <https://link.aps.org/doi/10.1103/PhysRevResearch.3.013063>
  4. Jules Tilly, Glenn Jones, Hongxiang Chen, Leonard Wossnig, and Edward Grant. Computation of molecular excited states on ibm quantum computers using a discriminative variational quantum eigensolver. *Phys. Rev. A*, 102:062425, Dec 2020. URL <https://link.aps.org/doi/10.1103/PhysRevA.102.062425>

## Chapter 2

# Variational quantum algorithms

|       |                                       |    |
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In this chapter, we introduce several subroutines that are widely used in variational quantum algorithms. These algorithms have gained significant attention in recent years due to their potential in solving complex computational problems.

One of the key contributions that paved the way for the development of these algorithms is the variational quantum eigensolver (VQE) [15]. VQE is a method to prepare the lowest energy eigenstate, called the ground state, for a second-quantized Hamiltonian. Since then, numerous variational quantum algorithms have been introduced that either use VQE as a subroutine or are inspired by it [27].

In this chapter, we start by introducing VQE. We then describe its extension to excited states (Section 2.2), and the technique to target specific symmetry-preserving sectors in the excitation spectrum (Section 2.3). We end this section with a variational linear algebra solver based on VQE (Section 2.4). These algorithms are used in other chapters, where we apply them to specific problems.

## 2.1 Variational quantum eigensolver (VQE)

Here we give an overview of the VQE algorithm, outlining the problem it tries to solve and the general steps. More detailed information will be provided in subsequent sections. The problem VQE tries to solve is the following: given a Hamiltonian  $H$  in its second-quantized form,

$$H = \sum_{ij} h_{ij} c_i^\dagger c_j + \sum_{ijkl} h_{ijkl} c_i^\dagger c_j^\dagger c_k c_l, \quad (2.1)$$

VQE finds a quantum circuit  $U_{\text{GS}}$  that approximately prepares the ground state  $|\text{GS}\rangle$  of  $H$ , i.e.  $U_{\text{GS}}|0\rangle = |\text{GS}\rangle$ . Here,  $i, j, k, l$  are generic indices for the relevant quantum degrees of freedom.  $c^{(\dagger)}$  are the annihilation (creation) operators on the corresponding degree of freedom.

This ground state problem finds wide usage in many applications when the ground state dominates the statistical ensemble. This dominance is evident at low temperatures ( $T \rightarrow 0$ ), but it can also persist at room temperature. In systems following a Boltzmann distribution, the ground state dominates when the energy gap between ground and excited states is much larger than  $k_B T$ , where  $k_B$  is the Boltzmann constant. This is true for most chemical systems<sup>1</sup>. Also, as we will see later in Section 2.2, VQE algorithm can be naturally extended to study the excited states of the Hamiltonian. Finally, after obtaining the circuit  $U_{\text{GS}}$ , we can measure the expectation values of various observables of interest on the prepared eigenstate.

VQE approaches this ground state problem using the Rayleigh-Ritz variational principle [37, 38], which states that for any normalised state  $|\psi\rangle$ <sup>2</sup>:

$$\langle \psi | H | \psi \rangle \geq E_{\text{GS}}. \quad (2.2)$$

The equality holds only if  $|\psi\rangle$  is the ground state (up to degeneracies). This principle forms the basis for almost all of the traditional quantum chemistry.

Next, VQE proposes to use an ansatz circuit  $U(\theta)$  parameterised by real numbers  $\theta$ , to approximate the exact  $U_{\text{GS}}$  by minimising the expectation value

<sup>1</sup>Strictly speaking, in chemical systems, the Born-Oppenheimer approximation is often employed to separate the total molecular energy into translational, rotational, vibrational, and electronic contributions. Since chemical properties are primarily influenced by the electronic part, one focus on solving the electronic Hamiltonian. The energy gap discussed in the main text refers specifically to that of the electronic Hamiltonian [36, Chapter 14].

<sup>2</sup>While the original references are difficult to access, its proof is ubiquitous in quantum mechanical textbooks, e.g. Griffiths [39, §7].

$\langle 0| U^\dagger(\theta) H U(\theta) |0\rangle$  measured using the quantum computer. After a sufficient amount of experimentation with different ansatz circuits and optimisations, one can be relatively confident that the final optimised ansatz circuit actually prepares the ground state for the given Hamiltonian. In detail, VQE can be divided into several different stages:

1. **Mapping.** The second-quantized Hamiltonian  $H$  is mapped to a linear combination of Pauli operators,

$$H = \sum_i w_i P_i. \quad (2.3)$$

Here  $P_i$  are different Pauli operators and  $w_i$  are their corresponding coefficients. Such a mapping could be achieved, for example, using the Jordan-Wigner transformation (Section 2.1.1). Note that, for simplicity, we use the same symbol  $H$  for both the second-quantized Hamiltonian and its mapped form.

2. **Ansatz.** Next, we choose a parameterised circuit  $U(\theta)$ , called an ansatz. It serves as a guess for parametrising a subspace containing the ground state. We will mention a few ansatzes that we used in Section 2.1.2.
3. **Measurement and error mitigation.** The expectation value of the mapped Hamiltonian with respect to a state prepared by the ansatz circuit  $U(\theta) |0\rangle$  can be easily measured on the quantum computer (Section 2.1.3). Additionally, many techniques exist to mitigate the systematic errors of measured values on near-term quantum hardware. These techniques will be discussed in Section 2.1.4.
4. **Optimisation.** Finally, we optimise the measured energy,

$$E(\theta) = \langle 0| U^\dagger(\theta) H U(\theta) |0\rangle,$$

using classical optimisation algorithms. We will describe the optimisation algorithms we used in Section 2.1.5.

Notably, different from quantum algorithms developed for a fault-tolerant algorithm, VQE could potentially be less sensitive to the level of noise present on near-term quantum computers. This resilience results from a combination of improvements from the hardware-specific design of the

ansatz, the relatively short depth of the ansatz circuit found by experimenting, and various post-processing steps (error mitigation, subspace methods) to reduce the noise. This ability to quickly incorporate such improvements makes VQE an attractive approach for near-term intermediate-scale quantum technologies (NISQ), a term coined in 2018 [14].

Note that optimisation is a trial-and-error process that involves many heuristic techniques. In Section 2.1.5 we will discuss the typical behaviour of the particular optimisation algorithm we used, and mention a general method to check if the optimised result is an eigenstate of the Hamiltonian. Once we reach a certain level of confidence about the result, we can declare that the best ansatz we have tried so far, with its optimised parameters  $\theta^*$ , approximates the ground state:

$$U_{\text{GS}} \approx U_{\text{best}}(\theta^*) \quad (2.4)$$

### 2.1.1 Mapping

First, we detail the mapping algorithm, Jordan-Wigner transformation, we used for VQE. While Jordan-Wigner has some undesirable effects, which are discussed at the end of this section, this mapping proved to be good enough for experiments described in other chapters.

Jordan-Wigner transformation is originally proposed in 1928 [40]. It was used to study spin- $\frac{1}{2}$  systems since it maps the ladder operators for spin  $\{S_n^\pm, S_m^z\}$  found in certain Heisenberg models to fermionic operators  $\{c_n^\dagger, c_m\}$ , which make the original problem easier to analyse analytically.

In quantum computation, however, the Jordan-Wigner transformation is used in the reversed direction to map fermionic operators  $\{c_n^\dagger, c_m\}$  to ladder operators, which has a representation in terms of Pauli operators which can be easily measured on quantum computers. Note that the energy spectrum is preserved in this transformation. Additionally, we can evolve any quantum state by  $e^{iHt}$  using quantum simulation algorithms after this mapping [17].

**Definition.** Jordan-Wigner transformation can be defined directly as follows. Assume we have fermionic operators  $\{c_n^\dagger, c_m\}$  on  $N$  fermionic modes ( $n, m \in \{1, 2, \dots, N\}$ ), then the transformation is defined by:

$$\begin{aligned} c_n &\mapsto Z_1 Z_2 \cdots Z_{n-1} \left( \frac{X_n + iY_n}{2} \right), \\ c_n^\dagger &\mapsto Z_1 Z_2 \cdots Z_{n-1} \left( \frac{X_n - iY_n}{2} \right). \end{aligned} \quad (2.5)$$

Note that when  $n = 1$ , no  $Z$  operator appears in the mapped operator above.

We briefly verify that the mapped operators follow the expected fermionic anti-commutation relations. For convenience, we use the following shorthand notations:

$$\begin{aligned}\sigma^- &\equiv \frac{X + iY}{2} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = |0\rangle\langle 1|, \\ \sigma^+ &\equiv \frac{X - iY}{2} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} = |1\rangle\langle 0|,\end{aligned}\tag{2.6}$$

Then, we consider two cases of the anti-commutation relations: (1) When  $n \neq m$ , terms like  $\{c_n, c_m\}$ ,  $\{c_n^\dagger, c_m^\dagger\}$ , and  $\{c_n, c_m^\dagger\}$ , after mapping, contain  $\sigma^\pm Z + Z\sigma^\pm$ , which is 0 after a straightforward calculation. (2) When  $n = m$ ,  $\{c_n, c_n\}$  and  $\{c_n^\dagger, c_n^\dagger\}$  are mapped to zero due to containing  $(\sigma^\pm)^2 = 0$ . Also,  $\{c_n, c_n^\dagger\}$  after mapping is  $\sigma^- \sigma^+ + \sigma^+ \sigma^-$  which obviously equals the identity matrix. Thus, after mapping,

$$\begin{aligned}\{c_n, c_m\} &= \{c_n^\dagger, c_m^\dagger\} = 0, \text{ and} \\ \{c_n, c_m^\dagger\} &= \delta_{nm}.\end{aligned}\tag{2.7}$$

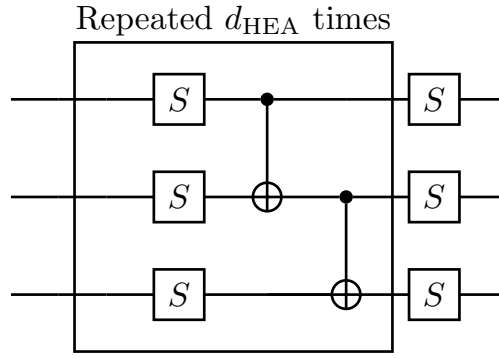
Note that the number operator is mapped as

$$c_n^\dagger c_n \mapsto \left( \frac{1 - Z_n}{2} \right) = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = |1\rangle\langle 1|.\tag{2.8}$$

**Beyond Jordan-Wigner.** An undesirable feature of this Jordan-Wigner transformation is that local terms in the fermionic Hamiltonian will be mapped to non-local Pauli operators, primarily due to the long  $Z_1 \cdots Z_{n-1}$  chain in Eq. (2.5). Non-local Pauli operators present two challenges on near-term hardware: measuring them would require multiple qubits, leading to more readout errors, and exponentiating such operators for time evolution would involve more qubits, leading to more noise. Therefore, many alternative mappings have been used to increase locality. For a review of those mappings, see Tilly et al. [27].

## 2.1.2 Ansatz

Here we introduce the ansatz we used in later chapters. Note that while the unitary coupled cluster (UCC) ansatz [25] is popular due to its connection with classical coupled cluster theory in quantum chemistry, practitioners have



**Figure 2.1:** An illustration of the hardware efficient ansatz for three qubits when CNOT is used as the two-qubit gate to create entanglement. Each  $S$  represents a series of parameterised single qubit gates, and different  $S$  on different positions have their own parameters independent of each other. The same pattern is repeated in several  $d_{\text{HEA}}$  layers, with each layer also having its own independent and adjustable parameters.

found it difficult to implement on quantum hardware due to the relatively large circuit depths and complicated connectivity requirements. In this thesis, we almost exclusively used the Hardware-efficient ansatz (HEA) introduced below. Many other ansatzes have been developed and enjoyed wider usage in more recent experiments. These ansatzes have been reviewed in Tilly et al. [27].

**Hardware-effect Ansatz (HEA).** Kandala et al. [20] proposes constructing the ansatz in a layer-wise manner, where each layer consists of single-qubit gates followed by two-qubit gates. The two-qubit gates are native to the hardware, allowing for efficient implementation. Motivated by the success of their experimental demonstration, we adopted their idea and used a layer-wise ansatz throughout this thesis, which we refer to as the Hardware-Efficient Ansatz (HEA). The key difference we introduced is a relaxation of the restriction on the two-qubit gates used, allowing them to be selected based either on efficiency for simulation or on efficiency for a hardware if hardware experiments are required. We detail this HEA in the following and illustrate in Figure 2.1 a parameterised circuit constructed based on HEA with  $d_{\text{HEA}}$  layers on three qubits.

The elementary unit of HEA is a single layer of a set of parameterised single qubit gates ( $S$  in the circuit diagram) on all qubits, followed by a series of two-qubit gates, sometimes called *entanglers*, connecting all the nearest neighbour qubits. Each such set of gates  $S$  consists of potentially multiple single qubit gates. While there are no strict rules on which gates are used,

each  $S$  typically contains three single qubit rotations about three different axes. For example,  $S$  can have an  $R_x$  first, followed by an  $R_y$  and then an  $R_z$ . Each rotation gate in  $S$  has its own parameter that can be adjusted independently. Similarly, each  $S$  in the circuit diagram contains its own set of parameters independent of others.

There are no strict rules on the entanglers as well. When not limited by the quantum hardware or the classical simulator, we typically use CNOT or CZ as the entangler. If we need to preserve symmetry, we may use symmetry-preserving two-qubit gates such as those used in Gard et al. [41]. If there are two-qubit gates native to the hardware, they can be used as the two-qubit gate. In most cases, we use parameter-free two-qubit gates to focus the optimisation problem on these single-qubit gates mentioned above.

After choosing the single qubits gates and the entanglers, we construct a parameterised circuit of a certain number of layers determined heuristically. The number of layers of HEA used is called the *depth* of the ansatz. For example, we can use a trial-and-error process in which we start with an initial depth for HEA, construct the parameterised circuit, run a few experiments to determine if the depth is big enough, and finally increase or decrease the depth of HEA if we are not satisfied with the experiments' result.

Increasing the depth of HEA is supposed to improve its capability to parameterise arbitrary unitary matrices, hence we are more likely to obtain the ground state preparation unitary when we optimise the circuit. On the other hand, we often decrease the depth when the benefit from a prior increase is reduced by problems related to either the increasing hardware noise or optimisation which becomes difficult with more parameters.

Note that sometimes we may consider an ansatz which applies only a single layer of  $S$  on all qubits without appending the entanglers. Such an ansatz is said to be *transversal* as it does not entangle and hence does not spread errors from different qubits. It has been extensively in experiments in Chapter 3.

**Barren plateau problems in HEA.** While we can easily adapt the Hardware-efficient ansatz to suit the hardware and simulation constraints, we may encounter a “barren plateau” problem when more qubits or higher depth is required for its application. In general, the barren plateau problem describes the phenomenon in which a parameterised circuit, when its parameters are randomly initialised, tends to land in a region where optimisation is difficult.

Specifically, when randomly initialised, the expectation value computed from this circuit will likely be constant throughout the neighbouring region in its hyperparameter space. An equivalent statement is that the gradient of the expectation value with respect to any parameter of this circuit will likely be zero after a random initialisation [28, 42].

There are several explanations [28, 43, 44] for the barren plateau phenomenon. Here, we present an interpretation based on the concentration of measure. Specifically, the barren plateau phenomenon arises from the probabilistic concentration of expectation values.

The concentration of measure [45, 46] is a general phenomenon where a statistical variable that depends on many other random variables converges to a constant probabilistically. This can be characterised by the following large deviation inequality where  $f$  is the statistical variable,  $m_f$  is typically its mean, and  $\varepsilon$  is the deviation. On the right-hand side, we assume a form  $c_1 \exp(-c_2 \varepsilon^2)$  with parameters  $c_1$  and  $c_2$ , which holds for many distributions<sup>3</sup>:

$$\text{Prob} \{ |f - m_f| \geq \varepsilon \} \leq c_1 \exp(-c_2 \varepsilon^2).$$

Moreover, when the parameter  $c_2$  depends on the number of variables, i.e., the dimension of the distribution, this phenomenon is referred to as a dimension-dependent concentration.

VQE may suffer from a dimension-dependent concentration. Specifically, if the unitary  $U(\theta)$  resulting from a random initialisation of the parameter  $\theta$  approximates the Haar distribution (in particular, if it forms a 2-design [48]), the cost function of VQE, i.e., the energy of the Hamiltonian  $H$ , satisfies the deviation inequality:

$$\text{Prob} \left\{ \left| \langle U | H | U \rangle - \frac{\text{Tr}(H)}{2^n} \right| \geq \varepsilon \right\} \leq 2 \exp(-c'_2 2^n \varepsilon^2),$$

where the specific constant  $c'_2$  can be found in Mele [48, Example 54]. Similarly, the concentration of the gradient would follow from the concentration of the cost function [42]. Notably, gradient concentration implies that upon initialisation, the parameter lands in a neighbourhood where the cost function remains nearly constant. As a result, any optimiser that relies on local information will struggle to navigate this “barren plateau”.

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<sup>3</sup>Interested readers may appreciate its wide applicability from Table 5.2 of Aubrun and Szarek [47].

Technically, the extent to which the randomly initialised ansatz circuit approximates a unitary 2-design is called its expressibility [49]. However, an explicit formula for the expressibility of a specific ansatz (such as HEA) is not available. Instead, numerical studies are typically used to assess whether an ansatz exhibits the barren plateau phenomenon, as has been done in McClean et al. [28].

Fortunately, we have not suffered from the barren plateau problem in the experiments we conducted, which we attribute to the fact that the number of qubits in our experiments is relatively small. We want to highlight that many cleverly designed ansatz and training techniques, as reviewed by Larocca et al. [50], may avoid this problem in practice. For example, for applications of VQE to the Fermi-Hubbard model, the Hamiltonian variational ansatz (HVA) has gained popularity [51, 52] and appears to suffer only mild barren plateau problems [53]. On the other hand, initialisation techniques have been explored in the literature to mitigate the barren plateau problem [29, 54]. It is worth noting that a similar problem called the “vanishing gradient” has happened in the classical machine learning literature. Many tricks [55, 56] have been developed in that community which effectively mitigated the vanishing gradient problem.

### 2.1.3 Measuring the expectation value

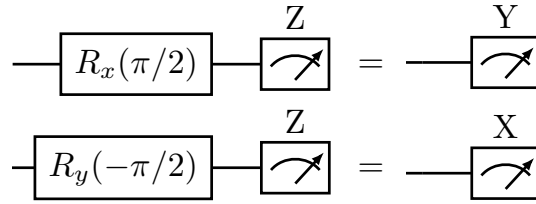
Here we describe the method to measure the expectation value of a Hamiltonian  $H$  on quantum computers. We assume that  $H$  has already been mapped such that it is a linear combination of various  $n$ -qubit Pauli operators,

$$H = \sum_i w_i P_i, \quad (2.9)$$

where  $w_i \in \mathbb{R}$  are the coefficients, and  $P_i$  are  $n$ -qubit Pauli operators. Then, the expectation of  $H$  can be obtained by summation,

$$\langle H \rangle = \sum_i w_i \langle P_i \rangle. \quad (2.10)$$

Therefore, we only need a method to measure the expectation value of any  $n$ -qubit Pauli operators  $P$ . If an  $n$ -qubit Pauli operator  $P$  contains only either Pauli- $Z$  matrices or the identity matrices, then measuring it is trivial on a quantum computer since both  $Z$  and identity matrix are diagonal in the computation basis. On the other hand, when there are non- $Z$  constituents in



**Figure 2.2:** Two circuit identities used for measuring Pauli- $X$  and  $Y$ . By adding rotation gates prior to measurement in the  $Z$  basis, one could effectively measure Pauli- $X$  or  $Y$ .

$P$ , we can use the following identities<sup>4</sup> for measurement:

$$\begin{aligned} R_x(-\pi/2)ZR_x(\pi/2) &= Y, \\ R_y(\pi/2)ZR_y(-\pi/2) &= X. \end{aligned} \quad (2.11)$$

This leads to<sup>5</sup> the circuit identities in Fig. 2.2, which is used for measuring non- $Z$  constituents in the  $n$ -qubit Pauli operator.

Note that it is easy to mistakenly use the wrong sign for the angles for rotation gates prior to the measurements (Fig. 2.2). Since,

$$\begin{aligned} R_x(\pi/2)ZR_x(-\pi/2) &= -Y, \\ R_y(-\pi/2)ZR_y(\pi/2) &= -X, \end{aligned} \quad (2.12)$$

all the weights  $w_i$  before Pauli operator  $P_i$  will be flipped if there are an odd number of  $X$  or  $Y$  inside  $P_i$ . This has happened unnoticed in the literature<sup>6</sup>.

### 2.1.3.1 Variance check

Here we comment on a way to check whether the state, prepared by the ansatz of the parameters at a certain iteration of VQE, is close to an eigenstate of the Hamiltonian.

It is well known that the variance of the Hamiltonian operator  $H$  with respect to any of its eigenstates is zero. This fact can be used to verify if an optimised state is the eigenstate of the VQE Hamiltonian. We call this a variance check on the state and illustrate the detailed steps in the following.

First, we show that the variance of the measured energy,  $\text{Var}(E)$ , is not

<sup>4</sup>Occasionally, other identities are used, such as  $SHZ(HS^\dagger) = Y$ ,  $HZH = X$ ,  $(\sqrt{X}R_z(\pi/2))^\dagger Z(\sqrt{X}R_z(\pi/2)) = X$ . This allows for flexibility when the native gate set on the quantum computer is limited.

<sup>5</sup>Note that  $\langle\psi|U^\dagger ZU|\psi\rangle = \langle\psi|A|\psi\rangle$  if  $U^\dagger ZU = A$ .

<sup>6</sup>For example, Section V of Supplementary Information of Kandala et al. [20].

expected to be zero. For simplicity, we may assume that the measurement statistics of each Pauli operator are independent of each other, which leads to, using the usual statistical formula for combining variances:

$$\text{Var}(E) = \sum_i w_i^2 \text{Var}(P_i) = \sum_i w_i^2 (\langle P_i^2 \rangle - \langle P_i \rangle^2). \quad (2.13)$$

Here, since we normally don't expect an eigenstate of the Hamiltonian, to be also an eigenstate of each of its constituent Pauli operators, at least one of the variances of Pauli operators  $\text{Var}(P_i)$  is greater than zero. Hence, the variance of the measured energy is usually greater than zero:

$$\text{Var}(E) > 0. \quad (2.14)$$

Instead, we should measure the variance operator  $O_{\text{VAR}}$ , which follows  $\langle O_{\text{VAR}} \rangle = \langle (H - \langle H \rangle)^2 \rangle = \langle H^2 \rangle - \langle H \rangle^2$ . This variance can be measured using the standard techniques mentioned. The measured expectation value should be close to zero, up to certain statistical fluctuations, if the prepared state is approximately an eigenstate of the Hamiltonian  $H$ .

### 2.1.4 Error mitigation for measurements

Here, we mention the technique of error mitigation, which is used to reduce the systematic error introduced by the imperfect nature of near-term quantum computers.

A large swarm of error mitigation methods have been proposed, and the author has reviewed this extensively in Tilly et al. [27]. Here, we want to briefly discuss two mitigation methods: an extrapolation-based error mitigation method (Section 2.1.4.1), and a method to mitigate readout errors (Section 2.1.4.2). They are among the most straightforward error mitigation methods and have been used in experiments in Chapter 4 and in Chapter 5.

#### 2.1.4.1 Extrapolation-based error mitigation

Extrapolation-based error mitigation is based on the simple idea that the measurement result  $\langle O \rangle$  on a quantum computer, is usually affected by the level of noise on it. If the level of noise can be characterised by a parameter or several parameters  $\varepsilon$ , then we may use the data  $\{\langle O \rangle(\varepsilon_i)\}$  to extrapolate the exact noiseless expectation value  $\langle O \rangle_{\text{exact}}$ . Here,  $\{\langle O \rangle(\varepsilon_i)\}$  are the measurement results of the same observable on the same circuit, but with different levels of noise  $\varepsilon_i$ .

Importantly, we assume that we can manipulate the quantum device, such that we can obtain the data. In practice, several proposals exist to achieve this. For example, the noise on a quantum computer could be adjusted by changing the control pulse of each individual quantum gate [57], in which case the  $\varepsilon$  will be the scaling parameter of that pulse. In other cases, we often artificially insert identity gates, e.g. pairs of CNOT gates [58, 59], in which case we may assume a linear relationship between  $\varepsilon$ , and the number of CNOT pairs inserted.

**Linear and exponential extrapolation.** Given the data  $\{\langle O \rangle(\varepsilon_i)\}$ , there are many methods to extrapolate and obtain the exact noise-free  $\langle O \rangle_{\text{exact}}$ . Two commonly used methods are linear and exponential extrapolation, which will be described here. Other methods have been summarised by the author in Tilly et al. [27].

In linear extrapolation, the noise of the quantum computer is modelled by a linear expression

$$\langle O \rangle(\varepsilon) \approx \langle O \rangle(\varepsilon|\alpha) = \langle O \rangle_{\text{exact}} + \alpha\varepsilon. \quad (2.15)$$

Here  $\alpha$  is the parameter of the model, and together with  $\langle O \rangle_{\text{exact}}$ , they are determined by fitting the experimental data  $\{\langle O \rangle(\varepsilon_i)\}$ . On the other hand, in exponential extrapolation, the model is

$$\langle O \rangle(\varepsilon) \approx \langle O \rangle_{\text{exact}} + A \exp(-B\varepsilon), \quad (2.16)$$

where  $A$  and  $B$  are the parameters.

The choice of the linear or exponential model depends on the specific details of the experiment. A rule of thumb is that when the number of noisy gates  $N_g$  is small (shallow circuit), linear extrapolation is favoured, and when  $N_g$  is large, exponential extrapolation performs better. Theoretical analysis of this rule is available in Endo et al. [60], while an empirical study comparing different methods is available in Majumdar et al. [61].

Additionally, it helps to conduct sanity check experiments to decide which error mitigation method to use. A sanity check can be done, for example, by measuring certain constants preserved by the symmetry of the VQE ansatz [41], or by studying a toy problem of similar circuit depth and structure, but with known results available.

### 2.1.4.2 Readout error mitigation

Here we touch on a simple readout error mitigation method, used in experiments in Chapter 5. Other methods are available in recent reviews [27, 62].

First, we note that readout error mitigation methods often implicitly treat state preparation and measurement (SPAM) errors. It is impractical to distinguish between these two sources of errors, state preparation and measurement, without certain tomography processes which do not scale in practice. Thus, while the two errors come from different processes, the methods discussed here can be viewed as readout error mitigation or SPAM mitigation depending on the situation.

SPAM errors play a significant role in NISQ quantum computation. To ensure accurate results in VQE experiments, it is crucial to address the impact of these errors through simple mitigation strategies. This becomes particularly important when measuring Pauli operators with many non-identity elements in the tensor product.

To address them, it is common to conduct preliminary calibration experiments on the quantum computer to acquire a measurement calibration matrix  $T$  (to be introduced below). This matrix establishes the connection between the likelihood of measuring each state given that the system is prepared in another state. Calibration matrix  $T$  in turn enables the mitigation of SPAM errors in the outputs of the quantum computer. Note that  $T$  models measurement errors using a conventional Markov process, which may be justified under specific quantum measurement model [63].

We also note that the choice of mappings other than the Jordan-Wigner transformation can directly affect the readout errors and hence the accuracy of the VQE algorithm. This has been discussed in Tilly et al. [27, §8.5].

**Measurement calibration matrix.** The measurement calibration matrix is defined element-wise by

$$T_{i,j} = \text{Probability (measured state } i | \text{prepared in state } j), \quad (2.17)$$

where  $i, j \in \{0, 1, \dots, 2^N - 1\}$  label the computational basis states on the  $N$  qubits that we want to measure.

The calibration matrix  $T$  could be measured on a quantum computer, by first preparing the state in the corresponding basis  $j$  using  $X$  gates, and measuring the resulting quantum computation in the computational basis. However, since the size of  $T$  grows exponentially with respect to the number

of measured qubits, it can rapidly become too expensive to obtain. There are many approaches to mitigate this scaling problem [27, §8.5].

**Correcting measurement outcomes.** Having obtained the calibration matrix  $T$  periodically during the experiment, the measurement outcome can be corrected using  $T$  by the following procedure.

First, we note that applying the inverse of  $T$  to the measurement outcomes is impractical:  $T$  is not guaranteed to be invertible. Also, the generalised (Moore–Penrose) inverse of  $T$  may produce unphysical quantities, as although  $T$  satisfy the required statistical properties (i.e.  $T_{i,j} \geq 0$  and  $\sum_i T_{i,j} = 1$ ), the generalised inverse may not.

Thus, we approach the inverse problem by solving the following optimisation problem as proposed in Gambetta et al. [64]:

$$\vec{x}^* = \underset{\vec{x}}{\operatorname{argmin}} |\vec{y} - T\vec{x}|^2, \quad (2.18)$$

subject to the constraint of  $\sum_i x_i = \sum_i y_i$  and,  $x_i \geq 0$ . Here,  $\vec{y}$  is the vector of raw measurement outcomes and  $\vec{x}$  is the vector of error-mitigated measurement outcomes. The  $i$ -th position of each vector stores the number of occurrences of the measurement outcome in state  $i$ . The vector norm is defined as  $|\vec{v}| = \sqrt{\vec{v} \cdot \vec{v}}$ . The constraint ensures that the optimised value  $\vec{x}^*$  is physical. In our experiments, this is usually done using the Sequential Least Squares Programming (SLSQP) module available from SciPy [65].

Obviously, the size of the matrix  $T$  and the vectors  $\vec{x}$  and  $\vec{y}$  scales exponentially with  $N$ , the number of qubits measured, making the above optimisation-based method impractical for large  $N$ . Several schemes have been proposed to alleviate this problem, and detailed discussions are available in Section 8.5 of Tilly et al. [27].

### 2.1.5 Optimisation using RotoSolve

Many optimisation algorithms have been applied to VQE problems with success. These include basic simplex-based algorithms such as the Nelder-Mead algorithm [66] or complex gradient based algorithms<sup>7</sup>. Other VQE or quantum machine learning specific algorithms have been proposed and reviewed in Tilly et al. [27].

Here we focus on the RotoSolve algorithm [68–70]. RotoSolve is useful for VQE experiments on small-scale problems [27], and have been used

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<sup>7</sup>See Chapter 12 of Zhang et al. [67] for many examples.

throughout experiments in other chapters. We describe this algorithm below. In RotoSolve, we assume that the function to be minimised, here also called the loss function, is sinusoidal in each of its arguments  $\theta_i$ :

$$L(\dots, \theta_i, \dots) = A \sin(\theta_i + B) + C. \quad (2.19)$$

This assumption is often satisfied for a wide variety of problems. In particular, for VQE problems where the ansatz's parameterised gates are all Pauli rotation gates, the energy measured  $E(\theta)$  will have the form in Eq. (2.19)<sup>8</sup>.

Assuming Eq. 2.19 is satisfied, then the optimum for a single parameter has a closed-form expression: Let  $L_i$  denote  $L$  when all parameters, except for  $\theta_i$ , are fixed. Then,

$$\begin{aligned} \theta_i^* &\equiv \arg \min_{\theta_i} L_i(\theta_i) \\ &= \phi - \frac{\pi}{2} - \arctan2(\phi_1, \phi_2) + 2\pi k, k \in \mathbb{N}, \end{aligned} \quad (2.20)$$

where

$$\begin{aligned} \phi_1 &= 2L_i(\phi) - L_i\left(\phi + \frac{\pi}{2}\right) - L_i\left(\phi - \frac{\pi}{2}\right), \\ \phi_2 &= L_i\left(\phi + \frac{\pi}{2}\right) - L_i\left(\phi - \frac{\pi}{2}\right). \end{aligned} \quad (2.21)$$

Here  $\phi$  could be any real number, and  $k$  is an arbitrary integer. The function  $\arctan2$  is a variant of  $\arctan$  which returns an angle that correctly deals with certain edge cases [71, 72]. In practice, we found it useful to use the value of  $\theta$  from a prior iteration (or the initial value) to set  $\phi$ .

Of course, optimising  $L_i$  for each  $i$  is different from optimising  $L$  for all parameters. In practice, we carry out iterations of RotoSolve on all parameters, and in each iteration, we often sequentially optimise all parameters one by one<sup>9</sup>. Specifically, in one iteration, we start with the first component  $\theta_1$  and find the optimal value  $\theta_1^*$  for it while fixing all other components. Then we move on to the next component  $\theta_2$  and find the optimal  $\theta_2^*$ . We repeat the same procedure until all components of  $\theta$  have gone through one optimisation.

Since each parameter is optimised in the context when other parameters

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<sup>8</sup>For proof, see for example, Appendix A of Ostaszewski et al. [69].

<sup>9</sup>Theoretically, it is unclear whether we should optimise all parameters sequentially as in  $\theta_1, \theta_2, \dots$ , or randomly. In practice, the sequential method often works.

are fixed, the optimisation of one parameter may invalidate (making it non-optimal) other previously optimised parameters. Fortunately, in practice, a few iterations of RotoSolve are enough to find a global optimum  $\theta^*$  for all parameters for VQE problems on only a few qubits and with relatively shallow ansatz.

**Stopping criterion.** Commonly, VQE optimisation is stopped when the energy function converges after several iterations. In particular, we observed in the experiments conducted in other chapters, that when the RotoSolve algorithm is used, the loss function typically converges quickly after only a few iterations, where one iteration consists of one optimisation for each  $\theta_i$  in the overall parameters  $\theta$ . We summarise our experience in these rules: First, set a maximum iteration for the RotoSolve algorithm according to budgetary constraint (e.g., the experiment needs to be finished in one hour etc.), and after running RotoSolve with the specified maximum iteration, confirm that the loss function drops quickly after the first few iterations, and use the parameter among the last few iterations corresponding to the best loss function value. Second, if the loss function does not drop quickly, check other factors which might have prevented optimisation. For example, whether the loss function satisfies the form in Eq. (2.19), whether hardware noise or programmatic errors exist and prevent optimisation. In short, tuning the details of the RotoSolve algorithm appears to be fruitless.

## 2.2 Excited state VQE

Here we discuss an extension of the basic VQE algorithm. We show how to use VQE to find the excited states of a Hamiltonian. Several methods [73, 74] have been proposed for this problem. Here we only discuss the method in Higgott et al. [73]<sup>10</sup>, since it uses the same principle as in another method we discussed in Sec. 2.3: adding penalty terms to the VQE energy function. This method will be used in experiments in Chapter 5.

Higgott et al. [73] proposes an extension to VQE, which finds all the first  $k$  eigenstates of a Hamiltonian, as sorted by their increasing energy eigenvalues.

<sup>10</sup>We seldom use the method proposed in Nakanishi et al. [74]. The method proposed there requires using the same ansatz  $U(\theta)$  to prepare all the desired eigenstates, which is disadvantageous since, in practice, different ansatzes are required to better suit different eigenstates. For example, low-energy excited states are less entangled and usually require fewer quantum gates to prepare. That method in [74] therefore puts extra requirements on the *expressibility* of the ansatz and is as a result less noise-friendly. The method discussed in the main text does not suffer from the same problem.

Specifically, assuming the first  $k - 1$  eigenstate states have been found, it finds the  $k$ -th eigenstate by penalising the current ansatz's overlap with all previously found eigenstates. For example, assume we have found

$$|\psi_i\rangle = U_i |0\rangle, i = 1, \dots, k - 1, \quad (2.22)$$

which are eigenstates of  $H$  with increasing eigenvalues  $E_i \leq E_{i+1}$ . Then, assume we want to use the ansatz  $U(\theta)$  to prepare the next eigenstate, i.e. we set  $|\psi_i\rangle = U(\theta) |0\rangle$ , Higgott et al. [73] proposes to optimise the following loss function:

$$L(\theta) = \langle \psi_k(\theta) | H | \psi_k(\theta) \rangle + \sum_{i=1}^{k-1} \beta_i |\langle \psi_i | \psi_k(\theta) \rangle|^2, \quad (2.23)$$

where  $\psi_i = U_i |0\rangle$ ,  $\psi_k(\theta) = U(\theta) |0\rangle$ ,  $\{\beta_i\}$  are a series of hyperparameters that should be chosen to ensure that the minimal of  $L$  corresponds to the desired eigenstate. Alternatively, this method can be considered as VQE on the modified Hamiltonian below

$$H' = H + \sum_{i=1}^{k-1} \beta_i |\psi_i\rangle\langle\psi_i|. \quad (2.24)$$

**Choosing  $\beta_i$ .** To ensure that the minimum of Eq. 2.23 corresponds to the desired  $k$ -th eigenstate,  $\beta_i$  must satisfy the following relations:

$$\beta_i > E_k - E_i, i = 1, \dots, k - 1, \quad (2.25)$$

which ensures that, e.g.,  $E_i + \beta_i$  is higher than  $E_k$ . Therefore, previous eigenstate  $|\psi_i\rangle$  does not minimise the energy of the modified Hamiltonian  $H'$  in Eq. 2.24.

However,  $E_k$  is not known before optimisation. Therefore, the condition in Eq. 2.25 becomes impractical. In practice, we may start optimisation with an initially guessed  $\beta_i$ , and observe the measured value of the overlap term  $|\langle \psi_i | \psi_k(\theta) \rangle|^2$ . If this overlap term does not vanish below a certain numerical threshold, we would further increase the penalty  $\beta_i$ . Note that, this increase need to be balanced against making  $\beta_i$  too large, which would caused numerical instability in optimisation by diminishing the other contribution towards the cost function, i.e. the energy from the original Hamiltonian  $H$ .

**Measuring the overlap term.** The second overlap term in Eq. (2.23) can be easily measured on a quantum computer. Since,

$$\langle \psi_i | \psi_k(\theta) \rangle = \langle 0 | U_i^\dagger U(\theta) | 0 \rangle, \quad (2.26)$$

$|\langle \psi_i | \psi_k(\theta) \rangle|^2$  is just the probability of measuring all 0 on all qubits, on the state prepared by the circuit  $U_i^\dagger U(\theta) | 0 \rangle$ . One practical problem is that, simply reverting individual gates in  $U_i$  will not automatically implement the inverse unitary  $U_i^\dagger$ . That is, if  $U_i = g_1 \cdots g_n$ , then  $U_i^\dagger$  may not be  $g_n^\dagger \cdots g_1^\dagger$  due to systematic noise in near-term quantum computers. Higgott et al. [73] proposes to use another ansatz, denoted  $U'_i(\theta)$ , for approximating  $U_i^\dagger$  by maximising the following quantity:

$$|\langle 0 | U'_i(\theta) U_i | 0 \rangle|^2. \quad (2.27)$$

## 2.3 Preserving symmetry in VQE

Sometimes we want to obtain the eigenstate of a Hamiltonian which preserves a specific symmetry. We will see an example in Chapter 5. There are in general two ways to achieve this: using symmetry preserving ansatz or adding penalty terms. In the first method, we initialise quantum computers in a state in the desired symmetry sector but could be easily prepared with minimal gates. Then one uses symmetry preserving ansatzes for VQE on this initial state, which guarantees that the optimised state will also be another eigenstate with the desired symmetry. Details of this method can be found in Section 6.2.3 of Tilly et al. [27].

On the other hand, here we discuss the method to preserve symmetry using constrained optimisation with penalty terms. We detail this method in the following<sup>11</sup>, since it has been used in Chapter 5.

Assume that the symmetry we want to preserve is represented by the observable  $S$ , and we want the ground state in VQE to be also an eigenstate of  $S$  with the eigenvalue  $s_t$ . To achieve this, we modify the Hamiltonian in VQE from  $H$ , to:

$$H' = H + \beta(S - s_t)^2. \quad (2.28)$$

Similar to the case of excited state VQE, the hyperparameter  $\beta$  needs to be chosen wisely. Unfortunately, an explicit guarantee as in Eq. 2.25 is not

<sup>11</sup>We note that although this algorithm has been very briefly mentioned in McClean et al. [18], Benfenati et al. [75], and used by us in Rungger et al. [30], we know no detailed discussions about it in the literature.

available, since no information about eigenstates of  $H$  is known before we optimise. However, the practical method of gradually increasing  $\beta$  (see Section 2.2) can be employed here. In addition, after optimisation, we can directly check two conditions to further improve our confidence that the symmetry has been preserved after optimisation:

1. measure  $\langle S \rangle$  and confirm it is close to  $s_t$ ,
2. Use the method mentioned when discussing the stopping criteria in Sec. 2.1.5 to check that the optimised result prepares an eigenstate of both  $H$  and  $S$ .

We note that an alternative method [76] exists, which optimise the following loss function

$$L(\theta) = \langle H \rangle + \beta(\langle S \rangle - s_t)^2. \quad (2.29)$$

However, optimising this function introduces an extra burden compared with the previous method: the previous method enjoys the advantage of being a pure VQE problem, and hence optimisation techniques such as RotoSolve (see Sec. 2.1.5) can be applied.

## 2.4 Variational linear algebra solver

Here we want to discuss another extension of the VQE algorithm. It uses VQE to solve linear algebra problems on quantum computers. They were proposed in [77, 78] and used in experiments in Chapter 4. Here, we first prove two related theorems and then show how the two theorems can be employed in a quantum algorithm to solve linear algebra problems.

**Theorem 2.1** (Multiplication by a matrix). *Let  $H_A$  be given by*

$$H_A = \mathbb{1} - \frac{A|b\rangle\langle b|A^\dagger}{||A|b\rangle||^2}, \quad (2.30)$$

where  $|b\rangle$  is an arbitrary vector (not necessarily normalised) and  $A$  is an  $n \times n$  matrix. Let  $|\psi\rangle$  denote the normalised ground state of  $H_A$ . Then  $|\psi\rangle$  corresponds to an eigenvalue 0, and it is proportional to  $A|b\rangle$ , i.e.  $|\psi\rangle \propto A|b\rangle$ .

*Note, here the norm of the vector  $A|b\rangle$  is denoted as  $||A|b\rangle||$ , which is equal to  $\sqrt{\langle b|A^\dagger A|b\rangle}$ .*

*Proof.* First, we note that  $H_A$  is Hermitian, which can be checked easily. Then we show that  $H_A$  is positive semi-definite, i.e. its ground state has eigenvalue

0: Define  $|u\rangle = A|b\rangle/||A|b\rangle||$ , which is normalised. Then  $H_A$  can be rewritten as  $H_A = \mathbb{1} - |u\rangle\langle u|$ , hence it is positive semi-definite. Note that  $H_A$  geometrically projects states to the vector space complement to that spanned by  $|u\rangle$ .

Finally, we turn to the normalised state  $|\psi\rangle$ . Since  $\langle\psi|H_A|\psi\rangle = 0$ , this implies that  $\langle\psi|\psi\rangle = ||\langle\psi|u\rangle||^2$ . Since  $\langle\psi|\psi\rangle$  is one by definition, we have  $|\psi\rangle \propto |u\rangle$ , i.e.  $|\psi\rangle \propto A|b\rangle$ .  $\square$

**Theorem 2.2** (Inversion of a matrix). *Let  $H_{A^{-1}}$  be*

$$H_{A^{-1}} = A^\dagger (\mathbb{1} - |b\rangle\langle b|) A, \quad (2.31)$$

where  $|b\rangle$  is a normalised vector<sup>12</sup>, and  $A$  is any non-singular  $n \times n$  matrix such that the inverse  $A^{-1}|b\rangle$  exists.

The normalised ground state, denoted  $|\psi\rangle$ , of  $H_{A^{-1}}$  will be such that

$$A|\psi\rangle \propto |b\rangle, \text{ or equivalently } |\psi\rangle \propto A^{-1}|b\rangle. \quad (2.32)$$

*Proof.* First, again,  $H_{A^{-1}}$  is Hermitian which can be checked easily. Also,  $H_{A^{-1}}$  is positive semi-definite, which we show here: For any  $|\phi\rangle$ ,

$$\langle\phi|H_{A^{-1}}|\phi\rangle = \langle\phi'|(\mathbb{1} - |b\rangle\langle b|)|\phi'\rangle,$$

where  $|\phi'\rangle = A|\phi\rangle$ . The right-hand side of the equation above is obviously non-negative since  $\mathbb{1} - |b\rangle\langle b|$  is a projector which projects any vector to the subspace complement to that spanned by  $|b\rangle$ . Hence,  $H_{A^{-1}}$  is positive semi-definite.

Next, when  $|\psi\rangle$  is such that  $\langle\psi|H_{A^{-1}}|\psi\rangle = 0$ , then  $A|\psi\rangle$  lies in the kernel of the projector above, in other words  $|\phi'\rangle = A|\psi\rangle \propto |b\rangle$ , or  $|\psi\rangle \propto A^{-1}|b\rangle$ .  $\square$

**Using these theorems for quantum algorithms.** In quantum algorithms, typically we have access to a normalised state  $|b\rangle$  prepared by a circuit,  $|b\rangle = U_b|0\rangle$ , which will satisfy conditions for both theorems. So we can use VQE algorithms, where the Hamiltonian  $H$  is either  $H_A$  or  $H_{A^{-1}}$ , to find a circuit which prepares a state  $|\psi\rangle$  that is proportional to either  $A|b\rangle$  or  $A^{-1}|b\rangle$ . In the first case, we effectively solved a matrix multiplication problem using VQE. In the second case, we effectively solved the linear problem  $Ax = b$  using VQE.

A problem that remains unexplained here, is how to measure expectation values of  $H_A$  or  $H_{A^{-1}}$  on quantum computers using the techniques outlined

<sup>12</sup>This is different from the previous theorem.

in Sec. 2.1.3. This problem is usually resolved in practical cases where  $A$  can be mapped to a linear summation of Pauli operators, and with the help of quantum circuits similar to the Hadamard-test (Appendix C.2) to compute overlaps. We will see an explicit example in Chapter 4.

We finally note that, for the VQE algorithms based on these two theorems, the theoretical minimum of the Hamiltonian is known to be zero. Two points need to be clarified here. First, if we could obtain a zero energy value from measurement, we can affirm that the optimisation has succeeded and produced the desired circuits. However, work done in Bravo-Prieto et al. [79] showed that the energy value tends to be amplified as  $1/\kappa^2$  where  $\kappa$  is the condition number of the problem matrix  $A$ . This can be problematic if  $A$  is near-singular, as will happen in the problem we faced in Chapter 4.

## Chapter 3

# Discovering logical gates for error-correcting codes

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In this chapter, we delve into the topic of quantum error-correcting codes, which play a vital role in safeguarding quantum computation against errors caused by noise. A potential challenge in this field is the design of logical operations for these error-correcting codes. To address this, we propose an automated procedure that generates logical operations by utilising variational quantum algorithms. Our procedure allows for the discovery of logical gates based on known encoding and correcting procedures. By employing variational circuits to learn both the logical gates and their corresponding physical operations, our procedure can automatically identify logical gates from analytically designed error-correcting codes. The procedure can in principle be extended to applications on error-correcting codes obtained through numerical optimisations.

Here, we applied our procedure to quantum codes using classical simulation. By applying our procedure to small-size quantum error-correcting codes ranging from four to fifteen qubits, we are able to automatically discover the

majority of logical gates documented in current literature (as of 2019). Notably, our procedure produced logical gates that are not found in the current literature for the  $[[5,1,2]]$  code, the  $[[6,3,2]]$  code, the  $[[8,3,2]]$  code, and the  $[[10,1,2]]$  code.

**Chapter outline.** In this chapter, we first give a brief introduction to the basics of quantum error correction in Section 3.1. We detail the problem of finding logical gates in Section 3.2 and review a few related results from the literature in there. We then present our procedure in Section 3.3, and we assess our procedure by providing proof of concept experiments on a classical computer where we apply the procedure to several CSS and non-CSS codes in Section 3.4. The logical gates we found using our procedure are also detailed there. Among these gates we obtained, there are several new logical gates which to our knowledge have not appeared previously in the literature. We discuss in detail these new logical gates in Section 3.4.2. Finally, we make several comments on the benefits and disadvantages of our procedure in Section 3.5, where we for example discuss the scalability of our procedure.

### 3.1 Error correction on quantum computer

The development of quantum error-correcting codes (QEC) is motivated by the presence of quantum errors caused by interactions with the external environment. These interactions vary in nature and depend on the specific hardware architecture. Here we introduce the basic ideas for protecting computations against such errors. We only discuss as far as is sufficient for presenting our procedure. More detailed treatments of quantum error correction are available in Nielsen and Chuang [1] and Gottesman [80].

A quantum code protects information by encoding it into a certain linear subspace, called the *code space*, of the large Hilbert space. The *encoding* can for example be achieved by a quantum gate which transforms a  $k$ -qubit quantum state containing useful information into the larger Hilbert space of  $n$ -qubits. The rest of the  $n - k$  qubits are initialised at  $|0\rangle$  and becomes entangled with the  $k$  qubits after encoding. The computational basis for the original  $k$  qubits, after the encoding, forms a new computational basis for the computation inside the code space. Elements of the new basis are thus called *code words*. In this way, we constructed  $k$  *logical qubits* using the code words on top of the  $n$  *physical qubits*.

To be resilient against error, the encoding should be designed such that,

typically errors transform the resulting code space into other subspaces that are distinct (i.e. orthogonal) to each other and the original code space. Detection and correction of these typical errors are thus possible. The projectors to all these orthogonal subspaces form a projective measurement. In principle, we can then perform such projective measurement to identify the error, and apply the inverse operation according to the error identified, to reverse its effect. Note that, we call quantum codes designed as such *quantum error-correcting codes*. However, if different errors lead to subspaces that may not be different from each other, but still remain orthogonal to the code space, then we can at least detect such errors. Such quantum codes are called *quantum error-detecting codes*. Since the same quantum code may be able to correct certain errors and detect others, it can be called error-correcting for some errors, and error-detecting for others. An example of an error-detecting is available in the Appendix A (page 132).

**Local errors and distance of a code.** Next, we discuss errors which have the form of a Pauli operator on  $n$  qubits. An  $n$ -qubit Pauli operator  $P$  of weight  $t$  is the Kronecker product of  $n$  single qubit matrices in  $\{I, X, Y, Z\}$ , and only  $t$  of these matrices are non-identity ( $\neq I$ ). Such operators are also called weight  $t$  local errors. Pauli errors are important due to its simplicity, and also because a randomised compilation technique exists to convert coherent or correlated noises into stochastic Pauli noise [81] consisting of Pauli errors.

Such local errors allow us to introduce the concept of distance. The *distance*  $d$  of a quantum code is the minimum weight  $t$  such that one code word is mapped by a weight  $t$  error to another code word. Any error whose weight  $t$  satisfies  $2t < d$  can be corrected (see a proof in [82, §2.3]). With this we can introduce the  $[[n, k, d]]$  notation: for a quantum code which encodes  $k$  qubits using  $n$  qubits and has a distance  $d$ , it is called a  $[[n, k, d]]$  code.

**Stabiliser code and logical gates.** Note that, the number of different weight  $t$  Pauli errors on  $n$ -qubits grows exponentially, and therefore the number of possible errors that can be corrected for a distance  $d$  code also grows exponentially. *Stabiliser formalism* [80] describes an efficient way to deal this situation. Codes that can be described in this formalism are called *stabiliser codes*. While we do not go into details of stabiliser codes (a brief look is available in the Appendix A), we outline the general idea. In a stabiliser code, the code space is fixed by a group of commuting operators called *stabilisers*. This group can be generated by a few elements called generators. These

generators can be measured and the results, called a *syndrome*, collectively determine whether a correctable error has happened and how to correct for it. Further, *decoding* algorithms are needed to predict the correct error from the syndrome. Note that, an expensive but conceptually simple decoding algorithm exists and is called a lookup-table decoder. In this case, all possible correctable errors for a distance  $d$  code are obtained by efficient Clifford gate simulators, and their corresponding syndromes are recorded in a table. Such a table (of exponential size) can be used to identify errors, i.e. for decoding.

Finally, we mention *logical gates*. Logical gates are isomorphisms on the code space, i.e. they map code words to code words. Logical gates need to be implemented on the  $n$  physical qubits using quantum gates, which we call the *physical operations* corresponding to the logical gate. Importantly, if errors on quantum computers resulted in such physical operations which perform logical gates, they would be impossible to detect. This explains the terminology “distance” as a code’s distance relates to the operators mapping code words to different code words.

In the case of stabiliser codes, the code space is fixed by stabilisers, i.e. code words are invariant under the action of stabilisers. There are operators in the Pauli group which commutes with every stabilisers, but does not leave the code words invariant, i.e., they are logical gates for stabiliser code. Such operators are called centralisers of the stabiliser group. However, logical gates based on centralisers are insufficient for universal quantum computation: for example, they can even be simulated efficiently using classical computers (Gottesman-Knill theorem).

## 3.2 The Problem for finding logical gates

In this chapter, we aim to address the problem of finding both the logical gate, denoted by  $g$ , and its shallow quantum circuit implementation, denoted as  $G$ , given a specific quantum code. For practical considerations, we will focus on stabiliser codes.

There are several approaches related to this. Assuming that we know how to implement a universal set of quantum gates on the logical qubits, then circuit compilers can be used to compile any quantum gate into a series of gates which contain only elements from the given universal gate set. There are such compilers which optimise the result to use less CNOT gates [83]. However, such an approach often creates implementations which involve circuits of exponential depth on the physical qubits. In contrast, we want to

focus on minimising the errors that are introduced during the application of gates. That is, we want shallow-depth circuit implementations involving applying only a few gates to the physical qubits to produce the desired logical gate. Ideally, we would want the implementation  $G$  to consist purely of quantum gates that act on individual qubits, so that errors do not spread across. Logical gates which can be implemented as such are referred to as *transversal gates*.

Finding shallow implementations of logical gates remains a challenge in the field of quantum error correction and has so far been done on a case-by-case basis. For example, using the *standard form* of the stabiliser group one can obtain implementations of logical Pauli gates easily. Also, quantum codes generated from classical codes using the CSS construction [84, 85] that encode a single logical qubit have a known fault-tolerant CNOT gate, which can be implemented on two copies of the same code by applying a CNOT between all pairs of physical qubits.

In this chapter, we propose to use variational quantum algorithms inspired methods to automate the process of finding physical implementations of logical gates. Since not all logical gates deserve a shallow implementation, we need to discover both the right logical gates to target and their shallow implementations. We want to clarify that, since transversal gates cannot form universal gate set [86], we do not aim to discover universal gate sets here.

### 3.2.1 Review of relevant literature

We review various related ideas in the literature. While few papers have addressed the same topic, various research teams have worked on either variational compiling or utilising machine learning to generate quantum error-correcting codes. We review these studies for two reasons: firstly, we employ analogous techniques and methods except for a different objective, and secondly, our interest in discovering logical gates stemmed from the challenges we encountered in discovering error-correcting codes through machine learning.

**Variational compiling.** Variational compiling uses parameterised circuits to find alternative implementations of unitary gates. It does so by optimising the parameters such that a certain distance measure between the parameterised circuit and the target unitary gate is minimised to zero, or equivalently, a certain measure of fidelity is maximised to one.

Here, we mention a widely used fidelity measure between two unitary

gates  $U$  and  $V$  called the *averaged gate fidelity*. It is calculated as the fidelity between outputs from  $U$  and  $V$ , averaged over all possible input quantum states:  $\int |\langle \psi | U^\dagger V | \psi \rangle|^2 d\psi$ . It has been shown that this fidelity is linked directly to the trace  $\text{Tr}(V^\dagger U)$  and the entanglement fidelity of the channel  $V^\dagger U$ , by a simple formula [87, Eq. (6)]. This formula is based on the equivalence between the averaged fidelity of a quantum channel and the entanglement fidelity of that channel [88].

This fidelity has inspired researchers to optimise ansatz circuits for alternative implementations of certain target unitary gates, which can be done through simulations on classical hardware if the Hilbert space is small enough. For example, researchers have aimed to find noise robust circuits [89], alternative encoding circuits for error-correcting codes [90], and more efficient circuits for computing state overlap [91]. One particular technique worth noting is proposed in Xu et al. [90], where the problem of finding circuits for preparing encoded logical states given a stabiliser code, is shown to be equivalent to the problem of preparing the ground state of a Hamiltonian, i.e. a VQE type problem. Understandably the Hamiltonian is closely related to the stabilisers of the given stabiliser code.

Additionally, the entanglement fidelity related  $\text{Tr}(V^\dagger U)$  enables a quantum circuit implementation, called the *Hilbert Schmidt test* [87], which computes the averaged gate fidelity and requires only applying  $U$  and a complex conjugate<sup>1</sup> of  $V$  on two different sets of qubit, CNOT gates connecting the two sets, and measurements on the computational basis (see Figure 4 of Khatri et al. [87]). Such gate fidelity measure based purely on quantum circuit provides hope that variational compiling can find more noise-robust implementations of targeted unitaries [92].

**Learning error-correcting gates.** Meanwhile, several groups have attacked the problem of generating error-correcting code using machine learning. There are two approaches. The first approach is based on finding the optimal encoding and recovering operators which maximise the fidelity of encoded quantum states under noise and after application these recovery operators [93–95]. These operators are represented using matrices satisfying the conditions for Kraus operators. This approach has been experimented on a few qubits but has obtained limited success. They suffer from an implementation problem: the numeral matrices representing the encoding and

<sup>1</sup>See Appendix A of Khatri et al. [87] for the meaning and implementation of this complex conjugation.

recovery operators are often inefficient to implement using native hardware operations.

Another approach is based on learning quantum circuits to preserve quantum information. In this approach, a classical machine is used to optimise parameterised circuits [96] or even arbitrary gate sequences [97] to preserve quantum information. Using parameterised circuits requires similar techniques in variational compiling, while targeting arbitrary gate sequences requires the whole toolkit from the reinforcement learning literature. Unfortunately, learning error correction code in this case is extremely hard. The first hindrance is that we often need the full information of the quantum process (hence optimisation is done through classical simulation), leading to huge simulation costs on larger quantum systems. Another problem is that, assuming a weak level of noise, a naive strategy of doing nothing is not a bad solution in terms of the fidelity measure, and any further exploration beyond the native strategy requires a prolonged time to pass through a *wide barrier* before approaching any optima, as documented in Fösel et al. [97].

However, as we will show in the next few sections, the optimisation becomes much easier for a related problem: instead of finding a sequence of gates to preserve information, we aim to find logical gates given these preservation procedures.

### 3.3 Procedure for finding logical gates

In this section, we outline our procedure for finding logical gates as well as their shallow circuit implementation, given an error-correcting code. To accomplish this, we incorporate numerical optimisation of the variational circuit ansatzes for both the logical gates and their physical implementations. The idea is similar to that of variational compiling where one aims to find an alternative implementation of a target unitary  $U_{\text{target}}$ , by optimising a parameterised ansatz  $U_{\text{ansatz}}$  so that the two match each other: the averaged gate fidelity is one.

In our case, both the target and the ansatz are guessed by using two different parameterised circuits: an ansatz  $G(\theta_1)$  for a physical implementation, and another ansatz  $g(\theta_2)$ , for a logical gate. The guess for logical gate,  $g(\theta_2)$ , when initialised performs a random logical operation on qubits before the encoding. The guess for physical operations,  $G(\theta_1)$ , when initialised performs a random operation on encoded qubits.  $G(\theta_1)$  might potentially result in bringing code words outside the code space, and we use corrections in-

herited from the given error-correcting code,  $C$ , to bring the results back to the code space. Finally, we compute a certain distance function between the two guesses, which we call the loss function. We optimise this loss function so that the two guesses converge, i.e., the trail logical circuits  $G = G(\theta_1^*)$  would implement the logical gate  $g = g(\theta_2^*)$ , represented by the comparison circuit. Here we used the superscript  $\theta^*$  to denote the parameter value after optimisation.

For optimisation, we can use the RotoSolve algorithm discussed in Section 2.1.5 [69, 98]. This is because the building blocks of our variational circuits consist of single-qubit rotational gates and the CNOT gate.

In the following, we first describe how the procedure can be performed via simulation on a classical computer, and we describe its extension to quantum computers afterwards. An outline of our procedure is available in Figure 3.1.

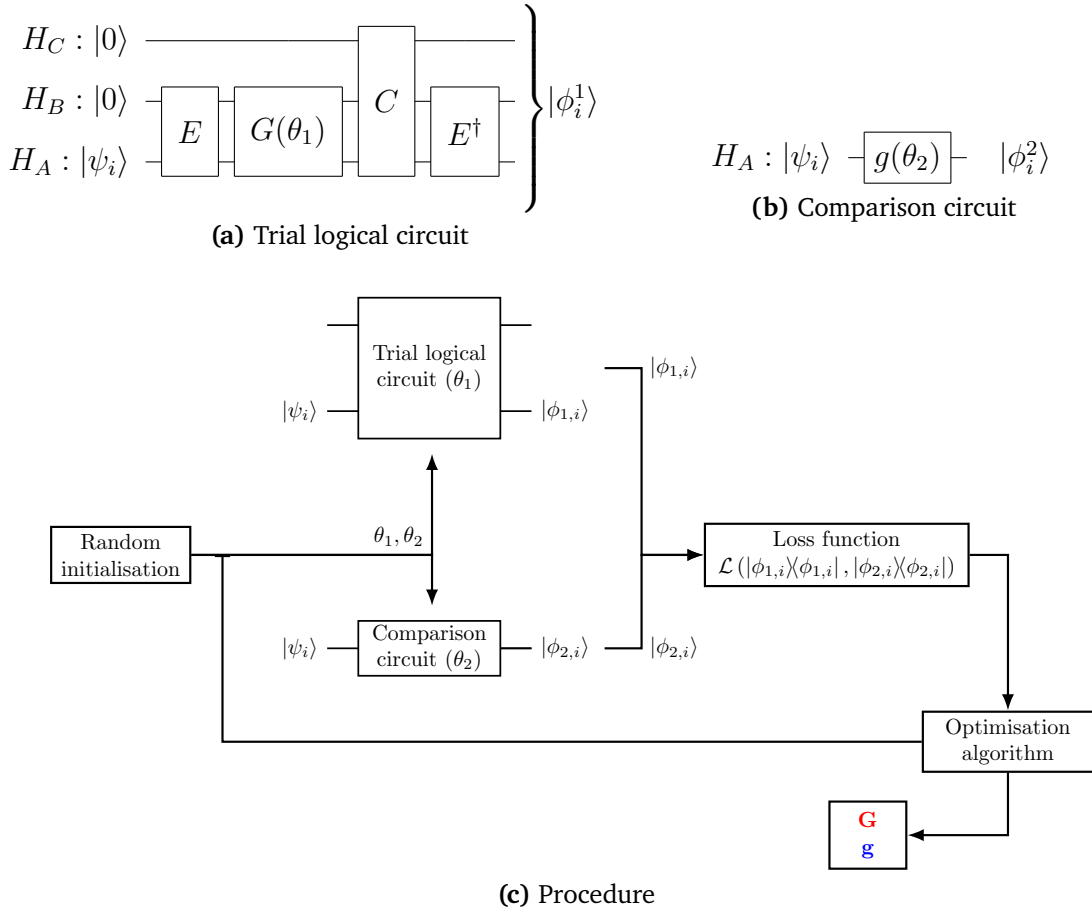


Figure 3.1: Learning logical gates

To learn logical gates, we optimise angles  $\theta_1$  and  $\theta_2$  such that the trial logical circuit and the comparison circuit perform effectively the same unitary as measured by our loss function  $\mathcal{L}$ . Inside the trial logical circuit (Fig. 3.1a), the encoding/inverse encoding circuit  $E/E^\dagger$  maps input states  $\{|\psi_i\rangle\}_1^N$  into/out of the logical space. The ansatz  $G(\theta_1)$  performs a series of physical operations and the circuit  $C$  performs the stabiliser measurements and minimum weight error correction. In the comparison circuit (Fig. 3.1b), only the ansatz circuit  $g(\theta_2)$  for the logical gate is performed. The whole procedure starts by randomly initialising the two angles in both circuits, and then a tomographically complete set of quantum states for  $H_A$ ,  $\{|\psi_i\rangle\}$ , are fed to the two circuits as inputs. Their outputs are gathered and fed to the loss function  $\mathcal{L}$ . The calculated loss function values are fed to a minimisation algorithm, which outputs new angles for  $\theta_1$  and  $\theta_2$ . Then we rerun the trial logical circuit and the comparison circuit again and the whole loop continues until  $\mathcal{L}$  is zero excluding float point errors, in which case the circuit  $G(\theta_1^*)$  will perform the logical gate  $g(\theta_2^*)$  on the encoded space. Here a superscript  $\theta^*$  denotes the optimised parameter for  $\theta$ .

**Simulating two circuits.** In experiments on a classical computer, our procedure involves simulating two circuits. The first is called *trial logical circuit*, and consists of initialisation to a certain input state  $|\psi_i\rangle$  (to be explained later), the encoding  $E$ , the physical operation  $G(\theta_1)$ , the correction  $C$ , and the inverse of the encoding  $E^{-1} = E^\dagger$ . The encoding and correction circuits

are given by the error-correcting code. This trial logical circuit simulates the effect of a random guess on operations on the physical qubits inside the code space.

Correction plays an important role here. While the randomly guessed physical operation may map logical states outside the code space, the correction  $C$  brings the resulting state back to the code space. This correction implements a projector on the code space. In standard error correction, the projection is accomplished by measuring the stabiliser generators and applying a correction based on the observed syndrome. Here the correction corresponds to the minimum weight error that is consistent with the measured syndrome.

Note that we can improve the simulation cost of the correction. Specifically, the syndrome measurement and its associated correction operations, are simulated classically without simulating the measurements (i.e. without collapsing the simulated quantum state), which is possible because the syndrome measurements are achieved using ancilla qubits whose state after the measurement can be discarded (see discussion on the loss function for the discarding). This is important to bring down computation costs since measurements involve operations difficult to simulate on a graphical processing unit (GPU).

Note that for the first trial logical circuit, the logical information is first stored in the Hilbert space  $H_A$ , later spread to the larger space of  $H_A \otimes H_B$  due to encoding  $E$ . The physical operation  $G(\theta_1)$  may spread information outside the code space, which is a subspace of  $H_A \otimes H_B$ . To bring information back to the code space, extra ancilla qubits are used for measurement and correction  $C$ , leading to a further enlarged Hilbert space  $H_A \otimes H_B \otimes H_C$ . Finally, the information is brought back to the Hilbert space  $H_A$  by the inverted encoding  $E^{-1}$ . This is illustrated in Figure 3.1a.

The second *comparison circuit* is more straightforward. It involves initialisation to the same state  $|\psi_i\rangle$  and applying the guessed logical gate  $g(\theta_2)$ . It is visualised in Figure 3.1b.

**Loss function.** In order to achieve our objective, we aim to find an optimal set of parameters for the ansatz circuits. They are optimal only when the physical operation circuit  $G(\theta_1)$ , along with the encoding, correction  $C$ , should collectively perform the same operation as the logical operation  $g(\theta_2)$  for all possible input quantum states. To achieve this, we first use a *tomo-*

graphically complete set  $\{|\psi_i\rangle\}$  of unitary operations on the Hilbert space  $H_A$  as input. Note that the result of any quantum channel on any quantum state can be determined by its result on a set of tomographically complete [99, §8.4.2] density matrices. The specific set of states used, described in Section 3.4.1, are eigenstates of Pauli matrices.

Then, we use the trace distance to measure the difference between the outputs from the two circuits. That is, we define the loss function  $\mathcal{L}$  as in,

$$\mathcal{L}(\theta_1, \theta_2) = \sum_i F[\text{Tr}_{H_B \otimes H_C} (|\phi_{1,i}\rangle\langle\phi_{1,i}|), |\phi_{2,i}\rangle\langle\phi_{2,i}|)]. \quad (3.1)$$

Here, input states  $\{|\psi_i\rangle\}$  is tomographically complete for  $H_A$ . The  $\phi_i^1$  and  $\phi_i^2$  represent the outputs of the trial logical circuit and comparison circuit respectively (see Fig. 3.1). The trace operation ( $\text{Tr}$ ) is performed over the Hilbert space  $H_B \otimes H_C$ , and effectively discarding the information still remains in ancilla qubits ( $H_C$ ), resulting in density matrices on  $H_A$ . Lastly, the distance between two quantum states  $\rho$  and  $\sigma$  is measured using the trace distance  $F(\rho, \sigma) \equiv \frac{1}{2} \text{Tr} [\sqrt{(\rho - \sigma)^2}]$ . Note that for pure states  $|\psi\rangle$  and  $|\phi\rangle$ , trace distance is closely related to state fidelity as  $1/2 \text{Tr} [\sqrt{(|\psi\rangle\langle\psi| - |\phi\rangle\langle\phi|)^2}] = \sqrt{1 - |\langle\psi|\phi\rangle|^2}$ . This relates our distance measure to the averaged gate fidelity introduced in our reviews (Section 3.2.1), except that we are targetting density matrices and make use of a tomographically complete set instead of averaging over Haar random states.

Our procedure is considered successful when the distance as measured by  $\mathcal{L}$  is reduced to zero, taking into account certain floating-point errors. In such cases, a logical gate for this code can be successfully identified by the current parameters,  $g = g(\theta_2^*)$ , with its physical operation realised by the optimised parameter for logical operations,  $G = G(\theta_1^*)$ .

**Adapting the procedure.** Our procedure offers flexibility by design. The ansatz employed for the physical operation can be customised to leverage the characteristics of a particular quantum computing architecture. In terms of logical gates, we utilise a variational ansatz, which allows for the automatic discovery of such gates based on a given quantum error-correcting code. Additionally, by fixing the ansatz to a specific logical gate, our procedure can target that gate specifically. As a result, our approach can be a straightforward choice for locating logical gates, particularly in the case of stabiliser codes, including non-CSS codes.

We also want to emphasise that our procedure can be applied to any code that can prepare encoded Pauli eigenstates and measure Pauli observables, i.e. it is not limited to qubit stabiliser codes and can be extended to the case of, for example, qudit codes. Of course, in these extended use cases, the ansatz and optimisation procedure might need to be adapted as well.

**Potential application on quantum computers.** Our method can either be adapted for running on a quantum computer to find logical gates for larger error-correcting codes where simulation is impossible, or to customise the ansatz for the specific quantum computing architecture. In this paragraph, we briefly outline such a possibility. The main difference from the classical implementation lies in the encoding and inverse encoding circuits. Instead of a (not fault-tolerant) encoding circuit, we require a reliable method for preparing encoded Pauli eigenstates on the quantum computer. Once we have these states, we can apply the logical operation ansatz in the same manner as the classical simulation-based implementation above. Additionally, we can optionally implement an error-correcting procedure after applying the logical operation ansatz. Finally, we need to replace the inverse encoding circuit by conducting logical measurements of the encoded states in the  $X$ ,  $Y$ , and  $Z$  bases. Using the outcomes of these measurements, we can perform logical state tomography [99, 100] on the output states. We can then compare the tomography results with the unencoded states obtained through classical simulation, using an analogous loss function to the one shown in equation Eq. (3.1).

It is useful to mention that the implementation of our method on a quantum computer requires minimal resources, as long as the number of logical qubits in the code is not too large. The necessary subroutines for execution on the quantum computer include preparing encoded Pauli eigenstates, applying a physical operation ansatz, and measuring Pauli observables. Here, there is a potential bottleneck in our proposed method as the number of required logical Pauli measurements grows exponentially with the number of logical qubits. To address this, we can potentially redefine the distance function  $F$  in the loss function (Eq. (3.1)) to be the infidelity between two density matrices and utilising a swap test [101, 102] for calculating the loss function. Specifically, another error-corrected quantum computer can be used to prepare the unencoded states, and the loss function in Eq. (3.1) can be calculated using a cross-device swap test.

## 3.4 Results for numerical experiments on small codes

We have used our procedure for a few quantum codes of  $n$  from 4 to 15 by numerical simulations on classical computers. In this section, we first provide a detailed description of the experimental configurations for all these experiments. We then summarise the resulting logical gates in Table 3.1. Notably, the result contains instances where non-Pauli logical gates were discovered through our experiment. We finally give a detailed description of the results of the experiments for the following codes:  $[[8,3,2]]$ ,  $[[8,2,2]]$ ,  $[[6,3,2]]$ , and  $[[7,1,3]]$  (also known as the surface code with a twist). The exact form of the new logical gates we found and their implementation using physical operators are deposited online [103].

### 3.4.1 Experiment configurations

Here we describe the ansatz, optimisation, simulation software, and the details about the tomographically complete set we used in the experiment.

**Ansatz for this experiment.** In general, we use the transversal ansatz for physical operations for most of the simulation experiments shown in Table 3.1, except for a few cases mentioned later. The choice of a transversal ansatz is motivated by its inherent fault tolerance. It involves using three single qubit rotation gates on each physical qubit, with each rotation gate having its own adjustable angle independent of the others. Fig. 3.2 provides an illustration of the transversal ansatz.

Note that transversal ansatz restricts the possible logical gates that would be found by the procedure. The quantum codes tested in this section can at least detect arbitrary one-qubit errors. But this ability to detect local errors means that all possible logical gates implemented using transversal physical operations (including their compositions!) form a finite discrete set due to a well-known no-go theorem [86]. Naturally, this set cannot cover the whole continuous set of all possible unitaries and is thus insufficient for universal quantum computation. Additional elements such as gate teleportation are needed for universal computation. Note also that one can form a universal gate set using very few common quantum gates (see discussions in Ross and Selinger [119], Shi [120], and Forest et al. [121]). Therefore, most likely very few logical gates could be found using a transversal ansatz.

For the parameterization of the logical gate  $g(\theta_2)$ , we adopt an ansatz

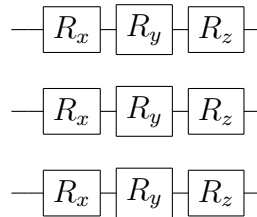
**Table 3.1:** Summary of quantum gates found by our procedure using classical simulation for a variety of small quantum error-correcting codes.

| Code<br>(Name and ref.)                                                                                                                                                           | Logical gate found by the procedure |                                                                                                                                                                                            |                                              |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
|                                                                                                                                                                                   | Pauli gates <sup>[a]</sup>          | Non-Pauli gates <sup>[b]</sup>                                                                                                                                                             | Percentage of non-Pauli gates <sup>[c]</sup> |
| [[4, 1, 2]] [104]<br>[[8, 2, 3]] [105]<br>[[8, 3, 3]] [106–108]<br>[[11, 5, 3]] [107]<br>[[12, 6, 3]] [105]<br>[[13, 7, 3]] [105]<br>[[14, 8, 3]] [105]<br>[[15, 7, 3]] [84, 107] | ✓                                   | None.                                                                                                                                                                                      | 0 %                                          |
| [[4, 2, 2]] [109, 110]                                                                                                                                                            | ✓                                   | CNOT                                                                                                                                                                                       | N/A                                          |
| [[5, 1, 2]] [111]                                                                                                                                                                 | ✓                                   | <b>S<sup>†</sup>, H</b>                                                                                                                                                                    | 71.4%                                        |
| [[5, 1, 3]] [112, 113]<br>(Five-qubit code)                                                                                                                                       | ✓                                   | $e^{i\pi/4}SH$ ,<br>$e^{i3\pi/4}XHX S^\dagger$ ,<br>$e^{-i3\pi/4}XHX S$ ,<br>$e^{-i\pi/4}HS^\dagger$ ,<br>$e^{-i\pi/4}S^\dagger H$ ,<br>$e^{-i3\pi/4}SXHX$ ,<br>$e^{i3\pi/4}S^\dagger XHX$ | 88.0%                                        |
| [[6, 3, 2]] [105]                                                                                                                                                                 | ✓                                   | <b>H<sub>12</sub>CZ<sub>12</sub>H<sub>12</sub></b>                                                                                                                                         | 60.0%                                        |
| [[7, 1, 3]] [84, 85]<br>(Steane code)                                                                                                                                             | ✓                                   | Generators for the group generated by $H$ and $S$                                                                                                                                          | N/A                                          |
| [[7, 1, 3]] [114]<br>(Surface code with a twist)                                                                                                                                  | ✓                                   | $e^{-i3\pi/4}SXHX$ ,<br>$e^{i3\pi/4}S^\dagger XHX$                                                                                                                                         | 11.1%                                        |
| [[8, 2, 2]] [115]<br>(Projective plane 2D colour code)                                                                                                                            | ✓                                   | $CZ, H^{\otimes 2}CZH^{\otimes 2}, H^{\otimes 2}SWAP$                                                                                                                                      | 20.0%                                        |
| [[8, 3, 2]] [116, 117]                                                                                                                                                            | ✓                                   | <b>CZ<sub>12</sub>, CZ<sub>13</sub>, CZ<sub>23</sub>, CCZ</b>                                                                                                                              | 71.4%                                        |
| [[10, 1, 2]] [111]                                                                                                                                                                | ✓                                   | <b>S<sup>†</sup>, T<sup>†</sup></b>                                                                                                                                                        | 51.6%                                        |

<sup>[a]</sup> Inside the column “Pauli gates”, a check mark (✓) indicates that we found a set of logical gates enough to generate all Pauli gates. <sup>[b]</sup> New logical gates are highlighted with **bold blue** text. Their details (parity check matrix of the code and the physical operations for the logical gate) are provided in OpenQASM [118] format online [103]. <sup>[c]</sup> In the column “Percentage of non-Pauli gates” we present the percentage of non-Pauli gates found among all optimisation runs. Repeated discoveries of the same logical gate count towards this percentage. Entries labelled to “N/A” are cases where the targeted logical gate were restricted to specific gates. See Section 3.3 for details.

that parameterises an arbitrary unitary transformation on the Hilbert space  $H_A$ . Note that the number of qubits in  $H_A$  is just  $k$ . When  $k = 1$ , the ansatz simply employs the three single qubit rotation gates mentioned as down in a transversal ansatz. When  $k = 2$ , the circuit depicted in Fig.2 in Shende et al. [122] is used. For  $k = 3$ , we employ a circuit parameterization for arbitrary three-qubit unitary gates, which is obtained using the Quantum Shannon Decomposition available as the QSD subroutine in the Universal Q Compiler [83]. This parameterization consists of 17 CNOT gates and 69 single qubit rotation gates, each with its own independently adjustable rotation angle. The visualisation for the circuit parameterization, with the 69 parameters labelled from 0 to 68, is available online [123]. For  $k > 3$ , we select only the first three qubits and apply the aforementioned three qubit ansatz. This choice is driven by the exponential growth in the number of possible ansatz circuits, which renders experimentation infeasible if we continue to use the same QSD subroutine.

The rationale for our choice of ansatz for different values of  $k$  is to give our procedure our maximum freedom in discovering logical gates. Consequently, the ansatz for the logical gate should always be able to parameterise arbitrary unitary gate on  $k$  qubits, without making any assumptions about the code's structure or the fault-tolerant logical gates it admits. In addition, our parameterization relies on a gate library consisting of single-qubit rotation gates and fixed two-qubit gates. This choice enables us to leverage the RotoSolve optimisation algorithm. Note that while we anticipate obtaining similar results with alternative parametrisations, provided they satisfy the condition required by the RotoSolve algorithm, we have not experimentally confirmed this.



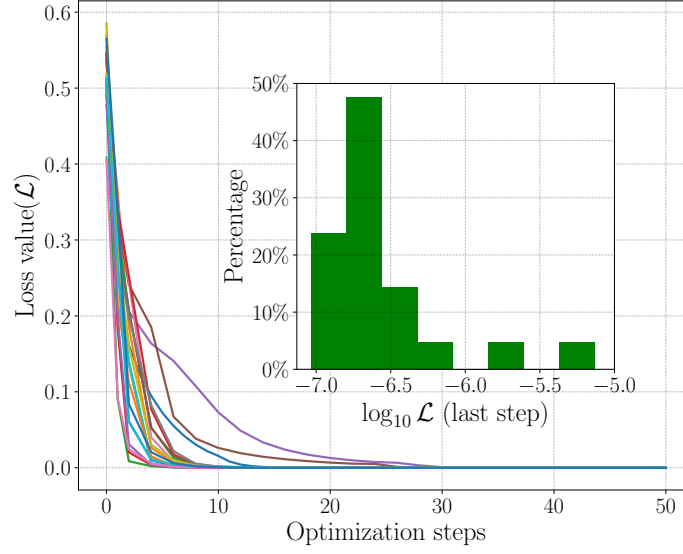
**Figure 3.2:** An illustration of a transversal ansatz on three qubits. It is formed by using three single qubit rotation gates ( $R_p = e^{-i\theta p/2}$ , where  $p$  is one of the Pauli matrices in  $\{X, Y, Z\}$ .) on each physical qubit. Each rotation gate has its own angle that can be adjusted independently of other rotation gates.

**Special cases for the ansatz.** For the experiments conducted on the  $[[4, 2, 2]]$  code, the encoder used only encodes one of the two logical qubits, therefore we focused the experiment on the only encoded qubit. Also, during a preliminary proof of concept experiment, we specifically targeted the logical gate CNOT for the  $[[4, 2, 2]]$  code in the experiment. In this preliminary experiment, we encoded a pair of physical qubits in two copies of the  $[[4, 2, 2]]$  code, with each code containing one encoded qubit. For the physical operations, we employed an ansatz parametrising arbitrary two-qubit gates [122], while acting on corresponding physical qubits in the different codes. With this ansatz, we successfully identified a transversal logical CNOT gate between the codes. The physical implementation of this logical gate involved also corresponds to CNOT gates between corresponding qubits in the different copies of the code. This preliminary experiment demonstrates the capability of our method to identify gates that operate between separate codes.

For the  $[[5, 1, 2]]$  code, we employed a non-transversal, yet fault-tolerant ansatz for the logical gate. Specifically, we utilised a transversal ansatz for the first three qubits, while employing a parameterization of arbitrary two-qubit unitary gates for the final two qubits. This choice of ansatz was motivated by the code's structure, as described in Vasmer and Kubica [111]. The fact that we were able to identify logical gates, including non-Pauli gates (see Table 3.1), using this tailored ansatz, demonstrates the effectiveness of our procedure with ansatz circuits designed for specific codes.

Similarly, for the  $[[10, 1, 2]]$  code, we employed a non-transversal, yet fault-tolerant ansatz for the logical gate. Specifically, we utilised a parameterization for arbitrary three-qubit unitary gates on qubits numbered 4, 6, and 8, obtained through the QSD decomposition in UniversalQCompiler [83, 123]. For the remaining qubits, we employed a transversal ansatz, as described earlier.

**Optimisation method used.** To minimise the function  $\mathcal{L}$ , we employed the algorithm RotoSolve [69], which has been introduced in Section 2.1.5. In our numerical experiments, the optimisation tends to be quite easy, without the common worry of being trapped into a local minimum. To confirm this, we generated a plot depicting the values of the loss function  $\mathcal{L}$  (Eq. (3.1)) for 21 optimisation runs for the  $[[5, 1, 2]]$  code. Each run started with random and independent initial parameters. The result is shown in Figure 3.3. From the figure, it is evident that the loss function rapidly converged towards its mini-



**Figure 3.3:** Loss function value as optimisation progress step-by-step for the experiments for the code  $[[5,1,2]]$ . The inset plot shows the percentage distribution of the logarithm of loss function value  $\mathcal{L}$  in the last optimisation step.

mum value under the RotoSolve optimisation algorithm. Notably, the values of the loss function after the final step fell within the range of  $(10^{-7}, 10^{-4})$ . For comparison, we calculated the loss function  $\mathcal{L}$  using the decomposition of known logical gates found in the literature, yielding  $\mathcal{L} \in (10^{-7}, 10^{-6})$ . Thus, we can conclude that optimisation has been successful when  $L \approx 10^{-7}$  or  $10^{-6}$ .

Further, in the case of  $L \in (10^{-5}, 10^{-4})$ , we believe that if the optimisation process were allowed to continue with additional iterations, the loss function could be further reduced. However, in this situation, we adopted a different, more economical approach. We halted the algorithm, extracted the last optimised angles to obtain numerical matrices for logical  $g$  and physical  $G$ , and determined the closest analytical expressions for them. Subsequently, we evaluated the loss function  $\mathcal{L}$  using these analytical expressions. Typically, we were able to find analytical expressions using common Clifford gates, which resulted in a loss function value of almost exact 0 when computed on simulators. This outcome demonstrates that further optimisation process in the last few iterations primarily corrects for numerical errors.

**Tomographically complete set used.** For calculating the cost function in Eq. (3.1), we need to provide the initial tomographically complete set of quantum states, since for any quantum channel  $\mathcal{E}$ , its action on any quantum state  $\rho$  can be uniquely determined by its action on a tomographically complete set  $\{|\psi_i\rangle\}$  [99].

If  $\{|\psi_i\rangle\}$  is such that the set  $\{|\psi_i\rangle\langle\psi_i|\}$  spans the vector space of all density matrices, then  $\{|\psi_i\rangle\}$  is a valid tomographically complete set. In our case, we only need a set  $\{|\psi_i\rangle\langle\psi_i|\}$  of one-qubit states such that it spans all  $2 \times 2$  density matrices, since for  $2^n \times 2^n$  density matrices representing  $n$ -qubit states, tensor products of the same set of one-qubit states will span the vector space of all  $n$ -qubit states.

Now we try to construct such a set from well-known quantum states. Consider the vector space of one-qubit density matrices as a subspace of  $\mathbb{C}^4$ . When a set  $S_1 = \{|\psi_i\rangle\langle\psi_i|\}$  spans the whole vector space  $\mathbb{C}^4$ , clearly it will also cover the smaller subspace of one-qubit density matrices. Then, it suffices to check whether the elements of the set, when considered as vectors in  $\mathbb{C}^4$ , could span the whole vector space  $\mathbb{C}^4$ .

Now we describe the six states, which we call the *six Bloch states*, and we use as the initial states for our experiments. We will show that they are tomographically complete. These states are:

$$S_1 = \{|0\rangle\langle 0|, |1\rangle\langle 1|, |+\rangle\langle +|, |-\rangle\langle -|, |+i\rangle\langle +i|, |-i\rangle\langle -i|\}. \quad (3.2)$$

It is easy to check that the six Bloch states, considered as six 4-dimensional vectors, span the whole  $\mathbb{C}^4$  space, by checking the rank of the matrix formed by putting each of the six density matrices into a column of the matrix. We can therefore use tensor products of the six Bloch states to construct the tomographically complete set for inputs to our loss function Eq. (3.1).

We note that with hindsight, we could have reduced the size of the tomographically complete set for  $n$ -qubits from  $6^n$  to  $4^n$ , by keeping only four out of the six Bloch states, i.e.  $\{|0\rangle\langle 0|, |1\rangle\langle 1|, |+\rangle\langle +|, |+i\rangle\langle +i|\}$ . It is easy to check that these remaining four states also span  $\mathbb{C}^4$ .

**Simulation software.** For performance purposes, we used the proprietary simulation code written by the author for Rahko Ltd.

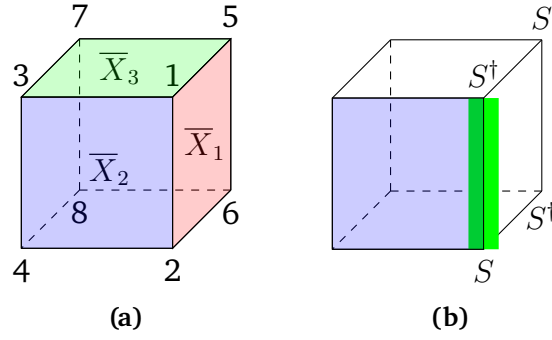
### 3.4.2 Notable logical gates found on small codes

Here we highlight a few resulting codes from Table 3.1, which will be interesting for readers familiar with quantum error correction codes.

**[[8,3,2]]:** The [[8,3,2]] code, which is the smallest non-trivial 3D colour code [116, 117]. We found three transversal Clifford gates for this code, which to the best of our knowledge has not been previously documented in the literature:  $CZ_{12}$ ,  $CZ_{13}$ , and  $CZ_{23}$ . Here  $CZ_{ij}$  represents a logical  $CZ$  gate acting on logical qubits  $i$  and  $j$ . Our optimisation procedure provides a straightforward implementation of these gates, which we will elaborate in the following.

Logical gates in the [[8,3,2]] code can be understood geometrically. Suppose we place qubits on the vertices of a cube, as shown in Fig. 3.4. The stabiliser group of the [[8,3,2]] code can be generated by Pauli- $X$  operators acting on every qubit, alongside operators that are associated with the faces of the cube, i.e., for each face,  $f$ , we have an operator  $\Pi_{v \in f} Z_v$ , where  $Z_v$  denotes a Pauli- $Z$  operator acting on the qubit on vertex  $v$ . With this definition, the logical  $X$  operators of the code are associated with the faces of the cube ( $\bar{X}_1 = \Pi_{v \in f} X_v$  etc.) and the logical  $Z$  operators are associated with the edges of the cube. We note that opposite faces support logical  $X$  operators that act on the same encoded qubit, and the corresponding logical  $Z$  operators are supported on the edges linking these faces.

To implement a logical  $\bar{CZ}_{ij}$  gate, we apply  $S = \text{diag}(1, i)$  and  $S^\dagger$  gates in an alternating pattern to the vertices of a face that supports  $\bar{X}_k$ . We show this operator in Fig. 3.4b and we denote it by  $U$ . We now show that  $U$  implements a  $\bar{CZ}_{ij}$  gate: namely it maps  $\bar{X}_i$  to  $\bar{X}_i \bar{Z}_j$ , and has no effect on all other logical operators. First, we note that all operators that consist exclusively of Pauli- $Z$  operators are unaffected by  $U$  because  $S$  and  $Z$  commute. Therefore,  $Z$ -type stabilisers and logical  $Z$  operators are mapped to themselves by  $U$ . Next, we consider operators that contain Pauli- $X$ , which are transformed by  $S$  as follows:  $SXS^\dagger = Y = iXZ$ . Consider the  $X$  stabiliser of the code ( $X$  on all the qubits). It is straightforward to see that  $U$  maps this operator to a product of itself and a  $Z$  stabiliser associated with the face where we applied  $U$  (the alternating pattern causes the factors of  $i$  and  $-i$  to cancel). Finally, consider  $\bar{X}_i$  (the blue face in Fig. 3.4b). This operator is mapped to a product of itself and a  $\bar{Z}_j$  operator with support on an edge that links the  $\bar{X}_j$  faces (the green edge in Fig. 3.4b). Similarly,  $\bar{X}_j$  is mapped to  $\bar{X}_j \bar{Z}_i$ .

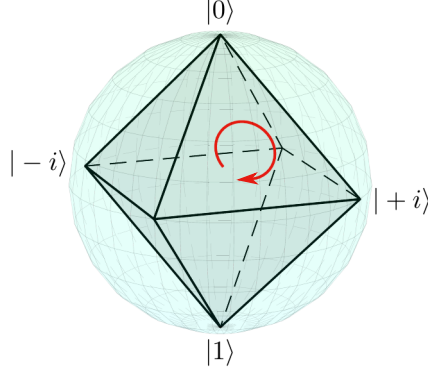


**Figure 3.4:**  $CZ$  gates in the  $[[8,3,2]]$  code. Qubits are placed on the vertices of the cube. In (a), we highlight three logical  $X$  operators of the code, namely  $\bar{X}_1 = X_1 X_2 X_5 X_6$  (red face),  $\bar{X}_2 = X_1 X_2 X_3 X_4$  (blue face), and  $\bar{X}_3 = X_1 X_3 X_5 X_7$  (green face). Three corresponding logical  $Z$  operators are:  $\bar{Z}_1 = Z_1 Z_3$ ,  $\bar{Z}_2 = Z_1 Z_5$ , and  $\bar{Z}_3 = Z_1 Z_2$ . (b) Suppose that we apply  $S$  and  $S^\dagger$  to the qubits on the red face. This unitary maps  $\bar{X}_2$  (blue face) to  $\bar{X}_2 \bar{Z}_3$  (blue face and green edge). Likewise,  $\bar{X}_3$  is mapped to  $\bar{X}_3 \bar{Z}_2$ . Therefore, this unitary implements a logical  $\bar{CZ}_{23}$  gate.

**Transversal non-Clifford gates** Furthermore, our procedure successfully identified a transversal implementation of the non-Clifford gate  $CCZ$  for the  $[[8,3,2]]$  code. While this gate has been previously discovered [116, 117], it is noteworthy that our procedure found it without any prior knowledge about the code's structure. Implementing Clifford gates and Pauli measurements in error-correcting codes is often straightforward. Note that although these operations can be simulated classically, they can achieve universality by incorporating a single non-Clifford gate. However, codes with transversal non-Clifford gates are rare, and understanding their structure is an active area of research [115, 116, 124–126]. Most constructions of code families with non-Clifford gates exploit specific structures of the code family, such as tri-orthogonality [127–129]. Therefore, our procedure can potentially discover fault-tolerant non-Clifford gates for codes whose structure is currently unknown. In particular, the structure of non-CSS codes with non-Clifford gates remains poorly understood, and our procedure could provide valuable insights in this area with improved computational resources.

**$[[8,2,2]]$ :** We successfully found non-Pauli gates for codes related to 2D colour codes: the  $[[8,2,2]]$  code (2D colour code defined on a projective plane [115]) and the  $[[6,3,2]]$  code (sub-code of a 2D colour code defined on a hexagon [130]). For the  $[[8,2,2]]$  code, we found the gates that we would expect to find in a 2D colour code with two logical qubits: logical

CZ implemented by transversal  $S$  and  $S^\dagger$ , logical  $H^{\otimes 2}CZH^{\otimes 2}$  implemented by transversal  $\sqrt{X}^\dagger$  and logical  $H^{\otimes 2}\text{SWAP}$  implemented by transversal  $H$ . And we found that the  $[[6,3,2]]$  code inherits one of the transversal gates of its parent 2D colour code (the  $[[6,4,2]]$  code [130]):  $CZ_{12}$  implemented by transversal  $S$  and  $S^\dagger$ .



**Figure 3.5:** An illustration of the Clifford gate  $K_{1,1,1} = e^{i\pi/4}SH$ . This gate maps  $|0\rangle \rightarrow |+i\rangle$ ,  $|+i\rangle \rightarrow |+\rangle$ , and  $|+\rangle \rightarrow |0\rangle$  (up to global phases). It can be understood geometrically by considering an octahedron embedded in the Bloch sphere. The operation performed by the gate is a clockwise rotation of  $2\pi/3$  around the axis normal to the face marked with a red circle. The  $|+\rangle$  state is at the front unlabelled vertex of the octahedron.

**[[7,1,3]]:** Our procedure also found transversal Clifford gates for the  $[[7,1,3]]$  surface code with a twist. As was noted in [114], this code has transversal implementations of the octahedral Clifford gates, which are defined as follows:

$$K_{x,y,z} = \exp\left\{i\frac{\pi}{3\sqrt{3}}(xX + yY + zZ)\right\}, \quad (3.3)$$

where  $x, y, z \in \{-1, 1\}$  (See Fig. 3.5 for an illustration of  $K_{1,1,1}$ ). These gates cyclically permute the Pauli operators. We can decompose these octahedral gates into products of more familiar Clifford gates as follows:

$$\begin{aligned} K_{1,1,1} &= e^{i\pi/4}SH, & K_{-1,-1,-1} &= e^{-i\pi/4}HS^\dagger, \\ K_{1,-1,1} &= e^{i\pi/4}HS, & K_{-1,1,-1} &= e^{-i\pi/4}S^\dagger H, \\ K_{1,1,-1} &= e^{i3\pi/4}XHX S^\dagger, & K_{-1,-1,1} &= e^{-i3\pi/4}SXHX, \\ K_{-1,1,1} &= e^{-i3\pi/4}XHX S, & K_{1,-1,-1} &= e^{i3\pi/4}S^\dagger XHX. \end{aligned} \quad (3.4)$$

The specific gates we found in our experiments were transversal realisations of  $K_{-1,-1,1}$  and  $K_{1,-1,-1}$ .

### 3.5 Discussion and Conclusion

In this section, we will explore several interesting characteristics of our approach and highlight possible avenues for future research.

**Fault-tolerance of the logical gate.** In general, the ansatz for logical circuit  $G$  (Fig. 3.1) is not guaranteed to be fault-tolerant. To restore the quantum state to the code space after the logical circuit, we employ an additional stabiliser measurement and minimal weight error correction procedure,  $C$ , in our simulation. To make it suitable for a quantum computer, it requires a version of  $C$  using a fault-tolerant error correction protocol [131–134] followed by a recovery operator based on the error syndrome.

Interestingly, throughout our numerical experiments, we find that our procedure yields fault-tolerant logical gates  $G$  when the loss function is minimised. This has been verified by computing the effect of  $G$  on the logical states.

Nevertheless, it is possible that, for other error correction codes, there may be cases where the combined action of  $G$  and  $C$  forms a logical gate (after optimisation), even if  $G$  alone does not. Regrettably, we have not encountered such an example in this study. But we argue that the combined action of  $G$  and a fault-tolerant version of  $C$  may be considered a valid fault-tolerant logical gate, since in almost all constructions of fault-tolerant computations, error correction is applied immediately after a logical gate.

**Resource estimate of our method.** Our method has limitations when it comes to scaling to larger codes. To calculate the loss function in equation Eq. (3.1), we either need to perform full process tomography of the encoded logical space, which requires a minimum of  $\mathcal{O}(4^k)$  quantum circuits, or find a way to implement a noise-resilient cross-device swap test between two quantum computers.

Despite these limitations, we believe that our method has potential applications. Many families of quantum codes, such as surface codes, colour codes, and quantum Reed-Muller codes, have a constant and mostly small value of  $k$  [135, 136]. Even for codes with a growing  $k$  as the number of physical qubits  $n$  increases, there is often a symmetry that relates the logical operators. This means that we only need to apply our method to a single representative with respect to this symmetry, effectively reducing  $k$  in proportion to the size of the symmetry group.

Therefore, the cost of our method is benign when  $k$  is small and can be

further mitigated by running the circuits in parallel. It is worth noting here that in the classical simulations we conducted, one round of optimisation usually converged in less than 5 hours when running on a commercially available GPU (Nvidia RTX 2080 Ti).

Additionally, since the transversal ansatz is likely to be used as the ansatz for logical gates, we expect a linear growth in the number of parameters that need to be optimised with respect to the number of physical qubits  $n$  in Eq. (3.1).

**Potential future research.** Our optimisation approach benefits from the utilisation of the RotoSolve [69] optimisation algorithm. This algorithm relies on the sinusoidal functional dependency of the loss function in Eq. (3.1), which is a result of the parameterised rotation gates used in our ansatz circuits (see Section 3.4.1 for more details and numerical performance). However, it is important to note that adapting our procedure for specific quantum computation hardware may require a different set of parameterised gates, which could render the use of RotoSolve invalid. In such cases, alternative optimisation algorithms will be necessary to find the logical gates.

Furthermore, we suggest considering methods to penalise Pauli gates or other logical gates that have been previously discovered during the optimisation of the loss function in Eq. (3.1). Again, introducing such a penalty would violate the requirements of the RotoSolve algorithm, and therefore, we hesitated to explore this approach in our experiment. For reference, we have estimated the percentage of non-Pauli gates that we would expect to observe. To do so, we assume that all gates have equal probability, and compared the number of Pauli gates generated with the total number of logical gates generated. The details are shown in Table 3.1. Although the table shows that the majority of gates found were Pauli gates, we note that the number of experiments conducted for each code is relatively small (less than or equal to 40), which limits our ability to ascertain that there is a bias towards finding Pauli gates (Table 3.1).

Finally, an interesting avenue for future research would be to apply our procedure to explore quantum codes that have not been extensively studied, such as qudit quantum codes. Qudits are the  $d$ -dimensional analogues of qubits, and the stabiliser formalism can be extended to prime (or prime power) qudits [137]. This means that it would be straightforward to generalise our procedure to these cases. There has been considerably less research

on implementing logical gates in qudit stabiliser codes compared to qubit stabiliser codes, so it is possible that we may discover more unknown fault-tolerant gates in the qudit context. Additionally, it is worth noting that our procedure is not limited to stabiliser codes and can be applied to non-stabiliser codes, such as the codes described in Cross et al. [138].

In conclusion, we have presented a method to automatically discover logical gates using shallow quantum circuits for a given quantum error-correcting code. The design of the logical gate can be customised to different quantum computing architectures in order to exploit their advantages. Our results demonstrate that the proposed method successfully identifies existing logical gates from the current literature for various error-correcting codes. Moreover, it generates novel logical gates for the  $[[5,1,2]]$  code, the  $[[6,3,2]]$  code, the  $[[8,3,2]]$  code, and the  $[[10,1,2]]$  code. Although the procedure is simulated using classical methods, we have also proposed an extension of this method for quantum computers. We are hopeful that an implementation of this procedure on near-term quantum computers is possible, particularly for error-correcting codes on a small number of physical qubits.

## Chapter 4

# Variational quantum algorithms for Green functions

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In this chapter, we explore the application of variational quantum algorithms (VQA) in addressing the technical challenges in the field of material science and quantum chemistry. Specifically, we investigate how VQAs can be utilised to compute the Green function, which plays a crucial role in understanding the dynamics of quantum mechanical models. Despite the widespread usage of the Green function, its exact computation is challenging due to the exponential increase in cost as the problem size grows. Many quantum algorithms offer a promising solution to overcome this exponential cost. In this chapter, we concentrate on variational quantum algorithms for solving this Green function problem.

**Chapter outline.** We begin by introducing the Green function and its various applications in Section 4.1. In this section, we will provide an overview of basic concepts and formulae, while leaving detailed explanations to the Appendix B. In Section 4.2, we outline the challenges associated with the computation of the Green function and explore several quantum algorithms proposed to mitigate this cost. The algorithms discussed include both non-variational and variational ones. Subsequently, in Section 4.3, we present our variational algorithms based on the correction vector. As a demonstration, we conducted numerical experiments based on our algorithm on simulated quantum computers (Section 4.4). Finally, we examine potential issues with our method and point to opportunities for further research in Section 4.5.

## 4.1 Green function

Green functions form the basis of quantum mechanical studies of many-particle systems. The idea originates from studying point-like sources for various electromagnetic fields [139] and has since been abstracted into mathematical studies of partial differential equations. Later it was extended to field theories and then to many-particle systems. See the seminal paper of Martin and Schwinger [140] (reprinted [141, p.423]) and the references cited therein.

Numerous other fields thus define their own Green functions [142, 143]. Such wide applications make the definition of the Green function sometimes confusing. Here, we limit our scope below with a precise but perhaps unintuitive definition of the Green function. We provide in Appendix B the proofs of the formulae presented below.

**Definition.** We restrict our attention to the Green function of fermionic particles at zero-temperature. We focus on the zero-temperature to align with most of the literature reviewed (Section 4.2) and also our algorithm in Section 4.3.

We begin with a fermionic Hamiltonian  $H$  written in the second quantisation formalism,

$$H = \sum_{\alpha\beta} h_{\alpha\beta} c_{\alpha}^{\dagger} c_{\beta} + \frac{1}{2} \sum_{\alpha\beta\gamma\delta} g_{\alpha\beta\gamma\delta} c_{\alpha}^{\dagger} c_{\beta}^{\dagger} c_{\delta} c_{\gamma} - \mu \sum_{\alpha} c_{\alpha}^{\dagger} c_{\alpha}. \quad (4.1)$$

Here  $\alpha, \beta, \gamma, \delta$  represent different fermionic modes (including orbitals and spins),  $c_{\alpha}$  is the fermionic annihilation operator,  $c_{\alpha}^{\dagger}$  is the fermionic creation operator,  $h_{\alpha\beta}$  and  $g_{\alpha\beta\gamma\delta}$  are one- and two-body matrix elements between the

corresponding degrees of freedom, and  $\mu$  is the chemical potential.

Then, the *retarded Green function* in time is defined as the expectation value:

$$G_{\alpha\beta}^R(t) = -i\Theta(t)\langle\{c_\alpha(t), c_\beta^\dagger(0)\}\rangle, \quad (4.2)$$

where  $\Theta(t)$  is the Heaviside step function:  $\Theta(t) = 1$  if  $t \geq 0$ , or 0 otherwise. The operators evolve as  $o(t) = e^{iHt} o e^{-iHt}$ , where  $o$  is either  $c_\alpha$  or  $c_\beta^\dagger$ .

Also, the *imaginary-time Green function* can be defined as the expectation value of two operators evolved in the imaginary-time:

$$G_{\alpha\beta}(\tau) = -\Theta(\tau)\langle c_\alpha(\tau) c_\beta^\dagger(0) \rangle + \Theta(-\tau)\langle c_\beta^\dagger(0) c_\alpha(\tau) \rangle \quad (4.3)$$

where the operators evolved in imaginary-time are  $o(\tau) = e^{\tau H} o e^{-\tau H}$  for  $o = c_\alpha / c_\beta^\dagger$ .

Here, we should clarify what *expectation value* means in the definitions above. Since we focus on the zero-temperature scenario, the physical system will be the pure ground state of the Hamiltonian  $H$ . Denoting this ground state as  $|\psi_0\rangle$ , we can then explicitly rewrite the expectation values above for the retarded  $G^R(t)$  and the imaginary-time  $G(\tau)$  into:

$$\begin{aligned} G_{\alpha\beta}^R(t) &= -i\Theta(t)\langle\psi_0|c_\alpha(t)c_\beta^\dagger(0) + c_\beta^\dagger(0)c_\alpha(t)|\psi_0\rangle, \\ G_{\alpha\beta}(\tau) &= -\Theta(\tau)\langle\psi_0|c_\alpha(\tau)c_\beta^\dagger|\psi_0\rangle + \Theta(-\tau)\langle\psi_0|c_\beta^\dagger c_\alpha(\tau)|\psi_0\rangle, \end{aligned} \quad (4.4)$$

Next, we present the expansion in an eigenbasis and the Fourier transform of the two Green functions. We first deal with the retarded Green function, then deal with the imaginary-time Green function. We finally summarise the Fourier transform of the two Green functions in a unified expression. The presentation is limited to the results only and leaves the detailed calculations to the Appendix B.

**Retarded Green function.** Here a set of eigenstates  $\{|\psi_m\rangle\}_m$  of the Hamiltonian  $H$  is used to expand the retarded Green function. Let  $E_m$  denotes the corresponding energies ( $H|\psi_m\rangle = E_m|\psi_m\rangle$ ), then we can use rewrite  $G^R(t)$  in this basis as:

$$\begin{aligned} G_{\alpha\beta}^R(t) &= -i\Theta(t) \sum_m \left( e^{-it(E_m - E_0)} \langle\psi_0|c_\alpha|\psi_m\rangle \langle\psi_m|c_\beta^\dagger|\psi_0\rangle \right. \\ &\quad \left. + e^{it(E_m - E_0)} \langle\psi_0|c_\beta^\dagger|\psi_m\rangle \langle\psi_m|c_\alpha|\psi_0\rangle \right), \end{aligned} \quad (4.5)$$

which when  $\alpha = \beta$ , simplifies into:

$$G_{\alpha}^{\text{R}}(t) = -i\Theta(t) \sum_m \left( e^{-it(E_m - E_0)} |\langle \psi_m | c_{\alpha}^{\dagger} | \psi_0 \rangle|^2 + e^{it(E_m - E_0)} |\langle \psi_m | c_{\alpha} | \psi_0 \rangle|^2 \right). \quad (4.6)$$

We can Fourier transform this Green function into the frequency domain. However, an ordinary Fourier transform would not work due to the oscillating factor of the form  $e^{iEt}$ . In order to formally evaluate the Fourier transform of  $G^{\text{R}}(t)$ , a regularisation factor of  $\lim_{\eta \rightarrow 0^+} e^{-\eta|t|}$  is included. The regularised Fourier transform will take the form

$$G_{\alpha\beta}^{\text{R}}(\omega) = \int_{-\infty}^{\infty} G_{\alpha\beta}^{\text{R}}(t) e^{i\omega t - \eta|t|} dt, \quad (4.7)$$

where the regularisation can be understood as a small imaginary shift of the target frequency  $\omega$ . The real-frequency retarded Green function is then obtained as:

$$G_{\alpha\beta}^{\text{R}}(\omega) = \sum_m \left( \frac{\langle \psi_0 | c_{\alpha} | \psi_m \rangle \langle \psi_m | c_{\beta}^{\dagger} | \psi_0 \rangle}{\omega + i\eta - (E_m - E_0)} + \frac{\langle \psi_0 | c_{\beta}^{\dagger} | \psi_m \rangle \langle \psi_m | c_{\alpha} | \psi_0 \rangle}{\omega + i\eta + (E_m - E_0)} \right), \quad (4.8)$$

where we similarly used the eigenbasis  $\{|\psi_m\rangle\}$  of  $H$ , and  $E_0$  is the energy of the ground state  $|\psi_0\rangle$ . This is called the *Lehmann representation* of the retarded Green function. When  $\alpha = \beta$ , it simplifies into

$$G_{\alpha}^{\text{R}}(\omega) = \sum_m \frac{|\langle \psi_m | c_{\alpha}^{\dagger} | \psi_0 \rangle|^2}{\omega + i\eta - (E_m - E_0)} + \frac{|\langle \psi_m | c_{\alpha} | \psi_0 \rangle|^2}{\omega + i\eta + (E_m - E_0)}. \quad (4.9)$$

**Imaginary-time Green function.** Similarly, we can introduce the corresponding representations to the imaginary-time Green function using the eigenbasis, which leads to:

$$\begin{aligned} G_{\alpha\beta}(\tau) = & -\Theta(\tau) \sum_m e^{-\tau(E_m - E_0)} \langle \psi_0 | c_{\alpha} | \psi_m \rangle \langle \psi_m | c_{\beta}^{\dagger} | \psi_0 \rangle \\ & + \Theta(-\tau) \sum_m e^{\tau(E_m - E_0)} \langle \psi_0 | c_{\beta}^{\dagger} | \psi_m \rangle \langle \psi_m | c_{\alpha} | \psi_0 \rangle. \end{aligned} \quad (4.10)$$

When  $\alpha = \beta$ , it simplifies into,

$$G_{\alpha}(\tau) = \sum_m -\Theta(\tau) e^{-\tau(E_m - E_0)} |\langle \psi_m | c_{\alpha}^{\dagger} | \psi_0 \rangle|^2 + \Theta(-\tau) e^{\tau(E_m - E_0)} |\langle \psi_m | c_{\alpha} | \psi_0 \rangle|^2. \quad (4.11)$$

We can also Fourier transform this representation into,

$$G_{\alpha\beta}(i\omega_n) = \sum_m \frac{\langle\psi_0|c_\alpha|\psi_m\rangle\langle\psi_m|c_\beta^\dagger|\psi_0\rangle}{i\omega_n - (E_m - E_0)} + \frac{\langle\psi_0|c_\beta^\dagger|\psi_m\rangle\langle\psi_m|c_\alpha|\psi_0\rangle}{i\omega_n + (E_m - E_0)}, \quad (4.12)$$

where  $\omega_n = (2n+1)\pi/\beta$  for  $n \in \mathbb{N}$ , and  $\beta$  is a fictitious inverse-temperature introduced during the Fourier transform. We emphasise two things here. First, the inverse temperature term and the frequency are actually motivated by the Green function defined for finite temperature. Second, the frequencies are called Matsubara frequencies and  $G_{\alpha\beta}(i\omega_n)$  as a whole is also called (zero-temperature) Matsubara functions. Lastly, the  $n$  here is unrelated to the  $m$  in the eigenbasis  $|\psi_m\rangle$ .

**Unifying different expressions.** Finally, we introduce a simple expression which unifies the different representations discussed above. It is clear that the Fourier transform of retarded Green function and imaginary-time Green function can be both expressed by a single expression of the form:

$$G_{\alpha\beta}(z) = \sum_m \frac{\langle\psi_0|c_\alpha|\psi_m\rangle\langle\psi_m|c_\beta^\dagger|\psi_0\rangle}{z - (E_m - E_0)} + \frac{\langle\psi_0|c_\beta^\dagger|\psi_m\rangle\langle\psi_m|c_\alpha|\psi_0\rangle}{z + (E_m - E_0)}. \quad (4.13)$$

In the case of  $z = i\omega_n$ ,  $G_{\alpha\beta}(z)$  reduces to the Matsubara Green function, and when  $z = \omega + i\eta$ ,  $G_{\alpha\beta}(z)$  reduces to the Fourier transform of retarded Green function. Also, we can equivalently write the expression above succinctly as:

$$G_{\alpha\beta}(z) = \langle\psi_0|c_\alpha \frac{1}{z - (H - E_0)} c_\beta^\dagger|\psi_0\rangle + \langle\psi_0|c_\beta^\dagger \frac{1}{z + (H - E_0)} c_\alpha|\psi_0\rangle, \quad (4.14)$$

## 4.2 Computing Green function

Computing the Green function is computationally expensive in general. First, based on the formulae outlined in the previous section, the Green function contains contributions from all eigenstates of the given Hamiltonian, and therefore, intuitively one needs to diagonalise the Hamiltonian for its eigenstates and energies first to compute the Green function. Solving this Hamiltonian problem is costly as the Hamiltonian's dimension grows exponentially in the orbitals involved. In this section, we take a brief look at approaches developed for both classical and quantum computers for computing Green functions.

### 4.2.1 Algorithm on classical computers

Here we provide an outline of the different classical approaches that can be used to probe the excitation spectrum of a Hamiltonian. These approaches generate vectors (ground and excitation states) and matrices (effective Hamiltonian and other operators) related to the excitation spectrum of the given Hamiltonian, which can then be used to compute the Green function. We summarise their ideas and provide references for the more detailed literature.

**Exact diagonalisation (ED).** In ED, we attempt to diagonalise the Hamiltonian exactly or find a few important eigenstates and eigenvalues gradually. Among many ED approaches, the *Lanczos method* is often used for computing Green functions. Lanczos method applies the Hamiltonian repeatedly on a reference initial state to iteratively construct a basis, called *Lanczos basis*, so that the original Hamiltonian, when projected on this basis, becomes tri-diagonal. In principle, we can extract the eigenvalues and Green function from this basis exactly. In practice, we only have the first few basis vectors available from the first few iterations, and extract approximations of the eigenvalues and Green functions from them. Detailed discussion is available in Sandvik et al. [144].

**Quantum Monte Carlo (QMC).** QMC is based on the statistical approach to sample large probability distributions called Markov chain Monte Carlo (MCMC) [145]. Application of this method for studying classical Ising Models rests on the principle of *detailed balance* of the system at equilibrium. QMC extends MCMC methods on classical Ising models to their quantum versions and has favourable scaling compared to ED methods above. However, for fermionic problems without particular symmetries, the QMC approach suffers from a “sign problem” which is proven to be NP-Hard to resolve [146].

**Tensor Network.** Tensor networks [147] are particular variational ansatz for representing quantum states inspired by their local entanglement structure. A variety of such ansatz has been widely developed and tested in various different applications but with a mixed set of results [148]. Also, tensor network methods are notoriously difficult for real-time evolution, thus making it difficult to study dynamics. This is limited by the growth of the entanglement with time for real-time evolution, requiring increasingly larger bond dimensions and therefore more computational costs [149].

**Neural Quantum State (NQS).** NQS started as a way to apply the generative machine learning method Restricted Boltzmann Machine (RBM) to studying Ising models. It has recently been extended as neural network states with many generalisations [150, 151]. NQS is deeply connected with tensor network [152] as well as quantum circuits [153]. As an ansatz, it has been applied with success to the study of the ground state of certain many-body systems [150]. However, it suffers from convergence and stability issues, especially when doing the real-time evolution based on time-dependent variational principles (though recent developments [154] are available).

## 4.2.2 Algorithms on quantum computers

In contrast, although detailed benefits and costs of quantum Green function algorithms have not been analysed, it is widely hoped that quantum methods will provide future capabilities to compute Green functions without the exponential cost.

Inspired by this, numerous methods have been introduced in the literature for calculating the Green function on quantum computers. In this section, we review the relevant literature to provide a border perspective. Readers interested in our method could skip to Section 4.3. This review follows the following organisation: we separately discuss algorithms depending on whether they compute the Green function in the time or the frequency domain.

### 4.2.2.1 Algorithms in time domain

Here we review the method to compute the retarded Green function  $G^R(t)$  (Eq. (4.2)). The Green function in the time domain is explicitly:

$$\begin{aligned} & -i\Theta(t)\langle\{c(t), c^\dagger(0)\}\rangle \\ & = -i\Theta(t)\left(\langle\psi_0|e^{iHt}ce^{-iHt}c^\dagger|\psi_0\rangle + \langle\psi_0|c^\dagger e^{iHt}ce^{-iHt}|\psi_0\rangle\right). \end{aligned} \quad (4.15)$$

Assume we have obtained a quantum circuit representation of the ground state  $|\psi_0\rangle$  using for example the VQE algorithm. Then, after a Jordan-Wigner transformation, computing the Green function in time domain would be equivalent to evaluating terms of the following form on quantum computers:

$$\langle\psi_0|e^{iHt}P_1e^{-iHt}P_2|\psi_0\rangle, \text{ or } \langle\psi_0|P_1e^{iHt}P_2e^{-iHt}|\psi_0\rangle, \quad (4.16)$$

where  $P_1$  and  $P_2$  are  $n$ -qubit Pauli operator from the Jordan-Wigner transformation. Evaluating Eq. (4.16) has been discussed in Kreula et al. [155, 156],

Bauer et al. [157], Keen et al. [158], Endo et al. [159]. For simplicity, we review here briefly quantum algorithms to compute the first term, since the second term can be evaluated using similar strategies.

First, we find a circuit to simulate the time evolution  $e^{\pm iHt}$ . This can be done by either a Trotterization method or a variational method. Trotterization is a popular technique in quantum simulations of materials [27] which slices the overall time evolution into small time steps where each time step is approximated by first-order Trotter-Suzuki approximations [160, 161]. Such a method can be directly integrated for computing Green function [155–158].

However, Trotterization tends to create deep circuits and is therefore not friendly to near-term quantum computation. Endo et al. [159] suggests replacing the non-variational Trotterization by the variational quantum simulation algorithm [21, 162] which uses ansatz circuits to approximate time evolution. Specifically, McLachlan’s variational principle is used to relate the time evolution of the state to the evolution of the ansatz’s parameters. Coefficients in the equation governing this evolution can be evaluated on quantum computers [21].

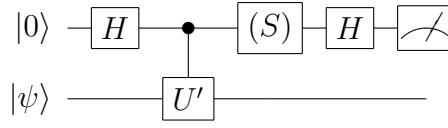
Next, having obtained such a circuit for time evolution simulation, we review different methods to evaluate the terms in the equations above (Eq. (4.16)). Note that, for convenience, we will denote the circuit which approximates the evolution  $e^{-iHt}$  as  $U_t$ .

In Bauer et al. [157], the terms can be further simplified as

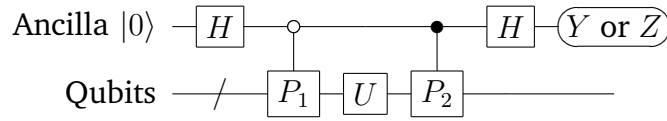
$$\langle \psi_0 | U_t^\dagger P_1 U_t P_2 | \psi_0 \rangle = e^{iE_0 t} \langle \psi_0 | P_1 U_t P_2 | \psi_0 \rangle, \quad (4.17)$$

where time evolution has been partly extracted into a phase factor, and the ground state energy  $E_0$  in the phase factor should be readily available from the ground state preparation procedure. The rest of the expression can then be easily computed with the standard Hadamard test described by the circuit in Fig. 4.1. Details are available in the appendix for various Hadamard tests (Appendix C).

However, requiring a controlled- $U_t$  gate inside the full quantum circuit isn’t ideal, which is made worse by the fact that  $U_t$  usually contains a significant amount of quantum gates. Therefore, a simplified circuit has been proposed in the literature [155, 156, 158, 159], which only requires executing controlled- $P_1$  and controlled- $P_2$  gates. Specifically, it is shown that the alternative circuit in Fig. 4.2 can be used to evaluate  $\langle \psi_0 | U_t^\dagger P_1 U_t P_2 | \psi_0 \rangle$  di-



**Figure 4.1:** Quantum circuit used to evaluate  $\langle \psi_0 | P_1 U_t P_2 | \psi_0 \rangle$  in Eq. (4.16). Here, when  $S$  gate is not executed, measuring the ancilla qubit in  $Z$  basis gives  $\text{Re} \langle \psi | U' | \psi \rangle$ . When  $S$  gate is executed, measurement instead produces  $-\text{Im} \langle \psi | U' | \psi \rangle$ . The gate  $U'$  is an abbreviation for  $U' = P_1 U_t P_2$ . Details of this circuit are discussed in the Appendix C.



**Figure 4.2:** Quantum circuit to measure  $\langle \psi_0 | U_t^\dagger P_i U_t P_j | \psi_0 \rangle$ . Upon measurement of the ancilla qubit in either  $Y$  or  $Z$  basis, the above circuit would give either the real or imaginary component of the desired value.

rectly. Proofs regarding this circuit are also available in the Appendix C.2.

#### 4.2.2.2 Algorithms in the frequency domain

Quantum algorithms for the Green function in the frequency domain can be largely divided into (i) direct methods, which compute the Green function in its Lehmann representation, (ii) those based on the classical Lanczos methods, and (iii) those based on the correction vector.

**Direct method.** Here we work with the unified representation expanded in the eigenbasis, i.e. the Lehmann representation (Eq. (4.13)). Sometimes only the diagonal part of the Green function is needed ( $\alpha = \beta$ ), which simplifies Eq. (4.13) into

$$G(z) = \sum_m \frac{|\langle \psi_m | c_\alpha^\dagger | \psi_0 \rangle|^2}{z - (E_m - E_0)} + \frac{|\langle \psi_m | c_\alpha | \psi_0 \rangle|^2}{z + (E_m - E_0)}. \quad (4.18)$$

We will start with a non-variational quantum algorithm. Wecker et al. [163] notes that in real frequency, the Fourier transform can be understood without the normalising factor  $\eta$ . Instead, we have Dirac  $\delta$  function as in:

$$\int_{-\infty}^{\infty} dt e^{i\omega t} \langle \psi_0 | A^\dagger(t) A(0) | \psi_0 \rangle = \sum_m |\langle \psi_m | A | \psi_0 \rangle|^2 \delta(\omega - (E_m - E_0)), \quad (4.19)$$

where  $A$  is either  $c_\alpha$  or  $c_\beta^\dagger$ . In general,  $G(z)$  becomes summation of  $\delta$ -functions weighted by factors  $|\langle\psi_m|c_\alpha^\dagger|\psi_0\rangle|^2$  or  $|\langle\psi_m|c_\alpha|\psi_0\rangle|^2$ .

Then, we can apply the quantum phase estimation algorithm repeatedly to the same initial state  $A|\psi_0\rangle$ . Repeated measurements would return each energy  $E_m$  with the probability  $|\langle\psi_m|A|\psi_0\rangle|^2$ . After a certain amount of repeated applications, the right-hand side of Eq. (4.19) can be obtained (up to certain numerical accuracies) and Fourier transformed back to the left-hand side.

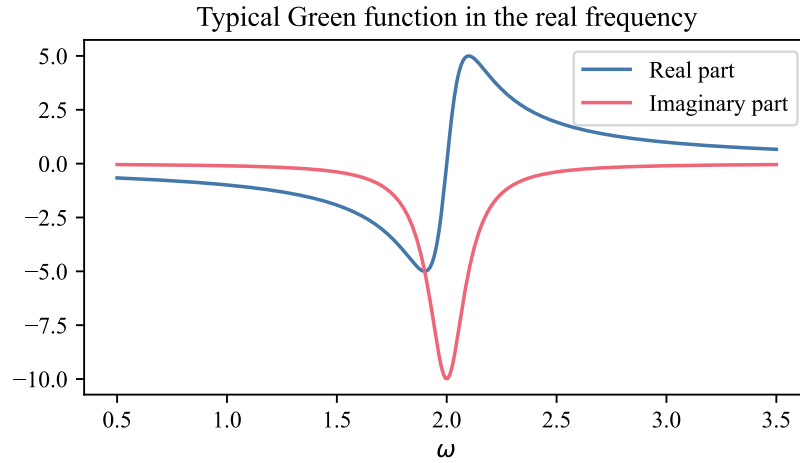
Note that preparation of the state  $A|\psi_0\rangle$  for phase estimation can be achieved with ground state preparation algorithm and the so-called linear combination of unitaries (LCU) subroutine<sup>1</sup>, since  $A$  is usually (after a Jordan-Wigner transformation) a sum of unitary operators.

Next, we turn to variational quantum algorithms. Excited-state VQE algorithms (see Section 2.2) can usually provide us with the energies of the first few excited states and quantum circuits to prepare the corresponding eigenstate. Endo et al. [159] notes that with excited state algorithm and a quantum circuit to measure the overlap elements  $|\langle\psi_m|c_\alpha^\dagger|\psi_0\rangle|^2$  and  $|\langle\psi_m|c_\alpha|\psi_0\rangle|^2$ , one could easily compute a few terms inside the summation in Eq. (4.18). However, it might not be practical to find all excited states using excited-state VQE algorithms, which means we will be missing terms in the summation. Nevertheless, in practice, the few terms from the few excited states usually give a good approximation of the Green function, even when theoretically it requires summing over all eigenstates of the Hamiltonian. This is particularly true in the real frequency case ( $z = \omega + i\eta$ ), where terms of the form

$$f(\omega) = \frac{a}{(\omega \pm b) + i\eta} \quad (4.20)$$

contribute to the Green function. Here  $a$  is usually a constant  $a \in \mathbb{R}$ , and  $b$  is the energy differences  $b = E_m - E_0$ . The real part is  $\text{Re } f = \frac{a(\omega \pm b)}{(\omega \pm b)^2 + \eta^2} \rightarrow \frac{a}{\omega \pm b} (\omega \neq b)$ , and imaginary part is  $\text{Im } f = \frac{-a\eta}{(\omega \pm b)^2 + \eta^2}$ . Both will have large magnitude only when  $\omega \approx \pm b$ , and will decay as a hyperbola (real part) or as an inverse square (imaginary part) as  $\omega$  moves away from  $\pm b$ . An illustration of this term  $f(\omega)$  is available in Figure 4.3. Therefore, the real frequency Green function typically has several peaks, centred around different energy differences  $\pm(E_{m_1} - E_{m_2})$ .

<sup>1</sup>A short discussion of this subroutine is available in Theorem 6 of Dervovic et al. [164].



**Figure 4.3:** Typical real frequency Green function. Here we plotted the real and the imaginary part of the function  $a/(\omega - b + i\eta)$  for  $a = 1$ ,  $b = 2$ , and  $\eta = 0.1$ . The Green function peaks at  $\omega = b$  and decays fast when  $\omega$  moves away from  $b$ .

**Lanczos method based.** Classical Lanczos method [165] is based on iteratively constructing an orthogonal basis for the Krylov subspace, which is a subspace formed by iteratively applying the matrix (Hamiltonian  $H$  in our case) to an initial vector. The Lanczos vectors are orthogonal to each other and iteratively constructed in this process. Their normalised version forms a basis and the matrix becomes tri-diagonal in this basis. Hence, various properties such as the eigenvalues and in our case the Green function can be computed iteratively in this process. Notably, the Green function in this basis has a continued fraction representation [166].

This idea has been taken to the quantum computer. First, Jamet et al. [167] considered the construction of Lanczos vectors variationally, that is the orthogonality is formed variationally by optimising a certain loss function. Then, Jamet borrowed the idea from quantum subspace expansion [168], where a subspace of the Hilbert space is constructed by diagonalising an effective Hamiltonian in this subspace. This effective Hamiltonian can be obtained by measuring the “matrix elements” of the Hamiltonian in the subspace on the quantum computer. This subspace is extended in Jamet et al. [169] for constructing the Krylov subspace and for the Lanczos method. Further developments for computation of the Green function based on this idea are available in Jamet et al. [170] and Irmejs and Santos [171].

**Correction vector based.** Tong et al. [172] discussed a method using a quan-

tum linear-algebra solver algorithm (the so-called HHL algorithm [6, 164]) to directly compute the unified expression Eq. (4.14) where the Green function contains terms of the form  $\langle\psi|c^\dagger(z \pm H)^{-1}c|\psi\rangle$ . A similar method has also been proposed in Cai et al. [173]. Essentially, they cast the Green function calculation problem into two parts. Firstly, one solves a quantum linear algebra problem using the HHL algorithm to find a correction vector  $|\chi\rangle$  (defined later in Eq. (4.22)). Then one computes an overlap of the form  $\langle\psi_0|c^{(\dagger)}|\chi\rangle$  using a Hadamard test-like quantum circuit. However, since the operator  $Q_\pm = (z \pm (H - E_0))$  (also see Eq. (4.22)) is non-Hermitian, a technique called “block-encoding” is needed in their methods to find a higher-dimensional unitary operator which replaces  $Q_\pm$  and on which they apply the HHL algorithm. This, when combined with the already NISQ unfriendly HHL algorithm, makes their approach unlikely to have near-term implementation.

We can readily observe that an alternative variational approach to the one above can be employed, using the variational linear algebra solver algorithm in Section 2.4. We will introduce this method in detail in the next section.

Further, we want to acknowledge here that our method has also been suggested briefly as an alternative variation algorithm in the work of Cai et al. [173], albeit with much fewer details and without numerical experiments.

### 4.3 A variational algorithm for Green function

In this section, we introduce our variational algorithm for computing the Green function. We will start with the definition of the correction vector and its physical meanings. We will then show that the correction vector is the ground state of a modified Hamiltonian, and discuss how to implement a VQE procedure for the modified Hamiltonian.

#### 4.3.1 Correction vector

Here we introduce the correction vector. Recall that the Green function in the frequency domain can be written as

$$G_{\alpha\beta}(z) = \langle\psi_0|c_\alpha \frac{1}{z - (H - E_0)} c_\beta^\dagger |\psi_0\rangle + \langle\psi_0|c_\beta^\dagger \frac{1}{z + (H - E_0)} c_\alpha |\psi_0\rangle. \quad (4.14 \text{ revisited})$$

We simplify the expression for the Green function by focusing only on the following form containing terms of interest:

$$G_{ij}(z) = \langle \psi_0 | V_i^\dagger \frac{1}{Q_\pm(z)} V_j | \psi_0 \rangle, \quad (4.21)$$

In the equation above, we have replaced the denominator  $z \pm (H - E_0)$  by  $Q_\pm(z)$ , where the sign depends on whether we want to compute the first or the second term in the Green function. We have also introduced  $V_{i(j)}$  as a replacement of the operators  $c_{\alpha(\beta)}^{(\dagger)}$  on the left and right side of the operator  $1/(z \pm (H - E_0))$ . The  $i(j)$  are generic labels and the relation between  $V_{i(j)}$  and the  $c_{\alpha(\beta)}^{(\dagger)}$  can be worked out easily once one determined which term to compute in the previous  $G_{\alpha\beta}$ .

The form  $G_{ij}$  in Eq. (4.21) also emphasise the connection of Green functions to dynamical correlation functions, which in many literatures [166, 174] are defined in the form  $G_{ij}$  but with  $V_i$  and  $V_j$  being generic operators not restricted to creation and annihilation operators as we have in our definition for the Green function. We will see below that, our algorithm can be applied to compute such generic dynamical correlation functions, as long as their  $V_{i(j)}$  can be transformed into a linear sum of Pauli operators.

Next, we further replace the right-hand side with a vector  $|\chi_j(z)\rangle$ , which is the *correction vector*:

$$\begin{aligned} G_{ij}(z) &= \langle \psi_0 | V_i^\dagger |\chi_j(z)\rangle, \\ |\chi_j(z)\rangle &= \frac{1}{Q_\pm(z)} V_j | \psi_0 \rangle. \end{aligned} \quad (4.22)$$

Note that correction vector  $|\chi_j(z)\rangle$  is usually unnormalised, and when  $Q_\pm(z) \rightarrow 0$ , it is not well-defined. Expression above are often further simplified by defining a state  $|V_i\rangle \equiv V_i | \psi_0 \rangle$ , such that  $G_{ij}(z) = \langle V_i | \chi_j(z) \rangle$ .

Physically, the correction vector captures the system's response to external perturbations. Historically, Hallberg [166] extended the DMRG technique to the study of dynamics, by computing dynamical correlation functions. Specifically, DMRG was used to target the first few Lanczos vectors to iteratively tri-diagonalise the Hamiltonian, and afterwards, the dynamical correlation functions can be computed via a continued fraction representation. By targeting, we mean that those Lanczos vectors were kept as the *target states* in between iterations of DMRG and were thus well described by DMRG. Kühner and White [174] noticed that targetting Lanczos vectors for

dynamical correlation functions produced bad results on the high-energy region. A correction was made there by instead targeting the correction vectors introduced here. This technique was further developed into the so-called dynamical DMRG [175, 176] and used for studying dynamical properties (see e.g. Bohr [177]).

### 4.3.2 A variational algorithm for correction vector

We noticed that the variational quantum linear algebra algorithms mentioned in Section 2.4 can be used to compute the Green function in Eq. 4.21. We detail such an algorithm below.

We assume that a standard VQE algorithm has been used to give a circuit preparing the ground state  $|\psi_0\rangle = U_{\text{GS}}|0\rangle$ , and thus states like  $|V_i\rangle = V_i|\psi_0\rangle$  can in principle be prepared on a quantum computer. Then, we focus on solving the linear equation below which comes from the definition of correction vector in Eq. (4.22):

$$Q_{\pm}(z)|\chi_j(z)\rangle = |V_j\rangle. \quad (4.23)$$

Since in general  $|V_j\rangle$  and  $|\chi_j\rangle$  are not normalised as  $V_j$  and  $Q_{\pm}(z)$  are usually non-unitary, we define their normalised version as we are discussing quantum circuits. The two normalised versions  $|\tilde{\chi}_j\rangle$  and  $|\tilde{V}_j\rangle$  are such that

$$|\tilde{\chi}_j\rangle \propto |\chi_j\rangle, |\tilde{V}_j\rangle \propto |V_j\rangle. \quad (4.24)$$

The result in Section 2.4 shows that we can run a VQE on another Hamiltonian defined below:

$$H'(z, j) = Q_{\pm}^{\dagger}(z) \left( \mathbb{1} - |\tilde{V}_j\rangle\langle\tilde{V}_j| \right) Q_{\pm}(z), \quad (4.25)$$

and upon optimisation this VQE would give us a circuit  $U_j(\theta^*)$  which prepares the state  $|\tilde{\chi}_j(z)\rangle$ . Ignoring the subscripts and the dependency on  $z$  or  $j$ , we provide a brief explanation of  $H'$  here: The projector  $\left( \mathbb{1} - |\tilde{V}_j\rangle\langle\tilde{V}_j| \right)$  has a ground state  $|\tilde{V}_j\rangle$  of eigenvalue 0. The operators around this projector link the ground state  $|\psi'\rangle$  of the modified Hamiltonian Eq. (4.25) to this ground state of the projector since:

$$\begin{aligned} \langle\psi'| H' |\psi'\rangle &= \langle\psi'| Q^{\dagger} \left( \mathbb{1} - |\tilde{V}_j\rangle\langle\tilde{V}_j| \right) Q |\psi'\rangle = 0 \\ \iff Q |\psi'\rangle &\propto |\tilde{V}_j\rangle \propto |V_j\rangle. \end{aligned}$$

Thus,  $|\psi'\rangle$  is the correction vector we seek, and finding the ground state of the modified Hamiltonian is equivalent to solving the linear equation.

It helps to write down the energy in this VQE optimisation in detail, leading to the following energy function, labelled  $g(\theta)$ , which needs to be minimised using VQE:

$$\begin{aligned}
g(\theta) &= \langle 0 | U_j^\dagger(\theta) H'(z, j) U(\theta) | 0 \rangle \\
&\quad \text{(Temporarily ignoring subscripts and dependencies)} \\
&= \langle 0 | U_j^\dagger Q^\dagger \left( 1 - |\tilde{V}\rangle\langle\tilde{V}| \right) Q U_j | 0 \rangle \\
&= \langle 0 | U_j^\dagger Q^\dagger Q U_j | 0 \rangle - \langle 0 | U_j^\dagger Q^\dagger \frac{|V_j\rangle\langle V_j|}{\langle V_j | V_j \rangle} Q U_j | 0 \rangle \\
&\quad \text{(Restoring the ignored details)} \\
&= \langle 0 | U_j^\dagger(\theta) Q^\dagger(z) Q(z) U_j(\theta) | 0 \rangle \\
&\quad - \frac{1}{\langle V_j | V_j \rangle} |\langle V_j | Q(z) U_j(\theta) | 0 \rangle|^2
\end{aligned} \tag{4.26}$$

Terms in this energy function can be computed using the quantum computer (see next section). Let us assume that after optimisation, we have found the circuit  $U_j(\theta^*)$  which prepares  $|\tilde{\chi}_j\rangle \propto |\chi_j\rangle$ , we can recover the unnormalised correction vector  $|\chi_j\rangle$  as follows. Assume that the two are related by

$$|\chi_j(z)\rangle = \gamma_j(z) |\tilde{\chi}_j(z)\rangle, \tag{4.27}$$

where  $\gamma_j(z)$  captures the unnormalised vector's magnitude. Using the definition of correction vector, which solves the linear equation below,

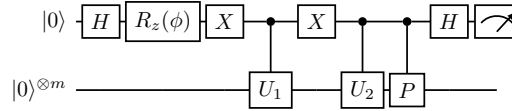
$$\begin{aligned}
Q_\pm(z) \underbrace{\gamma_j(z) U_j(\theta^*) | 0 \rangle}_{|\chi_j(z)\rangle} &= |V_j\rangle \\
\stackrel{\text{leads to}}{\implies} \gamma_j(z) &= \frac{\langle V_j | V_j \rangle}{\langle V_j | Q_\pm(z) U_j(\theta^*) | 0 \rangle}.
\end{aligned} \tag{4.28}$$

Therefore, overall we have<sup>2</sup>,

$$|\chi_j(z)\rangle = \gamma_j(z) U_j(\theta^*(z)) | 0 \rangle, \tag{4.29}$$

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<sup>2</sup>While drafting this thesis, we notice a typo in Eq. 15 of our published work [178]. The Eq. 4.29 here is the corrected one.



**Figure 4.4:** Quantum circuit for measuring the overlap term  $\langle 0|U_1^\dagger P U_2|0\rangle$ . The system consists of  $m + 1$  qubits:  $m$ -qubits for the unitaries  $U_1$ ,  $U_2$  etc., and one ancilla qubit. When the angle  $\phi = 0$ , measuring the ancilla qubit in  $Z$  basis samples  $\text{Re}\langle 0|U_1^\dagger P U_2|0\rangle$ , whereas if  $\phi = \pi/2$ , measurement samples  $-\text{Im}\langle 0|U_1^\dagger P U_2|0\rangle$ . Detailed analysis of this circuit is available in Sec. C.4.

and the full expression for Green function  $G_{ij}$  (Eq. 4.21) is

$$G_{ij}(z) = \gamma_j(z) \underbrace{\langle 0|U_{\text{GS}}^\dagger V_i^\dagger}_{\langle V_j^\dagger|} \underbrace{U_j(\theta^*(z))|0\rangle}_{|\tilde{x}_j(z)\rangle}, \quad (4.30)$$

where  $\gamma_j(z)$  is given by Eq. 4.28 above.

### 4.3.3 Implementation on a quantum device

Now we discuss the implementation of the above computations in a quantum computer. We show how the various terms in the energy function (Eq. (4.26)), the factor  $\gamma_j(z)$  (Eq. 4.28), and the  $G_{ij}$  (Eq. 4.30) can be computed.

Note that using standard VQE procedures,  $V_{i(j)}$  and  $Q_\pm(z)$  can always be mapped to a linear sum of Pauli operators (Section 2.1.1). Therefore, symmetric expectation values of Pauli operators can be computed in a straightforward fashion. These include, for example, the first term of the energy function, since

$$\begin{aligned} & \langle 0|U^\dagger(\theta)Q_\pm^\dagger(z)Q_\pm(z)U(\theta)|0\rangle \\ &= \sum_{a,b} w_{Q,a}^* w_{Q,b} \langle 0|U^\dagger(\theta)P_a P_b U(\theta)|0\rangle, \end{aligned} \quad (4.31)$$

where  $Q_\pm(z)$  has been mapped to a linear sum of Pauli operators, as in  $Q_\pm(z) = \sum_a w_{Q,a} P_a$  for some  $w_{Q,a} \in \mathbb{C}$ . Note that the product of two Pauli operators is another Pauli operator times a constant.

However, calculating overlaps between different states requires a more involved effort. These include the calculations of  $\gamma_j(z)$ , the  $G_{ij}$ , and the second term contributing to the energy function  $g(\theta)$ . All of these contributions can be simplified into computing the terms of the following form on a quan-

tum computer:

$$\langle 0|U_1^\dagger P U_2|0\rangle, \quad (4.32)$$

where  $U_1$  and  $U_2$  are two different and potentially parameterised quantum circuits, and  $P$  is an  $n$ -qubit Pauli operator. We compute the overlap term in Eq. 4.32 using a circuit from Endo et al. [159]. This circuit is a variant of the Hadamard test. It is illustrated in Figure 4.4 and proved in the Appendix C.2.

## 4.4 Numerical experiments

Next, we provide details on a proof-of-concept experiment for our variational algorithm for the Green function. The experiment was done on a classical computer emulating a near-term quantum computer with a noise model. The systems we looked at are the Hydrogen molecule ( $H_2$ ) and the lithium hydride (LiH).

Our interest in chemical systems are motivated by several post Hartree-Fock methods. It has been observed that mean-field methods such as Hartree-Fock and the density functional theory are insufficient for describing many systems. Approaches improving upon Hartree-Fock are generally referred to as post Hartree-Fock methods. Among them, many approaches are based on computing (or approximating) the Green functions of the chemical system. Unfortunately, the computational cost of Green functions has limited its applications. However, this limitation may be alleviated with the combination of Green function embedding theories [179] and the advent of quantum computers.

In this section, we start by describing the details of the setup of the experiments. These include setting up the molecular Hamiltonian, choosing the ansatz, optimisation tricks, noise model for the quantum computer employed, and the companion error mitigation method. We then present results on hydrogen and lithium hydride separately.

### 4.4.1 Experimental setup and the noise model

**Green function.** In our numerical experiments, we concentrated on assessing the accuracy of the single-particle spectrum of the systems. To achieve this, we calculated the diagonal part of the Green function as defined by the formula below.

$$G(z) = \sum_{\alpha} \text{Im} (G_{\alpha\alpha}(z)) . \quad (4.33)$$

That is, the scalar Green function  $G(z)$  corresponds to the trace of the imaginary part of the Green function matrix in Eq. 4.14. Note also that the real part can in theory be obtained from the imaginary part using the Kramers-Kronig relations [180].

**Hamiltonian setup.** The two systems we focused on are hydrogen ( $\text{H}_2$ ) and lithium hydride ( $\text{LiH}$ ). The chemistry software library we used is PySCF [181, 182], which is standard among many VQE calculations. PySCF provides convenient code for (i) setting the so-called basis set, (ii) computing the one- and two-body matrix elements ( $h_{\alpha\beta}$  and  $g_{\alpha\beta\gamma\delta}$ ) in the second quantised Hamiltonian (Eq. (4.1)) from this basis set, and (iii) carry out a simple Hartree-Fock calculation using this basis set as the atomic orbitals. The result of the Hartree-Fock calculation will be used as the reference mean-field value, since electron-electron correlations are ignored in Hartree-Fock.

The basis set is needed for both the Hartree-Fock (HF) calculation and for computing the one- and two-body matrix elements. Given a basis set, it specifies the form of the variational ansatz used in the HF calculation. The result of the HF calculation also specifies the form of orbitals used in the second quantisation. Thus the basis set determines the Hilbert space dimension of our second quantised Hamiltonian. The basis set we choose is the STO-3G, whose radial component for electronic orbitals has the form of a linear combination of three parameterised Gaussian functions. Once a basis set is chosen, PySCF computes the one- and two-body matrix elements automatically. The resulting second quantised Hamiltonian is mapped by us using the Jordan-Wigner transformation (see Section 2.1.1) to linear combinations of Pauli operators. Note that STO-3G is useful in that it has been picked as the standard in many VQE experiments. Its usefulness might vary if more sophisticated molecules or correlations are considered. Details are available in related VQE reviews [27, 183].

Note that we further restricted our simulations to conserve spin symmetry. This constraint treats both spin channels as identical, avoiding the breaking of single-particle spin symmetry.

Next, for the lithium hydride system, we introduced an active space embedding method. This is both a practical choice and a test of whether a similar reduction of the number of orbitals (size of the Hilbert space) can be done for near-term quantum hardware. On the practical side, a full simulation of lithium hydride would demand 12 qubits, which would make it impractical

for our experiments based on numerical simulations, whereas the embedded approach reduces this requirement to only four qubits.

This active space embedding method involves a complete active space (CAS) partitioning of the molecular orbitals of the molecular system. It can incorporate a potentially noisy Green function, computed by the quantum computer, with a mean-field Green function that can be efficiently computed using classical computing resources (Hartree-Fock method). This embedded approach to studying correlated dynamics is a common strategy when dealing with complex systems where obtaining a full solution is impractical. Instead, researchers identify and study a subspace of critical correlated physics [157, 184–191]. Details of this embedding method are available as we discuss results from lithium hydride in Section 4.4.3.

**Ansatz.** For both parts of the algorithm (the ground state VQE and the correction vector part), we used the Hardware Efficient Ansatz (HEA) which has been detailed in Section 2.1.2. The number of repeated blocks (referred to as the depth  $d_{\text{HEA}}$  in Figure 2.1) was dynamically adjusted to ensure optimisation convergence while avoiding excessive parameterization.

For the correction vector problem, an additional technique is needed for the adjustment of the ansatz’s depth. Specifically, when optimising the energy function  $g(\theta)$  (Eq. (4.26)), we start with an initial estimate for  $d_{\text{HEA}}$  and increase it incrementally if the optimisation has not brought the energy significantly closer to zero, the theoretical minimum for  $g(\theta)$ . Furthermore, we observed that the ansatz depth found in this process varies across the values of frequencies  $z$  in  $G(z)$ : when the frequency  $z$  approached the poles of the Green function  $G(z)$  along the real-frequency axis, greater depth was typically required for convergence.

**Optimisation.** We applied the RotoSolve algorithm, as discussed in Section 2.1.5, to optimise both the ground state VQE algorithm and determine the correction vector. Also, since the energy function  $g(\theta)$  theoretically should have a minimum of zero, we continue the optimisation process until we achieve a small threshold value of  $\varepsilon > 0$ . In our experiment, this threshold is usually set at 0.05.

However, we observed that this threshold needs to be adjusted dynamically at certain frequencies. This is easily noticeable when the Green function obtained at those frequencies deviates significantly from the value at neighbouring frequencies. Moreover, we observed that the variations in this re-

quired optimisation threshold were closely linked to the condition number of the underlying modified Hamiltonian problem  $H'$ . A high condition number arises when  $z$  is near the poles of the Green function on the real-frequency axis, in which case convergence tends to be slower, and a smaller threshold  $\varepsilon$  becomes necessary. A similar situation applies to the depth  $d_{\text{HEA}}$  required for the ansatz, as mentioned above.

Finally, we employed a warm-start technique to expedite the optimisation process: we reused the optimised values on neighbouring frequencies to initialise the angles and depth for the ansatz for finding the correction vector at the current frequency, as we do not usually expect the optimised parameters to differ significantly.

**Simulation and the noise model.** To mimic a quantum computer's behaviour in our study, we emulated the real-world performance on quantum devices: we conducted simulations for both the stochastic sampling noise using random number generators and the systematic noise from quantum computers using a two-qubit depolarization noise model. Note that we assumed the one-qubit errors could be ignored in our experiments due to the following reasons: (1) on current-generation quantum devices, the one-qubit error rate is at least one order of magnitude smaller than the two-qubit error rate; (2) our circuits, particularly those utilised for estimating overlap (see Fig. 4.4), include overall a substantial number of two-qubit gates.

The two-qubit depolarization noise is defined by the following channel, and is applied to the quantum state  $\rho$  after a two-qubit gate in the circuit:

$$\mathcal{N}(\rho) = (1 - p_2)\rho + \frac{p_2}{15} \left( \sum_{a,b} P_a^{(1)} P_b^{(2)} \rho P_a^{(1)} P_b^{(2)} \right) - \frac{p_2}{15} \rho. \quad (4.34)$$

Here the superscript 1 (2) denotes the first (second) qubit on which the two-qubit gate acts. The  $a$  ( $b$ ) sums over all possible one-qubit Pauli matrices (including the identity). The noise parameter  $p_2$  is set to  $10^{-3}$ .

Here we note that three-qubit gates (CCX and CCZ) are needed for the controlled unitary gates in Figure 4.4. They were decomposed into two-qubit gates before running on our quantum simulator. Therefore, we did not model any noise for three-qubit gates and instead applied two-qubit errors to the gates after the decomposition. This better simulates the behaviour of actual quantum devices. Note that the decomposition is done using the work of Shende and Markov [192], and it is detailed in Appendix D.

**Error mitigation.** We employed the exponential extrapolation error mitigation approach discussed in Section 2.1.4 to address systematic noise. This is due to the expected high level of noise from the large amount of two-qubit gates, as the linear extrapolation approach would fail in this case. To do so, we initiated the circuit with the original noise level of  $p_2 = 10^{-3}$  and subsequently increased the noise in the simulation by setting  $p_2$  to twice its original value. Achieving this noise adjustment on a physical quantum device can be accomplished through various methods, as outlined in the literature [193]. We fitted the expectation values of measurements obtained at these distinct noise levels to the exponential model,  $E(p_2) = E(0)e^{-B p_2}$ , to extrapolate a noise-free estimation of  $E(0)$ .

Also, to mitigate the sampling noise, we repeated each circuit  $10^6$  times, averaging their outcomes to estimate all the observables. It is worth noting that we deliberately selected this large number,  $10^6$ , because exponential error mitigation typically amplifies the sampling variance in addition to the inherent sampling noise arising from quantum measurements [60].

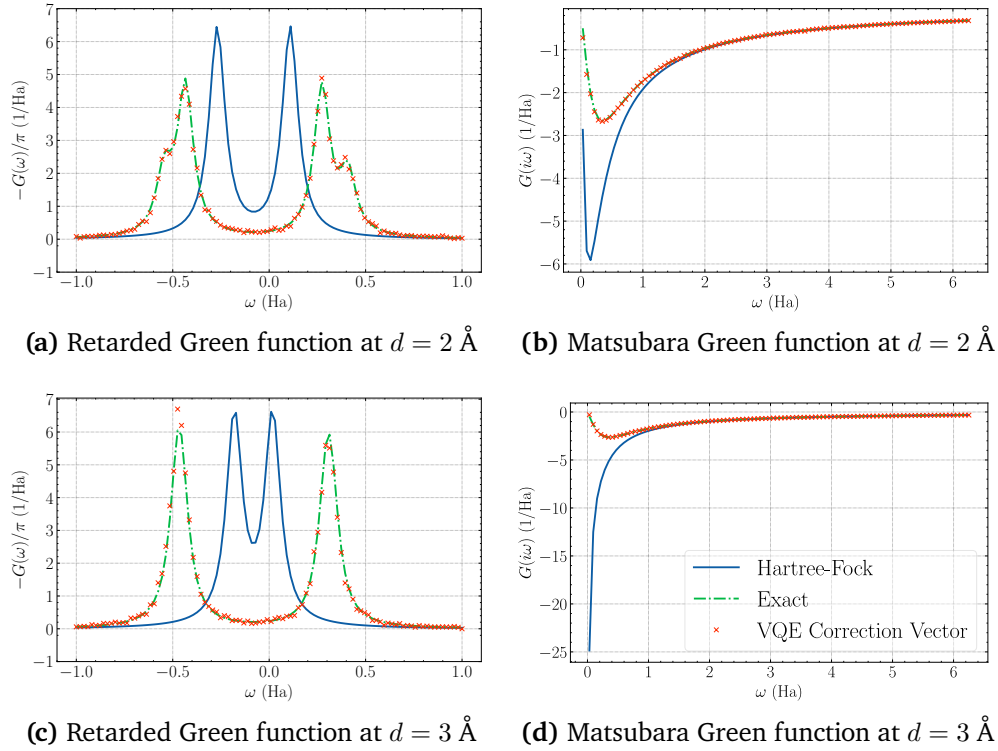
#### 4.4.2 Results for hydrogen

We now discuss the results and first focus here on the smaller hydrogen system ( $H_2$ ). We will discuss the more involved case of lithium hydride in the next section.

Numerical experiments were performed to calculate the Green function of the hydrogen molecule  $H_2$  at two different bond lengths  $d$ . The Green function was computed along both the real-frequency axis (with the constant  $\eta = 0.05$ ) and the Matsubara-frequency axis. In our calculations, we used Hartree to measure energy. One Hartree equals  $4.360 \times 10^{-18}$  Joules.

In both frequencies, the outcome from the quantum correction solver closely matched the exact value, as shown in Figure 4.5 where we have included two sets of reference data computed using classical computers: one computed via the exact inversion of the numerical matrix  $Q(z)$  using NumPy [194] (labelled as “Exact”), and the other computed using classical computers with a mean-field approximation (labelled as “Hartree-Fock”).

As depicted in the figure, our quantum correction vector solver reproduced the exact spectrum with high fidelity. Specifically, the difference along the Matsubara-frequency axis is barely noticeable in the graph, with a maximum error of  $0.22 \text{ Ha}^{-1}$  across the entire Matsubara-frequency range. On the real-frequency axis, differences primarily manifest in the transition am-



**Figure 4.5:** Matsubara Green function and retarded Green function calculated for  $\text{H}_2$  at different distances  $d$ .

Single-particle spectrum of  $\text{H}_2$  computed at two different bond lengths, on both the real-frequency and Matsubara axis. This spectrum is computed from the trace of the imaginary part of the retarded Green function, calculated via a quantum simulation of the VQE correction vector algorithm (VQE Correction Vector) and compared to the exact solution (Exact) and the uncorrelated mean-field limit (Hartree-Fock).

plitudes at the excitation energies (i.e. the poles of the Green function). This aligns with the fact that the condition number for  $Q(z)$  is unfavourable in those excitation energy regions. Indeed, the Green function on the real-frequency axis would rapidly diverge towards infinity as  $\eta \rightarrow 0^+$  at those energies. This affirms our approach to adapting the ansatz in these regions.

Physically, we can readily observe the impact of electron correlation by comparing our computed Green function at various bond distances  $d$  with the mean-field calculations. The influence of correlation and entanglement is more pronounced when the bond distance is stretched to  $d = 3 \text{ \AA}$ , and this stretching results in the opening of the charge gap. At bond distances that are less stretched, we observe correlation-induced substructures to a lesser degree.

Note that, the accuracy of our result, is comparable to that of Endo et al.

[159], where the same Green function for the same system was approached, by first performing (variational) time evolution to compute the Green function in the time domain, and then Fourier transform it to the frequency domain.

### 4.4.3 Results for lithium hydride

In this section, we present the outcomes of applying our variational algorithm for correction vector to a LiH molecule with a bond length of 3 Å. We will begin by commenting on the active space embedding approach we have employed and then present the numerical results.

**Active space approach.** For lithium hydride LiH, we have employed a ‘complete active space’ (CAS) method, which reduced the degree of freedom needed to be solved exactly. The method is common in quantum chemistry [195, 196], and in our case results in a four-qubit Hilbert space in which we solve for the correction vector. The method was coded by our collaborator Max Nusspickel using the PySCF library [181, 182]. Here we provide a qualitative description.

On a higher level, we can view this method as expressing the ground state wave function and the correction vector as tensor products of a highly entangled quantum state within CAS molecular orbitals with a simple product (Slater determinant) state of the remaining *core* orbitals. The core orbitals are assumed to be uncorrelated and therefore restricted to a single Slater determinant. The CAS space is simply chosen to span the highest-energy occupied and lowest-energy unoccupied single-particle orbitals which arise from a prior mean-field Hartree-Fock calculation. Thus, the CAS space is expected to contain the important quantum fluctuations required to describe the correlated physics of the system.

The method results in an effective space CAS Hamiltonian whose energy is shifted by a constant, resulting from the static mean-field Coulomb and exchange potentials from the core electrons. The CAS Hamiltonian is produced using PySCF package in a black-box fashion and resides in the active space of four qubits. The correction vector problem and the Green function are thus calculated in this CAS space of reduced size.

To obtain the Green function of the entire system, we must combine the Green function calculated in the CAS space, with the (non-interacting) Green function of the inactive core. There are two ways to combine them: the formally correct approach, which we termed the ‘Dyson’ approach, and the practically motivated ‘Non-Dyson’ approach. We detail them below.

Note that in both approaches, the full information of Green function as a matrix  $G_{\alpha\beta}$  is needed, as in the unified expression Eq. (4.14). This is in contrast with the definition in Eq. (4.33) and in the plotted figures, where we traced over the diagonal elements and looked at its imaginary part only.

In the formally correct approach, the combination is made using the self-energy of the active space: the Green function matrix ( $G_{\text{Full}}$ ) of the full CAS plus core system is obtained from combining the non-interacting Green function matrix of the full system ( $G_{0,\text{Full}}$ ) and the self-energy matrix of the CAS system ( $\Sigma_{\text{CAS}}$ ) using a Dyson equation:

$$G_{\text{Full}} = (G_{0,\text{Full}}^{-1} - \Sigma_{\text{CAS}})^{-1}.$$

Note that here the inversion is of the matrix (i.e., not an element-wise inversion). The self-energy of the CAS system only has components in the CAS space and can be computed from the similar Dyson equation in this CAS space by

$$\Sigma_{\text{CAS}} = G_{0,\text{CAS}}^{-1} - G_{\text{CAS}}^{-1}.$$

The free Green function matrix of both the CAS and the full system is computed from the inversion of the Fock matrix [195, 197], which is readily available from the prior mean-field Hartree-Fock calculation. The interacting Green function matrix of the CAS space is computed using our correction vector algorithm on the CAS Hamiltonian.

While this ‘Dyson’ approach is formally correct, it suffers several problems. First, it requires the calculation of the off-diagonal elements of the Green function, which contains noise from evaluating them on quantum computers. Further, the non-linear inversions used in the two Dyson equations will likely turn the noise from quantum computers into systematic error [198]. For this reason, we considered a simple approximation approach.

In this approximation approach, which we termed ‘Non-Dyson’, the CAS Green function matrix is directly substituted into the full Green function matrix. This can be described by the following formula using a projector  $P_{\text{CAS}}$  into the CAS space:

$$[G_{\text{Full}}]_{\alpha\beta} = [G_{0,\text{Full}}]_{\alpha\beta} + P_{\text{CAS}}([G_{\text{CAS}}]_{\alpha\beta} - [G_{0,\text{Full}}]_{\alpha\beta})P_{\text{CAS}}.$$

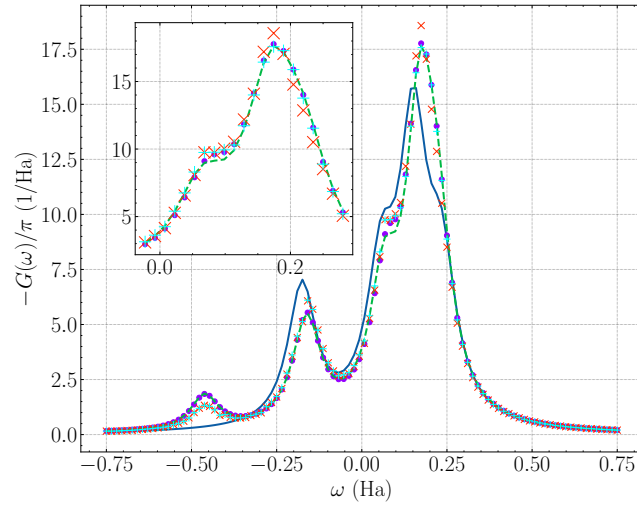
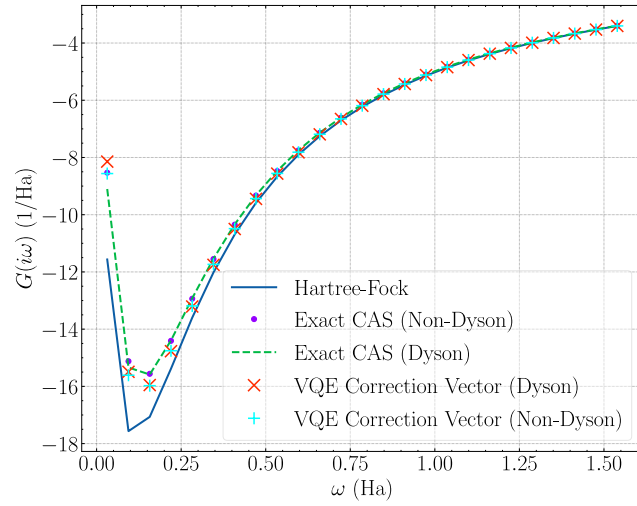
The ‘Non-Dyson’ approaches will become exact if the off-diagonal elements of the Fock matrix which couple the CAS and core spaces are zero. In this

case, the operations of matrix inversion and projection commute and the two approaches should be identical. While this is not generally the case, the Fock matrix is often block-diagonally dominant. Therefore, the Green function matrix of the ‘Non-Dyson’ approach becomes a good approximation to that of the ‘Dyson’ approach.

**Numerical result.** We plotted our data from numerical simulations in Figure 4.6 on page 90. It shows the Green function (its imaginary part after taking the trace, Eq. (4.33)) generated using either of two active space approaches for quantum correction vector solver (labelled as ‘VQE correction vector’). For reference, the exact values computed by solving the correction vector numerically using NumPy are also plotted for both approaches (labelled as ‘Exact’).

First, it is evident that when combining the Green function of the CAS space with the core, the results from both approaches are nearly identical. This outcome aligns with our expectations, particularly when considering the active space embedding we have chosen: In this embedding, we have selected the correlated subspace from canonical Hartree-Fock orbitals, and we anticipate minimal hybridisation between the active space and the core orbital in the final state.

Secondly, we observe that the quantum correction vector solutions for the embedded active space Green function once again accurately replicate the correlated spectrum, with only minor discrepancies stemming from the transition amplitudes at the excitation frequencies (poles). Also, in the vicinity of the poles, the precision of the ‘Non-Dyson’ method slightly outperforms the other approach, suggesting that the bias resulting from inverting the noisy Green function matrix has a more pronounced effect than neglecting the hybridisation between the active space and core orbitals within this specific set of active space orbitals. However, it is worth noting that both quantum methods slightly underestimate the low-energy satellite peak shown in Fig. 4.6a at approximately  $-0.45$  Ha.

(a)  $d = 3 \text{ \AA}$ , Retarded Green function(b)  $d = 3 \text{ \AA}$ , Matsubara Green function

**Figure 4.6:** Matsubara Green function and retarded Green function calculated for the lithium hydride molecule at distance  $d = 3$  angstrom ( $\text{\AA}$ ).

Single-particle spectral function for LiH on both the real-frequency and Matsubara axis. Results are shown from quantum simulation with the VQE correction vector (both ‘Dyson’ and ‘Non-Dyson’ approaches), exact classical solver (both with ‘Dyson’ and ‘Non-Dyson’ approaches), and the uncorrelated mean-field result (‘Hartree-Fock’). The inset in Figure 4.6a emphasises the region of the retarded Green function where there is a small though noticeable difference between the data obtained from the VQE correction vector method in the two different approaches.

## 4.5 Discussion and outlook

In this section, we discuss some future research directions related to our work here.

Our numerical result shows that variational algorithms can capture the

correlation effects and entanglement to high accuracy under noise. The result is based on experiments on small molecules and is conducted on a simulated quantum computer with a depolarising noise model. While results on small molecules are important to prove the validity of our method, experimenting on more challenging systems is crucial to examine the real-world benefit of our method. To do so, several challenges lie ahead.

First, on the scaling of the variational optimisation process. While using parameterised quantum circuits for simulations of both ground and excited state physics has been popular in recent years, the scalability of the optimisation process remains a source of debate in the literature [29, 199, 200]. In our case, we compound the complication involved since we are finding the ground state for a modified Hamiltonian  $H'(z, j)$  (Eq. (4.25)) rather than the original Hamiltonian. Still, we expect the scalability of our correction vector algorithm to be comparable to other VQE-based algorithms targeting excited states for computing the Green function. Our expectation is based on the following reasoning. The correction vector at each frequency contains contributions from all the excited states of the original Hamiltonian. On the real-frequency axis, this combination of states is dominated by the excited state corresponding to the nearest excitation energy to the current frequency. In this case, we expect the required ansatz complexity (depth, for example) to express the correction vector to be similar to the required ansatz complexity to describe these excited states. On the Matsubara axis, while we do not have a clear analytical picture we observe through our numerical experiments that the required ansatz depths are comparable to that of the real frequencies.

Second, on the sensitivity to noise. We note that the maximum circuit depth of quantum circuits required in our algorithm is at least twice that of a comparable VQE algorithm targeting excited states. Also, a significant number of controlled one-qubit and two-qubit gates are needed. These are needed for the calculation of several overlap terms based on Hadamard-like tests discussed in Section 4.3.3. Therefore, our algorithm is particularly sensitive to the levels of two-qubit gate noise in the quantum computer, which is why we have used the exponential extrapolation scheme (Section 2.1.4.1) for error mitigation in our simulated noisy experiment.

For the feasibility of implementing Green function algorithms on near-term quantum computers, currently the excited-state based algorithms for the frequency domain have enjoyed a few successful implementations, including one to be presented in the following chapter. However, a driving

force is perhaps the direct applications of frequency domain Green function in dynamical mean-field theory studies. It is possible that in the time domain the calculation of Green function is more noise-resilient due to fewer gates involved in the calculation (compare the circuit in Figure 4.2 and the expression in Eq. (5.31)) and the potentially more gate-efficient variational real-time simulation algorithms [162].

However, when the time domain Green function needs to be Fourier transformed into the frequency domain for the intended application, we shall remain conservative about their noise-resiliency. Typically, many data points in the time domain would be necessary for an accurate Fourier transform. Due to the increased computational cost, an interpolation scheme is often employed prior to the Fourier transform in the literature [201, Chapter 9]. Further research is needed to determine how well this interpolation scheme aligns with quantum algorithms.

Next, a likely future research direction is exploring the interaction of tensor networks and quantum circuits with a focus on improving the ansatz used in our correction vector. As commented, we intuitively believe that the ansatz used for parametrising correction vectors needs to capture the correlation effects of the physical system, especially on the real frequency axis. At the same time, several methods have been proposed to apply tensor network results into quantum circuits [202, 203], such as obtaining efficient circuit representation of a particular matrix-product state [204]. Intuitively, there can be a window of opportunity for quantum circuits to capture highly entangled states but with significantly fewer parameters even compared with matrix-product states [202]. Exploration of research in this direction has been demonstrated on simulated quantum computers for Lanczos method based quantum algorithms for Green function [167, 169, 170]. Here, we believe a similar result can be obtained for our correction vector algorithm.

Finally, we would like to credit the work of Hendry and Feiguin [176], which inspired our research. There, the problem of numerically finding the correction vector has similarly been turned into a different variational minimisation. This difference comes from treating the real and imaginary of the correction vector separately. Notably, they guided their optimisation using the so-called natural gradient which respects the geometry of the Hilbert space better than normal gradient descent optimisation. A quantum version of the natural gradient approach exists [205] and can therefore guide further applications of our algorithm when targeting larger systems.

## Chapter 5

# Dynamical mean-field calculations on quantum computers

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In the previous chapter, we reviewed various variational quantum algorithms to compute the Green function. The growing availability of quantum computers accessible via the cloud encourages us to consider a real-world implementation. In this chapter, we attempt to analyse the usefulness and feasibility of one of the reviewed algorithms on IBM’s superconducting quantum computers. Specifically, we apply it to a toy problem discussed in dynamical mean-field theory (DMFT), which at its core requires calculating the Green function for a particular impurity Hamiltonian containing a relatively small number of sites.

From a bigger perspective, DMFT is a mean-field theory for studying many-particle problems where strong correlation effects from electrons need to be carefully treated. Although it is a mean-field theory, it preserves a certain dynamical aspect of the problem and solves an effective quantum me-

chanical problem. It has been proven useful in the qualitative understanding of materials where other more traditional mean-field approaches failed.

For DMFT, quantum computers can play an important role. Specifically, we can reasonably expect that variational quantum algorithms running on near-term quantum computers can be used to compute the Green function for the impurity Hamiltonian (introduced later in Eq. (5.10)) in the DMFT of a moderate number of sites. We can thus target the Green function and the impurity Hamiltonian for the self-consistency loop at the heart of DMFT. As usual for many mean-field theories with a self-consistency loop, complex problems tend to require a larger effective impurity Hamiltonian and hence require an exponentially growing computation resource on classical computers. Therefore, more accurate DMFT calculations can hopefully be done using these variational quantum algorithms on near-term quantum computers.

Note that this chapter is based on the work done in collaboration with Ivan Rungger *et al.* and published on arXiv in 2019 [30], which has been cited by later publications discussing similar experiments (e.g. [206, 207]). The author of this thesis carried out experiments on IBM's quantum computers while other collaborators therein carried out numerical experiments on classical computers and the University of Maryland's ion-trap based quantum computers. Calculations done by the author are included in this thesis, and we briefly discuss other results available in the arXiv publication.

**Chapter outline.** In this chapter, we first provide a general background on dynamical mean-field theory in the preliminaries (Section 5.1). We then detail the formulae and simplifications needed for a proof-of-concept calculation using variational quantum algorithms for the solution of the impurity model in Section 5.2. The related experimental results we obtained from both classical-emulated quantum computers and IBM's quantum computers are detailed in Section 5.3, which notably contains results from a full DMFT self-consistency loop on the quantum computer. Finally, we comment on potential problems and future research directions in Section 5.4.

## 5.1 Preliminaries

First, we introduce the dynamical mean-field theory. We shall attempt to cover both the physical intuition in the theory and various often-seen formulae. Dynamical mean-field theory, abbreviated DMFT, is a mean-field theory for studying quantum mechanical effects of the Hubbard model on lattices.

The theory focuses on the Green function of a single site on the lattice, and approximates the environment's influence on this site by an effective field. It obtains a relation between a local quantity, the Green function on the site, and the lattice's global configuration, its (non-interacting) density of state (DOS). This relation is exact only in infinite dimensional lattices and is called the self-consistency condition in practice.

Notably, while the environment is described by a non-interacting field, DMFT allows for interactions on the local site and between the local site and the non-interacting environment.

In the following, we first introduce the Hubbard model and a Bethe lattice. We then describe the mean-field approximation where the lattice's dimension is taken to the infinity. The resulting self-consistency condition is presented attached with a local view of it from the on-site self-energy. We finally introduce the impurity Hamiltonian used to reproduce the self-consistency condition and its exact diagonalisation.

We focus on a conceptual description based on the review of Georges et al. [208]. While the formulae are written in the imaginary time  $\tau$  and with imaginary (i.e. Matsubara) frequency, we emphasize that the same formulae exist for the real frequency due to analytic continuation. That is, formulae written for one form should apply to the other form by a simple substitution of  $i\omega_n \leftrightarrow \omega + i0^+$ .

**Hubbard model on a lattice.** We start with the Hubbard model on a lattice, which is governed by the Hamiltonian  $H_{\text{Hubbard}}$  below. The lattice is here described by a graph where the nodes are sites where electrons reside and an edge connecting two nodes represents an interaction. The interaction strength are characterised by  $t_{ij}$ , and the summation  $\langle ij \rangle$  is taken over nearest neighbours, i.e. over the edges in the graph. Paired electrons of opposite spin on the same site experience a repulsion characterised by  $U$ .

$$H_{\text{Hubbard}} = - \underbrace{\sum_{\langle ij \rangle, \sigma} t_{ij} (c_{i\sigma}^\dagger c_{j\sigma} + c_{j\sigma}^\dagger c_{i\sigma})}_{\text{Hopping}} + \underbrace{U \sum_i n_{i\uparrow} n_{i\downarrow}}_{\text{Hubbard } U} - \underbrace{\mu \sum_{i\sigma} c_{i\sigma}^\dagger c_{i\sigma}}_{\text{Chemical potential}} \quad (5.1)$$

Let's introduce one particular lattice, the Bethe lattice. The Bethe lattice of dimension  $d_{\text{Bethe}}$  (which is also the coordination number of this lattice) can be built as follows. One starts with a single site, called the root, and connects it with  $d_{\text{Bethe}}$  new sites called leaves. Each of the new sites is further

considered as new roots and each is connected to a fresh new set of  $d_{\text{Bethe}} - 1$  new leaves. Note that the total number of neighbours of a newly created site is  $d_{\text{Bethe}}$  ( $d_{\text{Bethe}} - 1$  leaves plus one for parent root). This procedure is repeated infinitely, creating a shell-like structure where the initial site sits at the centre and other leaves and leaves of leaves spread out as concentric circles (shells) to infinity. Note that importantly, since the procedure is repeated infinitely and each site is connected to  $d_{\text{Bethe}}$  neighbours, all sites are indistinguishable from each other, and we can build the same Bethe lattice from any site<sup>1</sup>. Physics on the Bethe lattice is inherently translational invariant and has no boundary conditions.

**Mean-field approximation.** A mean-field description of the physics of such lattice can be achieved as follows. One first chooses a local site and attempts to describe the effective dynamics on this site. This involves integrating out the degrees of freedom on all other sites and leads to a relation between the local dynamics and the dynamics of the lattice. One then takes the lattice's dimension to infinity, which is equivalent to a mean-field description in reality. The relation between the local dynamics and the lattice's dynamics is simplified in this limit. To study real materials, this relation is satisfied by adding effective parameters in local dynamics and finding a self-consistent solution to this relation.

We focus on the mean-field picture on an infinite dimensional lattice below, and then detail the relation, now called a self-consistency condition. We focus on a conceptual description here and refer to Georges et al. [208, §III] for detailed derivations.

For Bethe lattice, taking the lattice dimension to infinity means taking the  $d_{\text{Bethe}} \rightarrow \infty$ . Therefore, it corresponds to considering a mean-field picture where each site is connected to an infinite amount of neighbours. In this process, the interaction strength (hopping terms) between sites needs to be scaled so that the hopping energy of a single site is of the same order as the local interaction energy. This leads to a  $t_{ij} \rightarrow t/\sqrt{d_{\text{Bethe}}}$  scaling<sup>2</sup>. This scaling ensures that various perturbation contributions to the Green function of higher orders in  $1/\sqrt{d_{\text{Bethe}}}$  vanishes and leads to the simple form of self-consistency condition.

<sup>1</sup>This distinguishes the Bethe lattice from the Cayley tree which has a well-defined centre and a boundary condition. See more at the comparison in Ostili [209].

<sup>2</sup>See [208, Appendix A] for details. Similar scaling laws apply to other lattices discussed there.

**Self-consistency condition.** DMFT assumes that the dynamics of a local site, labelled “ $s$ ”, can be described by the effective action  $S_{\text{eff}}$  as below:

$$S_{\text{eff}} = - \underbrace{\int_0^\beta d\tau \int_0^\beta d\tau' \sum_{\sigma} c_{s,\sigma}^\dagger(\tau) \mathcal{G}_0^{-1}(\tau - \tau') c_{s,\sigma}(\tau')}_{\text{Local effective field } \mathcal{G}_0 \text{ for the interaction with environment}} + U \int_0^\beta d\tau n_{s,\uparrow}(\tau) n_{s,\downarrow}(\tau) \quad (5.2)$$

Using the mean-field picture described above, one obtains <sup>3</sup> the following relation between the Green function of the local site and the non-interacting density of state (DOS) of the lattice:

$$G(i\omega_n) = \tilde{D}(i\omega_n + \mu - (\mathcal{G}_0^{-1}(i\omega_n) - G^{-1}(i\omega_n))) \quad (5.3)$$

$$\tilde{D}(\zeta) = \int_{-\infty}^{+\infty} d\varepsilon \frac{D(\varepsilon)}{\zeta - \varepsilon}. \quad (\text{Hilbert Transform}) \quad (5.4)$$

Here, we have implicitly taken the Fourier transform to Matsubara frequencies  $i\omega_n$ . The DOS is non-interacting due to the mean-field approximation on the lattice surrounding the local site. This non-interacting DOS  $D(\varepsilon)$  for infinite-dimensional lattice is available in many cases [208, §II.C]. Additionally, when an explicit formula of the inverse function  $R$  of the Hilbert Transform  $\tilde{D}(\zeta)$  is available, we often see an equivalent form of the self-consistency condition as below:

$$\mathcal{G}_0^{-1}(i\omega_n) = i\omega_n + \mu + G(i\omega_n)^{-1} - R[G(i\omega_n)], \quad (5.5)$$

$$\text{where } R(\tilde{D}(\zeta)) = \zeta. \quad (5.6)$$

**Self-consistent calculation.** We want to note that the self-consistency condition Eq. (5.5) and the action Eq. (5.2) form a closed system of equations which can lead to self-consistent calculations. Specifically, when our assumptions for the system break down, for example, when the lattice’s dimension is not infinite, or when the environment is finite and hence not translational invariant, we can still believe in a certain relation as in Eq. (5.5) holds, and find a self-consistent field  $\mathcal{G}_0$ .

For example, given a fixed density of state (fixed  $R$ ), we can (i) start with an initial guess for the on-site Green function  $G$ , (ii) compute the effective

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<sup>3</sup>See “the cavity method” in Georges et al. [208].

field  $\mathcal{G}_0$  using the self-consistency condition (Eq. (5.5)), (iii) compute the on-site Green function again but from the action  $S_{\text{eff}}$  where the  $\mathcal{G}_0$  (Eq. (5.2)) is substituted by its value obtained in (ii), and (iv) repeat steps (i-iii) but in (i) we use the  $G$  computed in a previous iteration of (iii). Such a method leads to a self-consistency loop which stops only when the  $G$  no longer changes across iterations. That is, the calculation becomes “self-consistent” in the sense that our final values for  $G$  agree with both Eq. (5.2) and Eq. (5.5).

**A local view from self-energy.** Here, we note that the derivation of self-consistency condition assumed a local self-energy. In fact, this local self-energy provides a different and sometimes more nature view of DMFT.

Specifically, we can assume that the self-energy in real space is highly local, which leads to a constant self-energy after the lattice Fourier transform to the  $k$ -space. We can derive the self-consistency condition from this perspective.

First, we note that normally in  $k$  space we expect a  $k$ -dependent relation between the on-site Green function and on-site self-energy (both in the  $k$  space) in Eq. (5.7). However, in DMFT we assume that self-energy losses the  $k$ -dependency as  $\Sigma(i\omega_n, k) = \Sigma(i\omega_n)$  (which is correct in infinite dimensions [208]). Then, after performing the inverse of the lattice Fourier transform ( $G(i\omega_n) = \sum_k G(i\omega_n, k)$ ) and changing the integration to energy using DOS  $D(\varepsilon)$ , we would have the Eq. (5.8) often seen in the literature. This would be the self-consistency condition (Eq. (5.3)) above if we identify the self-energy with the other terms as in Eq. (5.9).

$$G(i\omega_n, k) = \frac{1}{i\omega_n + \mu - \varepsilon_k - \Sigma(i\omega_n, k)}, \quad (5.7)$$

$$G(i\omega_n) = \int_{-\infty}^{+\infty} d\varepsilon D(\varepsilon) \frac{1}{i\omega_n + \mu - \varepsilon - \Sigma(i\omega_n)}, \quad (5.8)$$

$$\Sigma(i\omega_n) = \mathcal{G}_0^{-1}(i\omega_n) - G^{-1}(i\omega_n). \quad (5.9)$$

Note that, if we focus on self-energy only, the aforementioned self-consistency calculation can also be understood (as is often done in the literature) as starting with a guessed self-energy.

**Anderson impurity Hamiltonian.** Note that the above formulation is inconvenient as we would like a formulation using Hamiltonian in finite dimensional space for numerical purposes. For this, we turn our attention to an impurity Hamiltonian below, which also explains the common practice of calling

the chosen site an impurity and its environment the bath.

We focus on a single impurity Anderson model (SIAM) describing the interaction of a single impurity site and the bath<sup>4</sup>. In this Hamiltonian (Eq. (5.10)), electrons on the impurity site experience interactions between themselves, and can hop to and from a bath which is non-interacting. Integrating out the bath operators [208, §II.B], the non-interacting Green function of the impurity on this Hamiltonian has the form in Eq. (5.11) where part of it, the  $\Delta(i\omega_n)$ , is called the hybridisation function.

Note that in the equation below, the symbol  $a$  is temporarily used for operators on bath sites to distinguish it from  $c$  on impurity sites.

$$H_{\text{SIAM}} = \underbrace{\sum_{l\sigma} \epsilon_l a_{l\sigma}^\dagger a_{l\sigma}}_{\text{Bath}} + \underbrace{\sum_{l\sigma} V_l \left( a_{l\sigma}^\dagger c_{\text{imp},\sigma} + c_{\text{imp},\sigma}^\dagger a_{l\sigma} \right)}_{\text{Hopping}} - \underbrace{\mu \sum_{\sigma} c_{\text{imp},\sigma}^\dagger c_{\text{imp},\sigma} + U n_{\text{imp}\uparrow} n_{\text{imp}\downarrow}}_{\text{Impurity}} \quad (5.10)$$

$$G_{0,\text{imp}}^{-1}(i\omega_n) = i\omega_n + \mu - \underbrace{\sum_{l\sigma} \frac{V_l^2}{i\omega_n - \epsilon_l}}_{\Delta(i\omega_n)}. \quad (5.11)$$

It is believed [208, §II.B] that the form of  $G_{0,\text{imp}}$  above is general enough to reproduce the effective field  $\mathcal{G}_0$  of the mean-field equation. That is, in the self-consistency condition (Eq. (5.5)), the non-interacting  $G_{0,\text{imp}}$  replaces the field  $\mathcal{G}_0$  and the interacting  $G_{\text{imp}}$  replaces the on-site  $G(i\omega_n)$ . The parameters for this Hamiltonian (bath energy  $\epsilon_l$ , hopping constants  $V_l$ ) need to be determined self-consistently such that Eq. (5.5) holds after these replacements.

**Exact diagonalisation.** Finally, for making use of the wide range of methods developed for the solution of matrix problems, and also for (in our case) the application of variational quantum algorithms, the bath of infinite sites (the summation  $l$  are over an infinite amount of bath orbitals) are limited to finite sites. This is called the discretisation of the bath and is seen as an extra approximation. However, it is hoped that such projection from the continuum to the finite would not create a convergence problem for the self-consistency loop, and once the self-consistency condition is met, the finite-dimensional bath and the elements in the hopping matrix  $V_l$  provides an effective low-

<sup>4</sup>DMFT but with multiple impurity sites exist, and are sometimes discussed under the name of cluster DMFT [210, 211].

dimensional model for a single site in the lattice.

In general, methods to solve impurity problems are called impurity solvers, and exact diagonalisation methods are often applied in solving on classical computers. Obviously, the dimension of the Hilbert space of the impurity Hamiltonian  $H_{\text{SIAM}}$  grows exponentially with respect to the number of bath sites, and yet in practice, we would like to study models with the least discretisation error possible. This creates an interest in using quantum algorithms as the solver. Since the core object in DMFT is the Green function of the impurity Hamiltonian, the various methods for computation of Green functions discussed in Chapter 4 can be directly applied here, which may be called quantum impurity solvers.

Historically, many researchers have explored this possibility. Kreula [155, 156] and Keen [158] explored computing the Green function in the time domain using variational quantum algorithms and used the algorithm as impurity solver by Fourier transform the resulted Green function to the frequency domain. Instead, Bauer et al. [157] proposed an alternative Hilbert transform to take the time-domain Green function directly to the imaginary (Matsubara) frequency, where functions are usually a lot smoother. Jamet *et al.* proposed a string of work combining different ideas on quantum computers (variational compiling [167], quantum subspace extension [169], and forming quantum circuits [170]) to the Lanczos method based continued fraction representation of Green function for DMFT problems.

In our case, we are driven by the desire for a proof-of-concept demonstration on real quantum computers. Therefore, we focused on the real frequency where the Green function can be well-approximated from a few low-lying excited states, and made several simplifications based on symmetry arguments for the DMFT self-consistency condition. We were able to conduct such an experiment on IBM's superconductor-based quantum computers and revealed a series of potential problems that might appear in further applications of DMFT on quantum computers. We detail our simplifications and the data obtained in the following section.

## 5.2 Two-site DMFT on a quantum computer

For a proof-of-concept experiment on the quantum computer, we focused on a toy DMFT model discussed by Potthoff [212] and brought to the quantum computation community by Kreula et al. [155]. It uses a two-site impurity model to study the dynamics of Hubbard model on an infinite-dimensional

|       | Meaning                       |
|-------|-------------------------------|
| $c_1$ | $c_{\text{imp}, \uparrow}$    |
| $c_2$ | $c_{\text{bath}, \uparrow}$   |
| $c_3$ | $c_{\text{imp}, \downarrow}$  |
| $c_4$ | $c_{\text{bath}, \downarrow}$ |

**Table 5.1:** Subscripts used for the two-site DMFT formulae.

This subscript makes a direction comparison with the Jordan-Wigner transformed Hamiltonian easier. It is illustrated here for the annihilation operator.

Bethe lattice through the DMFT self-consistency condition. The impurity model has only one impurity site and one bath site, and hence this is called a two-site DMFT. Note that in the following, we have set the scaled interaction strength  $t = 1$  for simplification (see Section 5.1, page 96).

We are interested in the model for two reasons. First, it is inspiring physically. Potthoff [212] shows that this model gives good qualitative results for known limits. Second, as we will show later, it is small enough for currently (as of 2019) available quantum computers, although (as shown later) significant simplifications were needed. Therefore, this 2-site model provides a promising ground for the proof-of-principle.

Here we will first reiterate a few formulae in explicit forms which will be followed by details of various simplifications made.

### 5.2.1 Two-site DMFT

The SIAM Hamiltonian discretised to two sites are written below, where we used the indices detailed in Table 5.1. In the Hamiltonian, the repulsion  $U$  is inherited from the Hubbard model on the original lattice, and  $\mu$  is the chemical potential.  $\epsilon_b$  and  $V$  are model parameters, with the former denoting the effective energy on the bath site and the latter indicating the energy associated with the hopping process between the impurity and the bath.

$$\begin{aligned}
 H_{\text{2site}} = & \underbrace{U c_1^\dagger c_1 c_3^\dagger c_3}_{\text{Pauli repulsion}} - \underbrace{\mu(c_1^\dagger c_1 + c_3^\dagger c_3)}_{\text{Impurity (chemical potential)}} + \underbrace{\epsilon_b(c_2^\dagger c_2 + c_4^\dagger c_4)}_{\text{Bath}} \\
 & + V \underbrace{(c_1^\dagger c_2 + c_2^\dagger c_1 + c_3^\dagger c_4 + c_4^\dagger c_3)}_{\text{Interaction between impurity and bath}},
 \end{aligned} \tag{5.12}$$

The real-frequency (retarded) Green function is written explicitly in Eq. (5.13), where the concept of particle states (subscript  $p$ ) and hole states (subscript  $h$ ) are introduced for elegant notations. The particle states are

eigenstates having one more electron than the ground state, and hole states are eigenstates having one less electron than the ground state. The summation  $\sum_{n_p}$  ( $\sum_{n_h}$ ) is over all the particle (hole) states. The differences  $\Delta E$  are the differences between the corresponding energies ( $E_{n_h}$  for hole states and  $E_{n_p}$  for particle states) and the ground state energy  $E_{\text{GS}}$ . The order of the two is made such that the denominators in the Green function have a similar form of  $(z - \Delta E)$ .

$$G(z)_{ij} = \sum_{n_h} \underbrace{\frac{|\langle \psi_{n_h} | c_i | \psi_{\text{GS}} \rangle|^2}{z - \Delta E_{n_h}}}_{\text{Hole contributions}} + \sum_{n_p} \underbrace{\frac{|\langle \psi_{n_p} | c_i^\dagger | \psi_{\text{GS}} \rangle|^2}{z - \Delta E_{n_p}}}_{\text{Particle contributions}}, \quad (5.13)$$

where  $z = \omega + i0^+$ ,  $i(j) \in \{1, 2, 3, 4\}$ .

$$\Delta E_{n_h} = E_{\text{GS}} - E_{n_h}, \quad (5.14)$$

$$\Delta E_{n_p} = E_{n_p} - E_{\text{GS}}. \quad (5.15)$$

With the spin-flip symmetry in the impurity Hamiltonian (Hamiltonian remains unchanged when exchanging  $1 \leftrightarrow 3$  and  $2 \leftrightarrow 4$ ), we focus only on the spin-up channel and compute the impurity Green function  $G_{\text{imp}}$  in this sector:

$$G_{\text{imp}} \equiv G_{11}. \quad (5.16)$$

The non-interacting Green function of the impurity site with a discretised hybridisation function is written explicitly in real-frequency as below<sup>5</sup>:

$$G_{0,\text{imp}}(\omega) = (\omega + i0^+ + \mu - \Delta(\omega))^{-1}, \quad (5.17)$$

$$\Delta(\omega) = \frac{V^2}{\omega - \epsilon_b}. \quad (5.18)$$

**Self-consistency conditions.** Importantly, there are only two adjustable parameters for the self-consistency loop in this two-site SIAM Hamiltonian: the interaction  $V$  and bath energy  $\epsilon_b$ . Therefore, only two equations are needed to fix them. This has been done in Potthoff [212] where the self-consistency condition is simplified into the two conditions below:

$$n_{\text{imp}} = n_{\text{lat}}, \quad (5.19)$$

$$V^2 = z. \quad (5.20)$$

---

<sup>5</sup>Note that the sum over spin in hybridisation in Eq. (5.11) is no longer needed as we focus on one spin channel only.

The first one relates the occupation of the impurity  $n_{\text{imp}}$  to the occupation of the original lattice  $n_{\text{lat}}$ . The occupations number is computed from integration of the density of state (DOS):  $n = \int_{-\infty}^0 \text{DOS}(\omega) d\omega$ . The impurity DOS is  $\text{DOS}_{\text{imp}}(\omega) = -\frac{2}{\pi} \text{Im} G_{\text{imp}}(\omega + i0^+)$ , and the lattice DOS is  $-\frac{2}{\pi} \text{Im} G(\omega + i0^+)$ , where the lattice Green function is obtained by Eq. (5.8)<sup>6</sup>.

The second one is obtained by inspecting the small  $\omega$  limit of interacting  $G_{\text{imp}}$ . Here,  $z$  is called the quasi-particle weight and can be calculated by<sup>7</sup>:

$$z = \frac{1}{1 - \left. \frac{d \text{Re}[\Sigma(\omega)]}{d\omega} \right|_{\omega=0}} = \frac{1}{1 - \frac{\text{Im}[\Sigma(i0^+)]}{0^+}}, \quad (5.21)$$

Notably, the self-consistency condition can be further simplified if we introduce a particle-hole symmetry to the Hamiltonian. This can be done by setting  $\mu = U/2$  and  $\epsilon_b = 0$ , and ensures the first self-consistency condition in Eq. (5.19) is satisfied automatically [155, 212]. Hence, the parameters for the impurity Hamiltonian are now limited to just one, the interaction  $V$ .

**Self-consistency loop.** The self-consistency loop is explicitly carried out as follows. We begin with a certain initial parameter for  $V$ , which specifies the form of the impurity Hamiltonian  $H$  in Eq. (5.12). We solve this Hamiltonian to compute the Green function of the impurity site  $G(\omega)$  given the impurity Hamiltonian. Meanwhile, the non-interacting Green function is obtained with Eq. (5.17). Then with the self-energy computed from the Dyson equation (Eq. (5.9)), we can get the quasi-particle weight  $z$  from Eq. (5.21). Next, we will compare and see if the computed quasi-particle weight is compatible with the  $V$  under the second self-consistency condition in Eq. (5.20). If they agree, the self-consistency loop is terminated. Otherwise, we propose a new value for  $V$ , by mixing  $V$  via the rule below. It is determined with a heuristic mixing constant  $c_{\text{mix}}$ :

$$V_{\text{new}} = (1 - c_{\text{mix}})V_{\text{previous}} + c_{\text{mix}}\sqrt{z}, \quad (5.22)$$

New value  $V_{\text{new}}$  computed from the equation above is then fed as input to the next iteration of the self-consistency loop, until such procedure converges.

<sup>6</sup>Note that we need to replace the Matsubara frequency  $i\omega_n$  in Eq. (5.8) by  $\omega + i0^+$ , use the non-interacting DOS of Bethe lattice  $(1/2\pi)\sqrt{4 - \epsilon^2}$  as the  $D(\epsilon)$ , and replace  $\Sigma$  by  $\Sigma_{\text{imp}}$  there.

<sup>7</sup>Eq. (5.21) assumes that when  $\omega \rightarrow 0$ ,  $\Sigma \approx a + b(\omega + i0^+)$  for certain  $a, b \in \mathbb{R}$  [212, eq. (16)].

### 5.2.2 Computation on quantum computers

Now we discuss the details of computing real-frequency Green functions on quantum computers. We rewrite the Lehmann representation (Eq. (5.13)) as below:

$$G(z) = \sum_{n_h} \frac{\lambda_{n_h}}{z - \Delta E_{n_h}} + \sum_{n_p} \frac{\lambda_{n_p}}{z - \Delta E_{n_p}}, \quad (5.23)$$

where we introduce the notation  $\lambda$ , which is the square of the absolute value of the overlap amplitudes of the corresponding eigenstates ( $\psi_{n_h}$  for hole state and  $\psi_{n_p}$  for particle state) and the ground state ( $\psi_{\text{GS}}$ ), separated by the corresponding creation or annihilation operator on the impurity site of the spin-up channel (subscript 1, see Table 5.1):

$$\begin{aligned} \lambda_{n_h} &= |\langle \psi_{n_h} | c_1 | \psi_{\text{GS}} \rangle|^2, \\ \lambda_{n_p} &= |\langle \psi_{n_p} | c_1^\dagger | \psi_{\text{GS}} \rangle|^2. \end{aligned} \quad (5.24)$$

**Jordan-Wigner transformation.** As usual, the impurity Hamiltonian is transformed by Jordan-Wigner transformation into Eq. (5.25)<sup>8</sup>. After the introduction of particle-hole symmetry, it is further simplified to Eq. (5.26).

$$\begin{aligned} H &= \frac{U}{4} Z_1 Z_3 + \left( \frac{\mu}{2} - \frac{U}{4} \right) (Z_1 + Z_3) - \frac{\epsilon_2}{2} (Z_2 + Z_4) \\ &\quad + \frac{V}{2} (X_1 X_2 + Y_1 Y_2 + X_3 X_4 + Y_3 Y_4). \end{aligned} \quad (5.25)$$

$$H_{\text{ph}} = \frac{U}{4} Z_1 Z_3 + \frac{V}{2} (X_1 X_2 + Y_1 Y_2 + X_3 X_4 + Y_3 Y_4). \quad (5.26)$$

We note that one often reorders indices or redefines the Jordan-Wigner transformation to minimise the weights of Pauli operators. Here, since the impurity site of upper-spin has subscript 1 (Table. 5.1), the original Jordan-Wigner transformation mapping naturally minimised the number of  $Z$  operators in the definition of  $c_1^{(\dagger)}$ , and hence simplifies calculations of the  $\lambda$  in Eq. (5.24).

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<sup>8</sup>Constant terms are ignored here. Relevant calculations are available in a Jupyter notebook deposited online [213].

**Overlap computations.** The Jordan-Wigner transformation above implies that the overlap  $\lambda$ s are equal to the following terms:

$$\begin{aligned}\lambda_{n_h} &= \left| \left\langle \psi_{n_h} \left| \frac{X_1 + iY_1}{2} \right| \psi_{\text{GS}} \right\rangle \right|^2, \\ \lambda_{n_p} &= \left| \left\langle \psi_{n_p} \left| \frac{X_1 - iY_1}{2} \right| \psi_{\text{GS}} \right\rangle \right|^2.\end{aligned}\tag{5.27}$$

To compute such terms, we notice that first, the Jordan-Wigner transformation effectively relates the Pauli operators with creation and annihilation operators (Eq. (5.28)). Meanwhile, overlap containing certain creation or annihilation operators are zero (Eq. (5.29))<sup>9</sup>.

$$c_1^\dagger + c_1 = X_1, \quad c_1 - c_1^\dagger = iY_1.\tag{5.28}$$

$$\langle \psi_{n_h} | c_1^\dagger | \psi_{\text{GS}} \rangle = 0, \quad \langle \psi_{n_p} | c_1 | \psi_{\text{GS}} \rangle = 0.\tag{5.29}$$

Combining the above Eq. (5.28) and Eq. (5.29), we can show that the overlaps can be equivalently computed as:

$$\begin{aligned}\lambda_{n_h} &= |\langle \psi_{n_h} | X_1 | \psi_{\text{GS}} \rangle|^2, \\ \lambda_{n_p} &= |\langle \psi_{n_p} | X_1 | \psi_{\text{GS}} \rangle|^2.\end{aligned}\tag{5.30}$$

Note that we can also equivalently use Pauli  $Y_1$  instead of Pauli  $X_1$  in both equations. The advantage of the above expressions in Eq. (5.30), is such that if the ground states  $\psi_{\text{GS}}$  and the eigenstates  $\psi_n$  are both prepared by unitary circuits, say  $U_{\text{GS}}$  and  $U_n$ , then we can express the overlaps as simple as

$$|\langle \psi_n | X_1 | \psi_{\text{GS}} \rangle|^2 = |\langle 0 | U_n^\dagger X_1 U_{\text{GS}} | 0 \rangle|^2,\tag{5.31}$$

which is just the probability of measuring  $|0\rangle$  state after the circuit  $U_n^\dagger X_1 U_{\text{GS}}$  is executed on an initial state of  $|0\rangle$ .

**Eigenstate preparations on quantum computers.** Finally, we show how the various eigenstates mentioned previously for computing the Green functions can be prepared on the quantum computer. Several VQE based methods have been proposed to find those excited states [27]. We take directly the method proposed by Higgott et al. [73], which helps find the first excited state (and

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<sup>9</sup>Since creation/annihilation operators are linear maps in the Fock space from eigenspaces of lower/higher electron numbers to higher/lower electron numbers. Overlap of states from different eigenspaces naturally vanishes.

recursively all other excited states) of a Hamiltonian. In it, one uses VQE to find the ground state of the modified Hamiltonian  $H'$  which penalises the variational ansatz for being close to the ground state:

$$H' = H + \beta_{\text{energy}} U_{\text{GS}} |0\rangle \langle 0| U_{\text{GS}}^\dagger, \quad (5.32)$$

The coefficient  $\beta_{\text{energy}}$  quantifies the penalty. Theoretically, the penalty  $\beta_{\text{energy}}$  should be chosen such that contributions from the ground state towards measured energy  $\langle H' \rangle$  is higher than the contributions due to this penalty term from the first excited states. Hence,  $\beta_{\text{energy}}$  should be at least greater than the energy differences between the ground state and the first excited state. In practice though, the penalty term is chosen heuristically, not too small but also not too large to avoid causing numerical instability in optimisation.

Further steps are required to ensure that the prepared excited states belong to the eigenspace of the expected number of electrons, i.e. they are proper particle states or hole states. This can be achieved with further penalty terms [18]<sup>10</sup>, where we insert an additional modification to the Hamiltonian, to penalise the deviation from the target number of electrons ( $n_t$ ):

$$H'' = H' + \beta_{\text{electron}} (N - n_t)^2, \quad (5.33)$$

where the coefficient  $\beta_{\text{electron}}$  represents the penalty, and the operator  $N$  is the total electron number operator of both impurity orbitals and bath orbitals (and should be Jordan-Wigner transformed accordingly). If the number of electrons in the ground state is  $n_{\text{GS}}$ , then the targeted electron number  $n_t$  is  $n_{\text{GS}} + 1$  for particle states and  $n_{\text{GS}} - 1$  for hole states.

Finally, it is necessary to carry out an eigenstate check by measuring the variance of both the Hamiltonian and the electron number operator, as mentioned in Sec. 2.1.3.1. This is due to the tricky problems involved in fixing the electron number using the above penalty method. Specifically, first, there is no mathematical guarantee that an eigenstate of the modified Hamiltonian  $H''$  will be also an eigenstate of its two constituents  $H'$  and  $(N - n_t)^2$ .<sup>11</sup> Secondly and more importantly, it is possible that a certain linear combination of two eigenstates of  $H'$ , having electron number of e.g.  $n_t - 1$  and  $n_t + 1$ , can satisfy penalty term better than the targetted eigenstate of electron

<sup>10</sup>A more efficient method based on particle-hole preserving ansatz [41] has been used in Ehrlich et al. [206].

<sup>11</sup>Even though here we have  $H'$  commutes with  $N$  and then with  $(N - n_t)^2$ .

number  $n_t$ , which is more problematic. The problem arises since we have no knowledge of the energy spectrum and particle number spectrum of the Hamiltonian  $H$  itself, and therefore cannot provide guarantees on a given penalty  $\beta_{\text{electron}}$ . Therefore, it is important to check the particle number by measuring the number operator after a VQE optimisation.

### 5.2.3 Simplifications from symmetry

Here we report simplifications on the Green function based on symmetry arguments to reduce the computations required. The Hamiltonian observes particle number and spin- $z$  symmetry<sup>12</sup>, where the number operator  $N$  and the spin- $z$  operator  $S_z$  are written in Pauli operators (after Jordan-Wigner transformation) below. Clearly, contributions (i.e. expectations of Pauli  $Z_i$  for  $i = \{1, 2, 3, 4\}$ ) to the total particle number and total spins are related, leading to restrictions on the total spin number once a certain total particle number is given.

$$N = 2 - \frac{1}{2}(Z_1 + Z_2 + Z_3 + Z_4), \quad (5.34)$$

$$S_z = \frac{1}{2}(-Z_1 - Z_2 + Z_3 + Z_4). \quad (5.35)$$

Thus, for a finite-dimensional system, one can analyse different particle number sectors ( $\langle N \rangle = 1, 2, 3, \dots$ ) and define sub-algebras of the algebra of creation and annihilation operators in those sectors. These sub-algebras would be inspired by the restricted total spin numbers in these sectors. The analysis was carried out by our colleagues in Rungger et al. [30] and verified independently in Appendix E.

However, it is questionable if a similar analysis can be done for general impurity Hamiltonians. In fact, the analysis by our colleagues effectively “diagonalised” our four-qubit impurity Hamiltonian into different blocks of specific particle numbers. In general, we expect a factorial number of such blocks to exist for the general case, which would make the analysis impractical.

Similarly, our verification in Appendix E effectively solves the four-qubit impurity Hamiltonian analytically. Therefore, it can be said that the success of our proof-of-concept demonstration of a self-consistency loop relies heavily on such analytical block diagonalisation.

In the following, we summarise the simplifications resulting from this analysis. We also use a particular set of subscripts as in  $\lambda_{h,1}$  and  $\lambda_{h,2}$ . The

<sup>12</sup>We can numerically check the commutation rules  $[H, N] = [H, S_z] = 0$  and  $[N, S_z] = 0$ .

$\{h, 1\}$  ( $\{h, 2\}$ ) corresponds to the first (second) hole states found via the analysis and has no relation to the first (second) qubit.

**Simplifications on Green function formula.** The analysis leads to simplification of the Green function computation where only contributions from the hole sector are needed. The particle sector contributions are labelled in the simplification below, but their energies and overlaps are computed from hole sector contributions using the particle-hole symmetry. The simplified Green function is:

$$G(z) = \lambda_{h,1} \left( \underbrace{\frac{1}{z - \Delta E_{h,1}}}_{\text{Hole}} + \underbrace{\frac{1}{z + \Delta E_{h,1}}}_{\text{Particle}} \right) + \underbrace{\left( \frac{1}{2} - \lambda_{h,1} \right)}_{\text{Sum rule}} \left( \underbrace{\frac{1}{z - \Delta E_{h,2}}}_{\text{Hole}} + \underbrace{\frac{1}{z + \Delta E_{h,2}}}_{\text{Particle}} \right). \quad (5.36)$$

Here  $\Delta E_{h,1/2}$  are the corresponding energy differences of  $E_{h,1/2}$ . A sum rule below is also used for the formula above<sup>13</sup>.

$$\sum_{n=n_p, n_h} \lambda_n = 1. \quad (5.37)$$

Another sum rule, which comes from forcing the divergence of the interacting and non-interacting Green function at  $\omega + i0^+ \rightarrow i0^+$  to cancel each other (see Appendix E.1.4), gives the following simplified formula for the overlap:

$$\lambda_{h,1} = \frac{(\Delta E_{h,1})^2 (V^2 - (\Delta E_{h,2})^2)}{2V^2 ((\Delta E_{h,1})^2 - (\Delta E_{h,2})^2)}. \quad (5.38)$$

We would like to comment here that the sum rule for overlaps (Eq. (5.37)) exists for general Hamiltonians, and hence a simplified formula of the form Eq. (5.36) is possible for general impurity Hamiltonians if particle-hole symmetry can be introduced. However, computing the overlap from energies (as in Eq. (5.38)) relies on the simple four-qubit Hamiltonian we have, and would be impossible for general systems where overlaps from other symmetry sectors would exist on the right-hand side of Eq. (5.38).

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<sup>13</sup>This sum rule reflects the anti-commutation relation  $c^\dagger c + c c^\dagger = 1$ .

**Simplifications on the Hamiltonians.** Additionally, the analysis in different sectors of particle number operators leads to simplified Hamiltonians valid only for the corresponding sectors of fixed particle numbers. They are possible by introducing sub-algebras in such sectors. From the analysis, we knew that the ground state exists in a subspace of two electrons and that the hole states exist in a subspace of one-electron. We only need the Hamiltonians defined by those sub-algebras on these two subspaces. Running VQE on the two-electron subspace Hamiltonian would give the ground state energy. Running (excited state) VQE on the one-electron subspace would give the energies of two hole states. Together, this would be sufficient for the computation of overlap (Eq. (5.38)) and hence for the Green function (Eq. (5.36)).

For the one-electron subspace, the 4-qubit Hamiltonian can be effectively reduced to a one-qubit Hamiltonian:

$$H_{\text{one-electron}} = \frac{U}{4}Z + VX, \quad (5.39)$$

whose two energies eigenstates corresponds to the  $E_{h,1}$  and  $E_{h,2}$  in Eq. (5.36). For the two-electron subspace, the 4-qubit Hamiltonian can be effectively reduced to a 2-qubit Hamiltonian:

$$H_{\text{two-electron}} = \frac{U}{4}Z_1Z_2 + V(X_1 + X_2), \quad (5.40)$$

whose ground state energy corresponds to ground state  $E_{\text{GS}}$  of the 4-qubit Hamiltonian.

### 5.2.4 Ansatz

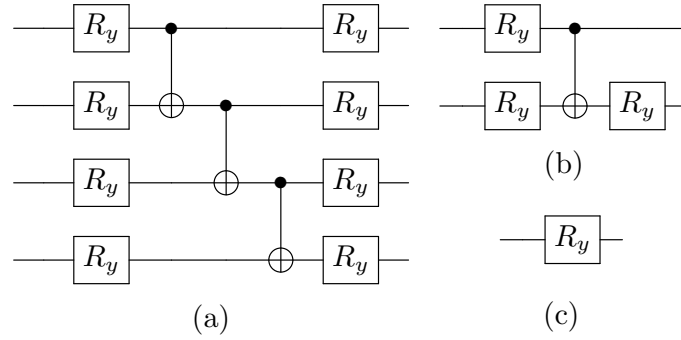
Let us digress a bit to introduce the ansatz we used to prepare the ground and excited states. The ansatzes discussed below are illustrated in Figure 5.1.

For the 4-qubit case, which we will still investigate to probe the validity of preparing different eigenstates of specific particle numbers for computation of the Green function without the simplifications made above, we used a single-layer HEA ansatz where the single-qubit rotations are restricted in the  $y$ -axis, i.e.  $R_y$  gates. This is inspired by the fact that real-valued (in  $\mathbb{R}$ ) wavefunctions can often be found for the system<sup>14</sup>.

For simplified Hamiltonians, a corresponding reduction in the ansatz is needed. Here, for the one-electron Hamiltonian, we used a simple one-

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<sup>14</sup>In this specific case, the Hamiltonian is real-valued.



**Figure 5.1:** VQE Ansatz for the reduced Hamiltonians. Here, (a) illustrates the ansatz used for the full 4-qubit Hamiltonian (Section 5.2.2). (b) and (c) illustrate the two ansatz used for the reduced Hamiltonians (Section 5.2.3). (b) is for its two-qubit sub-problem, and (c) is for its one-qubit sub-problem. Note that, although we haven't shown any angles in these illustrations, each rotation gate  $R_y$  has its own free-parameter  $\theta$  which can be adjusted independently of other rotation gates.

parameter ansatz. For the two-electron Hamiltonian, we used another simple 3-parameter ansatz. Both are visualized in Figure 5.1.

Note that for the one-qubit problem, a trick is available which allows us to obtain both energies  $E_{h,1}$  and  $E_{h,2}$  with just one VQE optimisation. Since here the Hilbert space dimension is 2, the excited state with the energy of, say  $E_{h,2}$ , can be obtained by simply constructing a new state orthogonal to another state corresponding to the energy  $E_{h,1}$ . Orthogonalisation can be achieved by, after obtaining an optimised angle  $\theta^*$  for the VQE problem using the ansatz mentioned above, substituting this angle by  $\theta^* + \pi$  in that ansatz (Figure 5.1 (c)).

## 5.3 Experiments

Now we turn attention to actual implementations of the methods above on either simulated or real quantum computers. We provide a short description of the quantum computers we used in IBM in Appendix E.2. In Section 5.3.1 we show data which indicate that errors from IBM's quantum computers do not provide sufficient fidelity for completing a DMFT loop with sufficient accuracy without the simplifications, i.e. for the four-qubit Hamiltonian  $H_{2\text{site}}$  (Eq. (5.12)). However, using the reduced Hamiltonians from Section 5.2.3, the noise from the quantum computers becomes acceptable. In Section 5.3.2, we detail the results from a full DMFT self-consistency loop on a quantum computer and show that it converged with the resulted parameter values in

good agreement with the theory.

### 5.3.1 Four qubit VQE

Here we focus on computing the Green function when restricted to the case of  $U = 4$  and  $V = 0.7454$ , where notably 0.7454 is the value of  $V$  where the self-consistency loop should converge theoretically [30, 212].

First, we use a simulated quantum computer to run the VQE algorithms for preparing both the ground and excited states, and for computing the overlap  $\lambda$ . That is, we implement the methods outlined in Section 5.2.2. For optimisation, we used the RotoSolve algorithm (Section 2.1.5) for the ground and excited state Hamiltonian. For the excited state, the penalties were set as  $\beta_{\text{energy}} = 5$  and  $\beta_{\text{electron}} = 80$  (Eq. (5.32) and Eq. (5.33)). Each circuit execution was repeated 10,000 times to reduce statistical fluctuations from measurements.

Then, having obtained quantum circuits to prepare those states, we executed them on an IBM quantum computer. We aimed to test the reliability of quantum computers in computing those values while avoiding the cost of running full VQE optimisations. Each circuit on the quantum computer was repeated 4,096 times. We also used SPAM corrections to reduce the influence from readout errors. The SPAM matrices needed were obtained before each circuit ran on the same quantum computer, and each circuit required for SPAM matrices was executed 1,000 times. This SPAM corrections scheme is outlined in Section 2.1.4.2.

**Results.** The energies and overlaps computed both from the simulation and from the IBM quantum computer are summarised in Table 5.2 on page 112. There, we only included values for the ground state and the hole states, which are sufficient for Green function computation (particle-hole symmetry). Also, in this table, the energies and overlap- $\lambda$ s are labelled slightly differently: the first subscript 1 indicates that they are hole states (since  $\langle N \rangle = 1$  for them), and the second subscript simply differentiates them one-by-one.

Let us provide several comments on these results. First, we focused on energies and overlaps. Based on simulations, we see that the variational methods in combination with the RotoSolve algorithm are capable of preparing the ground and excited states to a high degree of accuracy. The reported errors (as percentage) for each value are small, at around 1% in most cases. The error reached 7% only in the case of  $\lambda_{1,0}$ . In absolute terms, the biggest error

|                      | Exact  | Simulation | E% | Raw QPU | E%  | SPAM Cor. | E% |
|----------------------|--------|------------|----|---------|-----|-----------|----|
| $E_{\text{GS}}$      | -1.795 | -1.801     | 0  | -1.196  | 33  | -1.500    | 16 |
| $E_{1,0}$            | -1.247 | -1.245     | 0  | -0.810  | 35  | -1.111    | 11 |
| $E_{1,1}$            | -1.247 | -1.236     | 1  | -0.790  | 36  | -0.981    | 21 |
| $E_{1,2}$            | 1.247  | 1.246      | 0  | 1.001   | 20  | 1.025     | 18 |
| $E_{1,3}$            | 1.247  | 1.232      | 1  | 0.934   | 25  | 0.972     | 22 |
| $\lambda_{1,0}$      | 0.262  | 0.245      | 7  | 0.145   | 45  | 0.113     | 57 |
| $\lambda_{1,1}$      | 0      | 2.0E-6     |    | 9.52E-3 |     | 2.38E-3   |    |
| $\lambda_{1,2}$      | 0.238  | 0.237      | 0  | 0.189   | 20  | 0.159     | 33 |
| $\lambda_{1,3}$      | 0      | 6.0E-6     |    | 1.05E-2 |     | 4.23E-3   |    |
| $\text{Im } G(i0^+)$ | -0.174 | -0.159     | 9  | -0.202  | 16  | -0.147    | 16 |
| $z$                  | 0.522  | 0.134      | 74 | -0.168  | 132 | 7.99E-2   | 85 |

**Table 5.2:** Experimental data obtained on the IBM quantum computer for four-qubit VQE: Here the circuits obtained from numerical simulations have been tested on IBM quantum computers (Section 5.3.1). These circuits computed the energies and overlap  $\lambda$ s for a few hole states. They were obtained through numerically simulated VQE on a classical computer (“Simulation”). They were then executed on the IBM quantum computer `ibmq_16_melbourne`. The resulted energies and  $\lambda$ s from the numerical simulation and from quantum computers were reported in “Raw QPU” and “SPAM Cor.”. The Green function  $\text{Im } G(i0^+)$  and quasi-particle weight  $z$  were computed using formulae in Section 5.2.2 and with  $0^+ = 0.1$ .

The energies are labelled  $\{1, i\}$ , where 1 indicates the particle number, and  $i$  indicates the order in which they are obtained in the iterative VQE excited state algorithm.

In the column “Exact”, we report the exact values for the corresponding quantities computed from numerical diagonalizing the Hamiltonian matrix. In the column “Simulation”, we report the values computed from numerical simulations, where each circuit are simulated with 10,000 repetitions. And in “Raw QPU”, we report raw values computed from executing the circuits on the IBM quantum computer, each run for 4096 repetitions. At the same time, the corresponding values after the SPAM corrections are reported in the column “SPAM Cor.”. Whenever appropriate, we calculated the error percentage, i.e. the differences of all computed values to their exact values as a percentage of the exact values, and the results were rounded to the nearest integer and are listed in the following columns, labelled as “E%”.

value is only at the order of  $10^{-2}$ , which could contribute to statistical sampling noise from measurements as we repeated each circuit 10,000 times<sup>15</sup>.

Based on quantum computers, we first notice that the SPAM correction reduced the percentage errors in most cases, especially for the calculation of energies. This confirmed the effectiveness of SPAM correction for relatively short depths circuits since the computation of energies requires roughly

<sup>15</sup>Note that  $1/\sqrt{10000} = 0.01$ .

half the circuit depths when compared with that of overlaps (Eq. (5.31)). However, for overlap computation, further error mitigation strategies may be needed. Also in general, the errors due to quantum computers are significantly higher than those expected from statistical fluctuations (in the range of 10% to 50%, compared against simulation numbers).

Second, we turn to the Green function evaluated at  $z = i0^+$  (with the broadening constant set as  $0^+ = 0.1$ ) and the quasi-particle weight, which are derived from energies and overlaps. Here, the Green function value calculation remains reasonably close to the exact value, for both the values from simulation and from quantum computers. However, the error increased significantly for  $z$ , the quasi-particle weight. We make three comments on this surge of error here: (i), there are two steps in the calculation of quasi-particle weight which involve the inversion of small numerical values and therefore amplify errors: the calculation of self-energy from the Dyson equation, and the calculation of quasi-particle weight from self-energy (Eq. (5.21)). (ii), the huge error percentage in  $z$  from simulations, means that the self-consistency loop which should converge<sup>16</sup> would not converge after this step, implying that even under a simulated environment, the self-consistency may converge wrongly or not converging at all. (iii), the fact that  $z$  could be negative (for values computed from quantum computers) violates an implied condition,  $z \geq 0$ , from the self-consistency condition (Equation 5.20), stopping the self-consistency loop entirely.

In summary, the DMFT self-consistency loop is impractical without the simplifications in Section 5.2.3. In fact, we would note that these simplifications were actually motivated to regularise the negative numerical values in quasi-particle weight  $z$ . Specifically, the self-consistency condition Eq. (5.20) implies a condition on overlaps and energies, which leads to the simplified formula for overlap in Eq. (5.38). We see how they have improved the result in the next section.

### 5.3.2 Full self-consistency loop on IBM

In Appendix E.1.4, we analysed theoretically the behaviour of self-energy (i.e., the behaviour of non-interacting and interacting Green functions) around the critical point of  $0 + i0^+$ . The result has been discussed in Section 5.2.3. Combined with the simplified Hamiltonians, we report below results from quantum computers. Specifically, we first report a test run at the converging point

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<sup>16</sup>We started with  $V$  being the theoretically converging value.

|                      | Exact  | Simulation | QPU (SPAM Cor.) | Error (%) |
|----------------------|--------|------------|-----------------|-----------|
| $E_{\text{GS}}$      | -1.795 | -1.795     | -1.823          | 2         |
| $E_{h,1}$            | -1.247 | -1.247     | -1.248          | 0         |
| $E_{h,2}$            | 1.247  | 1.247      | 1.248           | 0         |
| $\text{Im } G(i0^+)$ | -0.174 | -0.174     | -0.175          | 1         |
| $z$                  | 0.556  | 0.556      | 0.610           | 10        |

**Table 5.3:** Experimental data obtained on the IBM quantum computer for the reduced problem. Here the circuits obtained from numerical simulations have been tested on the IBM quantum computer `ibmq_london`. The setup is similar to that mentioned inside Table. 5.2, except that we have used the simplified Hamiltonians discussed in Section 5.2.3. The resulted energies and  $\lambda$ s from numerical simulation are reported in the column “Simulation”. The values from the quantum computer are reported in the column “QPU (SPAM Cor.)”, which have been corrected using SPAM. The relative error in percentage are reported in the columns following, titled “Error %”.

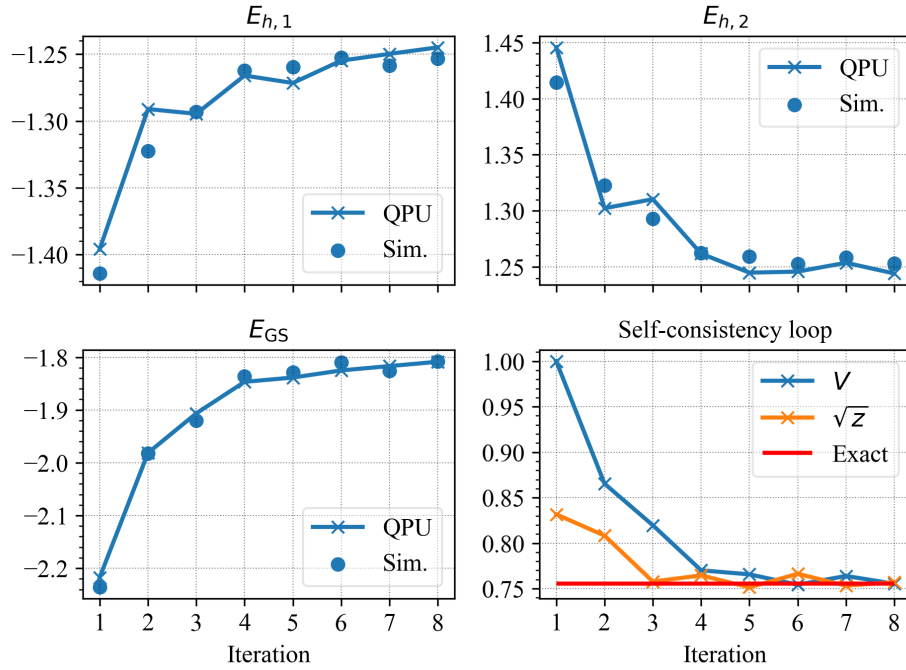
of  $V = 0.7454$ , and then detail a full DMFT self-consistency loop.

**Test run at converging  $V$ .** Similar to the previous experiments, we used optimised parameters obtained from simulated quantum computers and tested the resulting circuits on a quantum computer.

The result, available in Table 5.3, shows that the quantum computer reported nearly identical numbers as in simulated VQE runs for energies. Meanwhile, the error from quantum computers for quasi-particle weight  $z$ , which matters significantly for the self-consistency loop, is within 10% and in absolute terms limited to one decimal place.

**Full DMFT self-consistency loop.** Next, we report results from a full DMFT self-consistency loop on IBM quantum computers. The impurity Hamiltonian is initialised at  $U = 4$ , and we started with a guessed  $V$  of 1. The mixing parameter  $c_{\text{mix}}$  (Equation 5.22) was set at a conservative value of 0.8 to stabilise the convergence process. As before, the small constant for retarded Green function is set as  $0^+ = 0.1$ .

The experiments were executed on any quantum computers on IBM having the shortest queue at the time of running the experiment. Again, we used the SPAM mitigation scheme to mitigate readout errors (Section 2.1.4.2). In addition, each circuit was executed with 8192 shots. Each energy values were averaged from the last three executions of the same circuit, done in response to the limitation on the maximum repetition number allowed by the IBM service at the time of experiments.



**Figure 5.2:** Experimental data for self-consistency loop on 2-qubits. We report data for 8 iterations, in which the self-consistency calculation  $V$  converged to the exact values  $V$  as indicated in the plot. For each iteration, we reported both the energies ( $E_{GS}$ ,  $E_{h,1}$ ,  $E_{h,2}$ ) computed from VQE loops on the quantum computer (“QPU”), and the energies predicted from numerically diagonalizing the Hamiltonians using NumPy (“Sim.”). The two values are quite close throughout iterations. Finally, on the bottom right figure, we show the  $V$  and  $\sqrt{z}$  (square root of quasi-particle weight) calculated from those energies obtained from VQE on quantum computers. They quickly converged to the exact values predicted by the theory. The exact value is indicated by the red vertical line in the bottom right graph.

As expected from the accurate energy readings from the previous test case, the self-consistency loop we ran succeeded, and  $V$  converged to the exact value predicted theoretically. We plotted in Figure 5.2 energies computed from VQE on QPU, and the computed quasi-particle weight, across all 8 iterations of the self-consistency loop. The final converged value for  $V$  is 0.7567, which is close to the value of  $V = 0.7554$  from the previous iteration, indicating convergence as the difference is of the order of  $10^{-3}$ . They are also close enough to the exact value of 0.7454, indicating a successful DMFT self-consistency loop from the quantum computers. We can also appreciate this success in noting that the differences in  $V$  are indistinguishable from the plotted data in the figure.

|                         | Exact  | SPAM Cor. | E% |
|-------------------------|--------|-----------|----|
| $E_{\text{GS}}$         | -1.795 | -1.691    | 6  |
| $E_{1,0}$               | -1.247 | -1.178    | 6  |
| $E_{1,1}$               | -1.247 | -1.112    | 11 |
| $E_{1,2}$               | 1.247  | 1.144     | 8  |
| $E_{1,3}$               | 1.247  | 1.098     | 12 |
| $\lambda_{1,0}$         | 0.262  | 0.224     | 15 |
| $\lambda_{1,1}$         | 0      | 0.002     |    |
| $\lambda_{1,3}$         | 0.238  | 0.221     | 6  |
| $\lambda_{1,2}$         | 0      | 0.000     |    |
| $\text{Im } G(i\delta)$ | -0.174 | -0.171    | 2  |
| $z$                     | 0.522  | 0.329     | 37 |

**Table 5.4:** Experimental data obtained on University of Maryland’s ion-trap quantum computer.

The experiment executed the same circuits as discussed in Table 5.2. They were executed on an ion-trap based quantum computer by our colleagues in Maryland. The data here were copied from Table IV of Rungger et al. [30]. Additionally, we computed the imaginary part of retarded Green function ( $\text{Im } G(i\delta)$ ) and quasi-particle weight  $z$ , as well as the error percentages. They were computed using the same method discussed in Table 5.2.

## 5.4 Discussions and outlooks

Here we want to detail a few points that we believe are important for improving accuracies of future experiments. We will also comment on a recent (2024) experiment and discuss the improvements made there.

**Trade-off on ion-trap devices.** We note that ion-trap based quantum computers typically enjoy better fidelities than superconducting-based quantum computers while suffering from slower speed. A group of our colleagues from the University of Maryland has executed the same circuits on the four-qubit Hamiltonian for computing the energies and overlaps on their devices and reported better results than what we have shown in Table 5.2. The detailed data were replicated for convenience in Table 5.4. For most energies computed, ion-trap computer reduced error to around 10%, while on IBM we have around 10% to 20% error after the SPAM mitigation. Also, the error on overlap calculations was noticeably better.

However, since no simplification on the Hamiltonians was employed here, the computed quasi-particle weight  $z$  still suffered a 37% error and hinted at a failed DMFT self-consistency loop calculation under reasonable converging requirements between  $V^2$  and  $z$ .

A technical problem for the ion-trap based device is that they tend to be one or two orders of magnitude slower (hence no full DMFT loop experiments have been attempted on them) and also much more expensive. This might be resolved by future engineering improvements.

If we would like to stick to superconductors, then we believe significant post-processing using error mitigation would be needed. Many error mitigation techniques have sprung out in recent years, and reviewed extensively by the author in Tilly et al. [27]. In particular, we would like to refer the readers to a recent paper in 2023 [61] illustrating results from applying simple extrapolation-based error mitigation techniques to IBM's superconducting quantum computers.

**Non-physical self-energies.** Since the (interacting) Green function computed suffers from numerical inaccuracies, efforts are needed to prevent such inaccurate interacting Green function leading to non-physical self-energies computed from the Dyson equation.

To explain this, we digress for a short explanation of the physical meaning of self-energies. Starting with a non-interacting case where the Hamiltonian assumes a simple form of  $H = \sum_i (\varepsilon_i - \mu) c_i^\dagger c_i$ . In the (non-interacting) retarded Green function, only one of the hole terms contributes (see Eq. (5.13)). Thus, the diagonal part of the Green function simplifies into

$$G_{ii}(\omega) = \frac{1}{\omega + i0^+ - \zeta_i},$$

where the energy  $\zeta_i$  is understood as measured with respect to the chemical potential. After the (inverse) Fourier transform,  $G_{ii}(\omega)$  would contribute  $\Theta(t)e^{-i\zeta_i t}$  (see e.g. Eq. (B.26) in Appendix B). This can be interpreted from a field theory perspective [180, §7.3] as the amplitude for the propagation of a single-particle state. Once the interaction is added, the appearance of self-energy contributes roughly as (assuming  $\Sigma$  does not depend on  $i$ )

$$\frac{1}{\omega + i0^+ - \zeta_i - \Sigma}.$$

If further assuming a frequency independent self-energy  $\Sigma$ , then the amplitude in real time becomes (substituting  $\zeta_i \rightarrow \zeta_i + \Sigma$ ):

$$\Theta(t)e^{-it(\zeta_i + \text{Re } \Sigma) + t \text{Im } \Sigma}.$$

Hence, assuming  $\text{Im } \Sigma < 0$ , single-particle states decays as  $e^{t \text{Im } \Sigma}$  due to the interaction. Thus ensuring a finite  $\Sigma$  with negative  $\text{Im } \Sigma$  has important physical meanings.

Note that a negative  $\text{Im } \Sigma$  also ensures a positive quasi-particle weight  $z$ . Whereas all the negative values of  $z$  we encountered in certain numerical experiments could all be traced back to a positive  $\text{Im } \Sigma/0^+$  (see Eq. (5.21)).

In our paper [30], we have proposed that the divergent contributions of non-interacting and interacting Green functions must cancel each other for a non-divergent  $\Sigma$ . This leads to a series of equalities governing the energies and overlaps ( $\lambda$ ) for a general impurity model  $H_{\text{SIAM}}$ . The same formula but simplified to our two-site model is presented in Appendix E.1.4. In the paper, we proposed to enforce these equalities on energies and overlaps with a constrained minimisation approach.

However, Ehrlich et al. [206] reported numerical difficulties in applying the constrained minimisation approach proposed by our colleagues. We shared their experience in other experiments (not reported in our paper [30]). They then proposed a fitting approach, which approximates the self-energy at the zero frequency but is restricted to the particle-hole symmetric two-site DMFT case. The paper also reported that small numerical errors of the Green function, which may result simply from insufficient sampling, could easily break the particle-hole symmetric assumed in the model.

Finally, Selisko et al. [207] claimed a more robust and general solution. We discuss their experiment briefly below.

**Recent experiment.** In 2024, Selisko et al. [207] conducted a proof of concept experiment of DMFT on quantum computers for real materials. Here real implies that the DMFT orbitals were constructed from a prior density field theory (DFT) calculation and the DMFT is used to model the subspace of partially occupied d and f shells in transition metal and rare earth elements. Such an approach is often called DFT+DMFT in the literature. Besides incorporating DFT, several improvements are worth noting here.

First, while we focused on a variational approach for the computation of excited states, they opted for a subspace method. Specifically, they expressed the excitation operator from the ground state to excited states as a linear combination of creation and annihilation operators on the orbitals. The coefficients in the combination were obtained by solving classically a linear equation whose constituents were expressed as expectations of those creation

and annihilation operators on the ground state, and measured on the quantum computer. Excitation energies can then be extracted. This approach is named quantum equation of motion [214] due to its connection with the classical approach in quantum chemistry. It is similar to the quantum subspace expansion [168] and avoids repeated application of VQE optimisations as in our case.

Second, they chose to carry out the DMFT self-consistency loop in the Matsubara frequency where the Green function tends to be smoother (see, e.g. Figure 4.5 on page 86). They still need to compute the Green function in real frequency as in our case to obtain other quantities of interest (e.g. the spectral function and the interacting density of state).

In real frequency, the same problem appears where the divergent part of non-interacting and interacting Green functions need to cancel each other. Numerical inaccuracy in the interacting Green function computed failed such cancellation, and regularisation was needed to restore it. They used the approach developed in traditional DMFT literature [215] which redistributed the overlaps and energies until the self-energy is reasonable.

Finally, while they have remained on a superconducting-based quantum computer, they have included active error mitigation techniques (zero-noise extrapolation) for further mitigation of non-sampling noise from the quantum computer. With this, they were able to conduct 14 qubit experiments, which correspond to 6 bath sites.

## Chapter 6

# Conclusions and Outlooks

Here, we provide a short review of the results of this thesis in section 6.1. Then in section 6.2, we discuss the issues that need to be addressed when scaling the variational algorithms to larger problems in the future. We also discuss the potential of executing variational algorithms on error-corrected qubits. Finally, we comment on how the variational algorithms can be applied to real-world problems in section 6.3.

### 6.1 Review of results

In this review, we focus on a concise presentation of the idea used throughout the thesis. We then briefly mention the results.

**The Idea.** The overall idea has consistently been to recast the targeted problem in a form that can be resolved through a properly designed unitary circuit, which in practice is replaced by ansatz and optimised through minimising a loss function.

The ansatz used in this thesis is designed straightforwardly: it consists of layers of single qubit gates followed by two qubit entangling gates. The number of layers is gradually increased until the loss function can be optimised below a certain threshold.

The loss functions we used are inspired by the VQE form  $\langle U(\theta) | H | U(\theta) \rangle$ , where  $H$  is a Hermitian operator and  $|U(\theta)\rangle$  is a state prepared by an ansatz. For example, in finding implementations of logical gates (chapter 3), while we have used in the loss function the trace distance between two density matrices, our inspiration comes from a simple connection between trace distance and state fidelity on pure states (eq. (6.1)). Then, the averaged state fidelity can be viewed as a special VQE form loss function  $L$  where the targetted gate  $U_t$  corresponds to a series of Hermitian operators  $H_{t,j}$  and the same ansatz

acts on all states in a basis  $\{|i\rangle\}$ :

$$1/2 \operatorname{Tr} \left[ \sqrt{(|\psi\rangle\langle\psi| - |\phi\rangle\langle\phi|)^2} \right] = \sqrt{1 - |\langle\psi|\phi\rangle|^2} \quad (6.1)$$

$$\mathcal{L}(\theta) = \sum_{i,j} |\langle i| U^\dagger(\theta) U_t |j\rangle|^2 = \sum_{ij} \underbrace{\langle U_i(\theta)|}_{\langle i|U_i^\dagger(\theta)} \underbrace{H_{t,j}}_{U_t|j\rangle\langle j|U_t^\dagger} \underbrace{|U_i(\theta)\rangle}_{U_i(\theta)|i\rangle} \quad (6.2)$$

The loss functions in chapters 4 and 5 are more directly related to VQE: Chapter 4 uses the VQE-based variational linear algebra solver and chapter 5 uses the VQE extension to excited states.

**The results.** Using the same idea, we have conducted several proof-of-concept experiments. In chapter 3, we have notably generated a few logical gates for certain error-correcting codes which have not been reported in the literature at the time of our experimentation.

In chapters 4 and 5, we have proposed two variational algorithms for the computation of Green function on near-term quantum devices. The two variational algorithms for Green function each has its own merits. The one in chapter 4 can, in principle, compute the Green function for the whole frequency domain but requires executing a controlled-unitary gate where the unitary is the ansatz circuit itself, thus leading to significantly more two-qubit gates in the circuits. This is required for the loss function in variational linear algebra solver. The one in chapter 5 can only compute the Green function for a few frequencies close to the ground state energy, but does not incur additional controlled gates as the one in chapter 4.

Therefore, in chapter 4, we include numerical simulations by assuming that the noise on the hardware could be reduced to a certain level sufficient for successful error mitigation, and show that the algorithm performs well for computing the Green function of small diatomic molecules. On the contrary, in chapter 5, we use its algorithm to compute the Green function on real hardware and demonstrate the first DMFT self-consistent calculation for a two-site problem on quantum computers.

## 6.2 Scalability of the algorithms

Several factors have limited our experiments to small-scale problems. In this section, we examine these constraints and explore possible solutions. Additionally, we discuss the requirements for scaling variational algorithms to address larger problems in the future.

The method proposed in chapter 3 for finding logical gates is severely

limited as it scales exponentially with respect to the number of logical qubits in the given error-correcting code. Therefore, it can only be used for the study of codes encoding a small number of logical qubits. Looking ahead, recent explorations in Ruiz et al. [216] are interesting, where the authors applied reinforcement learning techniques to generate implementations of the circuit using fewer  $T$  gates. Notably, additional gadgets (representing known circuit equivalence templates) were introduced, which significantly improved minimisation performance.

The methods proposed in chapters 4 and 5 are based on VQE. Many factors constrain the scalability of VQE to larger problems. Here, we divide these factors into two parts and address them separately. The first part concerns the practical issues surfaced during our exploration. The second part concerns scalability issues that would be important for demonstrating a real advantage of variational quantum algorithms over algorithms on classical computers.

**Current practical constraints.** In the first part, we address the constraints due to the quantum hardware currently available. These include the low speed and the drifting of the hardware. We propose addressing these constraints using subspace methods.

The operating speed of quantum computers available is insufficient for real-world applications. Variational algorithms often require frequent access to the hardware. However, in our experience, a significant amount of execution time on cloud-accessible quantum computers is not spent on circuit execution. According to one report [206], this waste is about 75%. This is made worse by the fact that many of the algorithms we used in the thesis required recursive use of VQE optimisations. That is, VQE is often used as a subroutine contained in a higher level of the algorithm.

An illuminating observation can be made here. If measured in terms of how many circuits quantum computers can execute per second, quantum computers currently operate at the megahertz level<sup>1</sup>, which would be insufficient even if we ignored the impact of noise. If repeated sampling of multiple observables is required, quantum computers can easily slow down to the kilohertz level or worse. Tilly et al. [27] has estimated that for studying a small molecule with realistic assumptions, VQE runtime could last for years. The

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<sup>1</sup>Gate sequence time cost for superconducting computers is, for example, visualised in certain dynamic decoupling experiments. Tripathi et al. [217] shows one gate roughly costs  $\mathcal{O}(20)$  nanoseconds, which is about 50 megahertz. Alternatively, Ehrlich et al. [206] estimates that the circuit initialisation cost is about  $150 \mu s$ , which is about seven megahertz. A theoretical estimation of circuit execution time on silicon qubits is available in Cai [218].

overhead mainly comes from the number of operators requiring measurements exploding due to an increased chemical active space. Of course, this runtime may be significantly reduced if parallelism is possible.

The second practical constraint is the drifting of the hardware. Specifically, at the time of experimentation for chapter 5, we had to execute the experiment in a very short time window for statistical consistency, ideally right after the device itself had undergone a calibration<sup>2</sup>. This problem has received widespread attention in the community. See for example Agarwal et al. [219, Figure 7] which plotted the drifting behaviour of an idle qubit in different time steps across an hour.

Based on this, we propose using subspace methods [168]. Such methods would start with an initial point and then extract useful information from a subspace generated around this initial point. They generally require fewer interactions with the quantum computer but more classical post-processing steps. Many papers have followed this approach [214, 220–227].

Depending on the application, the subspace is generated by acting various operators on the state  $|\psi\rangle$ , which in most cases is the ground state prepared by a VQE. For quantum chemistry-inspired methods, one applies the creation and annihilation operators to the state, and solves a generalised eigenvalue problem classically from the resulted data to extract information about excitations [214, 222]. To perform imaginary-time evolution, a parameterisation of the imaginary-time evolution operator is used, and the parameters are determined by solving a linear equation classically [220]. Other cases extend the traditional Krylov subspace methods [224, 225]. Most of these methods are based on well-studied approaches having a long history either in quantum chemistry Stair et al. [222] (and Refs. [47–52] therein) or in numerical linear algebra [165].

**Scalability to near-term devices.** Here we discuss the scalability issues in future devices. We outline the steps necessary to scale up variational quantum algorithms and how can one examine their advantage over classical approaches.

First, the dependency on error mitigation on near-term devices limited the size of the problem that can be treated by variational quantum algorithms. To guarantee a successful error mitigation, the sampling cost is shown to grow exponentially with respect to the depth of the circuit [228]. Therefore, we

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<sup>2</sup>Unfortunately, no estimates on the variance of data were conducted at that time.

cannot expect variational algorithms to tackle problems of increasingly large size without error correction<sup>3</sup>. This however has not prevented the success of error mitigation on problems of a moderate size, which has been demonstrated experimentally [23, 24].

To find the problem size which can be successfully tackled by variational quantum algorithms, we believe a practical approach is necessary in which one gradually reproduces the best-known classical results of increasingly large sizes. Note that there exist datasets of molecules whose energies have been determined to high accuracy, but whose sizes are beyond the reach of an exact calculation on classical computers [229, VIII.C]. One needs to follow such datasets as the benchmark for the performance of variational quantum algorithms.

Then, we need a shallow ansatz. The algorithms in chapters 4 and 5 depend on the ability of the ansatz to capture the targeted problem. Given the constraint from the error mitigation as mentioned previously, we require that a shallow ansatz should have at most linear scaling in the number of qubits. While this is a strict requirement, several proposals exist for both chemistry problems and lattice Hamiltonians [27]. However, we note that these proposed linear ansatzes are linear only in one layer of the ansatz. Since multiple layers may be necessary for accurate results, the exact number of layers required needs to be examined as one moves to larger problems.

Next, we need to adjust the ansatz to avoid the barren plateau problem. A general intuition is provided in a recent review [50] which states that if the ansatz is designed to explore the Hilbert space uniformly, it is likely to suffer from the barren plateau problem. Similarly, the ground state of many interesting Hamiltonians exhibits an area-law entanglement [230], rather than the volume-law entanglement for typical quantum states [231]. This also suggests that an ansatz does not need to explore the whole Hilbert space.

Therefore, we believe that the key in avoiding the barren plateau is to bias the ansatz towards a specific problem where the ansatz is designed to only explore the relevant part of the Hilbert space. Proposals [232, 233] which gradually increase the size of ansatz in small steps through reinforcement learning appear to be promising, although more research is needed to ensure that such an ansatz would not be trapped in local minima.

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<sup>3</sup>Here we assume that the depth of the circuit would scale at least linear on the size of the problem.

**Prospects on long-term devices.** While the prior discussion has focused on noisy qubits, variational algorithms may be useful for error-corrected devices. For example, VQE can be used to prepare a state having a large overlap with the ground state. After the preparation, one could switch to the quantum phase estimation algorithm for a precise estimation of the ground state energy.

One challenge in executing variational algorithms on error-corrected devices is the implementation of rotation gates. Single qubit rotation gates may in fact consume the bulk of circuit execution time and their optimisation in the context of VQE has been researched recently [234].

In this case, it is also worth examining the benefit of using ansatz with parameterised rotations against the benefit of using ansatz made up of discrete gates optimised in a combinatorial manner. In the latter case, prior investigations where reinforcement learning is used to optimise the ansatz structure may be applied [232, 233].

## 6.3 Real-world problems

Finally, we discuss how the algorithms proposed in the thesis can be applied to real-world problems. We would like to illustrate this with the molecular docking problem. Molecular docking [235] is used in the pre-clinical phase of drug development to identify and optimise small molecules (called ligands) which would bind to the target macromolecule (typically a protein). By predicting promising candidates computationally, molecular docking minimises the number of compounds that need to be synthesized and tested experimentally.

An important part of molecular docking software is the so-called scoring function, which would rank energetically the favorability of a range of small molecules (and different geometric poses of the same molecule) proposed by a random sampling algorithm. Typical scoring functions in commercial software are based on heuristic formulae due to the intractability of solving the general electronic Hamiltonian problem.

We foresee two approaches to improve the accuracy of scoring functions using variational quantum algorithms. The first one is to substitute the scoring function with a certain VQE-based algorithm to directly estimate the energy. However, this approach would create huge computation costs due to the large number of candidate molecules encountered in pre-clinical studies. The second and more practical approach is to identify a few favourable candidates using traditional high throughput scoring functions, and then use vari-

ational algorithms for refining the selected results. Note that an embedding approach would be needed to isolate a small but chemically important region of the combined ligand-target complex. This smaller region would then be studied by variational algorithms on near-term devices. Typical embedding approaches include the active space embedding [196] or the DFT+DMFT approach [236].

## Appendix A

# Quantum Error Correction

Here we attempt an introduction to error correction. We cover enough concepts so that this chapter is self-contained, but do not attempt anything deep into the error correction theory that is insignificant for the procedure that we are proposing. We refer the series readers to Chapter 10 in Nielsen and Chuang [1].

The development of quantum error-correcting codes (QEC) is motivated by the presence of quantum errors caused by interactions with the external environment. These interactions vary in nature and strength depending on the specific hardware architecture. The classical idea of error correction depends on *redundancy*. That is, if we can introduce redundancy in the data, and if we can assume that errors tend to be local, then local errors tend to result in local disruption of the message which can be recovered due to the redundancy. Classical codes typically introduce redundancy by copying the data, which is impossible for quantum computation due to the no-clone theorem [237].

To get away from the no-clone theorem, *quantum codes* were invented to encode the original message into a subspace of Hilbert space, called the *code space*. In this case, we can protect and correct errors during transmission if the code is cleverly designed, such that the errors transform this code space into a set of distinct (i.e. orthogonal) subspaces. Intuitively, if we assume that the noise is local, then when the code space is physically distributed, we can engineer it so that the different subspaces due to the noise are distinct. Finally, we can correct the error by measuring projectors on these distinct subspaces, identifying the error by the measurement result, and applying quantum operations to reverse the effect brought by the noise.

The method mentioned above can be expensive in practice since the num-

ber of correctable errors can grow exponentially with respect to the number of qubits, due for example, to different ways in which single-qubit errors can combine to form multi-qubit errors. However, for general quantum code, we still perform measurements, called *syndrome measurements*, on certain observables. Their results are called the *syndromes*, and the process of identifying the error from these syndromes is called *decoding*.

We need to confess here that the above discussions is only valid after a diagonalisation process. Since, if a pair of errors  $E_i$  and  $E_j$  send the code space into orthogonal subspaces, there is no guarantee that an arbitrary linear combination of them will have similar properties. In general, the following theorem specifies the necessary and sufficient condition for a particular code to be able to correct a set of errors.

**Theorem A.1** (Quantum error-correction condition). *Let  $P$  be the projector into the code space of a quantum code, and assume that the error process under consideration is described by the Kraus operators  $\{E_i\}$ <sup>1</sup>. The necessary and sufficient conditions for the existence of an error correction operator which corrects all such errors, is that*

$$PE_i^\dagger E_j P = \alpha_{ij} P, \quad (\text{A.1})$$

where  $\alpha$  is a Hermitian matrix.

Proof of this theorem is available in Nielsen and Chuang [1, page 436, Theorem 10.1].

Note that, after a diagonalisation of the matrix  $\alpha$ , we can find another set of Kraus operators  $\{F_k\}$  for the same error process, related to the original operators by  $F_k = \sum_i u_{ik} E_i$ . Here  $u$  comes from the diagonalisation process and satisfies  $d = u^\dagger \alpha u$  for a diagonal matrix  $d$ . Then we have

$$PF_k^\dagger F_l P = (F_k P)^\dagger (F_l P) = d_{kl} P. \quad (\text{A.2})$$

This equation captures the intuition that different error leads to the distinct orthogonal subspaces which we mentioned earlier, as  $d_{kl}$  is zero when  $k \neq l$ . In addition, it is also a statement on any orthogonal basis for the code space as well. If  $\{|\psi_i\rangle\}$  forms an orthogonal basis for the code space, its elements are called *code words*. For any two code words we have  $\langle \psi_i | PF_k^\dagger F_l P | \psi_j \rangle =$

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<sup>1</sup>Note that the Kraus operators for the noise in the theorem are not necessarily trace-preserving in order to cover more general scenarios.

$\langle \psi_i | F_k^\dagger F_l | \psi_j \rangle = d_{kl} \langle \psi_i | P | \psi_j \rangle = d_{kl} \delta_{ij}$ , thus,

$$\langle \psi_i | F_k^\dagger F_l | \psi_j \rangle = d_{kl} \delta_{ij}. \quad (\text{A.3})$$

Together, we call Theorem A.1, and the related equations A.1 to Eq. A.3, collectively as the *error-correcting condition* for a code  $P$  which to correct errors  $E_a$  (or  $F_a$ ).

Note that, there will certainly be “errors” which transform the code space into itself. Such errors thus can never be detected. But if we can engineer such errors, we will be effective performing logical operations on the code space. Therefore, operators representing such errors are called *logical gates*.

**Local errors and distance of a code.** Here we want to focus on errors which are local, and introduce the distance of a code which captures how the code can correct errors which are local.

We can focus on local errors due to the *discretisation* of the errors: Assume a code corrects for an error channel  $\mathcal{E}$  described by  $\mathcal{E}(\rho) = \sum_i E_i \rho E_i^\dagger$ . If another error channel  $\mathcal{F}(\rho) = \sum_j F_j \rho F_j^\dagger$ , is related with  $\mathcal{E}$  by  $F_j = \sum_i m_{ji} E_i$ , then the code also corrects for the error channel  $\mathcal{F}$  [1, Section 10.3.1, page 438]. Discretisation ensures that when discussing error-correcting code, we can focus on a set of specific error operators  $E_i$ , and without regard to their coefficients in the channel.

*Local errors* are best described using Pauli operators. For an  $n$ -qubit Pauli operator of weight  $t$ , we mean an operator  $P = \sigma_1 \otimes \cdots \otimes \sigma_n$  which is the Kronecker product of  $n$  such matrices  $\sigma_i \in I, X, Y, Z$ , and only  $t$  of these matrices are not identity  $I$ . Such operators capture a local error  $\rho \mapsto P \rho P^\dagger$  which affects  $t$  qubits.

Then, we can discuss codes which can *correct* any such local errors of weight from  $1, 2, \dots$  to  $T$ . That is, the errors  $F_a$  and  $F_b$  in the condition (Eq. A.3) will have such weights from 1 to  $T$ . Therefore, their product  $F_a^\dagger F_b$  will have weights 1 to  $2T$ , where we implicitly included the identity matrix as an error that can be trivially corrected. Hence, for codes that are designed to correct such errors, any weight  $t$  Pauli operator of  $t = 1 \cdots 2T$  should satisfy the error-correcting condition.

If the error-correcting condition is satisfied for any Pauli operator of weight up to  $d - 1$ , the code is called to have *distance*  $d$ . It can therefore correct the error of weight  $t$  up to  $\lfloor (d - 1)/2 \rfloor^2$ . For example, a distance 4

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<sup>2</sup> $\lfloor d \rfloor$  is the largest integer not greater than  $d$ .

code can correct any 1-qubit errors.

Note that, the condition is a statement on code words (Eq. A.3). We therefore have an equivalent definition of distance. For a code to have distance  $d$ ,  $d - 1$  is the maximum weight of Pauli operators for which the condition holds, or  $d$  is the minimum weight for which the condition does *not* hold. Here, the distance intuitively represents a “physical distance” between the code words, so that a weight  $d$  Pauli operator  $P$  connects one code words, say  $\psi_1$ , to another code word  $\psi_2$ , in the sense that  $\langle \psi_1 | P | \psi_2 \rangle \neq 0$  (violates Eq. A.3).

Finally, we can introduce the  $[[n, k, d]]$  notation. For a quantum code which uses  $n$  qubits to encode  $k$  qubits (the dimension of the code space is  $2^k$ ) with a distance  $d$ , it is called a  $[[n, k, d]]$  code.

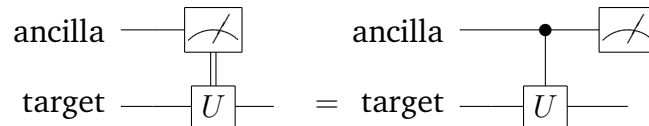
**Stabiliser codes.** Here we provide a brief note on stabiliser codes which have been used in our experiments. Stabiliser codes are designed to encode  $k$  logical qubits in  $n$  physical qubits. The information is stored in the code space  $V_s$ , which is the shared  $+1$  eigenspace (i.e. the intersection of several  $+1$  eigenspaces) of many commuting  $n$ -qubit Pauli operators  $p_1, p_2, \dots$ . We include all such operators  $p$  acting trivially on the code space  $V_s$  into a set  $S$ . This set forms an Abelian group under multiplication, and is called the *stabiliser group*.  $p \in S$  is called a stabiliser. The code space  $V_s$  is therefore fixed by the stabiliser group  $S$ . Furthermore, Pauli operators which commute with elements of the stabiliser group are called the *centralisers*, denoted by  $C(S)$ . Centralisers therefore preserve the code space  $V_s$ . Certainly  $S \subset C(S)$ . However, there are elements in  $C(S) \setminus S$  (“\” means “set minus”) that commute with stabilisers but are not inside the stabiliser group  $S$ . For  $c$  in  $C(S) \setminus S$ ,  $c$  must then act non-trivially on the code space  $V_s$  and are logical gates for the code.

Stabiliser codes detect errors in the following manner. We do syndrome measurement by measuring stabilisers (technically, we only measure the group generators to be efficient). Different errors result in different kinds of syndrome:

1. If an error  $e \in S$ , then  $e$  does not modify the code space, and the syndrome will contain all  $+1$ . Such errors can be ignored.
2. If an error  $e \in C(S) \setminus S$ , then  $e$  is equivalent to a logical gate and acts non-trivially on the code space. However, the syndrome is still all  $+1$  and hence such errors cannot be detected.

3. If  $e$  is a general  $n$ -qubit Pauli error that is not a centraliser, then its syndrome will contain  $-1$ . Stabiliser theory shows that error correction is possible for this situation, and decoding algorithms are developed to infer the right correction from the syndrome.

Here we mention a specific decoding algorithm called the *lookup-table decoder*. This algorithm first obtain a table of the corresponding syndromes for all possible  $n$ -qubit errors. This table will have features mentioned in the three points above, and will be of size exponential with respect to the number of qubits (which is why lookup-table decoder is not practical for larger codes). Still, each row of this table can be obtained efficiently on classical computers by simulating stabiliser circuits consisting of only Clifford gates and computational basis measurements (Gottesman-Knill theorem). Using this table, we can identify the error from syndrome measurement, and perform the operation reversing the error to correct it. Note that sometimes different errors result in the same syndrome. In this case, stabiliser theory shows that these errors are only different up to a stabiliser, and applying the reverse operation of any one of these errors can correct all cases. In practice, we often apply the reverse operation of minimum weight among these operations.



**Figure A.1:** An illustration of the principle of deferred measurements. The unitary gate  $U$  conditioned on the readout of the ancilla qubit (left-hand-side) is equivalent to the controlled gate followed by the deferred measurement (right-hand-side).

The lookup-table decoder can be simulated on classical computers without including measurements but with quantum gates controlled by additional ancilla qubits. We explain briefly this possibility. First, syndrome measurement can be implemented using ancilla qubits and results in measurements on them, i.e. the syndrome (See for example, Figure 10.13 on page 473 of Nielsen and Chuang [1]). Next, the correction applies reversing operations conditioned on the syndrome. Therefore, lookup-table decoder results can be simulated using ancilla qubits with gates controlled by the readout of ancilla qubits. However, such gates are equivalent to controlled gates controlled by the ancilla and followed by measurement at the end of the circuit, which is sometimes called the *principle of deferred measurements* (See Fig. A.1, also

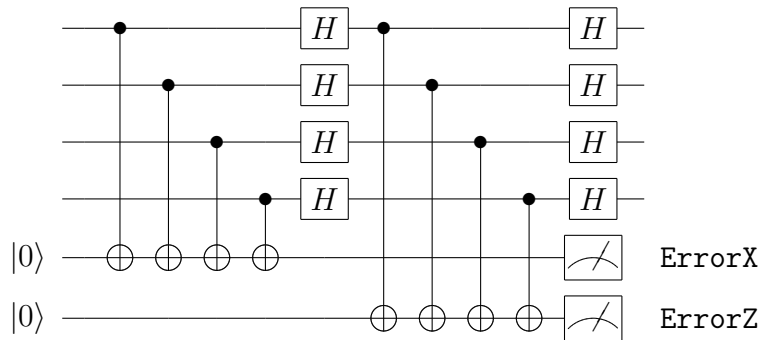
Exercise 4.34 on page 188 of Nielsen and Chuang [1]). If we do not care about the ancilla and trace them out, we can further skip the simulation of these deferred measurements. In summary, the lookup-table decoder can be simulated with unitary operations conditioned on ancilla qubits which we do not care about.

**[[4,2,2]] code.** Here we provide as an example a simple error-detecting code, [[4,2,2]] [238], which uses four physical qubits to encode two logical qubits. We will add a bar over symbols to distinguish a logical qubit (or operation), such as  $|\overline{00}\rangle$ , from a physical qubit (or operation), such as  $|0000\rangle$ . With this notation, the code words for this [[4,2,2]] code are:

$$\begin{aligned} |\overline{00}\rangle &= \frac{1}{\sqrt{2}} (|0000\rangle + |1111\rangle), \\ |\overline{01}\rangle &= \frac{1}{\sqrt{2}} (|1100\rangle + |0011\rangle), \\ |\overline{10}\rangle &= \frac{1}{\sqrt{2}} (|1010\rangle + |0101\rangle), \\ |\overline{11}\rangle &= \frac{1}{\sqrt{2}} (|0110\rangle + |1001\rangle). \end{aligned}$$

The stabilisers are  $XXXX$  and  $ZZZZ$ . Logical gates are  $\overline{X}_1 = XIXI$ ,  $\overline{X}_2 = XXII$ ,  $\overline{Z}_1 = ZZII$ , and  $\overline{Z}_2 = ZIZI$ .

This code has a distance of 2. It can detect any one-qubit errors, although it cannot correct them. To see this, notice that any code word has an even parity. For example,  $|0000\rangle$  gives  $0 \oplus 0 \oplus 0 \oplus 0 = 0$ , where we use  $\oplus$  for binary



**Figure A.2:** A simple circuit to detect errors for the [[4,2,2]] code [239]. The logical qubits reside in the first four qubits, while the last two qubits are initialised in  $|0\rangle$  and are used as data qubits to store the output of checks. The check ErrorX (or ErrorZ) being 1 implies an  $X$  (or  $Z$ ) error.

addition. Similarly,  $|0101\rangle$  gives  $0 \oplus 1 \oplus 0 \oplus 1 = 0$ . Therefore, an  $X$  error on any qubits would turn it into having an odd parity. This change be detected and is realised in the first part of the circuit in Figure A.2, where if the output  $\text{ErrorX}$  is one, it signifies that an  $X$  error has happened.

Similarly, a  $Z$  error on any qubit would add a relative phase in the two constituents of any code word. For example, turning  $(|0000\rangle + |1111\rangle)$  to  $(|0000\rangle - |1111\rangle)$ . This change can be detected by collecting the measurements in the  $\{|+\rangle, |-\rangle\}$  basis<sup>3</sup> from all qubits. This is realised in the second part of the circuit in Figure A.2, where if the output  $\text{ErrorZ}$  is one, it signifies that a  $Z$  error has happened.

Finally, we note that while this code detects errors, it cannot correct them as it does not know the location of the errors. Note also that  $[[4,1,2]]$  is a sub-code of this code, where only  $|\overline{00}\rangle$  and  $|\overline{01}\rangle$  are used as its code words [240].

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<sup>3</sup> $|\pm\rangle = (|0\rangle \pm |1\rangle)/\sqrt{2}$ .

## Appendix B

# Green function

Green function is widely used in many-body physics. Here in Appendix B.1, we review the related concepts and provide references to the general literature. Then, in Appendix B.2 we derive several key equations for the zero-temperature retarded Green function we used in the main thesis.

It is worth noting that while the Green function is named after the British mathematical physicist George Green, the Green function introduced here is not directly related to the one he used in the study of classical electrodynamics. To distinguish between the two, we will refer to the Green function in classical electrodynamics as the *mathematical Green function*, whereas the one defined here will be called the *many-body Green function*, or simply *Green function*.

The mathematical Green function is primarily used in the study of partial differential equations commonly encountered in classical electrodynamics. It characterizes a system's response to a unit impulse, i.e. a delta-function. Although the many-body Green function is defined differently, under the framework of linear response theory, it also describes the system's response to external perturbations, drawing an analogy to the mathematical Green function.

Also, Green function is sometimes referred to as the Green's function. While the later one has been gaining popularity (see a comparison in Wright [241]), we used the shorter phrase Green function throughout the thesis.

## B.1 Many-body Green function

The many-body Green function is defined as the expectation value of the commutator or anti-commutator of a pair of time-evolved operators, as introduced in Appendix B.1.1. In many-body physics, the Green function provides a fundamental framework for describing various observables, resulting in a vast

body of literature covering diverse aspects. Here, we highlight two specific examples. First, the retarded Green function describes a system's response to small external perturbations within the framework of linear response theory, as discussed in Appendix B.1.2. Second, the spectral function, given by the imaginary part of the retarded Green function, is directly linked to the intensity of photoelectrons detected in photoemission experiments, as detailed in Appendix B.1.3.

A systematic diagrammatic approach is available for computing the Green function  $G$  of an interacting Hamiltonian from the Green function  $G_0$  of the corresponding non-interacting Hamiltonian. Different levels of approximation within this framework allow for the computation of  $G$  from  $G_0$  with varying degrees of accuracy. As we do not explore this approach in detail, we refer the reader to the classical reference Mahan [242].

### B.1.1 General definition

In the main thesis, we have focused exclusively on the zero-temperature retarded Green function of the form

$$G^R(t - t') = -i\Theta(t - t')\langle\{c_\alpha(t), c_\beta^\dagger(t')\}\rangle, \quad (\text{B.1})$$

for certain quantum numbers  $\alpha$  and  $\beta$ . Here the annihilation operator is evolved by the Hamiltonian as  $c_\alpha(t) = e^{iHt}c_\alpha e^{-iHt}$ , and the creation operator evolves similarly.  $\Theta(t)$  is the Heaviside step function. The expectation value is with respect to the ground state of the Hamiltonian  $H$ .

In general, one may define the retarded Green function  $g_r$  of two operators  $X_1$  and  $X_2$  as [180, 243]:

$$g_r(X_1; X_2|t - t')_\zeta = -i\Theta(t - t')\langle[X_1(t), X_2(t)]_\zeta\rangle, \quad (\text{B.2})$$

where an operator  $X$  evolves in a Heisenberg picture:

$$X(t) = e^{iHt}X e^{-iHt}, \quad (\text{B.3})$$

and where  $\zeta = 1$  would give a commutator and  $\zeta = -1$  would give an anti-commutator:

$$[X_1(t), X_2(t')]_\zeta = X_1(t)X_2(t') - \zeta X_2(t')X_1(t). \quad (\text{B.4})$$

The choice of  $\zeta$  depends on the application. For example,  $\zeta = 1$  in the linear response theory (Eq. (B.5) below). The expectation average is obtained with respect to a certain ensemble. For example, in the canonical ensemble for an isolated system:

$$\langle X \rangle = \text{Tr}(X e^{-\beta H}) / \mathcal{Z}, \quad \mathcal{Z} = \text{Tr}(e^{-\beta H}),$$

where  $\beta = 1/k_B T$ ,  $T$  is temperature and  $k_B$  is the Boltzmann constant.

### B.1.2 Linear response theory

The commutator case of  $g_r$  is useful in the linear response theory, which we mention briefly here. Suppose a system described by the Hamiltonian  $H$  was initially prepared at a state  $\rho_0$ . An observable  $A$  would have an initial expectation value of  $\langle A \rangle_0 \equiv \text{Tr}(A \rho_0)$ . Then, suppose an external time-dependent perturbation  $V(t) = B f(t)$  was turned on, leading to a small change in the expectation value. Here  $f(t)$  is a scalar function which captures the time-dependency, and  $B$  is the operator representing the time-independent part. Up to first order in the perturbation, the change in expectation value of  $A$  is given by [243]:

$$-i \int_{-\infty}^t \langle [A(t), B(t')] \rangle_0 f(t') dt' = \int_{-\infty}^{\infty} g_r(A; B | t - t')_{\zeta=1} f(t') dt', \quad (\text{B.5})$$

where the expectation is with respect to the initial state  $\langle [A(t), B(t')] \rangle_0 \equiv \text{Tr}([A(t), B(t')] \rho_0)$ .  $B(t')$  is  $B(t') = e^{iHt'} B e^{-iHt'}$  and the perturbation's time-dependency remains in  $f(t')$ .

Eq. (B.5) forms the foundation of Kubo formalism, where quantities measured in experiments, such as the electrical conductivity and magnetic susceptibility, are expressed using Eq. (B.5) with appropriate operators replacing  $A$  and  $B$ . A detailed discussion is available in chapter 6 of [243].

### B.1.3 Spectral function and photoemission experiment

The anti-commutator case of  $g_r$  is useful in other scenarios. Here we mention in brief its connection to the photoemission experiments on crystal samples.

We are interested in the following retarded Green function:

$$G^R(t - t', k) = -i\Theta(t - t') \langle \text{GS} | \{c_k(t), c_k^\dagger(t')\} | \text{GS} \rangle. \quad (\text{B.6})$$

Here,  $k$  is the crystal momentum. When viewed from the general  $g_r$  in

Eq. (B.2), we have used the anti-commutator case  $\zeta = -1$ , set the two operators  $X_1 = c_k$  and  $X_2 = c_k^\dagger$ , and used the zero-temperature limit  $\beta = 1/k_B T \rightarrow \infty$ .

We can pick a basis  $\{|\psi_m^n\rangle\}$  which simultaneous diagonalise the Hamiltonian and the particle number operator. The energy for this state is denoted  $E_m^n$ , and the particle number is  $n$ . The ground state in this case would have energy  $E_0^N$ , particle number  $N$ , and be denoted  $|\text{GS}\rangle = |\psi_0^N\rangle$ .

In this basis, The retarded Green function can be Fourier transformed to the frequency domain, and be expanded as (see a proof in Appendix B.2.3):

$$G^R(\omega, k) = \sum_m \left( \frac{|\langle \psi_m^{N+1} | c_k^\dagger | \psi_0^N \rangle|^2}{\omega + i\eta - (E_m^{N+1} - E_0^N)} + \frac{|\langle \psi_m^{N-1} | c_k | \psi_0^N \rangle|^2}{\omega + i\eta - (E_0^N - E_m^{N-1})} \right). \quad (\text{B.7})$$

The imaginary part of  $G^R(\omega, k)$  is called the spectral function:

$$A(\omega, k) = -\frac{1}{\pi} \text{Im } G^R(\omega, k). \quad (\text{B.8})$$

Spectral function can be simplified using the following relation<sup>1</sup>:

$$\lim_{\eta \rightarrow 0^+} \frac{1}{x \pm i\eta} = \mp i\pi \delta(x) + \mathcal{P} \left( \frac{1}{x} \right),$$

where  $\mathcal{P}$  denotes the Cauchy principal value. This simplification leads to:

$$A^+(\omega, k) = \sum_m \left| \langle \psi_m^{N+1} | c_k^\dagger | \psi_0^N \rangle \right|^2 \delta(\omega - (E_m^{N+1} - E_0^N)) \quad (\text{B.9})$$

$$A^-(\omega, k) = \sum_m \left| \langle \psi_m^{N-1} | c_k | \psi_0^N \rangle \right|^2 \delta(\omega - (E_0^N - E_m^{N-1})) \quad (\text{B.10})$$

$$A(\omega, k) = A^+(\omega, k) + A^-(\omega, k) \quad (\text{B.11})$$

Spectral function obeys a sum rule:

$$\int_{-\infty}^{\infty} A(\omega, k) d\omega = 1 \quad (\text{B.12})$$

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<sup>1</sup>See a proof on page 77 of [243].

To show it, one notices that

$$\begin{aligned} \sum_m |\langle \psi_m^{N+1} | c_k^\dagger | \psi_0^N \rangle|^2 &= \langle \psi_0^N | c_k \left( \sum_m |\psi_m^{N+1}\rangle \langle \psi_m^{N+1}| \right) c_k^\dagger | \psi_0^N \rangle \\ &= \langle \psi_0^N | c_k c_k^\dagger | \psi_0^N \rangle, \end{aligned}$$

where the middle part  $\sum_m |\psi_m^{N+1}\rangle \langle \psi_m^{N+1}|$  can be extended to the whole eigenbasis and leads to a resolution of identity. Similarly,  $\sum_m |\langle \psi_m^{N-1} | c_k | \psi_0^N \rangle|^2 = \langle \psi_0^N | c_k^\dagger c_k | \psi_0^N \rangle$ . Thus,  $\int_{-\infty}^{\infty} A(\omega, k) d\omega = \langle \psi_0^N | c_k c_k^\dagger + c_k^\dagger c_k | \psi_0^N \rangle = 1$ .

Finally, spectral function can be measured in photoemission experiments such as the angle-resolved photoemission spectroscopy (ARPES) which has become popular in analysing the band structure of crystals. In photoemission experiments, a crystal sample is prepared, and a monochromatic radiation is incident on the material so that electrons in the inner bulk of the material can be excited due to the photoelectric effect. These photoelectrons escape in all directions into the vacuum and are then captured by an electron analyser. Under the so-called “sudden approximation”, the intensity measured is proportional to  $A^-(\omega, k)$  where  $\omega$  would contain contributions from the band structure and the photon’s energy. Similarly, an inverse photoemission experiment can be used to probe the particle contribution  $A^+(\omega, k)$ . There, highly energetic electrons are incident on the sample and decelerated. The photons produced from the deceleration are collected and analysed. Detailed discussions and mathematical derivations are available in Damascelli et al. [244] and Hüfner [245].

## B.2 Zero-temperature Green function

Here we focus on the Green function used in the main thesis. From the viewpoint of  $g_r$  in Eq. (B.2), we have taken the anti-commutator case and the zero-temperature limit to give the  $G^R(t)$  below.

Specifically, we will define the time-ordered Green function  $G^T(t)$ , the retarded Green function  $G^R(t)$ , and the advanced Green function  $G^A(t)$  in Appendix B.2.1. Here  $t \in \mathbb{R}$  is time and these Green functions are referred to as real-time Green functions. Due to similarities in them (which will be clear later), we focus only on the retarded Green function. We will expand the retarded Green function  $G^R(t)$  in an eigenbasis of the Hamiltonian in Appendix B.2.2, from which we can introduce its Fourier transform  $G^R(\omega)$  (Appendix B.2.3).

Then, we will introduce the analogous imaginary-time Green function  $G^T(\tau)$ , which replaces the real-time  $t$  by an imaginary-time  $-i\tau$  ( $\tau \in \mathbb{R}$ ). The Fourier transform of the imaginary-time Green function  $G(\tau)$  is called the Matsubara function and denoted by  $G(i\omega_n)$ . We will explain the meaning of  $n$  in  $\omega_n$  during the discussion (Appendix B.2.4). In Appendix B.2.5, we will introduce an expression which unifies the retarded Green function's Fourier transform and the Matsubara function. Finally, in ??, we mention the non-interacting retarded Green function and the self-energy briefly.

**Notation for Green function.** Since different Green functions mentioned above share the same symbol  $G$ , we distinguish them by their superscripts and their arguments, which are summarised as follows:

- $G^{T/R/A}(t)$ : the time-ordered, retarded and advanced Green functions in real-time  $t \in \mathbb{R}$ ,
- $G^R(\omega)$ : the Fourier transform of the retarded  $G^R(t)$ ,
- $G(\tau)$ : the imaginary-time Green function in imaginary-time  $i\tau$  where  $\tau \in \mathbb{R}$ ,
- $G(i\omega_n)$ : the Fourier transform of  $G(\tau)$ , which is also called the Matsubara Green function. The frequencies  $\omega_n$  are also called the Matsubara frequencies.

### B.2.1 Definitions

Let  $X_1$  and  $X_2$  be two physical operators obeying the fermionic anti-commutation rule:

$$\{X_1, X_2\} = 0, \quad (\text{B.13})$$

their time-ordered Green function in two different time  $t_1$  and  $t_2$  is defined by:

$$G^T(t_1, t_2) = -i\langle \mathcal{T}_t X_1(t_1) X_2(t_2) \rangle, \quad (\text{B.14})$$

In the main text (Section 4.1), the two operators are simply fermionic creation/annihilation operators on different channels:

$$X_1 = c_\alpha, \quad X_2 = c_\beta^\dagger. \quad (\text{B.15})$$

Here,  $X_i(t_i)$  is the time-dependent operator in the Heisenberg Picture:

$$X_i = e^{iHt} X_i e^{-iHt}, \quad i = \{1, 2\}, \quad (\text{B.16})$$

where  $H$  is the overall Hamiltonian, and the expectation value  $\langle \dots \rangle$  is taken with respect to the ground state  $\psi_0$  of the Hamiltonian  $H$  in our case, or the thermal average when studying finite temperature physics.  $\mathcal{T}_t$  is called the time-ordering operator and orders the operators afterwards in decreasing time. Explicitly,

$$G^T(t_1, t_2) = -i \begin{cases} \langle X_1(t_1)X_2(t_2) \rangle, & t_1 \geq t_2, \\ -\langle X_2(t_2)X_1(t_1) \rangle, & t_1 \leq t_2. \end{cases} \quad (\text{B.17})$$

Note that exchanging the two fermionic operators  $X_1$  and  $X_2$  gives the minus sign  $-1$  above. It will be shown later that the Green function above depends only on the time difference  $t_1 - t_2$ , therefore we can set  $t_1 = t$  and  $t_2 = 0$  without loss of generality:

$$G^T(t) = -i \langle \mathcal{T}_t X_1(t) X_2 \rangle. \quad (\text{B.18})$$

Next, we define retarded  $G^R(t_1, t_2)$  and advanced  $G^A(t_1, t_2)$  Green function as the expectation values of anti-commutators:

$$\begin{aligned} G^R(t_1, t_2) &= -i\Theta(t_1 - t_2) \langle \{X_1(t_1), X_2(t_2)\} \rangle, \\ G^A(t_1, t_2) &= +i\Theta(t_2 - t_1) \langle \{X_1(t_1), X_2(t_2)\} \rangle. \end{aligned} \quad (\text{B.19})$$

Similarly, they all depend only on time difference  $t_1 - t_2$ , and we can write:

$$\begin{aligned} G^R(t) &= -i\Theta(t) \langle \{X_1(t), X_2\} \rangle, \\ G^A(t) &= +i\Theta(-t) \langle \{X_1(t), X_2\} \rangle. \end{aligned} \quad (\text{B.20})$$

## B.2.2 Eigenbasis representation

We can rewrite the Green function in the eigenbasis of the Hamiltonian  $H$ . Here we demonstrate it explicitly for the retarded Green function of two observables in zero temperature. Starting from the expression of  $G^R(t_1, t_2)$  in Eq. (B.19):

$$\begin{aligned} G^R(t_1, t_2) &= -i\Theta(t) \langle \psi_0 | \{X_1(t_1), X_2(t_2)\} | \psi_0 \rangle \\ &= -i\Theta(t_1 - t_2) \langle \psi_0 | e^{iHt_1} X_1 e^{-iH(t_1-t_2)} X_2 e^{-iHt_2} \\ &\quad + e^{iHt_2} X_2 e^{iH(t_1-t_2)} X_1 e^{-iHt_1} | \psi_0 \rangle. \end{aligned} \quad (\text{B.21})$$

Using the eigenbasis  $\{|\psi_m\rangle\}$  of  $H$  ( $H|\psi_m\rangle = E_m|\psi_m\rangle$ ), we can express the first part of retarded  $G^R(t_1, t_2)$  above as:

$$\begin{aligned}
& \langle \psi_0 | e^{iHt_1} X_1 e^{-iH(t_1-t_2)} X_2 e^{-iHt_2} | \psi_0 \rangle \\
&= e^{iE_0(t_1-t_2)} \langle \psi_0 | X_1 e^{-iH(t_1-t_2)} X_2 | \psi_0 \rangle \\
&= \sum_m e^{iE_0(t_1-t_2)} \langle \psi_0 | X_1 | \psi_m \rangle e^{-iE_m(t_1-t_2)} \langle \psi_m | X_2 | \psi_0 \rangle \\
&= \sum_m e^{i(E_0-E_m)(t_1-t_2)} \langle \psi_0 | X_1 | \psi_m \rangle \langle \psi_m | X_2 | \psi_0 \rangle.
\end{aligned} \tag{B.22}$$

Similarly, the second part becomes:

$$\begin{aligned}
& \langle \psi_0 | e^{iHt_2} X_2 e^{iH(t_1-t_2)} X_1 e^{-iHt_1} | \psi_0 \rangle \\
&= \sum_m e^{i(E_m-E_0)(t_1-t_2)} \langle \psi_0 | X_2 | \psi_m \rangle \langle \psi_m | X_1 | \psi_0 \rangle.
\end{aligned} \tag{B.23}$$

Note that the expression above is only a function of time differences  $t_1 - t_2$ . Setting  $t_1 = t$ ,  $t_2 = 0$ , we can write

$$\begin{aligned}
G^R(t) = -i\Theta(t) \sum_m \bigg( & e^{-it(E_m-E_0)} \langle \psi_0 | X_1 | \psi_m \rangle \langle \psi_m | X_2 | \psi_0 \rangle \\
& + e^{it(E_m-E_0)} \langle \psi_0 | X_2 | \psi_m \rangle \langle \psi_m | X_1 | \psi_0 \rangle \bigg)
\end{aligned} \tag{B.24}$$

### B.2.3 Fourier transform

We want to Fourier transform the expression we obtained above. However, the oscillating factor of the form  $e^{iEt}$  in Eq. (B.24) does not vanish as  $t \rightarrow \pm\infty$ , which prohibits a naive application Fourier transform as the integration over time will not converge at infinity. Therefore, a modified Fourier transform is used. For a function  $f(t)$  in time, this modified Fourier transform is defined as

$$f(\omega) = \int_{-\infty}^{\infty} f(t) e^{i\omega t - \eta|t|} dt, \tag{B.25}$$

where  $\eta$  is a small real number. Then, for the function  $-i\Theta(t)e^{iEt}$ , its Fourier transform is,

$$\begin{aligned}
 -i\Theta(t)e^{-iEt} &\rightarrow -i \int_{-\infty}^{\infty} \Theta(t)e^{i(-E+\omega)t-\eta|t|} dt \\
 &= -i \int_0^{\infty} e^{i(\omega-E+i\eta)t} dt \\
 &= -i \frac{e^{i(\omega-E+i\eta)t} \Big|_{t \rightarrow \infty} - 1}{i(\omega - E + i\eta)} \\
 &= \frac{1}{\omega - E + i\eta},
 \end{aligned} \tag{B.26}$$

where the small number  $\eta$  ensures convergence. From the equations above, one can easily write down the Fourier transform of  $G^R(t)$  as:

$$\begin{aligned}
 G^R(\omega) = \sum_m \left( \frac{1}{\omega + i\eta - (E_m - E_0)} \langle \psi_0 | X_1 | \psi_m \rangle \langle \psi_m | X_2 | \psi_0 \rangle \right. \\
 \left. + \frac{1}{\omega + i\eta + (E_m - E_0)} \langle \psi_0 | X_2 | \psi_m \rangle \langle \psi_m | X_1 | \psi_0 \rangle \right).
 \end{aligned} \tag{B.27}$$

Note that  $G^R(\omega)$  is also referred to as the *Lehmann representation* of the retarded Green function in the literature. Finally, we can look at the individual parts forming the retarded Green function. Note that for example,

$$\sum_m \frac{|\psi_m\rangle \langle \psi_m|}{\omega + i\eta \pm (E_m - E_0)}$$

is just the spectral decomposition of the operator

$$\frac{1}{\omega + i\eta \pm (H - E_0)}.$$

Therefore, we can write the retarded Green function in frequency as,

$$\begin{aligned}
 G^R(\omega) = \langle \psi_0 | X_1 \frac{1}{\omega + i\eta - (H - E_0)} X_2 | \psi_0 \rangle \\
 + \langle \psi_0 | X_2 \frac{1}{\omega + i\eta + (H - E_0)} X_1 | \psi_0 \rangle.
 \end{aligned} \tag{B.28}$$

### B.2.4 Imaginary-time Green function

The analogy of Green function in imaginary-time is  $G^T(t)$ :

$$\begin{aligned} G(\tau_1, \tau_2) &= -\langle T_\tau X_1(\tau_1) X_2(\tau_2) \rangle \\ &= - \begin{cases} \langle X_1(\tau_1) X_2(\tau_2) \rangle, & \tau_1 \geq \tau_2 \\ -\langle X_2(\tau_2) X_1(\tau_1) \rangle, & \tau_2 > \tau_1 \end{cases}, \end{aligned} \quad (\text{B.29})$$

where similarly the operator  $T_\tau$  orders operators in decreasing  $\tau$ . Additionally, the imaginary Green function is assigned a periodic condition which is motivated in the context of finite (non-zero) temperature studies<sup>2</sup>. Here, we simply state that:

- $G(\tau)$  is a periodic function with a periodicity of  $2\beta$ , where  $\beta$  is the inverse temperature ( $1/T$ ). Therefore, we only need to consider  $G(\tau)$  for  $\tau \in [-\beta, \beta]$ .
- For fermionic operators  $\{X_1, X_2\} = 0$ , we have  $G(\tau) = -G(\tau + \beta)$  when  $-\beta \leq \tau \leq 0$ .

Similar to the case of retarded Green function, we can express the imaginary-time Green function in the eigenbasis of the Hamiltonian, and Fourier transform it.

In an eigenstates  $\{|\psi_m\rangle\}$  of the Hamiltonian  $H$ , we have

$$\begin{aligned} G(\tau_1, \tau_2) &= -\langle T_\tau X_1(\tau_1) X_2(\tau_2) \rangle \\ &= -\Theta(\tau_1 - \tau_2) \langle \psi_0 | X_1(\tau_1) X_2(\tau_2) | \psi_0 \rangle \\ &\quad + \Theta(\tau_2 - \tau_1) \langle \psi_0 | X_2(\tau_2) X_1(\tau_1) | \psi_0 \rangle \\ &= -\Theta(\tau_1 - \tau_2) \sum_m e^{-(\tau_1 - \tau_2)(E_m - E_0)} \langle \psi_0 | X_1 | \psi_m \rangle \langle \psi_m | X_2 | \psi_0 \rangle \\ &\quad + \Theta(\tau_2 - \tau_1) \sum_m e^{(\tau_1 - \tau_2)(E_m - E_0)} \langle \psi_0 | X_2 | \psi_m \rangle \langle \psi_m | X_1 | \psi_0 \rangle \end{aligned} \quad (\text{B.30})$$

Therefore, imaginary-time Green function  $G(\tau_1, \tau_2)$  also depends only on the relative difference  $\tau_1 - \tau_2$ . Setting  $\tau_2 = 0$  and  $\tau_1 = \tau$ , we can write

$$\begin{aligned} G(\tau) &= -\Theta(\tau) \sum_m e^{-\tau(E_m - E_0)} \langle \psi_0 | X_1 | \psi_m \rangle \langle \psi_m | X_2 | \psi_0 \rangle \\ &\quad + \Theta(-\tau) \sum_m e^{\tau(E_m - E_0)} \langle \psi_0 | X_2 | \psi_m \rangle \langle \psi_m | X_1 | \psi_0 \rangle. \end{aligned} \quad (\text{B.31})$$

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<sup>2</sup>Details available in the context of non-zero temperature in Section 7.3 of [180].

Due to the simple periodic structure, the Fourier transform in imaginary-time is defined by:

$$f(\omega) = \int_{-\beta}^{\beta} f(\tau) e^{i\omega\tau} d\tau, \quad (\text{B.32})$$

where the upper and lower limit of integral goes to  $\pm\infty$  in zero temperature. Additionally, due to the anti-periodicity  $G(\tau) = -G(\tau + \beta)$  for  $-\beta \leq \tau \leq 0$ , one can show that imaginary-time Green function's Fourier transform is non-zero only at specific frequencies, called Matsubara frequencies<sup>3</sup>:

$$\omega_n = (2n + 1)\pi/\beta, \quad n \in \mathbb{N}, \quad (\text{B.33})$$

Then, using the fact that at those frequencies, the factor  $e^{i\omega_n\beta} = -1$ , we have

$$\begin{aligned} - \int_{-\beta}^{\beta} \Theta(\tau) e^{-E\tau} e^{i\omega\tau} d\tau &\stackrel{\omega \rightarrow \omega_n}{=} - \frac{-e^{-\beta E} - 1}{i\omega_n - E} \stackrel{\beta \rightarrow +\infty}{\rightarrow} \frac{1}{i\omega_n - E}, \\ \int_{-\beta}^{\beta} \Theta(-\tau) e^{E\tau} e^{i\omega\tau} d\tau &\stackrel{\omega \rightarrow \omega_n}{=} \frac{1 + e^{-\beta E}}{i\omega_n + E} \stackrel{\beta \rightarrow +\infty}{\rightarrow} \frac{1}{i\omega_n + E}, \end{aligned} \quad (\text{B.34})$$

Therefore, we can write the Fourier transform  $G(i\omega_n)$  in:

$$G(i\omega_n) = \sum_m \frac{\langle \psi_0 | X_1 | \psi_m \rangle \langle \psi_m | X_2 | \psi_0 \rangle}{i\omega_n - (E_m - E_0)} + \frac{\langle \psi_0 | X_2 | \psi_m \rangle \langle \psi_m | X_1 | \psi_0 \rangle}{i\omega_n + (E_m - E_0)} \quad (\text{B.35})$$

Finally, recognising that for example  $\sum_m |\psi_m\rangle \langle \psi_m| (i\omega_n - (E_m - E_0))^{-1}$  is the spectral decomposition of  $(i\omega_n - (H - E_0))^{-1}$ , we can again write the Fourier transform  $G(i\omega_n)$  as:

$$\begin{aligned} G(i\omega_n) &= \langle \psi_0 | X_1 \frac{1}{i\omega_n - (H - E_0)} X_2 | \psi_0 \rangle \\ &\quad + \langle \psi_0 | X_2 \frac{1}{i\omega_n + (H - E_0)} X_1 | \psi_0 \rangle. \end{aligned} \quad (\text{B.36})$$

Note that the Fourier transform of the imaginary-time Green function is also called a *Matsubara function*, which is why the frequencies  $\omega_n$  are called Matsubara frequencies.

### B.2.5 Unifying expression

Finally, the similarity between the Fourier transform of retarded Green function and the Fourier transform of the imaginary-time Green function (Mat-

<sup>3</sup>See discussions following Eq. 3.21 of [242, Chapter 3].

subara function) inspires us to use a unifying expression for them:

$$G(z) = \langle \psi_0 | X_1 \frac{1}{z - (H - E_0)} X_2 | 0 \rangle + \langle 0 | X_2 \frac{1}{z + (H - E_0)} X_1 | \psi_0 \rangle, \quad (\text{B.37})$$

where setting the  $z$  as  $z = \omega + i\eta$  gives the Fourier transform of retarded Green function, while setting the  $z$  as  $z = i\omega_n$  gives the Fourier transform of the imaginary-time Green function. Obviously, we can also write:

$$G(z) = \sum_m \frac{\langle \psi_0 | X_1 | \psi_m \rangle \langle \psi_m | X_2 | \psi_0 \rangle}{z - (E_m - E_0)} + \frac{\langle \psi_0 | X_2 | \psi_m \rangle \langle \psi_m | X_1 | \psi_0 \rangle}{z + (E_m - E_0)}, \quad (\text{B.38})$$

for an eigenbasis  $\{|\psi_m\rangle\}$  of the Hamiltonian:  $H |\psi_m\rangle = E_m |\psi_m\rangle$ .

## Appendix C

# Hadamard test circuit and its variants

### C.1 General idea

Here we illustrate the idea behind various Hadamard test. We want to compute the overlap  $\langle\alpha|\beta\rangle$  of two states  $|\alpha\rangle$  and  $|\beta\rangle$ . Note that it suffices to compute the real part,  $\text{Re}(\langle\alpha|\beta\rangle)$ , first. Then after adding a phase, the imaginary can be computed.

In this section, we assume that we can prepare an interference of  $|\alpha\rangle|0\rangle + |\beta\rangle|1\rangle$  between the two states. The adjective “interference” is used in analogy with the famous Stern-Gerlach interferometry experiment. Then, we first show that the  $X$  component relates to the overlap  $\text{Re}(\langle\alpha|\beta\rangle)$ . We then show that  $X$  component can be rotated to the  $Z$  axis using the Hadamard gate, which explains the terminology “Hadamard test”.

The  $X$  component of a state  $\alpha|0\rangle + \beta|1\rangle$ , where  $\alpha$  and  $\beta$  are temporarily just complex numbers, can be obtained by the following tracing operation on the density matrix  $\rho$  corresponding to this pure state:

$$\text{Tr}(\rho X) = \text{Tr} \left[ \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \begin{pmatrix} \alpha^* & \beta^* \end{pmatrix} X \right] = \alpha\beta^* + \beta^*\alpha = 2\text{Re}(\alpha\beta^*).$$

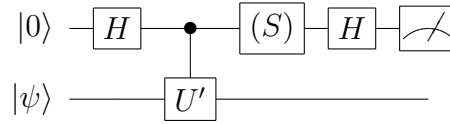
The above calculation can be generalised with the replacement  $\alpha \mapsto |\alpha\rangle$  and  $\beta \mapsto |\beta\rangle$ . The star (\*) need to be replaced by the dagger ( $\dagger$ ), leading to  $\text{Re}(\alpha\beta^*) \mapsto \text{Re}(\langle\beta|\alpha\rangle) = \text{Re}(\langle\alpha|\beta\rangle)$ . Thus, by measuring the  $X$  component on the second part in  $|\alpha\rangle|0\rangle + |\beta\rangle|1\rangle$ , we can obtain  $2\text{Re}(\langle\alpha|\beta\rangle)$ .

Measuring the  $X$  component can be realised on  $Z$  axis by the Hadamard gate. This is due to the equality  $HZH = X$ . From another view, on the

one-qubit Bloch sphere,  $H$  represents a rotation of  $\pi/2$  around the  $Y$ -axis, followed by a  $\pi$  rotation around the  $X$ -axis ( $H = X R_y(\pi/2)^1$ ). The projection of a vector to the  $X$ -axis is rotated into the  $Z$ -axis in this process.

Now we mention how the interference state can be prepared, and how to add a phase. The interference can usually be created by first creating the entangled pair  $|0\rangle + |1\rangle$  on an ancilla qubit by applying the Hadamard  $H$  to  $|0\rangle$ , and then applying controlled unitaries conditioned on the Hadamard to prepare the superposition  $|\alpha\rangle |0\rangle + |\beta\rangle |1\rangle$ . The phase can be added easily by the phase gate  $S$  on this ancilla.

## C.2 Hadamard test



**Figure C.1:** Figure 4.1 reproduced. Without  $S$  gate, measurement of  $Z$  in the ancilla qubit gives  $\text{Re} \langle \psi | U | \psi \rangle$ . With  $S$  gate, it produces  $-\text{Im} \langle \psi | U | \psi \rangle$  instead.

Here we introduce the circuit which evaluates the real the imaginary part of the overlap  $\langle \psi | U | \psi \rangle$ . The circuit has been discussed in Figure 4.1 (Section 4.2.2.1) and is visualized here again in Figure C.1. The calculations below simply confirm the idea illustrated in the previous section.

After the Hadamard gate and the controlled- $U$  gate, the quantum state is changed into:

$$\frac{1}{\sqrt{2}} (|0\rangle \otimes |\psi\rangle + |1\rangle \otimes U |\psi\rangle). \quad (\text{C.1})$$

When only the Hadamard gate is applied afterwards, the state before measurement is

$$\frac{1}{2} ((|0\rangle + |1\rangle) \otimes |\psi\rangle + (|0\rangle - |1\rangle) \otimes U |\psi\rangle). \quad (\text{C.2})$$

Measuring the first ancilla qubit in the  $Z$  basis gives

$$\begin{aligned} & \frac{1}{4} ((\langle \psi | + \langle \psi | U^\dagger) (|\psi\rangle + U |\psi\rangle) - (\langle \psi | - \langle \psi | U^\dagger) (|\psi\rangle - U |\psi\rangle)) \\ &= \text{Re} \langle \psi | U | \psi \rangle. \end{aligned} \quad (\text{C.3})$$

Similarly, when the  $S$  gate is applied before the Hadamard gate, the state

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<sup>1</sup>Here,  $R_y(\theta) = \exp(-i\theta Y/2)$ . The principal square root  $\sqrt{Y} = \frac{1+i}{\sqrt{2}} R_y(\pi/2)$ .

before measurement is

$$\frac{1}{\sqrt{2}} (|0\rangle \otimes |\psi\rangle + i |1\rangle \otimes U |\psi\rangle) \quad (\text{C.4})$$

When measured, it produces

$$\text{Re}(\langle\psi| iU |\psi\rangle) = -\text{Im}(\langle\psi| U |\psi\rangle) \quad (\text{C.5})$$

### C.3 Hadamard test variant 1

Here we explain the quantum circuit shown in Fig. 4.2. It is easy to see that the state before the measurement is

$$\begin{aligned} & \frac{1}{2} \left( \underbrace{(|0\rangle + |1\rangle)}_{\text{From } H|0\rangle} P_2 U |\psi\rangle + \underbrace{(|0\rangle - |1\rangle)}_{\text{From } H|1\rangle} \otimes U P_1 |\psi\rangle \right) \\ &= \frac{1}{2} (|0\rangle (P_2 U + U P_1) |\psi\rangle + |1\rangle (P_2 U - U P_1) |\psi\rangle). \end{aligned} \quad (\text{C.6})$$

Measuring the ancilla in  $Z$  basis gives:

$$\begin{aligned} & \frac{1}{4} \langle\psi| (U^\dagger P_2 + P_1 U^\dagger) (P_2 U + U P_1) |\psi\rangle \\ & - \frac{1}{4} \langle\psi| (U^\dagger P_2 - P_1 U^\dagger) (P_2 U - U P_1) |\psi\rangle \\ &= \frac{1}{4} (2 + 2R) - \frac{1}{4} (2 - 2R) \\ &= \text{Re}(\langle\psi| P_1 U^\dagger P_2 U |\psi\rangle). \end{aligned} \quad (\text{C.7})$$

Measuring the ancilla in  $Y$  basis gives<sup>2</sup>:

$$\begin{aligned} & -\frac{i}{4} \text{Tr}((P_2 U + U P_1) |\psi\rangle \langle\psi| (U^\dagger P_2 - P_1 U^\dagger)) \\ & + \frac{i}{4} \text{Tr}((P_2 U - U P_1) |\psi\rangle \langle\psi| (U^\dagger P_2 + P_1 U^\dagger)) \\ &= -\frac{i}{4} (\langle\psi| U^\dagger P_2 U P_1 - P_1 U^\dagger P_2 U |\psi\rangle) \\ & + \frac{i}{4} (\langle\psi| P_1 U^\dagger P_2 U - U^\dagger P_2 U P_1 |\psi\rangle) \\ &= -\frac{i}{4} (2i \text{Im} \langle\psi| U^\dagger P_2 U P_1 |\psi\rangle) + \frac{i}{4} (-2i \text{Im} \langle\psi| U^\dagger P_2 U P_1 |\psi\rangle) \\ &= \text{Im} \langle\psi| P_1 U^\dagger P_2 U |\psi\rangle. \end{aligned} \quad (\text{C.8})$$

---

<sup>2</sup>Note that  $Y = -i|0\rangle\langle 1| + i|1\rangle\langle 0|$

Equations C.7 and C.8 shows that the circuit correctly evaluates the real and imaginary part of  $\langle \psi | P_1 U^\dagger P_2 U | \psi \rangle$  when measured in  $Z$  and  $Y$  basis.

## C.4 Hadamard test variant 2

This is yet another variant of the Hadamard-test. Here we want to compute the overlap term:

$$\langle 0 | U_1^\dagger(\theta_1) P U_2(\theta_2) | 0 \rangle. \quad (\text{C.9})$$

The circuit is illustrated in Fig. C.2. In the following we will show that the outcome of that circuit is

$$\text{Re} \left( e^{i\phi} \langle 0 | U_1^\dagger P U_2 | 0 \rangle \right). \quad (\text{C.10})$$

Therefore, when  $\phi = 0$ , it samples  $\text{Re}(\langle 0 | U_1^\dagger P U_2 | 0 \rangle)$ , and when  $\phi = \pi/2$ ,  $e^{i\phi} = i$ , it samples  $-\text{Im}(\langle 0 | U_1^\dagger P U_2 | 0 \rangle)$ .

*Proof.* After  $H$  and  $R_z(\phi)$  gates, the state is initialized to

$$2^{-1/2}(|0\rangle + e^{i\phi}|1\rangle)|0\rangle.$$

We then perform a 0-controlled- $U_1$  gate using three gates:  $X$ , controlled- $U_1$ , and  $X$ . This resulted in the state

$$2^{-1/2}(|0\rangle U_1|0\rangle + e^{i\phi}|1\rangle|0\rangle).$$

We then perform normal 1-controlled  $U_2$  and  $P$  gates, leading to:

$$2^{-1/2}(|0\rangle U_1|0\rangle + e^{i\phi}|1\rangle P U_2|0\rangle).$$

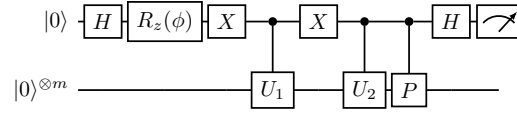
Finally, after the Hadamard gate and before measurement, the state of the  $m+1$  qubits is:

$$\begin{aligned} & 2^{-1}|0\rangle (U_1|0\rangle + e^{i\phi} P U_2|0\rangle) \\ & + 2^{-1}|1\rangle (U_1|0\rangle - e^{i\phi} P U_2|0\rangle). \end{aligned}$$

For convenience, define:

$$A = U_1|0\rangle, B = e^{i\phi} P U_2|0\rangle, \quad (\text{C.11})$$

and denote the probability of measuring 0(1) on the ancilla as  $p(0)$  ( $p(1)$ ). We



**Figure C.2:** Quantum circuit for measuring the overlap term  $\langle 0|U_1(\theta_1)PU_2(\theta_2)|0\rangle$ . The expectation value of the  $Z$ -measurement on the ancillary qubit yields Eq. C.10. Reproduction of Fig. 4.4.

have:

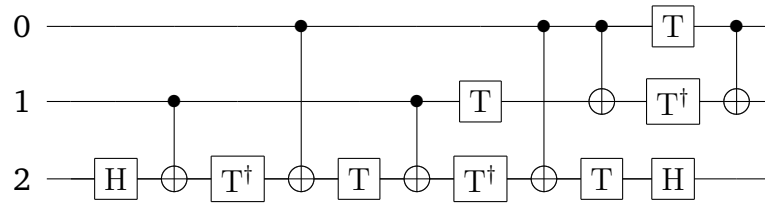
$$\begin{aligned}
 4p(0) &= (A^\dagger + B^\dagger)(A + B) = |A|^2 + A^\dagger B + B^\dagger A + |B|^2, \\
 4p(1) &= (A^\dagger - B^\dagger)(A - B) = |A|^2 - A^\dagger B - B^\dagger A + |B|^2, \\
 \Rightarrow \quad p(0) - p(1) &= \frac{1}{4}4 \operatorname{Re}(A^\dagger B) = \operatorname{Re}(e^{i\phi} \langle 0|U_1^\dagger P U_2|0\rangle)
 \end{aligned}$$

□

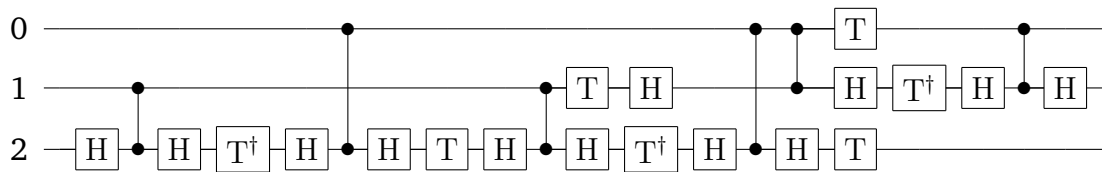
## Appendix D

# Circuit decomposition

Here we detail explicit the decomposition we used to implement CCX and CCZ gates using only two-qubit gates. The decomposition for CCX is illustrated in Fig. D.1 and for CCZ is illustrated in Fig. D.2.



**Figure D.1:** Circuit decomposition of CCX



**Figure D.2:** Circuit decomposition of CCZ

## Appendix E

# Other details for the two-site DMFT

### E.1 Analysis and reductions on the four-qubit Hamiltonian

**Eigenvalues of the 4-qubit Hamiltonian.** The 4-qubit Hamiltonian,

$$H = \frac{U}{4} Z_1 Z_3 + \frac{V}{2} (X_1 X_2 + Y_1 Y_2 + X_3 X_4 + Y_3 Y_4), \quad (5.26 \text{ revisited})$$

can be solved such that all its energies are known. Also, since  $H$  commutes with electron number  $N$  (Eq. (5.34)) and spin- $z$  operator  $S_z$  (Eq. (5.35)), the corresponding eigenvalues for  $N$  and  $S_z$  can also be enumerated. The calculation was carried out in a Mathematica notebook [246] and the result is shown in Table E.1. Similar calculations could be done without solving the Hamiltonian analytically, but by (i) examining the symmetry; (ii) focusing on different eigenspaces of electron number  $N$ ; and (iii) analysing relations between  $\langle N \rangle$  and  $\langle S_z \rangle$  (Rungger et al. [30, Appendix B]).

#### E.1.1 Mapping to 2-qubit Hamiltonians

Here we discuss how the 4-qubit Hamiltonian can be mapped to various 2-qubit Hamiltonians acting on subspaces representing distinct electron number sectors. Since we are only interested in the case of  $\langle N \rangle = 1$  (hole case) and  $\langle N \rangle = 2$  (ground state case), we will only mention these two cases here. The other cases' discussion is available in the Appendix B of Rungger et al. [30].

**Notations.** Since we deal with both operators on different subspaces and their representations on these subspaces, we introduce a new notation for clarity. We use  $\sigma_{x/y/z}$  to represent the three Pauli operator, which obeys the

| Index | Energy                           | $\langle N \rangle$ | $\langle S_z \rangle$ | Remark       | $E$             | $\lambda$       |
|-------|----------------------------------|---------------------|-----------------------|--------------|-----------------|-----------------|
| 1     | $-\frac{1}{4}\sqrt{U^2 + 64V^2}$ | 2                   | 0                     | Ground State | $E_{\text{GS}}$ |                 |
| 2     | $-\frac{1}{4}\sqrt{U^2 + 16V^2}$ | 3                   | 1                     | Particle     | $E_{p,1}$       | $\lambda_{p,1}$ |
| 3     | $-\frac{1}{4}\sqrt{U^2 + 16V^2}$ | 3                   | -1                    | Particle     |                 | 0               |
| 4     | $-\frac{1}{4}\sqrt{U^2 + 16V^2}$ | 1                   | 1                     | Hole         |                 | 0               |
| 5     | $-\frac{1}{4}\sqrt{U^2 + 16V^2}$ | 1                   | -1                    | Hole         | $E_{h,1}$       | $\lambda_{h,1}$ |
| 6     | $-\frac{U}{4}$                   | 2                   | 2                     |              |                 |                 |
| 7     | $-\frac{U}{4}$                   | 2                   | 0                     |              |                 |                 |
| 8     | $-\frac{U}{4}$                   | 2                   | -2                    |              |                 |                 |
| 9     | $\frac{U}{4}$                    | 4                   | 0                     |              |                 |                 |
| 10    | $\frac{U}{4}$                    | 2                   | 0                     |              |                 |                 |
| 11    | $\frac{U}{4}$                    | 0                   | 0                     |              |                 |                 |
| 12    | $\frac{1}{4}\sqrt{U^2 + 16V^2}$  | 3                   | 1                     | Particle     | $E_{p,2}$       | $\lambda_{p,2}$ |
| 13    | $\frac{1}{4}\sqrt{U^2 + 16V^2}$  | 3                   | -1                    | Particle     |                 | 0               |
| 14    | $\frac{1}{4}\sqrt{U^2 + 16V^2}$  | 1                   | 1                     | Hole         |                 | 0               |
| 15    | $\frac{1}{4}\sqrt{U^2 + 16V^2}$  | 1                   | -1                    | Hole         | $E_{h,2}$       | $\lambda_{h,2}$ |
| 16    | $\frac{1}{4}\sqrt{U^2 + 64V^2}$  | 2                   | 0                     |              |                 |                 |

**Table E.1:** Eigenvalues of the 4-qubit Hamiltonian  $H$ . Here we solved the Hamiltonian, listed the corresponding eigenvalues of eigenstates obtained (energies, electron numbers  $\langle N \rangle$ , and spin- $z$  numbers  $\langle S_z \rangle$ ), sorted by increasing energies. We also commented in the “Remark” column on whether this state is the ground state, the particle state, or the hole state. In the last two columns, we listed the mathematical symbols which have been used in the main text to represent the energy  $E$  or the overlap- $\lambda$ . Note that wherever the value does not interest us or is zero due to symmetry concerns, we leave it empty in the table.

usual relation  $\sigma_x \sigma_y = i \sigma_z$ . At the same time, the notations  $X_i/Y_i/Z_i$  continue to mean specific numerical Pauli matrices on qubit- $i$ .

We will add an overline to the operator after the mappings below. For example, if  $\sigma_x$  is the Pauli-X operator on original algebra, then  $\bar{\sigma}_x$  is an operator in the mapped subalgebra.

Note that, while we keep it our convention that the raise/creation operator increases electron number (as in  $c^\dagger$ ) and the lowering/annihilation operator decreases electro number (as in  $c$ ), confusion might arise when comparing formulae here and that in Rungger et al. [30]. For example, due to the Jordan-Wigner transformation used (Eq. (2.6)), the creation operator maps eigen-

states of Pauli- $Z$  matrix from eigenvalue 1 to  $-1$  (in qubit notation  $|0\rangle\langle 1|$ ). This matrix is sometimes denoted by  $\sigma^-$ , related to the sense of lowering qubit energy level. A similar problem exists for the lowering/annihilation operator.

**General strategy for mapping.** In general, we can find mappings of the algebras through two approaches. First, we can relate the Pauli- $X/Y/Z$  operators between the old and new algebra. Second, we can do the same for the Pauli- $Z$  and ladder operators. These two approaches are equivalent since Pauli- $X/Y$  operators and ladder operators are related together by:

$$\begin{aligned}\sigma^+ &= (\sigma_x + i\sigma_y)/2 & \sigma^- &= (\sigma_x - i\sigma_y)/2 \\ \sigma_x &= \sigma^+ + \sigma^- & \sigma_y &= i(\sigma^+ - \sigma^-)\end{aligned}\tag{E.1}$$

**Companion notebook.** Details of calculations were carried out in a companion Mathematica notebook [247].

### E.1.2 Mapping to one-electron subspace

In this case, since  $\langle N \rangle = 1$ , and using the knowledge from the Table. E.1,  $\langle S_z \rangle = 1/2$ . We define the subalgebra's Pauli operators by:

$$\begin{aligned}\bar{\sigma}_{z1} &= \frac{1}{2}(\sigma_{z1} + \sigma_{z2} + \sigma_{z3} + \sigma_{z4}) \\ \bar{\sigma}_{z2} &= \frac{1}{2}(\sigma_{z1} - \sigma_{z2} + \sigma_{z3} - \sigma_{z4})\end{aligned}\tag{E.2}$$

such that the first qubit determines the parity in total spin, and the second qubit determines the sum of parities in the first two and the last two orbitals. We can work out the eigenstates, and their corresponding eigenvalues of  $\sigma_{z1}$  to  $\sigma_{z4}$ , and the corresponding eigenvalues of the mapped  $\bar{\sigma}_{z1}$  and  $\bar{\sigma}_{z2}$ , in a table (Table E.2). This table helps us discover relations expressing the eigenvalues of  $\sigma_{z1}$  to  $\sigma_{z4}$  from eigenvalues of  $\bar{\sigma}_{z1}$  and  $\bar{\sigma}_{z2}$ . This translates to the following expressions of the operators:

$$\begin{aligned}\sigma_{z1} &= -\frac{1}{2}(1 - \bar{\sigma}_{z1})(1 - \bar{\sigma}_{z2}) + 1 \\ \sigma_{z2} &= -\frac{1}{2}(1 - \bar{\sigma}_{z1})(1 + \bar{\sigma}_{z2}) + 1 \\ \sigma_{z3} &= -\frac{1}{2}(1 + \bar{\sigma}_{z1})(1 - \bar{\sigma}_{z2}) + 1 \\ \sigma_{z4} &= -\frac{1}{2}(1 + \bar{\sigma}_{z1})(1 + \bar{\sigma}_{z2}) + 1.\end{aligned}\tag{E.3}$$

| No. | $\sigma_{z1}$ | $\sigma_{z2}$ | $\sigma_{z3}$ | $\sigma_{z4}$ | $\bar{\sigma}_{Z1}$ | $\bar{\sigma}_{Z2}$ |
|-----|---------------|---------------|---------------|---------------|---------------------|---------------------|
| 1   | -1            | 1             | 1             | 1             | -1                  | -1                  |
| 2   | 1             | -1            | 1             | 1             | -1                  | 1                   |
| 3   | 1             | 1             | -1            | 1             | 1                   | -1                  |
| 4   | 1             | 1             | 1             | -1            | 1                   | 1                   |

**Table E.2:** Corresponding eigenvalues of Pauli-Z operators before and after the mapping, for the one-electron case.

Next, ladder operators can be obtained from the Table E.2. The lowering operator on the first qubit should either exchange rows 1 to 3 and rows 2 to 4, or exchange rows 1 to 4 and 2 to 3. We choose the first one. Similarly, the lowering operator for qubit 2 is chosen to map row 1 to 2, and row 3 to row 4. This leads to the following relation:

$$\begin{aligned}\bar{\sigma}_1^+ &= \sigma_3^- \sigma_1^+ + \sigma_4^- \sigma_2^+ \\ \bar{\sigma}_2^+ &= \sigma_2^- \sigma_1^+ + \sigma_4^- \sigma_3^+.\end{aligned}\tag{E.4}$$

Together, Eq. (E.3) and Eq. (E.4) are sufficient to work out the mapped Hamiltonian in 2-qubits:

$$H_{\text{one-electron}} = \left(\frac{\mu}{2} + \frac{\epsilon}{2}\right) Z_2 + V X_2 + \left(\frac{\mu}{2} - \frac{U}{4} - \frac{\epsilon}{2}\right), \tag{E.5}$$

with its particle-hole symmetric version

$$H_{\text{one-electron,ph}} = \frac{U}{4} Z_2 + V X_2. \tag{E.6}$$

Note that, the fact that this reduced Hamiltonian only acts on one of the two qubits, reflects two things: first, there is only two distinct energies, due to the degree-2 degeneracy from spin symmetry; second, the first Pauli-Z operator is just the spin- $z$   $\bar{\sigma}_{z1} = -S_z$ , and spin- $z$  take either  $-1$  or  $1$  here.

**Ansatz for this Hamiltonian.** Since only one qubit is required for this Hamiltonian, the ansatz circuit used can be significantly simplified. In fact, we found that just one  $R_y$  rotation with a parameterised angle is sufficient to find the lowest energy ground state for this Hamiltonian. Also, the next eigenstate can be found by simply adding  $\pi$ . That is if the lowest energy eigenstate of  $H_{\text{one-electron,ph}}$  is found by VQE to be prepared by  $R_y(\theta_1) |0\rangle$ , then the next eigenstate is certainly prepared with  $R_y(\theta_1 + \pi) |0\rangle$ , as it is orthogonal to the

| No. | $\sigma_{z1}$ | $\sigma_{z2}$ | $\sigma_{z3}$ | $\sigma_{z4}$ | $S_z$ |
|-----|---------------|---------------|---------------|---------------|-------|
| 1   | -1            | -1            | 1             | 1             | 2     |
| 2   | -1            | 1             | -1            | 1             | 0     |
| 3   | -1            | 1             | 1             | -1            | 0     |
| 4   | 1             | -1            | -1            | 1             | 0     |
| 5   | 1             | -1            | 1             | -1            | 0     |
| 6   | 1             | 1             | -1            | -1            | -2    |

**Table E.3:** Corresponding eigenvalues of Pauli-Z operators and spin- $z$  operator, for the two-electron case.

previous one. The energies measured from this two eigenstates correspond to  $E_{h,1}$  and  $E_{h,2}$  in Table E.1.

### E.1.3 Mapping to two-electron subspace

In this case, we can enumerate the eigenvalues for two-electrons subspace in Table E.3. From the table, we notice that when spin- $z$  is 0 (which contains the ground state that we are interested), we always have  $\langle\sigma_{z1}\rangle = \langle\sigma_{z2}\rangle$ , and  $\langle\sigma_{z3}\rangle = -\langle\sigma_{z4}\rangle$ . This redundancy encourages us to define Pauli-Z in the 2-qubit space by:

$$\begin{aligned}\bar{\sigma}_{z1} &= \sigma_{z1} = -\sigma_{z2} \\ \bar{\sigma}_{z2} &= \sigma_{z3} = -\sigma_{z4}.\end{aligned}\tag{E.7}$$

Next, the Pauli-X operator normally flips eigenstates of Pauli-Z, so naturally we want  $\bar{\sigma}_{x1}$  to flip states in row 2 (3) of the Table E.3 above, to row 4 (5). And similarly for  $\bar{\sigma}_{x2}$  to flip row 2 (4) to row 3 (5). So naturally we have,

$$\begin{aligned}\bar{\sigma}_{x1} &= \sigma_{x1}\sigma_{x2} \\ \bar{\sigma}_{x2} &= \sigma_{x3}\sigma_{x4}\end{aligned}\tag{E.8}$$

Relations in Pauli-Z and X naturally lead to relations in Pauli-Y and other operators. Therefore, Eq. (E.7) and Eq. (E.8) are sufficient for us to work out the mapped Hamiltonian in the 2-electron subspace:

$$H_{\text{two-electron}} = \frac{U}{4}\bar{\sigma}_{z1}\bar{\sigma}_{z2} + V(\bar{\sigma}_{x1} + \bar{\sigma}_{x2}) + \left(\frac{\mu}{2} - \frac{U}{4} + \frac{\epsilon}{2}\right)(\bar{\sigma}_{z1} + \bar{\sigma}_{z2}), \tag{E.9}$$

and its particle-hole symmetric version:

$$H_{\text{two-electron,ph}} = \frac{U}{4} \bar{\sigma}_{z1} \bar{\sigma}_{z2} + V(\bar{\sigma}_{x1} + \bar{\sigma}_{x2}). \quad (\text{E.10})$$

#### E.1.4 Formula for overlap- $\lambda$ s

Note that the calculations done here are available as a companion Mathematica notebook [248].

While working through the eigenstates for the 4-qubit Hamiltonian, we can obtain the analytical formula for all overlap- $\lambda$ s:

$$\begin{aligned} \lambda_{h,1} &= \frac{1}{4} \left( 1 + \frac{32 - \frac{U^2}{V^2}}{\sqrt{1024 + \frac{U^4}{V^4} + \frac{80U^2}{V^2}}} \right) \\ \lambda_{h,2} &= \frac{1}{2} - \lambda_{h,1} \end{aligned} \quad (\text{E.11})$$

where we have used the subscripts and symbols defined in Table E.1.

Next, we can in addition obtain other rules relating the overlap- $\lambda$ s, the energy differences  $\Delta E$ , and impurity Hamiltonian's parameters. In fact, for the simple system we considered here, we can compute overlap- $\lambda$ s directly from energies. This is obtained by Ivan in Rungger et al. [30], and we detailed the steps below.

First, we note that the self-energy cannot diverge for the self-consistency loop. The self-energy is related to the quasi-particle weight  $z$  by Eq. (5.21). This weight itself should be positive, as implied in the second self-consistency condition (Eq. (5.20)). This leads to  $\text{Im}[\Sigma(i0^+)]/0^+ < 1$  and hence  $\text{Im}[\Sigma(i0^+)]$  should even approach zero faster than  $0^+$ .

To prevent  $\Sigma(i0^+)$  from diverging, we derive a series of equalities involving interacting and non-interacting Green functions. Specifically, note that the free Green function is:

$$G_0^{-1}(\omega) = \omega + i0^+ + \mu - \Delta(\omega). \quad (\text{5.17 revisited})$$

It diverges whenever the hybridisation  $\Delta(\omega) = V^2/\omega$  (Eq. (5.18)) diverges. Since the self-energy  $\Sigma$  does not diverge, as  $\omega$  approaches  $0+i0^+$ , the divergent part of the non-interacting and interacting Green functions must cancel each other exactly, so that self-energy, being the difference of the two, remains a reasonable value.

This leads to a series of equalities:

$$\begin{aligned}\lim_{\omega \rightarrow 0} G_0(\omega) &= \lim_{\omega \rightarrow 0} G(\omega) \\ \lim_{\omega \rightarrow 0} \partial_\omega G_0(\omega) &= \lim_{\omega \rightarrow 0} \partial_\omega G(\omega), \\ &\dots\end{aligned}\tag{E.12}$$

Using the simplified equation for the Green function in Eq. (5.36), we can verify that the first one is satisfied trivially, but the second one leads to,

$$\sum_{n=\{h1,h2,p1,p2\}} \frac{\lambda_n}{(\Delta E_n)^2} = \frac{1}{V^2}.\tag{E.13}$$

Next, using the sum rule and the particle-symmetry results discussed in Sec. 5.2.3, we have:

$$\lambda_{h,1} = \frac{(\Delta E_{h,1})^2 (V^2 - (\Delta E_{h,2})^2)}{2V^2 ((\Delta E_{h,1})^2 - (\Delta E_{h,2})^2)}.\tag{E.14}$$

## E.2 IBM Quantum computers

In the experiments, we used quantum computers (or QPUs for short) available from IBM. These computers are based on the superconducting transmon qubits [249]. All unitary gates in a quantum circuit are compiled into a few gates called the basic gate set. For IBM's QPU, the basic gate set includes the following gates: ECR, ID, RZ, SX, X. Here, ECR stands for echo cross-resonance (ECR) pulse [250, 251], which is used to implement any other two-qubit gates. ID stands for the identity gate. RZ is the  $z$ -rotation gate  $R_z$  and is implemented using software tricks (a “virtual gate”) [252]. X is the  $X$  gate and is implemented as twice of a  $\sqrt{X}$  gate. SX is the  $\sqrt{X}$  gate and is implemented as physical pulses. Taken together (and also with the measurement gate), three important parameters are useful to indicate the quality of an IBM QPU: the error rate for ECR, SX, and for the readout process. Although further parameters such as the  $T_{1/2}$  time, the amount of cross-talk is also important in specifying the quality.

QPUs from IBM are accessed through a cloud interface and are constantly updated in iterations. Each IBM QPU has a code name, which can be used to identify a QPU available at a certain laboratory of a certain generation. During our experiment, we have used a few QPUs provided by IBM, whose code names are: `ibmq_london`, `ibmq_essex`, `ibmqx2`, and `ibmqx`.

Unfortunately, as IBM began to retire QPUs belonging to prior generations, all the QPUs used in our experiment may have been retired and are no longer accessible. Although we did not record the exact calibration data and could not retrieve them due to their retirement at the time of writing, it is possible to find calibration data of the same or similar systems from the same generation from other publications [253].

Additionally, while IBM has continuously made progress in improving its quantum computers, we believe the quality of the QPUs we used in our experiment still compares favourably with that of the best available quantum computers at the time of writing. The main reason is that our results are based on short-depth circuits which have at most one-layer of two-qubit gates. Also, for the full DMFT calculations in Section 5.3.2, the circuits further require only up-to-two qubits, which enables us to use the best qubit pairs available on any QPUs available. Therefore, the most problematic error sources such as qubit connectivity (i.e., when one needs to execute a gate between far-away qubits) and circuit depth did not play a significant role in our experiments. Lastly, while much progress has been made to increase the number of qubits available in a single QPU, little generational progress has been made on improving the fidelity of single qubit gates [254].

# Glossary

## Chapter 1

|                       |                                                                      |               |
|-----------------------|----------------------------------------------------------------------|---------------|
| $ \text{GS}\rangle$   | The ground state.                                                    |               |
| $\theta^*$            | The superscript star (*) generally indicates an optimised parameter. |               |
| HEA, $d_{\text{HEA}}$ | Hardware efficient ansatz. $d_{\text{HEA}}$ is its depth.            | Section 2.1.2 |
| Jordan-Wigner         | The transformation algorithm we used.                                | Section 2.1.1 |
| RotoSolve             | The optimisation algorithm we used.                                  | Section 2.1.5 |
| VQE                   | Variational quantum eigensolver.                                     | Section 2.1   |

## Chapter 2

|                               |                                                                                                                                                                                                                                     |  |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| $\overline{X} (\overline{Z})$ | Physical operations which implement logical $X$ ( $Z$ ). Using the overline notation $\overline{O}$ for the physical operations implementing $O$ is common in the quantum error correction literature. Introduced in Section 3.4.2. |  |
| $C$                           | $C$ for correction denotes the combined operation of stabiliser measurement and decoding in a stabiliser code.                                                                                                                      |  |
| $E$                           | $E$ denotes the circuit used to encode quantum information into the logical code space.                                                                                                                                             |  |
| $G, g$                        | Small $g$ refers to a logical gate, and big $G$ refers to the physical implementation of the logical gate.                                                                                                                          |  |
| $H_A$                         | The Hilbert space of logical information.                                                                                                                                                                                           |  |
| $H_B, H_C$                    | Auxiliary Hilbert spaces for encoding $E$ and correction $C$ . See Figure 3.1.                                                                                                                                                      |  |

$[[n, k, d]]$  Notation for a quantum code.  $n$  is the number of physical qubits,  $k$  is the number of physical qubits, and  $d$  is code's distance. Introduced in Section 3.1.

### Chapter 3

$0^+$  or  $\eta$  The small broadening constant used for real frequency in the retarded Green function. Theoretically,  $\eta \rightarrow 0^+$ , while in numerics a small  $\eta$  is used.

$|\chi\rangle$  The correction vector. Discussed in Section 4.3.

$G(t), G(\tau), G(z)$  Depending on the argument, it is the retarded Green function in time ( $t$ ), imaginary time ( $\tau$ ), or their Fourier transforms ( $z$ ).

CAS Complete active space. See Section 4.4.3.

Lehmann repre. Lehmann representation is real frequency Green function expressed in an eigenbasis. See Eq. (4.8).

Matsubara freq. When  $z = i\omega_n$  where  $n = 1, 2, \dots$ . Sometimes referred to as the imaginary frequency.

Real frequency When  $z = \omega + i0^+$  or  $z = \omega + i\eta$ .

### Chapter 4

$0^+$  Broadening constant for the retarded Green function.

$\epsilon$  Used in this chapter for various energies.

$G, G_0, \Sigma$  Fixing a Hamiltonian,  $G$  is the (interacting) Green function,  $G_0$  the non-interacting Green function,  $\Sigma$  the self-energy.

DMFT Dynamical mean-field theory. See Section 5.1.

DOS Density of state.

Dyson equation  $\Sigma = G_0^{-1} - G^{-1}$ .

SIAM Single impurity Anderson model. See Section 5.1.

Subscript imp. The impurity site.

Subscript  $l$ . For summation over bath sites.

Subscripts  $\{h, 1\}$  ( $\{h, 2\}$ ). Introduced by the analysis (Section 5.2.3, Appendix E.1).

Subscripts  $n_p$  or  $n_h$ . For summation over particle  $n_p$  or hole  $n_h$  states.

# Bibliography

- [1] Michael A. Nielsen and Isaac L. Chuang. *Quantum Computation and Quantum Information: 10th Anniversary Edition*. Cambridge University Press, 2010.
- [2] Richard P. Feynman. Simulating physics with computers. *International Journal of Theoretical Physics*, 21(6–7):467–488, jun 1982. ISSN 1572-9575. URL <http://dx.doi.org/10.1007/BF02650179>.
- [3] P.W. Shor. Algorithms for quantum computation: discrete logarithms and factoring. In *Proceedings 35th Annual Symposium on Foundations of Computer Science*, pages 124–134, 1994.
- [4] Peter W. Shor. Polynomial-time algorithms for prime factorization and discrete logarithms on a quantum computer. *SIAM Journal on Computing*, 26(5):1484–1509, oct 1997. URL <https://doi.org/10.1137/s0097539795293172>.
- [5] Seth Lloyd. Universal quantum simulators. *Science*, 273(5278):1073–1078, aug 1996. ISSN 1095-9203. URL <http://dx.doi.org/10.1126/science.273.5278.1073>.
- [6] Aram W. Harrow, Avinatan Hassidim, and Seth Lloyd. Quantum algorithm for linear systems of equations. *Phys. Rev. Lett.*, 103:150502, Oct 2009. URL <https://link.aps.org/doi/10.1103/PhysRevLett.103.150502>.
- [7] IBM. IBM quantum roadmap, 2024. URL <https://www.ibm.com/roadmaps/quantum/>. [Accessed 08 March 2024].
- [8] Google. Google quantum computing milestones, 2024. URL <https://quantumai.google/qecmilestone>. [Accessed 08 March 2024].

- [9] Amazon Web Services (AWS). Amazon braket, 2024. URL <https://aws.amazon.com/braket/>. [Accessed 08 March 2024].
- [10] Frank Arute, Kunal Arya, Ryan Babbush, Dave Bacon, Joseph C. Bardin, Rami Barends, Rupak Biswas, Sergio Boixo, Fernando G. S. L. Brandao, David A. Buell, Brian Burkett, Yu Chen, Zijun Chen, Ben Chiaro, Roberto Collins, William Courtney, Andrew Dunsworth, Edward Farhi, Brooks Foxen, Austin Fowler, Craig Gidney, Marissa Giustina, Rob Graff, Keith Guerin, Steve Habegger, Matthew P. Harrigan, Michael J. Hartmann, Alan Ho, Markus Hoffmann, Trent Huang, Travis S. Humble, Sergei V. Isakov, Evan Jeffrey, Zhang Jiang, Dvir Kafri, Kostyantyn Kechedzhi, Julian Kelly, Paul V. Klimov, Sergey Knysh, Alexander Korotkov, Fedor Kostritsa, David Landhuis, Mike Lindmark, Erik Lucero, Dmitry Lyakh, Salvatore Mandrà, Jarrod R. McClean, Matthew McEwen, Anthony Megrant, Xiao Mi, Kristel Michielsen, Masoud Mohseni, Josh Mutus, Ofer Naaman, Matthew Neeley, Charles Neill, Murphy Yuezhen Niu, Eric Ostby, Andre Petukhov, John C. Platt, Chris Quintana, Eleanor G. Rieffel, Pedram Roushan, Nicholas C. Rubin, Daniel Sank, Kevin J. Satzinger, Vadim Smelyanskiy, Kevin J. Sung, Matthew D. Trevithick, Amit Vainsencher, Benjamin Villalonga, Theodore White, Z. Jamie Yao, Ping Yeh, Adam Zalcman, Hartmut Neven, and John M. Martinis. Quantum supremacy using a programmable superconducting processor. *Nature*, 574(7779):505–510, oct 2019. ISSN 1476-4687. URL <http://dx.doi.org/10.1038/s41586-019-1666-5>.
- [11] Han-Sen Zhong, Hui Wang, Yu-Hao Deng, Ming-Cheng Chen, Li-Chao Peng, Yi-Han Luo, Jian Qin, Dian Wu, Xing Ding, Yi Hu, Peng Hu, Xiao-Yan Yang, Wei-Jun Zhang, Hao Li, Yuxuan Li, Xiao Jiang, Lin Gan, Guangwen Yang, Lixing You, Zhen Wang, Li Li, Nai-Le Liu, Chao-Yang Lu, and Jian-Wei Pan. Quantum computational advantage using photons. *Science*, 370(6523):1460–1463, dec 2020. ISSN 1095-9203. URL <http://dx.doi.org/10.1126/science.abe8770>.
- [12] Feng Pan and Pan Zhang. Simulating the sycamore quantum supremacy circuits, 2021, arXiv:2103.03074.
- [13] Edwin Pednault, John A. Gunnels, Giacomo Nannicini, Lior Horesh,

- and Robert Wisnieff. Leveraging secondary storage to simulate deep 54-qubit sycamore circuits, 2019, arXiv:1910.09534.
- [14] John Preskill. Quantum Computing in the NISQ era and beyond. *Quantum*, 2:79, aug 2018. ISSN 2521-327X. URL <https://doi.org/10.22331/q-2018-08-06-79>.
- [15] Alberto Peruzzo, Jarrod McClean, Peter Shadbolt, Man-Hong Yung, Xiao-Qi Zhou, Peter J. Love, Alán Aspuru-Guzik, and Jeremy L. O’Brien. A variational eigenvalue solver on a photonic quantum processor. *Nature Communications*, 5(1):4213, 2014. ISSN 2041-1723.
- [16] M. Cerezo, Andrew Arrasmith, Ryan Babbush, Simon C. Benjamin, Suguru Endo, Keisuke Fujii, Jarrod R. McClean, Kosuke Mitarai, Xiao Yuan, Lukasz Cincio, and Patrick J. Coles. Variational quantum algorithms. *Nature Reviews Physics*, 3(9):625–644, aug 2021. URL <https://doi.org/10.1038/s42254-021-00348-9>.
- [17] Yudong Cao, Jonathan Romero, Jonathan P. Olson, Matthias Degroote, Peter D. Johnson, Mária Kieferová, Ian D. Kivlichan, Tim Menke, Borja Peropadre, Nicolas P. D. Sawaya, Sukin Sim, Libor Veis, and Alán Aspuru-Guzik. Quantum chemistry in the age of quantum computing. *Chemical Reviews*, 119(19):10856–10915, 2019. URL <https://doi.org/10.1021/acs.chemrev.8b00803>. PMID: 31469277.
- [18] Jarrod R McClean, Jonathan Romero, Ryan Babbush, and Alán Aspuru-Guzik. The theory of variational hybrid quantum-classical algorithms. *New Journal of Physics*, 18(2):023023, feb 2016. URL <https://doi.org/10.1088/1367-2630/18/2/023023>.
- [19] P. J. J. O’Malley, R. Babbush, I. D. Kivlichan, J. Romero, J. R. McClean, R. Barends, J. Kelly, P. Roushan, A. Tranter, N. Ding, B. Campbell, Y. Chen, Z. Chen, B. Chiaro, A. Dunsworth, A. G. Fowler, E. Jeffrey, E. Lucero, A. Megrant, J. Y. Mutus, M. Neeley, C. Neill, C. Quintana, D. Sank, A. Vainsencher, J. Wenner, T. C. White, P. V. Coveney, P. J. Love, H. Neven, A. Aspuru-Guzik, and J. M. Martinis. Scalable quantum simulation of molecular energies. *Physical Review X*, 6(3), jul 2016. ISSN 2160-3308. URL <http://dx.doi.org/10.1103/PhysRevX.6.031007>.

- [20] Abhinav Kandala, Antonio Mezzacapo, Kristan Temme, Maika Takita, Markus Brink, Jerry M. Chow, and Jay M. Gambetta. Hardware-efficient variational quantum eigensolver for small molecules and quantum magnets. *Nature*, 549(7671):242–246, 2017. ISSN 1476-4687.
- [21] Ying Li and Simon C. Benjamin. Efficient variational quantum simulator incorporating active error minimization. *Phys. Rev. X*, 7:021050, Jun 2017.
- [22] Abhinav Kandala, Kristan Temme, Antonio D. Córcoles, Antonio Mezzacapo, Jerry M. Chow, and Jay M. Gambetta. Error mitigation extends the computational reach of a noisy quantum processor. *Nature*, 567(7749):491–495, mar 2019. ISSN 1476-4687. URL <http://dx.doi.org/10.1038/s41586-019-1040-7>.
- [23] Youngseok Kim, Christopher J. Wood, Theodore J. Yoder, Seth T. Merkel, Jay M. Gambetta, Kristan Temme, and Abhinav Kandala. Scalable error mitigation for noisy quantum circuits produces competitive expectation values. *Nature Physics*, 19(5):752–759, feb 2023. ISSN 1745-2481. URL <http://dx.doi.org/10.1038/s41567-022-01914-3>.
- [24] T. E. O’Brien, G. Anselmetti, F. Gkritis, V. E. Elfving, S. Polla, W. J. Huggins, O. Oumarou, K. Kechedzhi, D. Abanin, R. Acharya, I. Aleiner, R. Allen, T. I. Andersen, K. Anderson, M. Ansmann, F. Arute, K. Arya, A. Asfaw, J. Atalaya, J. C. Bardin, A. Bengtsson, G. Bortoli, A. Bourassa, J. Bovaird, L. Brill, M. Broughton, B. Buckley, D. A. Buell, T. Burger, B. Burkett, N. Bushnell, J. Campero, Z. Chen, B. Chiaro, D. Chik, J. Cogan, R. Collins, P. Conner, W. Courtney, A. L. Crook, B. Curtin, D. M. Debroy, S. Demura, I. Drozdov, A. Dunsworth, C. Erickson, L. Faoro, E. Farhi, R. Fatemi, V. S. Ferreira, L. Flores Burgos, E. Forati, A. G. Fowler, B. Foxen, W. Giang, C. Gidney, D. Gilboa, M. Giustina, R. Gosula, A. Grajales Dau, J. A. Gross, S. Habegger, M. C. Hamilton, M. Hansen, M. P. Harrigan, S. D. Harrington, P. Heu, M. R. Hoffmann, S. Hong, T. Huang, A. Huff, L. B. Ioffe, S. V. Isakov, J. Iveland, E. Jeffrey, Z. Jiang, C. Jones, P. Juhas, D. Kafri, T. Khattar, M. Khezri, M. Kieferová, S. Kim, P. V. Klimov, A. R. Klotz, A. N. Korotkov, F. Kostritsa, J. M. Kreikebaum, D. Landhuis, P. Laptev, K.-M.

- Lau, L. Laws, J. Lee, K. Lee, B. J. Lester, A. T. Lill, W. Liu, W. P. Livingston, A. Locharla, F. D. Malone, S. Mandrà, O. Martin, S. Martin, J. R. McClean, T. McCourt, M. McEwen, X. Mi, A. Mieszala, K. C. Miao, M. Mohseni, S. Montazeri, A. Morvan, R. Movassagh, W. Mruczkiewicz, O. Naaman, M. Neeley, C. Neill, A. Nersisyan, M. Newman, J. H. Ng, A. Nguyen, M. Nguyen, M. Y. Niu, S. Omonije, A. Opremcak, A. Petukhov, R. Potter, L. P. Pryadko, C. Quintana, C. Rocque, P. Roushan, N. Saei, D. Sank, K. Sankaragomathi, K. J. Satzinger, H. F. Schurkus, C. Schuster, M. J. Shearn, A. Shorter, N. Shutty, V. Shvarts, J. Skruzny, W. C. Smith, R. D. Somma, G. Sterling, D. Strain, M. Szalay, D. Thor, A. Torres, G. Vidal, B. Villalonga, C. Vollgraff Heidweiller, T. White, B. W. K. Woo, C. Xing, Z. J. Yao, P. Yeh, J. Yoo, G. Young, A. Zalcman, Y. Zhang, N. Zhu, N. Zobrist, D. Bacon, S. Boixo, Y. Chen, J. Hilton, J. Kelly, E. Lucero, A. Megrant, H. Neven, V. Smelyanskiy, C. Gogolin, R. Babbush, and N. C. Rubin. Purification-based quantum error mitigation of pair-correlated electron simulations. *Nature Physics*, 19(12):1787–1792, oct 2023. ISSN 1745-2481. URL <http://dx.doi.org/10.1038/s41567-023-02240-y>.
- [25] Abhinav Anand, Philipp Schleich, Sumner Alperin-Lea, Phillip W. K. Jensen, Sukin Sim, Manuel Díaz-Tinoco, Jakob S. Kottmann, Matthias Degroote, Artur F. Izmaylov, and Alán Aspuru-Guzik. A quantum computing view on unitary coupled cluster theory. *Chemical Society Reviews*, 51(5):1659–1684, 2022. ISSN 1460-4744. URL <http://dx.doi.org/10.1039/D1CS00932J>.
- [26] Edward Grant, Marcello Benedetti, Shuxiang Cao, Andrew Hallam, Joshua Lockhart, Vid Stojevic, Andrew G. Green, and Simone Severini. Hierarchical quantum classifiers. *npj Quantum Information*, 4(1), dec 2018. ISSN 2056-6387. URL <http://dx.doi.org/10.1038/s41534-018-0116-9>.
- [27] Jules Tilly, Hongxiang Chen, Shuxiang Cao, Dario Picozzi, Kanav Setia, Ying Li, Edward Grant, Leonard Wossnig, Ivan Rungger, George H. Booth, and Jonathan Tennyson. The variational quantum eigensolver: A review of methods and best practices. *Physics Reports*, 986:1–128, 2022. ISSN 0370-1573. URL <https://www.sciencedirect.com/>

- science/article/pii/S0370157322003118. The Variational Quantum Eigensolver: a review of methods and best practices.
- [28] Jarrod R. McClean, Sergio Boixo, Vadim N. Smelyanskiy, Ryan Babush, and Hartmut Neven. Barren plateaus in quantum neural network training landscapes. *Nature Communications*, 9(1), nov 2018. URL <https://doi.org/10.1038/s41467-018-07090-4>.
- [29] Edward Grant, Leonard Wossnig, Mateusz Ostaszewski, and Marcello Benedetti. An initialization strategy for addressing barren plateaus in parametrized quantum circuits. *Quantum*, 3:214, dec 2019. URL <https://doi.org/10.22331/q-2019-12-09-214>.
- [30] I. Rungger, N. Fitzpatrick, H. Chen, C. H. Alderete, H. Apel, A. Cowntan, A. Patterson, D. Munoz Ramo, Y. Zhu, N. H. Nguyen, E. Grant, S. Chretien, L. Wossnig, N. M. Linke, and R. Duncan. Dynamical mean field theory algorithm and experiment on quantum computers, 2019, arXiv:1910.04735.
- [31] Hongxiang Chen, Max Nusspickel, Jules Tilly, and George H. Booth. Variational quantum eigensolver for dynamic correlation functions. *Phys. Rev. A*, 104:032405, Sep 2021. URL <https://link.aps.org/doi/10.1103/PhysRevA.104.032405>.
- [32] Hongxiang Chen, Michael Vasmer, Nikolas P. Breuckmann, and Edward Grant. Automated discovery of logical gates for quantum error correction (with supplementary (153 pages)). *Quantum Information and Computation*, 22(11&12):947–964, aug 2022. URL <https://doi.org/10.26421/qic22.11-12-3>.
- [33] Michael Broughton, Guillaume Verdon, Trevor McCourt, Antonio J. Martinez, Jae Hyeon Yoo, Sergei V. Isakov, Philip Massey, Ramin Halavati, Murphy Yuezhen Niu, Alexander Zlokapa, Evan Peters, Owen Lockwood, Andrea Skolik, Sofiene Jerbi, Vedran Dunjko, Martin Leib, Michael Streif, David Von Dollen, Hongxiang Chen, Shuxiang Cao, Roeland Wiersema, Hsin-Yuan Huang, Jarrod R. McClean, Ryan Babush, Sergio Boixo, Dave Bacon, Alan K. Ho, Hartmut Neven, and Masoud Mohseni. Tensorflow quantum: A software framework for quantum machine learning, 2021, 2003.02989. URL <https://arxiv.org/abs/2003.02989>.

- [34] Andrew Patterson, Hongxiang Chen, Leonard Wossnig, Simone Severini, Dan Browne, and Ivan Rungger. Quantum state discrimination using noisy quantum neural networks. *Phys. Rev. Res.*, 3:013063, Jan 2021. URL <https://link.aps.org/doi/10.1103/PhysRevResearch.3.013063>.
- [35] Jules Tilly, Glenn Jones, Hongxiang Chen, Leonard Wossnig, and Edward Grant. Computation of molecular excited states on ibm quantum computers using a discriminative variational quantum eigensolver. *Phys. Rev. A*, 102:062425, Dec 2020. URL <https://link.aps.org/doi/10.1103/PhysRevA.102.062425>.
- [36] Frank Jensen. *Introduction to computational chemistry*. John Wiley & Sons, Nashville, TN, 3 edition, nov 2016. ISBN 978-1-118-82598-3.
- [37] J. W. Rayleigh. In finding the correction for the open end of an organ-pipe. *Phil. Trans.*, 161(77), 1870.
- [38] W. Ritz. Über eine neue methode zur lösung gewisser variationsprobleme der mathematischen physik. *J. reine angew. Math*, 135:1–61, 1908.
- [39] David Jeffery Griffiths. *Introduction to Quantum Mechanics*. Pearson international edition. Pearson Prentice Hall, 2005. ISBN 9780131118928. URL <https://books.google.co.uk/books?id=-BsvAQAIAAJ>.
- [40] P. Jordan and E. Wigner. Über das paulische äquivalenzverbot. *Zeitschrift für Physik*, 47(9):631–651, Sep 1928. ISSN 0044-3328. URL <https://doi.org/10.1007/BF01331938>.
- [41] Bryan T. Gard, Linghua Zhu, George S. Barron, Nicholas J. Mayhall, Sophia E. Economou, and Edwin Barnes. Efficient symmetry-preserving state preparation circuits for the variational quantum eigensolver algorithm. *npj Quantum Information*, 6(1), jan 2020. ISSN 2056-6387. URL <http://dx.doi.org/10.1038/s41534-019-0240-1>.
- [42] Andrew Arrasmith, Zoë Holmes, M Cerezo, and Patrick J Coles. Equivalence of quantum barren plateaus to cost concentration and narrow gorges. *Quantum Science and Technology*, 7(4):045015, aug 2022.

- ISSN 2058-9565. URL <http://dx.doi.org/10.1088/2058-9565/ac7d06>.
- [43] M. Cerezo, Akira Sone, Tyler Volkoff, Lukasz Cincio, and Patrick J. Coles. Cost function dependent barren plateaus in shallow parametrized quantum circuits. *Nature Communications*, 12(1), mar 2021. ISSN 2041-1723. URL <http://dx.doi.org/10.1038/s41467-021-21728-w>.
- [44] Samson Wang, Enrico Fontana, M. Cerezo, Kunal Sharma, Akira Sone, Lukasz Cincio, and Patrick J. Coles. Noise-induced barren plateaus in variational quantum algorithms. *Nature Communications*, 12(1), nov 2021. ISSN 2041-1723. URL <http://dx.doi.org/10.1038/s41467-021-27045-6>.
- [45] Michel Talagrand. A new look at independence. *The Annals of Probability*, 24(1), jan 1996. ISSN 0091-1798. URL <http://dx.doi.org/10.1214/aop/1042644705>.
- [46] Michel Ledoux. *The concentration of measure phenomenon*. Mathematical Surveys and Monographs. American Mathematical Society, Providence, RI, feb 2005. ISBN 0821837923.
- [47] Guillaume Aubrun and Stanislaw J Szarek. *Alice and bob meet Banach*. Mathematical Surveys and Monographs. American Mathematical Society, Providence, RI, oct 2017. ISBN 978-1-4704-3468-7.
- [48] Antonio Anna Mele. Introduction to haar measure tools in quantum information: A beginner’s tutorial. *Quantum*, 8:1340, may 2024. ISSN 2521-327X. URL <http://dx.doi.org/10.22331/q-2024-05-08-1340>.
- [49] Zoë Holmes, Kunal Sharma, M. Cerezo, and Patrick J. Coles. Connecting ansatz expressibility to gradient magnitudes and barren plateaus. *PRX Quantum*, 3(1), jan 2022. ISSN 2691-3399. URL <http://dx.doi.org/10.1103/prxquantum.3.010313>.
- [50] Martin Larocca, Supanut Thanasilp, Samson Wang, Kunal Sharma, Jacob Biamonte, Patrick J. Coles, Lukasz Cincio, Jarrod R. McClean, Zoë Holmes, and M. Cerezo. A review of barren plateaus in variational quantum computing, 2024, arXiv:2405.00781.

- [51] Chris Cade, Lana Mineh, Ashley Montanaro, and Stasja Stanisic. Strategies for solving the fermi-hubbard model on near-term quantum computers. *Physical Review B*, 102(23), dec 2020. ISSN 2469-9969. URL <http://dx.doi.org/10.1103/PhysRevB.102.235122>.
- [52] Dave Wecker, Matthew B. Hastings, and Matthias Troyer. Progress towards practical quantum variational algorithms. *Physical Review A*, 92(4), oct 2015. ISSN 1094-1622. URL <http://dx.doi.org/10.1103/PhysRevA.92.042303>.
- [53] Roeland Wiersema, Cunlu Zhou, Yvette de Sereville, Juan Felipe Carrasquilla, Yong Baek Kim, and Henry Yuen. Exploring entanglement and optimization within the hamiltonian variational ansatz. *PRX Quantum*, 1(2), dec 2020. ISSN 2691-3399. URL <http://dx.doi.org/10.1103/PRXQuantum.1.020319>.
- [54] Tyler Volkoff and Patrick J Coles. Large gradients via correlation in random parameterized quantum circuits. *Quantum Science and Technology*, 6(2):025008, jan 2021. ISSN 2058-9565. URL <http://dx.doi.org/10.1088/2058-9565/abd891>.
- [55] "Wikipedia". "Rectifier (neural networks) — Wikipedia, the free encyclopedia", "2024". URL "[http://en.wikipedia.org/w/index.php?title=Rectifier%20\(neural%20networks\)&oldid=1218083418](http://en.wikipedia.org/w/index.php?title=Rectifier%20(neural%20networks)&oldid=1218083418)". "[Online; accessed 18-April-2024]".
- [56] "Wikipedia". "Residual neural network — Wikipedia, the free encyclopedia", "2024". URL "<http://en.wikipedia.org/w/index.php?title=Residual%20neural%20network&oldid=1219158991>". "[Online; accessed 18-April-2024]".
- [57] Kristan Temme, Sergey Bravyi, and Jay M. Gambetta. Error mitigation for short-depth quantum circuits. *Physical Review Letters*, 119(18), nov 2017. URL <https://doi.org/10.1103/physrevlett.119.180509>.
- [58] E. F. Dumitrescu, A. J. McCaskey, G. Hagen, G. R. Jansen, T. D. Morris, T. Papenbrock, R. C. Pooser, D. J. Dean, and P. Lougovski. Cloud quantum computing of an atomic nucleus. *Physical Review Letters*, 120(21), may 2018. URL <https://doi.org/10.1103/physrevlett.120.210501>.

- [59] Andre He, Benjamin Nachman, Wibe A. de Jong, and Christian W. Bauer. Zero-noise extrapolation for quantum-gate error mitigation with identity insertions. *Physical Review A*, 102(1), jul 2020. URL <https://doi.org/10.1103/physreva.102.012426>.
- [60] Suguru Endo, Simon C. Benjamin, and Ying Li. Practical quantum error mitigation for near-future applications. *Physical Review X*, 8(3), jul 2018. URL <https://doi.org/10.1103/physrevx.8.031027>.
- [61] Ritajit Majumdar, Pedro Rivero, Friederike Metz, Areeq Hasan, and Derek S Wang. Best practices for quantum error mitigation with digital zero-noise extrapolation, 2023, arXiv:2307.05203. URL <https://arxiv.org/abs/2307.05203>.
- [62] Zhenyu Cai, Ryan Babbush, Simon C. Benjamin, Suguru Endo, William J. Huggins, Ying Li, Jarrod R. McClean, and Thomas E. O’Brien. Quantum error mitigation. *Rev. Mod. Phys.*, 95:045005, Dec 2023. URL <https://link.aps.org/doi/10.1103/RevModPhys.95.045005>.
- [63] Michael R Geller. Rigorous measurement error correction. *Quantum Science and Technology*, 5(3):03LT01, jun 2020. URL <https://doi.org/10.1088/2058-9565/ab9591>.
- [64] Jay Gambetta, Matthew Treinish, Paul Nation, Paul Kassebaum, Diego M. Rodríguez, Qiskit-Bot, Salvador De La Puente González, Shaohan Hu, Kevin Krsulich, Laura Zdanski, Jessie Yu, David McKay, Juan Gomez, Lauren Capelluto, Travis-S-IBM, Julien Gacon, Lerongil, Rafey Iqbal Rahman, Steve Wood, , Drew, Joachim Schwarm, Luciano Bello, MELVIN GEORGE, Manoel Marques, Omar Costa Hamido, RohitMidha23, Sean Dague, Shelly Garion, Tigerjack, and Yuri Kobayashi. Qiskit/qiskit: Qiskit 0.25.0, April 2021.
- [65] Pauli Virtanen, Ralf Gommers, Travis E. Oliphant, Matt Haberland, Tyler Reddy, David Cournapeau, Evgeni Burovski, Pearu Peterson, Warren Weckesser, Jonathan Bright, Stéfan J. van der Walt, Matthew Brett, Joshua Wilson, K. Jarrod Millman, Nikolay Mayorov, Andrew R. J. Nelson, Eric Jones, Robert Kern, Eric Larson, C J Carey, İlhan Polat, Yu Feng, Eric W. Moore, Jake VanderPlas, Denis Laxalde, Josef

- Perktold, Robert Cimrman, Ian Henriksen, E. A. Quintero, Charles R. Harris, Anne M. Archibald, Antônio H. Ribeiro, Fabian Pedregosa, Paul van Mulbregt, and SciPy 1.0 Contributors. SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python. *Nature Methods*, 17: 261–272, 2020.
- [66] J. A. Nelder and R. Mead. A simplex method for function minimization. *The Computer Journal*, 7(4):308–313, jan 1965. URL <https://doi.org/10.1093/comjnl/7.4.308>.
- [67] Aston Zhang, Zachary C. Lipton, Mu Li, and Alexander J. Smola. Dive into deep learning, 2023, 2106.11342.
- [68] Ken M. Nakanishi, Keisuke Fujii, and Synge Todo. Sequential minimal optimization for quantum-classical hybrid algorithms. *Physical Review Research*, 2(4), oct 2020. URL <https://doi.org/10.1103/physrevresearch.2.043158>.
- [69] Mateusz Ostaszewski, Edward Grant, and Marcello Benedetti. Structure optimization for parameterized quantum circuits. *Quantum*, 5: 391, January 2021. ISSN 2521-327X.
- [70] David Wierichs, Josh Izaac, Cody Wang, and Cedric Yen-Yu Lin. General parameter-shift rules for quantum gradients. *Quantum*, 6:677, mar 2022. URL <https://doi.org/10.22331/q-2022-03-30-677>.
- [71] Wikipedia contributors. Atan2 — Wikipedia, the free encyclopedia, 2022. URL <https://en.wikipedia.org/w/index.php?title=Atan2&oldid=1117723779>. [Online; accessed 28-November-2022].
- [72] Numpy contributors. arctan2 manual, numpy, 2022. URL <https://numpy.org/doc/stable/reference/generated/numpy.arctan2.html>. [Online; accessed 28-November-2022].
- [73] Oscar Higgott, Daochen Wang, and Stephen Brierley. Variational quantum computation of excited states. *Quantum*, 3:156, jul 2019. URL <https://doi.org/10.22331/q-2019-07-01-156>.
- [74] Ken M. Nakanishi, Kosuke Mitarai, and Keisuke Fujii. Subspace-search variational quantum eigensolver for excited states. *Physical*

- Review Research*, 1(3), oct 2019. URL <https://doi.org/10.1103/physrevresearch.1.033062>.
- [75] Francesco Benfenati, Chiara Capecci, YoditG, Gino Bishop, and Leonardo Guidoni. Modifying hamiltonian and wave function for vqe in quantum chemistry · issue 37. qiskit camp europe 2019. github, 2019. URL <https://github.com/qiskit-community/qiskit-camp-europe-19/issues/37>.
- [76] Ilya G. Ryabinkin, Scott N. Genin, and Artur F. Izmaylov. Constrained variational quantum eigensolver: Quantum computer search engine in the fock space. *Journal of Chemical Theory and Computation*, 15(1):249–255, dec 2018. URL <https://doi.org/10.1021/acs.jctc.8b00943>.
- [77] Xiaosi Xu, Jinzhao Sun, Suguru Endo, Ying Li, Simon C. Benjamin, and Xiao Yuan. Variational algorithms for linear algebra. *Science Bulletin*, 66(21):2181–2188, nov 2021, arXiv:1909.03898. URL <https://doi.org/10.1016/j.scib.2021.06.023>.
- [78] Carlos Bravo-Prieto, Ryan LaRose, M. Cerezo, Yigit Subasi, Lukasz Cincio, and Patrick J. Coles. Variational quantum linear solver, 2019, arXiv:1909.05820.
- [79] Carlos Bravo-Prieto, Ryan LaRose, M. Cerezo, Yigit Subasi, Lukasz Cincio, and Patrick J. Coles. Variational Quantum Linear Solver. *Quantum*, 7:1188, nov 2023. ISSN 2521-327X. URL <https://doi.org/10.22331/q-2023-11-22-1188>.
- [80] Daniel Gottesman. Stabilizer codes and quantum error correction, 1997, quant-ph/9705052.
- [81] Joel J. Wallman and Joseph Emerson. Noise tailoring for scalable quantum computation via randomized compiling. *Physical Review A*, 94(5), nov 2016. ISSN 2469-9934. URL <http://dx.doi.org/10.1103/PhysRevA.94.052325>.
- [82] Daniel Gottesman. Stabilizer codes and quantum error correction, 1997, arXiv:quant-ph/9705052. URL <https://arxiv.org/abs/quant-ph/9705052>.

- 
- [83] Raban Iten, Oliver Reardon-Smith, Luca Mondada, Ethan Redmond, Ravjot Singh Kohli, and Roger Colbeck. Introduction to UniversalQ-Compiler, 2019, 1904.01072.
- [84] A. R. Calderbank and Peter W. Shor. Good quantum error-correcting codes exist. *Phys. Rev. A*, 54:1098–1105, Aug 1996.
- [85] Andrew Steane. Multiple-particle interference and quantum error correction. *Proc. R. Soc. Lond. A.*, 452(1954):2551–2577, 1996.
- [86] Bryan Eastin and Emanuel Knill. Restrictions on transversal encoded quantum gate sets. *Physical Review Letters*, 102(11), mar 2009. ISSN 1079-7114. URL <http://dx.doi.org/10.1103/PhysRevLett.102.110502>.
- [87] Sumeet Khatri, Ryan LaRose, Alexander Poremba, Lukasz Cincio, Andrew T. Sornborger, and Patrick J. Coles. Quantum-assisted quantum compiling. *Quantum*, 3:140, may 2019. ISSN 2521-327X. URL <http://dx.doi.org/10.22331/q-2019-05-13-140>.
- [88] Michael A Nielsen. A simple formula for the average gate fidelity of a quantum dynamical operation. *Physics Letters A*, 303(4):249–252, oct 2002. ISSN 0375-9601. URL [http://dx.doi.org/10.1016/S0375-9601\(02\)01272-0](http://dx.doi.org/10.1016/S0375-9601(02)01272-0).
- [89] Tyson Jones and Simon C. Benjamin. Robust quantum compilation and circuit optimisation via energy minimisation. *Quantum*, 6:628, jan 2022. ISSN 2521-327X. URL <http://dx.doi.org/10.22331/q-2022-01-24-628>.
- [90] Xiaosi Xu, Simon C. Benjamin, and Xiao Yuan. Variational circuit compiler for quantum error correction. *Physical Review Applied*, 15(3), mar 2021. ISSN 2331-7019. URL <http://dx.doi.org/10.1103/PhysRevApplied.15.034068>.
- [91] Lukasz Cincio, Yiğit Subaşı, Andrew T Sornborger, and Patrick J Coles. Learning the quantum algorithm for state overlap. *New Journal of Physics*, 20(11):113022, nov 2018. ISSN 1367-2630. URL <http://dx.doi.org/10.1088/1367-2630/aae94a>.

- 
- [92] Kunal Sharma, Sumeet Khatri, M Cerezo, and Patrick J Coles. Noise resilience of variational quantum compiling. *New Journal of Physics*, 22(4):043006, apr 2020. ISSN 1367-2630. URL <http://dx.doi.org/10.1088/1367-2630/ab784c>.
- [93] Robert L. Kosut, Alireza Shabani, and Daniel A. Lidar. Robust quantum error correction via convex optimization. *Phys. Rev. Lett.*, 100:020502, Jan 2008.
- [94] S. Taghavi, R. L. Kosut, and D. A. Lidar. Channel-optimized quantum error correction. *IEEE Transactions on Information Theory*, 56(3):1461–1473, March 2010.
- [95] Robert L. Kosut and Daniel A. Lidar. Quantum error correction via convex optimization. *Quantum Information Processing*, 8(5):443–459, Oct 2009. ISSN 1573-1332.
- [96] Peter D. Johnson, Jonathan Romero, Jonathan Olson, Yudong Cao, and Alan Aspuru-Guzik. Qvector: an algorithm for device-tailored quantum error correction, 2017, arXiv:1711.02249.
- [97] Thomas Fösel, Petru Tighineanu, Talitha Weiss, and Florian Marquardt. Reinforcement learning with neural networks for quantum feedback. *Phys. Rev. X*, 8:031084, Sep 2018.
- [98] Ken M. Nakanishi, Keisuke Fujii, and Synge Todo. Sequential minimal optimization for quantum-classical hybrid algorithms. *Phys. Rev. Research*, 2:043158, Oct 2020.
- [99] Michael A. Nielsen and Isaac L. Chuang. *Quantum Computation and Quantum Information: 10th Anniversary Edition*. Cambridge University Press, New York, NY, USA, 10th edition, 2011. ISBN 1107002176, 9781107002173.
- [100] K. Vogel and H. Risken. Determination of quasiprobability distributions in terms of probability distributions for the rotated quadrature phase. *Phys. Rev. A*, 40:2847–2849, Sep 1989.
- [101] Harry Buhrman, Richard Cleve, John Watrous, and Ronald de Wolf. Quantum fingerprinting. *Phys. Rev. Lett.*, 87:167902, Sep 2001.

- 
- [102] Daniel Gottesman and Isaac Chuang. Quantum digital signatures, 2001, quant-ph/0105032.
- [103] Hongxiang Chen. Supplementary materials of learned logical gates, 2023. Availalbe inside the archive “mlec\_supplementary\_material.tar.gz” in Chen [255].
- [104] Markus S. Kesselring, Fernando Pastawski, Jens Eisert, and Benjamin J. Brown. The boundaries and twist defects of the color code and their applications to topological quantum computation. *Quantum*, 2:101, oct 2018.
- [105] Markus Grassl. Bounds on the minimum distance of linear codes and quantum codes, 2007. URL <http://www.codetables.de>. Accessed on 2019-12-09.
- [106] Daniel Gottesman. Class of quantum error-correcting codes saturating the quantum hamming bound. *Phys. Rev. A*, 54:1862–1868, Sep 1996.
- [107] A. M. Steane. Simple quantum error-correcting codes. *Phys. Rev. A*, 54:4741–4751, Dec 1996.
- [108] A. R. Calderbank, E. M. Rains, P. W. Shor, and N. J. A. Sloane. Quantum error correction and orthogonal geometry. *Phys. Rev. Lett.*, 78:405–408, Jan 1997.
- [109] M. Grassl, Th. Beth, and T. Pellizzari. Codes for the quantum erasure channel. *Phys. Rev. A*, 56:33–38, Jul 1997.
- [110] Lev Vaidman, Lior Goldenberg, and Stephen Wiesner. Error prevention scheme with four particles. *Phys. Rev. A*, 54:R1745–R1748, Sep 1996.
- [111] Michael Vasmer and Aleksander Kubica. Morphing quantum codes, 2021, 2112.01446.
- [112] Raymond Laflamme, Cesar Miquel, Juan Pablo Paz, and Wojciech Hubert Zurek. Perfect quantum error correcting code. *Phys. Rev. Lett.*, 77:198–201, Jul 1996.
- [113] Charles H. Bennett, David P. DiVincenzo, John A. Smolin, and William K. Wootters. Mixed-state entanglement and quantum error correction. *Phys. Rev. A*, 54:3824–3851, Nov 1996.

- 
- [114] Theodore J. Yoder and Isaac H. Kim. The surface code with a twist. *Quantum*, 1:2, April 2017. ISSN 2521-327X.
- [115] Christophe Vuillot and Nikolas P. Breuckmann. Quantum pin codes, 2019, 1906.11394.
- [116] Aleksander Kubica, Beni Yoshida, and Fernando Pastawski. Unfolding the color code. *New Journal of Physics*, 17(8):083026, aug 2015.
- [117] Earl Campbell. The smallest interesting colour code, 2016. URL <https://earlrcampbell.com/2016/09/26/the-smallest-interesting-colour-code/>. Accessed on 2019-12-09.
- [118] Andrew W. Cross, Lev S. Bishop, John A. Smolin, and Jay M. Gambetta. Open quantum assembly language, 2017, 1707.03429.
- [119] Neil J. Ross and Peter Selinger. Optimal ancilla-free clifford+t approximation of z-rotations. *Quantum Information and Computation*, 2014, arXiv:1403.2975. URL <https://arxiv.org/abs/1403.2975>.
- [120] Yaoyun Shi. Both toffoli and controlled-not need little help to do universal quantum computation, 2002, arXiv:quant-ph/0205115. URL <https://arxiv.org/abs/quant-ph/0205115>.
- [121] Simon Forest, David Gosset, Vadym Kliuchnikov, and David McKinnon. Exact synthesis of single-qubit unitaries over clifford-cyclotomic gate sets. *Journal of Mathematical Physics*, 2015, arXiv:1501.04944.
- [122] Vivek V. Shende, Igor L. Markov, and Stephen S. Bullock. Minimal universal two-qubit controlled-not-based circuits. *Phys. Rev. A*, 69: 062321, Jun 2004.
- [123] Hongxiang Chen. Visualization of the ansatz for three-qubit quantum gates, 2020. Available as the file whose filename contains the identifier “23A26.DBAD” in Chen [255].
- [124] Aleksander Kubica and Michael E. Beverland. Universal transversal gates with color codes: A simplified approach. *Phys. Rev. A*, 91:032330, Mar 2015.

- [125] Michael Vasmer and Dan E. Browne. Three-dimensional surface codes: Transversal gates and fault-tolerant architectures. *Phys. Rev. A*, 100: 012312, Jul 2019.
- [126] Narayanan Rengaswamy, Robert Calderbank, Michael Newman, and Henry D. Pfister. On optimality of CSS codes for transversal t. *IEEE Journal on Selected Areas in Information Theory*, 1(2):499–514, aug 2020.
- [127] Sergey Bravyi and Jeongwan Haah. Magic-state distillation with low overhead. *Phys. Rev. A*, 86:052329, Nov 2012.
- [128] Earl T. Campbell and Mark Howard. Unified framework for magic state distillation and multiqubit gate synthesis with reduced resource cost. *Phys. Rev. A*, 95:022316, Feb 2017.
- [129] Earl T. Campbell and Mark Howard. Unifying gate synthesis and magic state distillation. *Phys. Rev. Lett.*, 118:060501, Feb 2017.
- [130] Ben Criger and Barbara M. Terhal. Noise thresholds for the  $[[4, 2, 2]]$  concatenated toric code. *Quantum Information and Computation*, 16 (15-16):1261–1281, apr 2016, 1604.04062. ISSN 15337146.
- [131] P.W. Shor. Fault-tolerant quantum computation. In *Proceedings of 37th Conference on Foundations of Computer Science*, pages 56–65, 1996.
- [132] A. M. Steane. Active stabilization, quantum computation, and quantum state synthesis. *Phys. Rev. Lett.*, 78:2252–2255, Mar 1997.
- [133] E. Knill. Quantum computing with realistically noisy devices. *Nature*, 434(7029):39–44, mar 2005.
- [134] Rui Chao and Ben W. Reichardt. Quantum error correction with only two extra qubits. *Phys. Rev. Lett.*, 121:050502, Aug 2018.
- [135] Earl T. Campbell, Barbara M. Terhal, and Christophe Vuillot. Roads towards fault-tolerant universal quantum computation. *Nature*, 549 (7671):172–179, sep 2017.
- [136] Barbara M. Terhal. Quantum error correction for quantum memories. *Rev. Mod. Phys.*, 87:307–346, Apr 2015.

- 
- [137] Daniel Gottesman. Fault-tolerant quantum computation with higher-dimensional systems. In Colin P. Williams, editor, *Quantum Computing and Quantum Communications*, pages 302–313, Berlin, Heidelberg, 1999. Springer Berlin Heidelberg. ISBN 978-3-540-49208-5.
- [138] Andrew Cross, Graeme Smith, John A. Smolin, and Bei Zeng. Code-word stabilized quantum codes. *IEEE Transactions on Information Theory*, 55(1):433–438, 2009.
- [139] John David Jackson. *Classical Electrodynamics*. John Wiley & Sons, Nashville, TN, 3 edition, jul 1998. ISBN 0-471-30932-X.
- [140] Paul C. Martin and Julian Schwinger. Theory of many-particle systems. i. *Phys. Rev.*, 115:1342–1373, Sep 1959. URL <https://link.aps.org/doi/10.1103/PhysRev.115.1342>.
- [141] K. A. Milton. *A quantum legacy : seminal papers of Julian Schwinger / editor Kimball A. Milton*. World scientific series in 20th century physics vol. 26. World Scientific, Singapore ;, 2000. ISBN 9810240066.
- [142] Andrew T. Boothroyd. 31 diffraction in the static approximation. In *Principles of Neutron Scattering from Condensed Matter*, pages 31–72. Oxford University Press, 07 2020. ISBN 9780198862314. URL <https://doi.org/10.1093/oso/9780198862314.003.0002>.
- [143] G. L. Squires. *Introduction to the Theory of Thermal Neutron Scattering*. Cambridge University Press, mar 2012. ISBN 9781107644069. URL <http://dx.doi.org/10.1017/CB09781139107808>.
- [144] Anders W. Sandvik, Adolfo Avella, and Ferdinando Mancini. Computational studies of quantum spin systems, 2010, arXiv:1101.3281. ISSN 0094-243X. URL <http://dx.doi.org/10.1063/1.3518900>.
- [145] Miranda Holmes-Cerfon. Lecture 3: Markov chains (ii): Detailed balance, and markov chain monte carlo (mcmc), 2019. URL [https://cims.nyu.edu/~holmes/teaching/asa19/handout\\_Lecture3\\_2019.pdf](https://cims.nyu.edu/~holmes/teaching/asa19/handout_Lecture3_2019.pdf). Accessed: 2023-09-30.
- [146] Matthias Troyer and Uwe-Jens Wiese. Computational complexity and fundamental limitations to fermionic quantum monte carlo simulations. *Physical Review Letters*, 94(17), may 2005. ISSN 1079-7114. URL <http://dx.doi.org/10.1103/PhysRevLett.94.170201>.

- 
- [147] Román Orús. Tensor networks for complex quantum systems. *Nature Reviews Physics*, 1(9):538–550, aug 2019. ISSN 2522-5820. URL <http://dx.doi.org/10.1038/s42254-019-0086-7>.
- [148] Mari Carmen Bañuls. Tensor network algorithms: A route map. *Annual Review of Condensed Matter Physics*, 14(1):173–191, mar 2023. ISSN 1947-5462. URL <http://dx.doi.org/10.1146/annurev-conmatphys-040721-022705>.
- [149] Ulrich Schollwöck. The density-matrix renormalization group in the age of matrix product states. *Annals of Physics*, 326(1):96–192, 2011. ISSN 0003-4916. URL <https://www.sciencedirect.com/science/article/pii/S0003491610001752>. January 2011 Special Issue.
- [150] Roger G. Melko, Giuseppe Carleo, Juan Carrasquilla, and J. Ignacio Cirac. Restricted boltzmann machines in quantum physics. *Nature Physics*, 15(9):887–892, jun 2019. ISSN 1745-2481. URL <http://dx.doi.org/10.1038/s41567-019-0545-1>.
- [151] Filippo Vicentini, Damian Hofmann, Attila Szabó, Dian Wu, Christopher Roth, Clemens Giuliani, Gabriel Pescia, Jannes Nys, Vladimir Vargas-Calderón, Nikita Astrakhantsev, and Giuseppe Carleo. NetKet 3: Machine Learning Toolbox for Many-Body Quantum Systems. *SciPost Phys. Codebases*, page 7, 2022. URL <https://scipost.org/10.21468/SciPostPhysCodeb.7>.
- [152] Jing Chen, Song Cheng, Haidong Xie, Lei Wang, and Tao Xiang. Equivalence of restricted boltzmann machines and tensor network states. *Phys. Rev. B*, 97:085104, Feb 2018. URL <https://link.aps.org/doi/10.1103/PhysRevB.97.085104>.
- [153] Xun Gao and Lu-Ming Duan. Efficient representation of quantum many-body states with deep neural networks. *Nature Communications*, 8(1), sep 2017. ISSN 2041-1723. URL <http://dx.doi.org/10.1038/s41467-017-00705-2>.
- [154] Markus Schmitt and Markus Heyl. Quantum many-body dynamics in two dimensions with artificial neural networks. *Phys. Rev. Lett.*, 125:100503, Sep 2020. URL <https://link.aps.org/doi/10.1103/PhysRevLett.125.100503>.

- [155] Juha M Kreula, Laura García-Álvarez, Lucas Lamata, Stephen R Clark, Enrique Solano, and Dieter Jaksch. Few-qubit quantum-classical simulation of strongly correlated lattice fermions. *EPJ Quantum Technology*, 3(1), aug 2016. URL <https://doi.org/10.1140/epjqt/s40507-016-0049-1>.
- [156] J. M. Kreula, S. R. Clark, and D. Jaksch. Non-linear quantum-classical scheme to simulate non-equilibrium strongly correlated fermionic many-body dynamics. *Scientific Reports*, 6(1), sep 2016. URL <https://doi.org/10.1038/srep32940>.
- [157] Bela Bauer, Dave Wecker, Andrew J. Millis, Matthew B. Hastings, and Matthias Troyer. Hybrid quantum-classical approach to correlated materials. *Phys. Rev. X*, 6:031045, Sep 2016. URL <https://link.aps.org/doi/10.1103/PhysRevX.6.031045>.
- [158] Trevor Keen, Thomas Maier, Steven Johnston, and Pavel Lougovski. Quantum-classical simulation of two-site dynamical mean-field theory on noisy quantum hardware. *Quantum Science and Technology*, 5(3):035001, apr 2020. URL <https://doi.org/10.1088/2058-9565/ab7d4c>.
- [159] Suguru Endo, Iori Kurata, and Yuya O. Nakagawa. Calculation of the green’s function on near-term quantum computers. *Phys. Rev. Research*, 2:033281, Aug 2020. URL <https://link.aps.org/doi/10.1103/PhysRevResearch.2.033281>.
- [160] David Poulin, M. B. Hastings, D. Wecker, N. Wiebe, Andrew C. Doberty, and M. Troyer. The trotter step size required for accurate quantum simulation of quantum chemistry. *Quantum Information and Computation*, page 361–384, apr 2015, 1406.4920. ISSN 1533-7146. URL <http://dx.doi.org/10.26421/QIC15.5-6-1>.
- [161] Hongzheng Zhao, Marin Bukov, Markus Heyl, and Roderich Moessner. Adaptive trotterization for time-dependent hamiltonian quantum dynamics using piecewise conservation laws. *Physical Review Letters*, 133(1), jul 2024. ISSN 1079-7114. URL <http://dx.doi.org/10.1103/PhysRevLett.133.010603>.

- 
- [162] Xiao Yuan, Suguru Endo, Qi Zhao, Ying Li, and Simon C. Benjamin. Theory of variational quantum simulation. *Quantum*, 3:191, oct 2019. URL <https://doi.org/10.22331/q-2019-10-07-191>.
- [163] Dave Wecker, Matthew B. Hastings, Nathan Wiebe, Bryan K. Clark, Chetan Nayak, and Matthias Troyer. Solving strongly correlated electron models on a quantum computer. *Phys. Rev. A*, 92:062318, Dec 2015. URL <https://link.aps.org/doi/10.1103/PhysRevA.92.062318>.
- [164] Danial Dervovic, Mark Herbster, Peter Mountney, Simone Severini, Naïri Usher, and Leonard Wossnig. Quantum linear systems algorithms: a primer, 2018, arXiv:1802.08227.
- [165] Gene H Golub and Charles F Van Loan. *Matrix Computations*. Johns Hopkins University Press, Baltimore, MD, 4 edition, feb 2013. ISBN 978-1-4214-0794-4.
- [166] Karen A. Hallberg. Density-matrix algorithm for the calculation of dynamical properties of low-dimensional systems. *Physical Review B*, 52(14):R9827–R9830, oct 1995. ISSN 1095-3795. URL <http://dx.doi.org/10.1103/PhysRevB.52.R9827>.
- [167] Francois Jamet, Abhishek Agarwal, Carla Lupo, Dan E. Browne, Cedric Weber, and Ivan Rungger. Krylov variational quantum algorithm for first principles materials simulations. *arXiv:2105.13298 [quant-ph]*, aug 2021, 2105.13298. URL <http://arxiv.org/abs/2105.13298>.
- [168] Jarrod R. McClean, Mollie E. Kimchi-Schwartz, Jonathan Carter, and Wibe A. de Jong. Hybrid quantum-classical hierarchy for mitigation of decoherence and determination of excited states. *Physical Review A*, 95(4):042308, apr 2017. ISSN 2469-9926, 2469-9934. URL <http://link.aps.org/doi/10.1103/PhysRevA.95.042308>.
- [169] Francois Jamet, Abhishek Agarwal, and Ivan Rungger. Quantum subspace expansion algorithm for Green’s functions, dec 2022, 2205.00094. URL <http://arxiv.org/abs/2205.00094>.
- [170] Francois Jamet, Connor Lenihan, Lachlan P. Lindoy, Abhishek Agarwal, Enrico Fontana, Baptiste Anselme Martin, and Ivan Rungger. An-

- derson impurity solver integrating tensor network methods with quantum computing, sep 2023, 2304.06587. URL <http://arxiv.org/abs/2304.06587>.
- [171] Reinis Irmejs and Raul A. Santos. Approximating dynamical correlation functions with constant depth quantum circuits, 2024, arXiv:2406.03204.
- [172] Yu Tong, Dong An, Nathan Wiebe, and Lin Lin. Fast inversion, preconditioned quantum linear system solvers, fast green’s-function computation, and fast evaluation of matrix functions. *Phys. Rev. A*, 104:032422, Sep 2021. URL <https://link.aps.org/doi/10.1103/PhysRevA.104.032422>.
- [173] Xiaoxia Cai, Wei-Hai Fang, Heng Fan, and Zhendong Li. Quantum computation of molecular response properties. *Phys. Rev. Research*, 2:033324, Aug 2020. URL <https://link.aps.org/doi/10.1103/PhysRevResearch.2.033324>.
- [174] Till D. Kühner and Steven R. White. Dynamical correlation functions using the density matrix renormalization group. *Physical Review B*, 60(1):335–343, jul 1999. URL <https://doi.org/10.1103/physrevb.60.335>.
- [175] Eric Jeckelmann. Dynamical density-matrix renormalization-group method. *Physical Review B*, 66(4), jul 2002. URL <https://doi.org/10.1103/physrevb.66.045114>.
- [176] Douglas Hendry and Adrian E. Feiguin. Machine learning approach to dynamical properties of quantum many-body systems. *Physical Review B*, 100(24), dec 2019. URL <https://doi.org/10.1103/physrevb.100.245123>.
- [177] Dan Bohr. *Quantum Transport in Strongly Correlated Systems: A Density Matrix Renormalization Group Study*. PhD thesis, Technical University of Denmark, Denmark, oct 2007. URL <https://orbit.dtu.dk/en/publications/quantum-transport-in-strongly-correlated-systems-a-density-matrix>.
- [178] Hongxiang Chen, Max Nusspickel, Jules Tilly, and George H. Booth. Variational quantum eigensolver for dynamic correlation functions.

- Physical Review A*, 104(3), sep 2021. ISSN 2469-9934. URL <http://dx.doi.org/10.1103/PhysRevA.104.032405>.
- [179] Qiming Sun and Garnet Kin-Lic Chan. Quantum embedding theories. *Accounts of Chemical Research*, 49(12):2705–2712, nov 2016. URL <https://doi.org/10.1021/acs.accounts.6b00356>.
- [180] A. Altland and B.D. Simons. *Condensed Matter Field Theory*. Cambridge books online. Cambridge University Press, 2010. ISBN 9780521769754. URL <https://www.cambridge.org/core/books/condensed-matter-field-theory/OA8DE6503ED868D96985D9E7847C63FF>.
- [181] Qiming Sun, Timothy C. Berkelbach, Nick S. Blunt, George H. Booth, Sheng Guo, Zhendong Li, Junzi Liu, James D. McClain, Elvira R. Sayfutyarova, Sandeep Sharma, Sebastian Wouters, and Garnet Kin-Lic Chan. PySCF: the python-based simulations of chemistry framework. *WIREs Computational Molecular Science*, 8(1):e1340, 2018. URL <https://onlinelibrary.wiley.com/doi/abs/10.1002/wcms.1340>.
- [182] Qiming Sun, Xing Zhang, Samragni Banerjee, Peng Bao, Marc Barbry, Nick S. Blunt, Nikolay A. Bogdanov, George H. Booth, Jia Chen, Zhi-Hao Cui, Janus J. Eriksen, Yang Gao, Sheng Guo, Jan Hermann, Matthew R. Hermes, Kevin Koh, Peter Koval, Susi Lehtola, Zhendong Li, Junzi Liu, Narbe Mardirossian, James D. McClain, Mario Motta, Bastien Mussard, Hung Q. Pham, Artem Pulkin, Wirawan Purwanto, Paul J. Robinson, Enrico Ronca, Elvira R. Sayfutyarova, Maximilian Scheurer, Henry F. Schurkus, James E. T. Smith, Chong Sun, Shi-Ning Sun, Shiv Upadhyay, Lucas K. Wagner, Xiao Wang, Alec White, James Daniel Whitfield, Mark J. Williamson, Sebastian Wouters, Jun Yang, Jason M. Yu, Tianyu Zhu, Timothy C. Berkelbach, Sandeep Sharma, Alexander Yu. Sokolov, and Garnet Kin-Lic Chan. Recent developments in the PySCF program package. *The Journal of Chemical Physics*, 153(2):024109, 2020. URL <https://doi.org/10.1063/5.0006074>.
- [183] Sam McArdle, Suguru Endo, Alán Aspuru-Guzik, Simon C. Benjamin, and Xiao Yuan. Quantum computational chemistry. *Reviews of*

- Modern Physics*, 92(1), mar 2020. URL <https://doi.org/10.1103/revmodphys.92.015003>.
- [184] Jules Tilly, P. V. Sriluckshmy, Akashkumar Patel, Enrico Fontana, Ivan Rungger, Edward Grant, Robert Anderson, Jonathan Tennyson, and George H. Booth. Reduced density matrix sampling: Self-consistent embedding and multiscale electronic structure on current generation quantum computers. *Physical Review Research*, 3(3), sep 2021. URL <https://doi.org/10.1103/physrevresearch.3.033230>.
- [185] He Ma, Marco Govoni, and Giulia Galli. Quantum simulations of materials on near-term quantum computers. *npj Computational Materials*, 6(1), jul 2020. URL <https://doi.org/10.1038/s41524-020-00353-z>.
- [186] Takeshi Yamazaki, Shunji Matsuura, Ali Narimani, Anushervon Saidmuradov, and Arman Zaribafiyani. Towards the practical application of near-term quantum computers in quantum chemistry simulations: A problem decomposition approach. *arXiv:1806.01305*, 2018.
- [187] Diksha Dhawan, Mekena Metcalf, and Dominika Zgid. Dynamical self-energy mapping (dsem) for quantum computing. *arXiv:2010.05441*, 2020.
- [188] Tyler Takeshita, Nicholas C. Rubin, Zhang Jiang, Eunseok Lee, Ryan Babbush, and Jarrod R. McClean. Increasing the representation accuracy of quantum simulations of chemistry without extra quantum resources. *Phys. Rev. X*, 10:011004, Jan 2020. URL <https://link.aps.org/doi/10.1103/PhysRevX.10.011004>.
- [189] Yongxin Yao, Feng Zhang, Cai-Zhuang Wang, Kai-Ming Ho, and Peter P. Orth. Gutzwiller hybrid quantum-classical computing approach for correlated materials. *Physical Review Research*, 3(1), feb 2021. URL <https://doi.org/10.1103/physrevresearch.3.013184>.
- [190] Max Rossmannek, Panagiotis Kl. Barkoutsos, Pauline J. Ollitrault, and Ivano Tavernelli. Quantum HF/DFT-embedding algorithms for electronic structure calculations: Scaling up to complex molecular systems. *The Journal of Chemical Physics*, 154(11):114105, mar 2021. URL <https://doi.org/10.1063/5.0029536>.

- [191] Saad Yalouz, Bruno Senjean, Jakob Günther, Francesco Buda, Thomas E O'Brien, and Lucas Visscher. A state-averaged orbital-optimized hybrid quantum–classical algorithm for a democratic description of ground and excited states. *Quantum Science and Technology*, 6(2):024004, jan 2021. URL <https://doi.org/10.1088/2058-9565/abd334>.
- [192] Vivek V. Shende and Igor L. Markov. On the cnot-cost of toffoli gates. *Quantum Info. Comput.*, 9(5):461–486, may 2009. ISSN 1533-7146.
- [193] Tudor Giurgica-Tiron, Yousef Hindy, Ryan LaRose, Andrea Mari, and William J. Zeng. Digital zero noise extrapolation for quantum error mitigation. 2020 IEEE International Conference on Quantum Computing and Engineering (QCE), Denver, CO, USA, 2020, 2020, arXiv:2005.10921.
- [194] Charles R. Harris, K. Jarrod Millman, Stéfan J. van der Walt, Ralf Gommers, Pauli Virtanen, David Cournapeau, Eric Wieser, Julian Taylor, Sebastian Berg, Nathaniel J. Smith, Robert Kern, Matti Pícus, Stephan Hoyer, Marten H. van Kerkwijk, Matthew Brett, Allan Haldane, Jaime Fernández del Río, Mark Wiebe, Pearu Peterson, Pierre Gérard-Marchant, Kevin Sheppard, Tyler Reddy, Warren Weckesser, Hameer Abbasi, Christoph Gohlke, and Travis E. Oliphant. Array programming with NumPy. *Nature*, 585(7825):357–362, sep 2020. URL <https://doi.org/10.1038/s41586-020-2649-2>.
- [195] Björn O. Roos, Peter R. Taylor, and Per E.M. Sigbahn. A complete active space scf method (casscf) using a density matrix formulated super-ci approach. *Chemical Physics*, 48(2):157–173, may 1980. ISSN 0301-0104. URL [http://dx.doi.org/10.1016/0301-0104\(80\)80045-0](http://dx.doi.org/10.1016/0301-0104(80)80045-0).
- [196] Jeppe Olsen. The CASSCF method: A perspective and commentary. *International Journal of Quantum Chemistry*, 111(13):3267–3272, may 2011. URL <https://doi.org/10.1002/qua.23107>.
- [197] K. Andersson. Different forms of the zeroth-order hamiltonian in second-order perturbation theory with a complete active space self-consistent field reference function. *Theoretica Chimica Acta*, 91(1–2): 31–46, apr 1995. ISSN 1432-2234. URL <http://dx.doi.org/10.1007/BF01113860>.

- [198] Nick S. Blunt, Ali Alavi, and George H. Booth. Nonlinear biases, stochastically sampled effective hamiltonians, and spectral functions in quantum monte carlo methods. *Physical Review B*, 98(8), aug 2018. URL <https://doi.org/10.1103/physrevb.98.085118>.
- [199] Alexey Uvarov, Jacob D. Biamonte, and Dmitry Yudin. Variational quantum eigensolver for frustrated quantum systems. *Physical Review B*, 102(7), aug 2020. URL <https://doi.org/10.1103/physrevb.102.075104>.
- [200] Samson Wang, Enrico Fontana, M. Cerezo, Kunal Sharma, Akira Sone, Lukasz Cincio, and Patrick J. Coles. Noise-Induced Barren Plateaus in Variational Quantum Algorithms, jul 2020, 2007.14384.
- [201] Eva Pavarini, editor. *The LDA+DMFT approach to strongly correlated materials*. "Forschungszentrum Jülich", sep 2011. ISBN 978-3-89336-734-4.
- [202] Fernando G.S.L. Brandão, Wissam Chemissany, Nicholas Hunter-Jones, Richard Kueng, and John Preskill. Models of quantum complexity growth. *PRX Quantum*, 2(3), jul 2021. ISSN 2691-3399. URL <http://dx.doi.org/10.1103/PRXQuantum.2.030316>.
- [203] Sheng-Hsuan Lin, Rohit Dilip, Andrew G. Green, Adam Smith, and Frank Pollmann. Real- and imaginary-time evolution with compressed quantum circuits. *PRX Quantum*, 2(1), mar 2021. ISSN 2691-3399. URL <http://dx.doi.org/10.1103/PRXQuantum.2.010342>.
- [204] Shi-Ju Ran. Encoding of matrix product states into quantum circuits of one- and two-qubit gates. *Physical Review A*, 101(3), mar 2020. ISSN 2469-9934. URL <http://dx.doi.org/10.1103/PhysRevA.101.032310>.
- [205] James Stokes, Josh Izaac, Nathan Killoran, and Giuseppe Carleo. Quantum natural gradient. *Quantum*, 4:269, may 2020. ISSN 2521-327X. URL <http://dx.doi.org/10.22331/q-2020-05-25-269>.
- [206] Jannis Ehrlich, Daniel Urban, and Christian Elsässer. Perspectives of running self-consistent dmft calculations for strongly correlated electron systems on noisy quantum computing hardware, 2023, arXiv:2311.10402.

- [207] Johannes Selisko, Maximilian Amsler, Christopher Wever, Yukio Kawashima, Georgy Samsonidze, Rukhsan Ul Haq, Francesco Tacchino, Ivano Tavernelli, and Thomas Eckl. Dynamical mean field theory for real materials on a quantum computer, 2024, arXiv:2404.09527.
- [208] Antoine Georges, Gabriel Kotliar, Werner Krauth, and Marcelo J. Rozenberg. Dynamical mean-field theory of strongly correlated fermion systems and the limit of infinite dimensions. *Reviews of Modern Physics*, 68(1):13–125, jan 1996. URL <https://doi.org/10.1103/revmodphys.68.13>.
- [209] M. Ostilli. Cayley trees and bethe lattices: A concise analysis for mathematicians and physicists. *Physica A: Statistical Mechanics and its Applications*, 391(12):3417–3423, jun 2012. URL <https://doi.org/10.1016/j.physa.2012.01.038>.
- [210] Thomas Maier, Mark Jarrell, Thomas Pruschke, and Matthias H. Hettler. Quantum cluster theories. *Reviews of Modern Physics*, 77(3):1027–1080, oct 2005. ISSN 1539-0756. URL <http://dx.doi.org/10.1103/RevModPhys.77.1027>.
- [211] Andreas Gleis. Cluster Dynamical Mean-Field + Numerical Renormalization Group Approach to Strongly Correlated Systems (Master thesis), 2019. URL [https://www.theorie.physik.uni-muenchen.de/lsvondelft/publications/pdf/190602\\_gleis.pdf](https://www.theorie.physik.uni-muenchen.de/lsvondelft/publications/pdf/190602_gleis.pdf).
- [212] Michael Potthoff. Two-site dynamical mean-field theory. *Physical Review B*, 64(16), oct 2001. URL <https://doi.org/10.1103/physrevb.64.165114>.
- [213] Hongxiang Chen. Jordan-wigner transformation for hamiltonians (note), 2023. Available as the file whose filename contains the identifier “23810.JWJW” in Chen [255].
- [214] Pauline J. Ollitrault, Abhinav Kandala, Chun-Fu Chen, Panagiotis Kl. Barkoutsos, Antonio Mezzacapo, Marco Pistoia, Sarah Sheldon, Stefan Woerner, Jay M. Gambetta, and Ivano Tavernelli. Quantum equation of motion for computing molecular excitation energies on a noisy quantum processor. *Physical Review Research*, 2(4), oct 2020. ISSN

- 2643-1564. URL <http://dx.doi.org/10.1103/PhysRevResearch.2.043140>.
- [215] Y. Lu, M. Höppner, O. Gunnarsson, and M. W. Haverkort. Efficient real-frequency solver for dynamical mean-field theory. *Physical Review B*, 90(8), aug 2014. ISSN 1550-235X. URL <http://dx.doi.org/10.1103/PhysRevB.90.085102>.
- [216] Francisco J. R. Ruiz, Tuomas Laakkonen, Johannes Bausch, Matej Balog, Mohammadamin Barekatain, Francisco J. H. Heras, Alexander Novikov, Nathan Fitzpatrick, Bernardino Romera-Paredes, John van de Wetering, Alhussein Fawzi, Konstantinos Meichanetzidis, and Pushmeet Kohli. Quantum circuit optimization with alphasolver, 2024, arXiv:2402.14396.
- [217] Vinay Tripathi, Huo Chen, Mostafa Khezri, Ka-Wa Yip, E.M. Levenson-Falk, and Daniel A. Lidar. Suppression of crosstalk in superconducting qubits using dynamical decoupling. *Phys. Rev. Appl.*, 18:024068, Aug 2022. URL <https://link.aps.org/doi/10.1103/PhysRevApplied.18.024068>.
- [218] Zhenyu Cai. Resource estimation for quantum variational simulations of the hubbard model. *Physical Review Applied*, 14(1), jul 2020. ISSN 2331-7019. URL <http://dx.doi.org/10.1103/PhysRevApplied.14.014059>.
- [219] Abhishek Agarwal, Lachlan P Lindoy, Deep Lall, François Jamet, and Ivan Rungger. Modelling non-markovian noise in driven superconducting qubits. *Quantum Science and Technology*, 9(3):035017, apr 2024. ISSN 2058-9565. URL <http://dx.doi.org/10.1088/2058-9565/ad3d7e>.
- [220] Mario Motta, Chong Sun, Adrian T. K. Tan, Matthew J. O’Rourke, Erika Ye, Austin J. Minnich, Fernando G. S. L. Brandão, and Garnet Kin-Lic Chan. Determining eigenstates and thermal states on a quantum computer using quantum imaginary time evolution. *Nature Physics*, 16(2):205–210, nov 2019. ISSN 1745-2481. URL <http://dx.doi.org/10.1038/s41567-019-0704-4>.

- [221] Robert M. Parrish and Peter L. McMahon. Quantum filter diagonalization: Quantum eigendecomposition without full quantum phase estimation, 2019, arXiv:1909.08925. URL <https://arxiv.org/abs/1909.08925>.
- [222] Nicholas H. Stair, Renke Huang, and Francesco A. Evangelista. A multireference quantum krylov algorithm for strongly correlated electrons. *Journal of Chemical Theory and Computation*, 16(4):2236–2245, feb 2020. ISSN 1549-9626. URL <http://dx.doi.org/10.1021/acs.jctc.9b01125>.
- [223] William J Huggins, Joonho Lee, Unpil Baek, Bryan O’Gorman, and K Birgitta Whaley. A non-orthogonal variational quantum eigensolver. *New Journal of Physics*, 22(7):073009, jul 2020. ISSN 1367-2630. URL <http://dx.doi.org/10.1088/1367-2630/ab867b>.
- [224] Kazuhiro Seki and Seiji Yunoki. Quantum power method by a superposition of time-evolved states. *PRX Quantum*, 2(1), feb 2021. ISSN 2691-3399. URL <http://dx.doi.org/10.1103/PRXQuantum.2.010333>.
- [225] Cristian L. Cortes and Stephen K. Gray. Quantum krylov subspace algorithms for ground- and excited-state energy estimation. *Physical Review A*, 105(2), feb 2022. ISSN 2469-9934. URL <http://dx.doi.org/10.1103/PhysRevA.105.022417>.
- [226] Katherine Klymko, Carlos Mejuto-Zaera, Stephen J. Cotton, Filip Wudarski, Miroslav Urbanek, Diptarka Hait, Martin Head-Gordon, K. Birgitta Whaley, Jonathan Moussa, Nathan Wiebe, Wibe A. de Jong, and Norm M. Tubman. Real-time evolution for ultracompact hamiltonian eigenstates on quantum hardware. *PRX Quantum*, 3(2), may 2022. ISSN 2691-3399. URL <http://dx.doi.org/10.1103/PRXQuantum.3.020323>.
- [227] Tom O’Leary, Lewis W. Anderson, Dieter Jaksch, and Martin Kiffner. Partitioned quantum subspace expansion, 2024, arXiv:2403.08868. URL <https://arxiv.org/abs/2403.08868>.
- [228] Ryuji Takagi, Suguru Endo, Shintaro Minagawa, and Mile Gu. Fundamental limits of quantum error mitigation. *npj Quantum Information*,

- 8(1), sep 2022. ISSN 2056-6387. URL <http://dx.doi.org/10.1038/s41534-022-00618-z>.
- [229] Sam McArdle, Suguru Endo, Alán Aspuru-Guzik, Simon C. Benjamin, and Xiao Yuan. Quantum computational chemistry. *Reviews of Modern Physics*, 92(1), mar 2020. ISSN 1539-0756. URL <http://dx.doi.org/10.1103/RevModPhys.92.015003>.
- [230] J. Eisert, M. Cramer, and M. B. Plenio. Colloquium: Area laws for the entanglement entropy. *Reviews of Modern Physics*, 82(1):277–306, feb 2010. ISSN 1539-0756. URL <http://dx.doi.org/10.1103/RevModPhys.82.277>.
- [231] Eugenio Bianchi, Lucas Hackl, Mario Kieburg, Marcos Rigol, and Lev Vidmar. Volume-law entanglement entropy of typical pure quantum states. *PRX quantum*, 3(3), jul 2022.
- [232] Yuxuan Du, Tao Huang, Shan You, Min-Hsiu Hsieh, and Dacheng Tao. Quantum circuit architecture search: error mitigation and trainability enhancement for variational quantum solvers, 2020, arXiv:2010.10217. URL <https://arxiv.org/abs/2010.10217>.
- [233] Shi-Xin Zhang, Chang-Yu Hsieh, Shengyu Zhang, and Hong Yao. Differentiable quantum architecture search. *Quantum Science and Technology* 7 045023 (2022), 2020, arXiv:2010.08561.
- [234] Hasan Sayginel, Francois Jamet, Abhishek Agarwal, Dan E Browne, and Ivan Rungger. A fault-tolerant variational quantum algorithm with limited t-depth. *Quantum Science and Technology*, 9(1):015015, nov 2023. ISSN 2058-9565. URL <http://dx.doi.org/10.1088/2058-9565/ad0571>.
- [235] Joseph M. Paggi, Ayush Pandit, and Ron O. Dror. The art and science of molecular docking. *Annual Review of Biochemistry*, 93(1):389–410, aug 2024. ISSN 1545-4509. URL <http://dx.doi.org/10.1146/annurev-biochem-030222-120000>.
- [236] Edward Linscott. *Accounting for Strong Electronic Correlation in Metalloproteins*. PhD thesis, University of Cambridge, 2019. URL <https://www.repository.cam.ac.uk/handle/1810/297751>.

- [237] W. K. Wootters and W. H. Zurek. A single quantum cannot be cloned. *Nature*, 299(5886):802–803, oct 1982. ISSN 1476-4687. URL <http://dx.doi.org/10.1038/299802a0>.
- [238] Error correction zoo, 2025. URL [https://errorcorrectionzoo.org/c/stab\\_4\\_2\\_2](https://errorcorrectionzoo.org/c/stab_4_2_2). [Accessed 07 February 2025].
- [239] Simon J. Devitt, Kae Nemoto, and William J. Munro. Quantum error correction for beginners. 2009, arXiv:0905.2794.
- [240] D. W. Leung, M. A. Nielsen, I. L. Chuang, and Y. Yamamoto. Approximate quantum error correction can lead to better codes. 1997, arXiv:quant-ph/9704002.
- [241] M. C. M. Wright. Green function or green's function? *Nature Physics*, 2(10):646–646, oct 2006. URL <https://doi.org/10.1038/nphys411>.
- [242] Gerald D. Mahan. *Many-Particle Physics*. Springer US, 2000. ISBN 9781475757149. URL <http://dx.doi.org/10.1007/978-1-4757-5714-9>.
- [243] Michael G. Cottam and Zahra Haghshenasfard. *Many-Body Theory of Condensed Matter Systems: An Introductory Course*. Cambridge University Press, jul 2020. ISBN 9781108488242. URL <http://dx.doi.org/10.1017/9781108762366>.
- [244] Andrea Damascelli, Zahid Hussain, and Zhi-Xun Shen. Angle-resolved photoemission studies of the cuprate superconductors. *Reviews of Modern Physics*, 75(2):473–541, apr 2003. ISSN 1539-0756. URL <http://dx.doi.org/10.1103/RevModPhys.75.473>.
- [245] Stefan Hüfner. *Photoelectron Spectroscopy*. Springer Berlin Heidelberg, 2003. ISBN 9783662092804. URL <http://dx.doi.org/10.1007/978-3-662-09280-4>.
- [246] Hongxiang Chen. Solving the four-qubit hamiltonian (note), 2023. Available as the file whose filename contains the identifier “23906.HYFA” in Chen [255].
- [247] Hongxiang Chen. Mapping to two qubits (note), 2023. Available as the file whose filename contains the identifier “23906.FJUD” in Chen [255].

- [248] Hongxiang Chen. Simplifications for the overlap calculations (note), 2023. Available as the file whose filename contains the identifier “23506.FUDI” in Chen [255].
- [249] Jay M. Gambetta, Jerry M. Chow, and Matthias Steffen. Building logical qubits in a superconducting quantum computing system. *npj Quantum Information*, 3(1), jan 2017. URL <https://doi.org/10.1038/s41534-016-0004-0>.
- [250] Petar Jurcevic, Ali Javadi-Abhari, Lev S Bishop, Isaac Lauer, Daniela F Bogorin, Markus Brink, Lauren Capelluto, Oktay Günlük, Toshinari Itoko, Naoki Kanazawa, Abhinav Kandala, George A Keefe, Kevin Kruslich, William Landers, Eric P Lewandowski, Douglas T McClure, Giacomo Nannicini, Adinath Narasgond, Hasan M Nayfeh, Emily Pritchett, Mary Beth Rothwell, Srikanth Srinivasan, Neereja Sundaresan, Cindy Wang, Ken X Wei, Christopher J Wood, Jeng-Bang Yau, Eric J Zhang, Oliver E Dial, Jerry M Chow, and Jay M Gambetta. Demonstration of quantum volume 64 on a superconducting quantum computing system. *Quantum Science and Technology*, 6(2):025020, mar 2021. URL <https://doi.org/10.1088/2058-9565/abe519>.
- [251] Sarah Sheldon, Easwar Magesan, Jerry M. Chow, and Jay M. Gambetta. Procedure for systematically tuning up cross-talk in the cross-resonance gate. *Phys. Rev. A*, 93:060302, Jun 2016. URL <https://link.aps.org/doi/10.1103/PhysRevA.93.060302>.
- [252] David C. McKay, Christopher J. Wood, Sarah Sheldon, Jerry M. Chow, and Jay M. Gambetta. Efficient  $z$  gates for quantum computing. *Physical Review A*, 96(2), aug 2017. URL <https://doi.org/10.1103/physreva.96.022330>.
- [253] Erdi ACAR and İhsan YILMAZ. COVID-19 detection on IBM quantum computer with classical-quantum transfer learning. *Turkish Journal of Electrical Engineering and Computer Sciences*, 29(1):46–61, jan 2021. URL <https://doi.org/10.3906/elk-2006-94>.
- [254] Chuanqi Xu and Jakub Szefer. Long-term analysis of the dependability of cloud-based NISQ quantum computers. In *Proceedings of*

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*the 18th International Conference on Availability, Reliability and Security*, pages 1–6. ACM, aug 2023. URL <https://doi.org/10.1145/3600160.3600192>.

- [255] Hongxiang Chen. Companion notebooks, November 2023. URL <https://zenodo.org/records/10181093>. DOI: 10.5281/zenodo.10181093.